

Power Allocations and (De)Activation of Antennas and Symbols for Energy Reduction in Sub-6 GHz Base Stations

Emanuele Peschiera

Dissertation presented in partial fulfillment
of the requirements for the degree of
Doctor of Engineering Technology (PhD)

Supervisors:
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Preface

This thesis contains the results of four years of research. During this time, I have lived abroad as a young adult—an experience that brought many discoveries and challenges. At this point, I can confidently say it has been highly enriching. I have no doubt that both work and life are shaped by collaborations and relationships, and I wish to thank the people who shared this journey with me.

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Barbara, and Andrea. I always feel your affection, proving that it spreads beyond geographical borders. To my girlfriend, Chiara: I feel extremely lucky to share every facet of my life with you. They say that ten decisions shape your life—you'll be aware of five about. You are one of those five.

Emanuele
Brussels, May 2026

Popularized Abstract

Mobile networks are ubiquitous and have had a transformative impact on our society. Each new generation has brought better connectivity in terms of higher speed, coverage, and reduced latency. This has led, however, to an increase in energy consumption. This growth in consumption is unsustainable if the telecommunications sector wants to follow the environmental targets and keep the operational expenses under control. Most of the energy consumption in mobile networks comes from the base stations (BSs) that provide service to the users in a certain area. The BSs of fifth-generation (5G) networks are more complex than those of previous generations because they are equipped with more antennas, use larger bandwidths, and perform more sophisticated signal processing tasks.

This thesis aims at developing algorithms and providing insights on how to significantly reduce the energy usage of the BSs while not sacrificing the delivered service. In general, a BS comprises multiple power amplifiers (PAs), used to amplify the signals such that to reach the users, analog and digital signal processing modules, and supply and cooling units. Each one is associated with a share of the total consumption. Moreover, a BS has multiple resources to allocate to the communication, such as time slots, frequency slots, antennas, and transmit power. The main objective of this thesis is to efficiently allocate one or more of these BS resources such that the energy usage in the BS modules is reduced. The key point is to make this allocation adaptive to the data traffic. Indeed, the BS often activates all its resources to be ready to sustain the peak requirements, but in most cases, this overuse of resources is not needed.

In the first part of the thesis, the models describing the communication and power consumption are defined and serve as foundations for the following developments. The power model incorporates features of state-of-the-art models and is applied to estimate the consumption of next-generation BSs. Next, the focus is on the optimized use of transmit power and number of activated antennas for energy reduction in BS that operate in the frequency band below

6 GHz. The general approach of this thesis is to use classical model-based signal processing tools with the goal of obtaining solutions expressible in simple mathematical form. In a single BS, such as one of a cellular network, we show that activating only one antenna brings reduction in the PAs consumption in some cases, in sharp contrast with conventional solution that activates all the antennas. In terms of BS consumption, the proposed strategy activates a part of the BS antennas, and allocates transmit power, as a function of the traffic. When considering multiple BSs that are cooperating, we find that the optimal strategy to minimize the PA consumption is to activate all the antennas of certain BSs, and completely deactivate the antennas of other BSs. In this way, the BSs with non-active antennas can be turned off, and this brings further reductions in terms of overall network consumption.

Next, the focus is on the joint selection of transmit power, number of activated antennas and activated time slots for the energy reduction of a sub-6 GHz BS. Under the considered assumptions, we show that this joint selection can be efficiently retrieved using low-complexity methods that are run infrequently in time. The quantification of the energy savings and comparison of techniques is based on data from on-site measurements of deployed BSs. We differentiate between BSs equipped with a few or several antennas, operating at different traffic levels and with different hardware functionalities. It is shown that in many cases, activating as few antennas as possible brings the largest energy savings compared to activating fewer time slots or reducing the transmit power.

Gepopulariseerde Samenvatting

Mobiele netwerken zijn alomtegenwoordig en hebben een transformatieve impact gehad op onze samenleving. Elke nieuwe generatie heeft gezorgd voor betere connectiviteit in termen van hogere snelheid, dekking en verminderde vertraging. Dit heeft echter geleid tot een toename van het energieverbruik. Deze groei in energieverbruik is onhoudbaar als de telecommunicatiesector de milieudoelstellingen wil halen en de operationele kosten onder controle wil houden. Het grootste deel van het energieverbruik in mobiele netwerken is afkomstig van de basisstations (BSs) die diensten leveren aan de gebruikers in een bepaald gebied. De BSs van vijfde generatie (5G) netwerken zijn complexer dan die van vorige generaties omdat ze zijn uitgerust met meer antennes, grotere bandbreedtes gebruiken en geavanceerdere signaalverwerkingstaken uitvoeren.

Deze thesis heeft als doel algoritmen en inzichten te ontwikkelen over hoe het energieverbruik van de BSs aanzienlijk kan worden verminderd zonder dat dit ten koste gaat van de geleverde dienst. Over het algemeen bestaat een BS uit meerdere versterkers die worden gebruikt om de signalen te versterken zodat ze de gebruikers bereiken, analoge en digitale signaalverwerkingsmodules, en voedings- en koeleenheden. Elk daarvan is verantwoordelijk voor een deel van het totale verbruik. Bovendien beschikt een BS over meerdere middelen die aan de communicatie kunnen worden toegewezen, zoals tijdslots, frequentieslots, antennes en zendvermogen. Het belangrijkste doel van dit proefschrift is om een of meer van deze BS-middelen efficiënt te alloceren, zodat het energieverbruik in de BS-modules wordt verminderd. Het belangrijkste punt is om deze allocatie aan te passen aan het dataverkeer. De BS activeert namelijk vaak al zijn middelen om klaar te zijn voor piekbehoeften, maar in de meeste gevallen is dit overmatig gebruik van middelen niet nodig.

In het eerste deel van deze thesis worden de modellen gedefinieerd die de

communicatie en het verbruik beschrijven en die als basis dienen voor de volgende ontwikkelingen. Het verbruiksmodel bevat kenmerken van state-of-the-art modellen en wordt toegepast om het verbruik van de volgende generatie BSs te schatten. Vervolgens ligt de focus op het geoptimaliseerde gebruik van zendvermogen en het aantal geactiveerde antennes voor energiebesparing in BSs die werken in de frequentieband onder 6 GHz. De algemene aanpak van deze thesis is het gebruik van klassieke, op modellen gebaseerde signaalverwerkingstools met als doel oplossingen te verkrijgen die in eenvoudige wiskundige vorm kunnen worden uitgedrukt. Voor één BS, zoals een van een mobiel netwerk, laten we zien dat het activeren van slechts één antenne in sommige gevallen leidt tot een vermindering van het versterker-verbruik. Dit is in schril contrast met de conventionele oplossing waarbij alle antennes worden geactiveerd. Wat het verbruik van de BS betreft, activeert de voorgestelde strategie een deel van de BS-antennes en wijst het zendvermogen toe als een functie van het verkeer. Wanneer we kijken naar meerdere BSs die samenwerken, zien we dat de optimale strategie om het versterker-verbruik te minimaliseren, is om alle antennes van bepaalde BSs te activeren en de antennes van andere BSs volledig te deactiveren. Op deze manier kunnen de BSs met niet-actieve antennes worden uitgeschakeld, wat leidt tot een verdere vermindering van het totale netwerkverbruik.

Vervolgens ligt de focus op de gezamenlijke selectie van zendvermogen, aantal geactiveerde antennes en geactiveerde tijdslots voor de energiebesparing van een sub-6 GHz BS. Onder de beschouwde assumpties tonen we aan dat deze gezamenlijke selectie efficiënt kan worden gevonden met behulp van methoden met een lage complexiteit die niet vaak worden uitgevoerd. De kwantificering van de energiebesparingen en de vergelijking van technieken is gebaseerd op gegevens van metingen ter plaatse van geïnstalleerde BSs. We maken onderscheid tussen BSs die zijn uitgerust met enkele of meerdere antennes, die op verschillende dataverkeersniveaus werken en met verschillende hardwarefunctionaliteiten. Er wordt aangetoond dat in veel gevallen het activeren van zo min mogelijk antennes de grootste energiebesparingen oplevert in vergelijking met het activeren van minder tijdslots of het verminderen van het zendvermogen.

Abstract

The increase in the energy consumption of fifth-generation (5G) base stations (BSs) is a growing concern in terms of environmental and economic sustainability. A reason for this increase is that wireless networks are often designed to meet the peak requirements, which have been constantly rising in each new network generation. In this thesis, we adopt an alternative approach and aim at operating the BSs such that their energy use is minimized for a fixed target quality-of-service (QoS). Specifically, we consider multi-antenna BSs using orthogonal frequency-division multiplexing (OFDM) over multiple symbols, and analyze the allocation of transmit power at the BS, number of utilized BS antennas, and number of utilized transmit symbols by the BS. The optimized resource allocation adaptively changes with the target QoS.

We model the energy consumption of the BS as due to the power amplifiers (PAs), analog and digital processing modules, and supply and cooling units. The consumption of each component is averaged over different states in a time frame, which are working mode and data transmission or reference signaling, micro-sleep, idle, and sleep modes. Moreover, we model the consumption of subcomponents (e.g., the digital-to-analog converters (DACs) as part of the analog processing). Consumed power is linearly proportional to consumed energy in our framework. The power and communications models are applied to prospective BSs operating in the upper mid-band (7–24 GHz) with thousands of antenna elements and large bandwidths. The results show that hybrid analog-digital beamforming achieves the best performance in bits per Joule.

The focus is then placed on the allocation of transmit power and (de)activation of antennas in a BS operating in a sub-6 GHz. The problem is first formulated as the design of digital precoders that minimize the PAs consumption under QoS constraints. We derived closed-form expressions for the precoders that depend on fixed-point equations on the transmit powers of the antennas. In single-carrier systems, the proposed solution utilizes only a few antennas, while the baseline approach utilizes all the antennas to maximize the array gain.

When many subcarriers are utilized, no antennas are deactivated from the PA consumption perspective. However, we show that under the considered assumptions on the channel, we can incorporate the consumption of the whole BS and minimize it subject to QoS and transmit power constraints. The proposed solution activates a subset of the antennas (dependently on the number of users) with uniform allocated power. The number of activated antennas is retrieved in closed form and depends on the long-term channel statistics. We then extended the precoding design problem for PA-power minimization to distributed cooperating BSs, in the case where many subcarriers are utilized. We find that the optimized strategy generally activates only a subset of the BSs. In these activated BSs, all the antennas are activated. To reduce the complexity of the initial solution, we propose an allocation based on random matrix theory (RMT), and show that it provides a reliable approximation of the optimal allocation. The RMT-based solution brings benefits in terms of consumption and complexity of the fronthaul (FH) links, used to connect the BSs with the central processing unit.

The last two chapters incorporate the (de)activation of transmit (OFDM) symbols. By considering a system with many subcarriers and a fixed precoder, we considered the problem of BS-consumption minimization subject to QoS and transmit power constraints. The transmit power is expressed in terms of the number of activated antennas and time slots using the QoS constraints and the assumptions on the channel. We then proved the convexity of the resulting two-dimensional objective function (relaxed as continuous) and a sufficient condition for the convexity of the optimization domain. The final proposed solution is retrieved by using a low-complexity algorithm and computing a rounding on the continuous variables. In the evaluations, the power model's parameters are adopted from a model in the literature validated on operator measurements of sub-6 GHz BSs, and the large-scale fading parameters are derived from data reported by the same sub-6 GHz BSs. The energy performance is evaluated for different BS configurations, varying the target QoS, and considering enabled or disabled power-saving modes in the PA and analog processing. We find that minimizing the number of activated antennas ("rush-to-mute") is often quasi-optimal in terms of energy consumption, and serving the maximum number of users is beneficial in terms of energy efficiency.

Beknopte samenvatting

De toename van het energieverbruik van vijfde generatie (5G) basisstations (BSs) is een groeiende zorg in termen van ecologische en economische duurzaamheid. Een reden voor deze toename is dat draadloze netwerken vaak zijn ontworpen om te voldoen aan de piekvereisten, die bij elke nieuwe netwerkgeneratie voortdurend zijn toegenomen. In deze thesis kiezen we voor een alternatieve benadering en streven we ernaar de BSs zo te laten werken dat hun energieverbruik wordt geminimaliseerd voor een vaste beoogde kwaliteit van dienstverlening. Concreet kijken we naar BSs met meerdere antennes die gebruikmaken van orthogonal frequency-division multiplexing (OFDM) over meerdere symbolen, en analyseren we de allocatie van zendvermogen aan de BS, het aantal gebruikte BS-antennes en het aantal gebruikte zendsymbolen door het BS. De geoptimaliseerde toewijzing van middelen verandert adaptief met de beoogde kwaliteit van dienstverlening.

We modelleren het energieverbruik van de BS als het gevolg van de versterkers, analoge en digitale verwerkingsmodules en voedings- en koeleenheden. Het verbruik van elke component wordt gemiddeld over verschillende toestanden in een tijdsbestek, namelijk de werkmodus en gegevensoverdracht of referentiesignalering, microslaap, inactieve modus en slaapmodus. Bovendien modelleren we het verbruik van subcomponenten (bijvoorbeeld de digitaal-naar-analoog converters (DAC's) als onderdeel van de analoge verwerking). Het verbruikte vermogen is in ons raamwerk lineair evenredig met de verbruikte energie. De vermogens- en communicatiemodellen worden toegepast op toekomstige BSs die werken in de bovenste middenband (7-24 GHz) met duizenden antenne-elementen en grote bandbreedtes. De resultaten tonen aan dat hybride analoog-digitale beamforming de beste prestaties levert in bits per joule.

De focus ligt dan op de toewijzing van zendvermogen en (de)activering van antennes in een BS die werkt in een sub-6 GHz. Het probleem wordt eerst geformuleerd als het ontwerp van digitale precoders die het versterkerverbruik minimaliseren voor een vaste kwaliteit van de dienstverlening. We hebbenuitdrukkingen afgeleid voor de precoders die afhankelijk zijn van vastepuntvergelijkingen op het zendvermogen van de antennes. In systemen met

één draaggolf maakt de voorgestelde oplossing slechts gebruik van enkele antennes, terwijl de basisbenadering alle antennes gebruikt om de array-winst te maximaliseren. Wanneer veel subcarriers worden gebruikt, worden er vanuit het oogpunt van versterker-verbruik geen antennes gedeactiveerd. We tonen echter aan dat we onder de beschouwde aannames over het kanaal het verbruik van de hele BS kunnen integreren en minimaliseren, rekening houdend met kwaliteits- en zendvermogen garanties. De voorgestelde oplossing activeert een subset van de antennes (afhankelijk van het aantal gebruikers) met uniform toegewezen vermogen. Het aantal geactiveerde antennes wordt in gesloten vorm bepaald en is afhankelijk van de langetermijnkanaalstatistieken. Vervolgens hebben we het precoding-ontwerpprobleem voor versterker-vermogensminimalisatie uitgebreid naar gedistribueerde samenwerkende BSs, in het geval dat veel subcarriers worden gebruikt. We stellen vast dat de geoptimaliseerde strategie over het algemeen slechts een subset van de BSs activeert. In deze geactiveerde BSs worden alle antennes geactiveerd. Om de complexiteit van de initiële oplossing te verminderen, stellen we een allocatie voor op basis van willekeurige-matrix-theorie en tonen we aan dat deze een betrouwbare benadering van de optimale allocatie biedt. Deze oplossing biedt voordelen op het gebied van verbruik en complexiteit van de fronthaul verbindingen, die worden gebruikt om de BSs met de centrale verwerkingseenheid te verbinden.

De laatste twee hoofdstukken behandelen de (de)activering van transmissiesymbolen (OFDM). Door een systeem met veel subcarriers en een vaste precoder te beschouwen, hebben we gekeken naar het probleem van het minimaliseren van het BS-verbruik, rekening houdend met kwaliteits- en transmissievermogens garanties. Het transmissievermogen wordt uitgedrukt in termen van het aantal geactiveerde antennes en tijdslots, waarbij gebruik wordt gemaakt van de kwaliteitsgarantie en de aannames over het kanaal. Vervolgens hebben we de convexiteit van de resulterende tweedimensionale objective functie en een voldoende voorwaarde voor de convexiteit van het optimalisatie domein aangetoond. De uiteindelijke voorgestelde oplossing wordt verkregen door gebruik te maken van een algoritme met lage complexiteit en door een afronding van de continue variabelen te berekenen. In de evaluaties worden de parameters van het vermogensmodel overgenomen uit een model in de literatuur dat is gevalideerd op basis van metingen van operators van sub-6 GHz BSs, en worden de parameters voor grootschalige fading afgeleid uit gegevens die door dezelfde sub-6 GHz BSs zijn gerapporteerd. De energieprestaties worden geëvalueerd voor verschillende BS-configuraties, waarbij de beoogde kwaliteit wordt gevarieerd en rekening wordt gehouden met ingeschakelde of uitgeschakelde energiebesparingsmodi in de versterker en analoge verwerking. We stellen vast dat het minimaliseren van het aantal geactiveerde antennes (“rush-to-mute”) vaak quasi-optimaal is in termen van energieverbruik, en dat het bedienen van het maximale aantal gebruikers gunstig is in termen van energie-efficiëntie.

Acronyms

μDTX micro-discontinuous transmission.

3GPP 3rd Generation Partnership Project.

4G fourth-generation.

5G fifth-generation.

6G sixth-generation.

AAU active antenna unit.

ADC analog-to-digital converter.

AFE analog front-end.

AI artificial intelligence.

AP access point.

ASIC application-specific integrated circuit.

BB baseband.

BS base station.

CDF cumulative distribution function.

CMOS complementary metal-oxide-semiconductor.

CoMP coordinated multi-point.

CQI channel quality indicator.

CSI channel state information.

CU centralized unit.

DAC digital-to-analog converter.

DBB digital baseband.

DL downlink.

DPD digital pre-distortion.

DSP digital signal processing.

DTX discontinuous transmission.

EB exabyte.

EE energy efficiency.

EU European Union.

FDD frequency-division duplexing.

FFT fast Fourier transform.

FH fronthaul.

FLOPs floating-point operations.

FPGA field-programmable gate array.

FR1 Frequency Range 1 (from 410 MHz to 7.125 GHz).

FR2 Frequency Range 2 (from 24.25 GHz to 71 GHz).

FR3 Frequency Range 3 (from 7.125 GHz to 24.25 GHz).

GaAs gallium arsenide.

GaN gallium nitride.

GB gigabyte.

GHG greenhouse gas.

GOPS giga operations per second.

HW hardware.

- i.i.d.** independently and identically distributed.
- ICT** information and communications technology.
- IFFT** inverse fast Fourier transform.
- IoT** Internet of Things.
- IUI** inter-user interference.
- LDPC** low-density parity-check.
- LNA** low-noise amplifier.
- LO** local oscillator.
- LoS** line-of-sight.
- LTE** Long-Term Evolution.
- MIMO** multiple-input multiple-output.
- MISO** multiple-input single-output.
- mmWave** millimeter wave.
- MNO** mobile network operator.
- MRT** maximum ratio transmission.
- MSE** mean square error.
- MtCO₂e** million tonnes of carbon dioxide equivalent.
- NR** New Radio.
- OFDM** orthogonal frequency-division multiplexing.
- OFDMA** orthogonal frequency-division multiple access.
- PA** power amplifier.
- PAPR** peak-to-average power ratio.
- QoS** quality-of-service.
- RAN** radio access network.

RF radio frequency.

RFSoc RF system-on-chip.

RMT random matrix theory.

RRU remote radio unit.

RU radio unit.

SDMA space-division multiple access.

SiGe silicon-germanium.

SINR signal-to-interference-plus-noise ratio.

SISO single-input single-output.

SNR signal-to-noise ratio.

SoC system-on-chip.

TDD time-division duplexing.

UL uplink.

UMi Urban Micro.

UPA uniform planar array.

ZF zero-forcing.

ZMCSCG zero-mean circularly-symmetric complex-Gaussian.

List of Symbols

Mathematical notation

a or A	Scalar
$\mathbf{a}$	Column vector
$\mathbf{A}$	Matrix
$[\mathbf{A}]_{i,j}$	Element (i, j) of $\mathbf{A}$
$\mathbb{N}, \mathbb{R}, \mathbb{C}$	Set of natural, real, and complex numbers
$\mathbb{R}^{N \times M}$	Set of real-valued matrices of size $N \times M$
$\mathbb{C}^{N \times M}$	Set of complex-valued matrices of size $N \times M$
j	Imaginary unit
$\Re\{a\}, \Im\{a\}$	Real part of a , imaginary part of a
$\mathbf{A}^\top$	Transpose of $\mathbf{A}$
$\mathbf{A}^*$	Complex conjugate of $\mathbf{A}$
$\mathbf{A}^\mathsf{H}$	Conjugate transpose (Hermitian) of $\mathbf{A}$
$\text{tr}(\mathbf{A})$	Trace of $\mathbf{A}$
$\mathbb{E}\{a\}$	Expectation of random variable a
$\text{diag}(\mathbf{a}) = \mathbf{D}_{\mathbf{a}}$	Diagonal matrix with diagonal equal to $\mathbf{a}$
$\text{diag}(\mathbf{A})$	Column vector equal to diagonal elements of $\mathbf{A}$
$\ \mathbf{a}\ $	L_2 -norm of $\mathbf{a}$
$\ \mathbf{A}\ $	Frobenius norm of $\mathbf{A}$

$\mathbf{A}^{-1}$	Matrix inverse of $\mathbf{A}$
$\mathbf{A}^n$	Multiplication of n $\mathbf{A}$ matrices
$\mathbf{A}^{\frac{1}{2}}$	Square root of $\mathbf{A}$, i.e., the only Hermitian positive semidefinite matrix satisfying $\mathbf{A}^{\frac{1}{2}} \mathbf{A}^{\frac{1}{2}} = \mathbf{A}$
$\mathbf{I}_N$	Identity matrix of size $N \times N$
$\mathbf{0}_N$	Column vector of size N with all zeros
$\mathcal{U}(a, b)$	Uniform distribution between a and b
$\mathcal{CN}(\mu, \sigma^2)$	Scalar complex-normal distribution with mean μ and variance σ^2
$\mathcal{CN}(\mathbf{0}_K, \mathbf{\Sigma})$	Vector zero-mean circularly-symmetric complex-normal distribution of size K with covariance matrix $\mathbf{\Sigma} \in \mathbb{C}^{K \times K}$
$\mathcal{CW}(\mathbf{A}, N, M)$	Complex-Wishart distribution with mean $\mathbf{A} \in \mathbb{C}^{N \times N}$ and M degrees of freedom
$\mathcal{CW}^{-1}(\mathbf{A}, N, M)$	Complex inverse-Wishart distribution with scale matrix $\mathbf{A} \in \mathbb{C}^{N \times N}$ and M degrees of freedom
$\mathcal{O}(g(a)) = f(a)$	Big O notation, as $a \rightarrow x$, if there exist positive numbers μ and λ such that $ f(a) \leq \lambda g(a) $ when $0 < a - x < \mu$
$\lceil a \rceil, \lceil a \rceil, \lfloor a \rfloor$	Round, ceil, and floor operator
$\lfloor a \rfloor$	Ceil-floor operator, selecting among the two closest integers to a , the one associated with the minimum value of the cost function under study
$\lfloor a, b \rfloor$	Two-dimensional ceil-floor operator, selecting among $(\lceil a \rceil, \lceil a \rceil)$, $(\lfloor a \rfloor, \lceil a \rceil)$, $(\lceil a \rceil, \lfloor a \rfloor)$, and $(\lfloor a \rfloor, \lfloor a \rfloor)$, the pair associated with the minimum value of the cost function under study
$\{a_n\}_{n=1}^N$ or $\{a_n\}$	Set $\{a_1, \dots, a_n, \dots, a_N\}$

Main system model notation

M, M_a	Total and active number of antennas/power amplifiers in downlink or antennas/low-noise amplifiers in uplink of base station
$M_{\text{RF}}, M_{\text{RF},a}$	Total and active number of radio-frequency chains in downlink and uplink of base station
$K_{\text{max}}, K_{\text{DL}}, K_{\text{UL}}$	Maximum number of users in downlink or uplink, number of active users in downlink and uplink
$Q_{\text{max}}, Q_{\text{DL}}, Q_{\text{UL}}$	Total number of data subcarriers in downlink or uplink, number of active data subcarriers in downlink and uplink
$N_{\text{DL}}, N_{a,\text{DL}}$	Total and active number of time slots in downlink
$N_{\text{UL}}, N_{a,\text{UL}}$	Total and active number of time slots in uplink
P_{cons}	Average consumed power by base station
$P_a, P_{\text{T},m}$	Transmit power of any active antenna/power amplifier of base station, transmit power of antenna/power amplifier m of base station
P_{max}	Maximum transmit power of one active antenna/power amplifier of base station
$P_{\text{T},\text{DL}}$	Total transmit power in downlink of base station
$p_{k,\text{DL}}, p_{k,\text{UL}}$	Power allocation for user k in downlink, transmit power of user k in uplink
T_{frame}	Frame duration
T	Orthogonal frequency-division multiplexing symbol period including cyclic prefix
$\tau_{\text{DL}}, \tau_{\text{DL},\text{sig}}$	Time ratio of downlink duration to frame duration, time ratio of downlink reference signaling duration to downlink duration
$\tau_{\text{UL}}, \tau_{\text{UL},\text{sig}}$	Time ratio of uplink duration to frame duration, time ratio of uplink reference signaling duration to uplink duration
$\overline{\ell}_{\text{DL}}, \overline{\ell}_{\text{UL}}$	Average physical resource load in downlink and uplink

$P_{\text{PA}}, \overline{P_{\text{PA}}}$	Power consumption of power amplifiers during working mode and data transmission, power consumption of power amplifiers averaged over the frame
$P_{\text{ana,DL}}, P_{\text{ana,UL}}$	Power consumption of analog processing in downlink and uplink during working mode
$\overline{P_{\text{ana}}}$	Power consumption of analog processing averaged over the frame
$P_{\text{dig,DL}}, P_{\text{dig,UL}}$	Power consumption of digital processing in downlink or uplink during working mode
$\overline{P_{\text{dig}}}$	Power consumption of digital processing averaged over the frame
$\delta_{\text{PA,micro}}, \delta_{\text{PA,idle}}, \delta_{\text{PA,sleep}}$	Power-reduction factor of power amplifier during micro-sleep (or discontinuous transmission), idle, and sleep mode
$\delta_{\text{ana,micro}}, \delta_{\text{ana,idle}}, \delta_{\text{ana,sleep}}$	Power-reduction factor of power amplifier during micro-sleep, idle, and sleep mode
$\delta_{\text{dig,micro}}, \delta_{\text{dig,idle}}, \delta_{\text{dig,sleep}}$	Power-reduction factor of digital processing during micro-sleep, idle, and sleep mode
$\alpha, \xi_{\text{PA}}, \eta_{\text{PA,max}}$	Power amplifier's consumption exponent, weight of static consumption, and maximal efficiency
$\mathbf{H}_{\text{DL}}, \mathbf{H}_{\text{UL}}$	Channel matrix in downlink and uplink for a single data subcarrier and active time slot
$\mathbf{W}_{\text{dig}}, \mathbf{W}_{\text{ana}}$	Digital and analog precoding matrices for a single data subcarrier and active time slot
$\mathbf{V}_{\text{dig}}, \mathbf{V}_{\text{ana}}$	Digital and analog combining matrices for a single data subcarrier and active time slot
$\sigma_{k,\text{DL}}^2, \sigma_{k,\text{UL}}^2$	Noise power in downlink and uplink of user k
ψ_k	Target signal-to-noise ratio in downlink for user k (Chapter 3 and Chapter 4)
R_k	Target normalized rate in downlink for user k (Chapter 5 and Chapter 6)

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Chapter 1

Introduction

1.1 Background and Motivation

Information and communications technology (ICT) permeates our daily life. The technological innovations that took place in the last 25 years contributed to the development of the current digital society. The COVID-19 pandemic has further accentuated how ICT is of fundamental importance to our professional and private spheres. Even though its benefits are evident, ICT has impacts on social and environmental levels that should not be overlooked. According to the Paris Agreement, global greenhouse gas (GHG) emissions must decline by 43% by 2030 to limit the global warming to 1.5 °C [1], and the ICT sector has set a clear objective of following this goal [2]. The ICT sector comprises data centers, networks (mobile and fixed), user devices, and Internet of Things (IoT). There exist different studies on the emissions and consumption of ICT. According to [3], ICT consumed 916 TWh of electricity and emitted 763 million tonnes of carbon dioxide equivalent (MtCO₂e) in 2020. These corresponded to about 4% of global electricity consumption and 1.4% of global GHG emissions [3]. The review paper [4] reported a higher ICT share of world emissions in 2020, in the range 1.8%–2.8%. The trend is clearly upward for the electricity consumption, which reached 1100 TWh in 2024 according to [5], and does not show the requested decrease in terms of GHG emissions. Fig. 1.1 illustrates this scenario based on the figures reported in [3, 5].

Alongside the steady growth in ICT energy consumption, the global data traffic is also constantly increasing. Fig. 1.1 shows that the total traffic in downlink (DL) and uplink (UL) has surpassed 170 exabyte (EB) per month in 2025, with

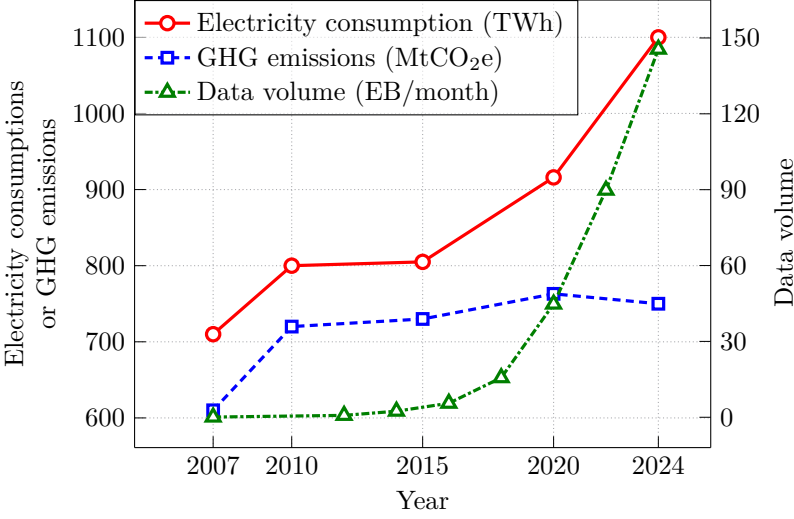


Figure 1.1: From 2007 to 2024, use-stage electricity consumption and GHG emissions of ICT sector (data up to 2020 are from [3], while data of 2024 are from [5]), and global data traffic in mobile networks (data from [5, 6, 7, 8]).

an annual increase of about 18%. With the proliferation of artificial intelligence (AI), the traffic to and from the data centers is expected to further increase [5]. Analysing different estimates for the ICT sector in 2020, the study [4] shows three breakdowns of the total ICT sector carbon footprint. Networks, user devices (including IoT), and data centers account for (i) 35%, 32%, and 33%, (ii) 22%, 37%, and 41%, or (iii) 24%, 57%, and 19% of the total. The last breakdown is the outcome of [3], which further reports that user devices have the largest impact on embodied GHG emissions—including manufacturing, transport, and disposal processes—representing 27% of the total in ICT. The use-stage GHG emissions are instead more spread out between user devices, networks, and data centers, representing the 30%, 20%, and 12% of the total, respectively [3]. In both networks and data centers, the use-stage electricity consumption is associated with roughly 80% of their total GHG emissions [3]. Therefore, to reduce GHG emissions, one can act on three levels: (i) the user devices during both production and operation, (ii) the networks during operation, and (iii) the data centers during operation. We focus here on the network segment, in particular on mobile networks, and we show in the next section when and how energy savings could be achieved.

More than 80% of the energy consumption of mobile networks comes from the radio access network (RAN), which comprises many base stations (BSs) in

different cells [9]. These BSs handle the communication between the users (or mobile terminals) and the core network. For mobile network operators (MNOs), it is essential to reduce the operational emissions to keep their expenses under control and comply with the climate objectives [10]. To achieve this goal, MNOs should have a twofold objective targeting their BSs and decreasing both the carbon intensity of their energy mix and the electricity they consume during operation. This thesis focuses on allocating transmit power and (de)activating antennas and transmit symbols at the BSs to reduce the *use-stage electricity consumption* in the RAN.

1.1.1 Energy Consumption of Radio Access Network

Globally, mobile networks consumed approximately 161 TWh of electricity in 2020, where 146 TWh are due to the RAN [3]. In the European Union (EU)-27, the consumption of mobile networks was estimated to be 29 TWh in 2020 [9]. Studies at the country level highlight an increase in consumption. In Belgium, the consumption by fourth-generation (4G) and fifth-generation (5G) networks was predicted to be around 200 GWh in 2025, compared to 110 GWh by 4G in 2020, corresponding to +81% [12]. In France, by considering all network generations, it was forecasted an increase from 2.37 TWh in 2021 to 3.15 TWh in 2024, corresponding to +33% [13].

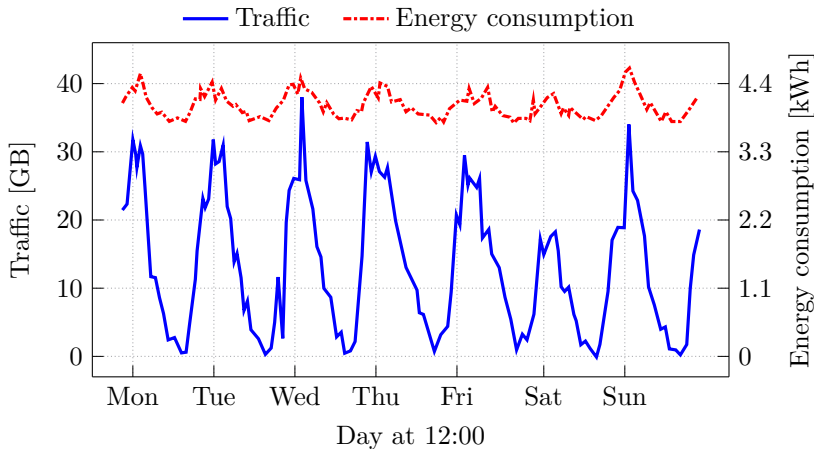


Figure 1.2: Data from [11] showing data traffic and energy consumption over a week of a BS running multiple radio access technologies, deployed in a dense urban area of Helsinki, Finland. Measurements windowed over an hour.

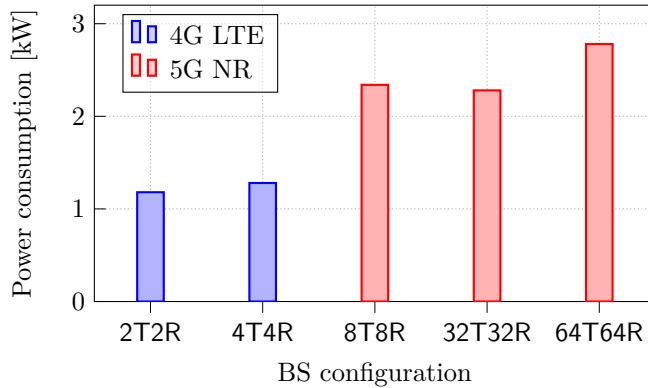


Figure 1.3: Data from [15] showing power consumption of different BS configurations at 30% of (physical resource) load, where an *MTMR* BS has M transmitting RF chains and M receiving RF chains.

A typical data traffic and energy consumption profile of a cellular BS is depicted in Fig. 1.2. The measurements are collected in Helsinki, Finland, from a BS of a Finnish MNO in a dense urban area [11]. The BS is running multiple radio access technologies. While the traffic varies greatly during a day, from about 30 gigabyte (GB) to almost 0 GB, the energy consumption shows little decrease. The energy consumption ranges from 3.76 kWh to 4.65 kWh, indicating that the lowest consumption is around 81% of the maximum one. This indicates that there is a significant static (traffic-independent) consumption. This behavior has been often observed in other studies on the topic, from recent MNO measurements in Belgium to older seminal works on green radio [12, 14]. In general, it is known that the energy consumption of BSs is not highly, or not hardly, proportional to the so-called *load*. The load of a BS is directly related to the ratio of time-frequency resources utilized to transmit or receive data.

An insightful comparison is the one among BS configurations running different technologies. Fig. 1.3 depicts the average power consumption of 4G-Long-Term Evolution (LTE) and 5G-New Radio (NR) BSs at a load of 30%, which is a common high load observed in deployed networks [12]. The power model is derived from measurements of an MNO in Belgium and documentation of manufacturers [15]. An *MTMR* configuration has M transmitting and M receiving RF chains, as well as M power amplifiers (PAs) and low-noise amplifiers (LNAs). The 4G BSs employ small-dimension multiple-input multiple-output (MIMO) and consume slightly more than 1 kW. The 5G BSs, instead, employ higher MIMO dimensions and consume more than 2 kW. The static (load-independent) consumption is larger than the load-dependent consumption in

all BS configurations, which is even more problematic in low-load situations (e.g., at 5% of load). These power numbers are in line with other independent evaluations, e.g., the study in [13] that reports in France a consumption of 1.96 kW for 4G BSs and 3.91 kW to 6.47 kW for 5G BSs, and the work in [16] that indicates for the Chinese market a consumption of 1.1 kW in 4G BSs and 4.3 kW in 5G BSs. These evaluations raise questions on the sustainability of new-generation BSs: although they are not directly comparable to previous-generation ones as their provided quality-of-service (QoS) or coverage is higher, evidence suggests that their absolute consumption is larger.

1.1.2 From Best Effort to Good-Enough Effort

Wireless networks have traditionally been designed and deployed to meet some peak performance (e.g., a peak data rate, a minimal latency, a maximal coverage) [17]. This is evident in plots such as Fig. 1.2: the BS is either in always-on mode or implements only a few energy-saving modes to be ready to provide the largest data rates, coverage, or other indicators. The reality is that most of the time, the network does not require that peak performance. Measurements from an MNO in Belgium show that even during peak traffic hours, the load is 8%–30% of the maximum allowed [12]. Many works in academia and industry have optimized the energy performance of BSs based on their energy efficiency (EE), measured in bit per Joule and defined as the throughput delivered by the BS divided by its power consumption [18]. However, maximizing the EE as the only goal does not enforce a target QoS and may yield undesirable operating points, e.g., low power consumption but low rates, or high rates but high power consumption. A different strategy is to fix the QoS requirement and minimize the absolute power consumption. This allows the transmission resources to be adapted to a given traffic level and prevents allocating excess resources when they are not needed. In turn, the trade-off between energy use and performance requirements becomes explicit.

A typical power consumption against load profile in a BS is illustrated in Fig. 1.4. A linear model of the consumption as a function of the load is used in several works [14, 19], although energy-saving modes at low loads imply a certain nonlinearity [15]. In a BS designed for best effort, the consumption does not decrease significantly at low loads but remains, e.g., 70% to 80% of the one at maximal load. An example of best-effort design is the initial deployment of massive MIMO, a cornerstone technology of 5G [20]. Using many antennas and RF chains (e.g., 32 or 64), massive MIMO focuses the energy towards the users via beamforming and enables spatial multiplexing. Despite lowering the required transmission power for a target QoS, the presence of many power-hungry PAs and RF chains makes massive MIMO most energy efficient in the saturation

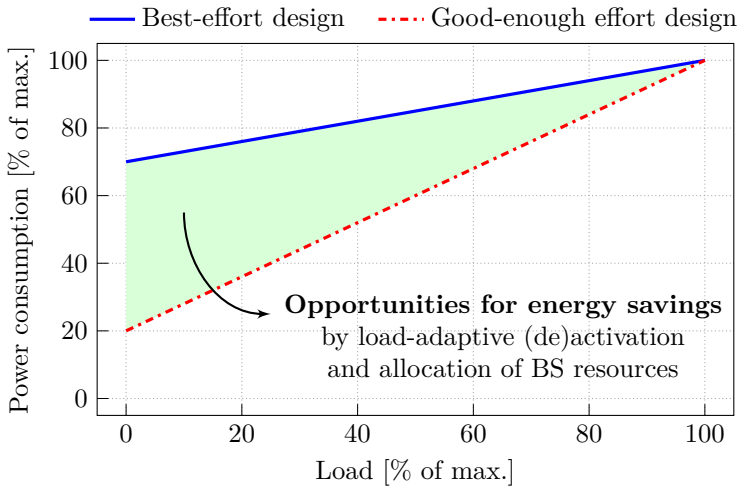


Figure 1.4: Power consumption versus load in a BS with widespread best-effort design and less widespread good-enough effort design.

regime [21]. In this regime, there are many users requesting large data rates. A good-enough effort BS, on the other hand, adapts its utilized resources to the load and achieves a larger proportionality of the consumption with respect to the load. Given the sector’s energy reduction targets, energy-saving techniques in BSs are becoming essential rather than a nice-to-have.

1.1.3 Multi-Domain Network Energy-Saving Techniques

One way to decrease the BS power consumption at medium to low loads is to deactivate hardware (HW) components (e.g., PAs or RF chains) or decrease their allocated power (e.g., lower transmit power per PA). These so-called *network energy-saving techniques* are receiving growing attention by academia [22, 23, 24, 25], BS vendors [26, 27, 28, 29], and standardization bodies [30, 31]. The 3rd Generation Partnership Project (3GPP) classifies network energy-saving techniques in four domains, i.e., time, frequency, space, and power. An illustration of these four domains is provided in Fig. 1.5. Initial evaluations by 3GPP have shown, to give an example, that energy-saving gains up to 48.2% can be achieved by adapting only the number of active BS antennas and PAs to the traffic [30]. Combining multiple domains is an active area of research and network trials, and is the focus of this thesis.

Let us briefly introduce the four energy-saving domains. In the time domain,

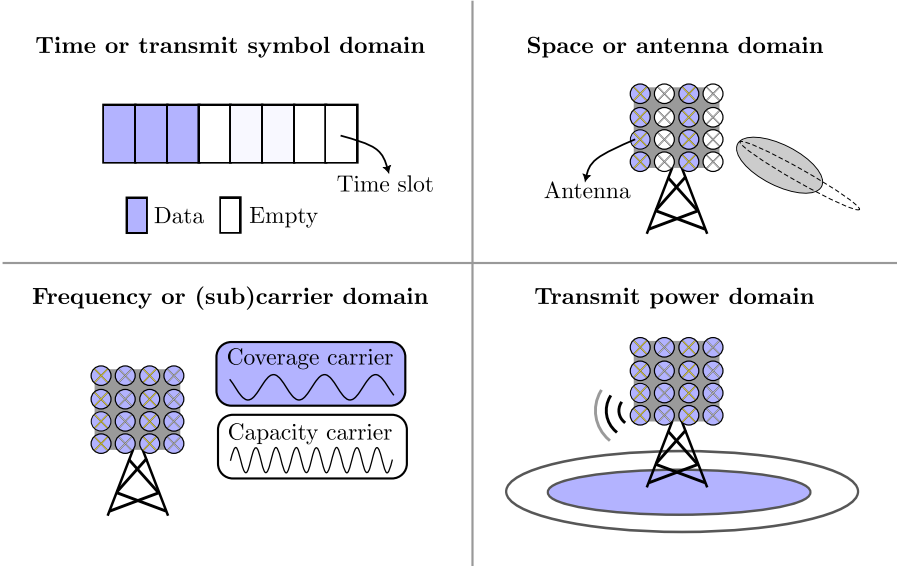


Figure 1.5: Illustration of network energy-saving techniques in the domains of time (symbol), space (antenna), frequency ((sub)carrier), and transmit power. Purple color means active resource, while white indicates a non-active resource.

a first approach to achieve energy savings is to design leaner protocols that require less frequent reference signaling. This is the case in 5G, which increased the period between reference signaling bursts compared to 4G [32]. From the data transmission perspective, information can be condensed in a few transmit symbols (that will have larger allocated power), leaving the other symbols empty. This allows the PAs to be switched off when the BS is not transmitting, a technique known as discontinuous transmission (DTX) since 4G and operating at the symbol-duration scale (i.e., milliseconds) [33]. Longer time intervals of inactivity allow the BS to enter deeper sleep modes, up to a duration of seconds, where more HW components are progressively deactivated [34]. The HW reactivation times should be considered when deeper sleep modes are implemented to prevent loss in coverage or provided QoS.

The space domain refers to the transceiving elements of a BS (RF chains, PAs, and antennas) as well as to entire BSs if multi-cell coordination is enabled. In the literature, works have focused on the load-aware (de)activation of antennas in massive MIMO and even of full BSs by handling users' handover [35, 36]. An example of space-domain energy-saving technique is antenna selection, where a subset of antennas is used for transmission or reception and the selection criterion can favor the reduction of consumed energy [37]. In a BS with fully-digital

beamforming, equipped with one RF chain per antenna and per PA, deactivating an antenna (also known as antenna muting [22]) enables the deactivation of its associated PA and RF chain with consequent reduction in energy consumption.

Power domain refers to energy-saving techniques that vary, instead, the total transmit power at the BS or the individual transmit powers at the BS antennas. These individual transmit powers can be optimized, e.g., to improve the PA efficiency—which is inversely proportional to its consumed power [38]. Indeed, it is known that the PA is most efficient when operating close to saturation (i.e., at large output powers) [39, 40]. Reducing the total transmit power at the BS is another option, which also reduces the overall consumption by the PAs when the output power is equal among the PAs. Certain designs of MIMO precoders, which are used to multiplex the users' symbols in DL and determine the transmitted signal by the BS, act as joint space-power energy-saving techniques. The precoding coefficients of each antenna can be selected such that to calibrate the transmit power or to deactivate an antenna (allocating zero transmit power).

In the frequency domain, entire carriers can be shut down when not needed [17]. For instance, a carrier at high frequency used to provide more capacity, may be activated only when large data rates are requested, while a carrier at low frequency used to guarantee the cell coverage, is kept active. Carrier shutdown algorithms operate at long time scales, i.e., seconds, and have been tested in deployed networks [26]. In orthogonal frequency-division multiplexing (OFDM), the broadband modulation format used since 4G networks, energy savings can also be targeted via subcarrier selection or allocation mechanisms [41].

1.2 Research Questions

In this thesis, we consider network energy-saving techniques in the domains of time, space, and power. Therefore, we consider the joint (de)activation of transmit symbols, antennas (and PAs/RF chains), and allocation of transmit power to minimize the energy use at the BS while satisfying the target QoS. We further address research gaps in the BS power modeling, in exploiting data or analysis from on-site measurements of deployed BSs, and in the application of energy-saving techniques to distributed BSs that are cooperating. Throughout the thesis, we address the following five research questions.

RQ1: How can existing base station power consumption models be extended and refined to include next-generation base stations?

RQ2: How can transmit power allocations and (de)activation of antennas and transmit symbols be optimized to cut base station energy use while still meeting quality-of-service targets?

RQ3: How can the RQ2 optimization framework be generalized to distributed and cell-free base stations?

RQ4: What practical gains and insights can be achieved when real-world measurements are used to design and evaluate transmit power allocations and (de)activation of antennas and transmit symbols?

RQ5: Which concrete advice can be given to mobile network operators to systematically reduce the energy consumption of their networks?

1.3 Thesis Outline and Contributions

The outline of this thesis is provided in Fig. 1.6. The thesis focuses on algorithms for the allocation of transmit powers and (de)activation BS antennas and transmit symbols to achieve energy reduction while satisfying the target QoS. The energy-saving schemes are designed for BSs operating in the sub-6 GHz band. We analyze one or multiple BSs with multiple subcarriers, users, antennas, and time slots, and particularize the scenario in certain chapters or sections. The objective of the thesis is to provide insights to MNOs that want to reduce the energy consumption of their deployed BSs.

We have introduced the context, the motivation and the research questions in the previous part of Chapter 1. In Chapter 2, we provide the communication and power consumption models. The communication is described in DL and UL and, despite making some simplifying assumptions, is kept general and allows the analysis, e.g., of hybrid analog-digital beamforming. The proposed power model generalizes the models in the literature proposed for sub-6 GHz and millimeter wave (mmWave) systems by combining subcomponent-level (e.g., a single digital-to-analog converter) power modeling in working mode with component-level (e.g., a full RF chain) power modeling averaged over the different operating modes in a frame. We discuss its application to upper mid-band (7–24 GHz) BSs with a very large number of antenna elements and large bandwidths. The use of this advanced power model in the optimizations and evaluations presented in the next chapters constitutes the main difference with respect to the state of the art and hence the main novelty of this thesis.

Chapter 3 focuses on a sub-6 GHz BS in a single time slot and optimizes its transmit power allocation and activation of antennas as the number of users changes. The problem is initially formulated as a precoding design one, and

differs from the literature in the use of a PA power model considering a non-constant PA efficiency. It is shown that non-uniform power allocation among antennas is beneficial when the number of users or subcarriers is low. The analysis is then extended to the full BS consumption and we show that, in a BS with many subcarriers, one can resort to the conventional precoding scheme (allocating power uniformly) among a number of active antennas, whose value is given in closed form.

Chapter 4 generalizes the power allocation and deactivation of antennas to multiple cooperating distributed BSs, each one equipped with multiple antennas. Retrieving the transmit power allocation is more challenging than in centralized BSs, and we leverage asymptotic equivalents to derive low-complexity solutions. In the studied scenario, the best strategy is to activate all the antennas of the active BSs, also called access points (APs) in the context of cell-free MIMO, and fully deactivate the remaining APs.

We include the multi-time slot scenario in Chapter 5, where the joint allocation of active antennas, active time slots, and transmit power is discussed. The initially complex problem is efficiently solved by exploiting convexity. This allows us to define a low-complexity optimization algorithm and showcase three simpler allocations, i.e., “rush-to-sleep” that minimizes the number of active symbols, “rush-to-mute” with minimal number of active antennas, and “awake-but-whisper” that minimizes the transmit power. Differently from the literature, the joint solution (leveraging the three domains) can be seamlessly and fairly compared to the three simpler allocations.

In Chapter 6, we utilize data from on-site measurements of an MNO in Belgium to evaluate the algorithms proposed in Chapter 5. These data are used indirectly to initialize the power model parameters, and directly to determine the large-scale fading channel parameters. We analyze different sub-6 GHz BSs configurations, using 4G or 5G technologies, and propose a variation of parameters to include, e.g., the micro-discontinuous transmission (μ DTX) feature. We show that rush-to-mute is often the best strategy to be used in BS with many antennas and RF chains (e.g., a 64T64R massive MIMO), while a joint optimization of active time slots, antennas, and transmit power provides the largest energy savings in BS with fewer antennas implementing μ DTX.

Chapter 7 concerns the valorization plan of the thesis, and discusses the market needs and implementations toward network energy savings, as well as a list of actionable recommendations for MNOs in order to reduce the energy consumption of their deployed BSs. In Chapter 8, we draw the conclusions and foresee future work.

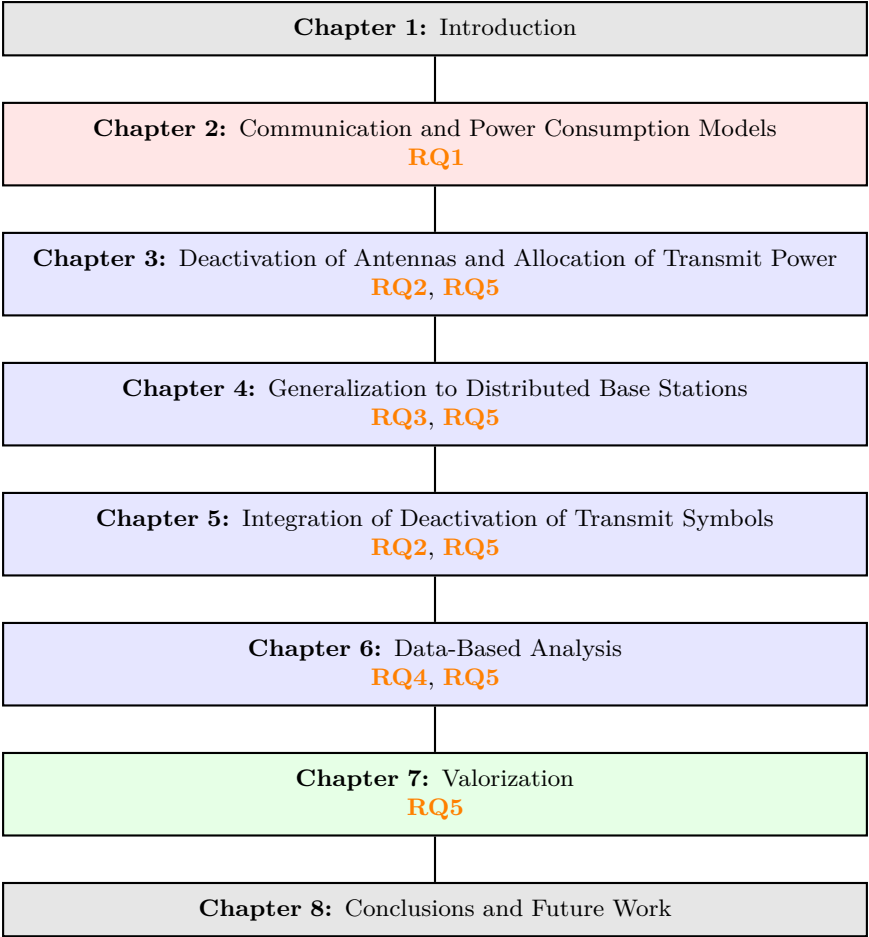


Figure 1.6: Thesis structure with chapters and addressed research questions.

Chapter 2

Communication and Power Consumption Models

Bibliographic Reference

This chapter is based in part on: E. Peschiera, S. Yun, Y. Lee, L. Van der Perre, and F. Rottenberg, “A parametric power model of upper mid-band (FR3)* base stations for 6G,” in *Proceedings of 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2026)*, Barcelona, Spain, 2026, pp. 21476-21480 [42].¹

*The upper mid-band, also known as Frequency Range 3 (FR3) within the 3rd Generation Partnership Project (3GPP), comprises the carrier frequencies from 7.125 GHz to 24.25 GHz. It is one of the main candidate frequency bands for sixth-generation (6G) networks, mainly to provide a target ten-fold capacity increase in urban areas [43, 44, 45, 46].

In this chapter, we describe the system model used in the thesis, covering the hardware (HW) devices included in the base station (BS) and their power consumption models, and the communication process between the BS and the users. We describe downlink (DL) and uplink (UL) scenarios and adopt a general approach to both communication and power models. In the next

¹The Python simulation code is available in open access at the following link: github.com/dramco-research/fr3-power-model-epeschiera-icassp26.

chapters, we particularize the models presented here to the scenario under study. The proposed power model is a generalization of existing ones in the literature, proposed for BSs working in the sub-6 GHz band, also known as Frequency Range 1 (FR1), or in the millimeter wave (mmWave) band, also known as Frequency Range 2 (FR2).² More details on the differences with respect to the state of the art are given at the beginning of Section 2.2. We discuss its application to prospective upper mid-band (or FR3) BSs in Section 2.3.

Throughout this thesis, we consider a BS equipped with M antennas and power amplifiers (PAs) in DL, and M antennas and low-noise amplifiers (LNAs) in UL. The BS has M_{RF} radio frequency (RF) chains in DL and M_{RF} RF chains in UL.³ The BS serves single-antenna users using space-division multiple access (SDMA), meaning that all users are served on the same time-frequency resources. The transmission and reception adopt orthogonal frequency-division multiplexing (OFDM). The maximum number of supported users is K_{max} , which we consider to be the same in DL and UL, while the number of active users is K_{DL} in DL and K_{UL} in UL. The DL takes place on a carrier frequency $f_{\text{c,DL}}$ and the UL takes place on a carrier frequency $f_{\text{c,UL}}$. Both DL and the UL utilize a bandwidth of B Hz and a subcarrier spacing of Δ_f Hz. The bandwidth is divided into Q_{IFFT} subcarriers, of which Q_{max} are available to transmit/receive data symbols. The OFDM symbol period, including cyclic prefix, is denoted by T . An illustration of the frequency-domain structure considered in this thesis is given in Fig. 2.1. A subcarrier is denoted as ‘active’ when it is allocated non-zero power. Within the bandwidth, there is a number of null subcarriers, and those can be arbitrarily placed in the bandwidth. There are no pilot subcarriers in an active OFDM symbol because we consider, as a simplifying assumption, that the reference signalling is separated in time from the data transmission/reception (cf. Fig. 2.2).

We consider a frame of T_{frame} seconds divided into DL and UL phases. The DL phase has a duration of $\tau_{\text{DL}}T_{\text{frame}}$ seconds, where $\tau_{\text{DL}} \in [0, 1]$, while the UL phase has a duration of $\tau_{\text{UL}}T_{\text{frame}}$ seconds, where $\tau_{\text{UL}} \in [0, 1]$. In DL (UL), there are N_{DL} (N_{UL}) available OFDM symbols,⁴ of which $N_{\text{a,DL}}$ ($N_{\text{a,UL}}$) are activated, i.e., carrying users’ data. By considering $i \in \{\text{DL}, \text{UL}\}$ as the variable representing either the DL or the UL, each phase is further subdivided in three phases:

²FR1 spans the carrier frequencies from 410 MHz to 7.125 GHz, while FR2 spans the carrier frequencies from 24.25 GHz to 71 GHz [47].

³In general, we refer to an “RF chain” as a signal chain from the inverse fast Fourier transform input to the phase shifter input in DL, and viceversa in UL (cf. Fig. 2.4). An RF chain includes, therefore, digital and analog modules. When discussing the digital processing modules specifically, we refer to a digital signal processing (DSP) chain as the part of the RF chain that goes from the inverse fast Fourier transform to the digital-to-analog converter in DL, and from the analog-to-digital converter to the fast Fourier transform in UL.

⁴In the thesis, we interchangeably use the terms OFDM symbol and time slot.

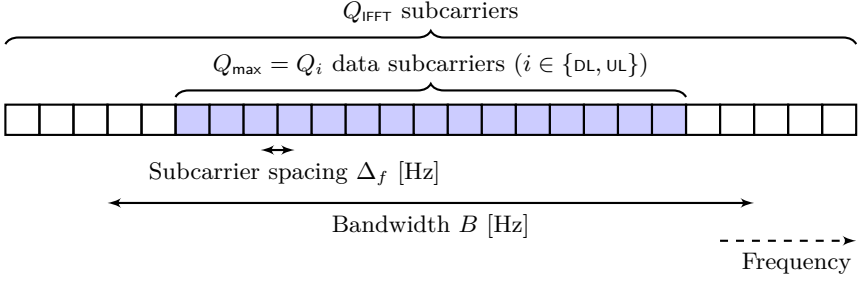


Figure 2.1: Frequency-domain structure considered in this thesis, applicable to both DL and UL. We consider that for an active user and active OFDM symbol, all data subcarriers are utilized, giving $Q_i[k, n] = Q_{\max}$.

- Reference signaling, lasting $\tau_i \tau_{\text{sig}, i}$ seconds, where $\tau_{\text{sig}, i} \in [0, 1]$ is the ratio of DL or UL signaling duration to DL or UL duration. In this phase, the BS and the users exchange synchronization, control, broadcast, and pilot signals used, e.g., to estimate channel state information (CSI) [32].
- Data transmission or reception, lasting $\tau_i(1 - \tau_{\text{sig}, i}) \frac{N_{a,i}}{N_i}$ seconds.
- No reference signaling and no data transmission or reception, lasting $\tau_i(1 - \tau_{\text{sig}, i})(1 - \frac{N_{a,i}}{N_i})$ seconds. This corresponds to micro-sleep mode, where the BS is active but is not transmitting or receiving any signal. In DL, this mode is also known as discontinuous transmission (DTX) [30].

A visual illustration of the time domain subdivision is given in Fig. 2.2.

We define the instantaneous physical resource load in DL or UL of user k and time slot n as the number of utilized data subcarriers over the number of available data subcarriers

$$\ell_i[k, n] = \frac{Q_i[k, n]}{Q_{\max}}, \quad i \in \{\text{DL}, \text{UL}\}. \quad (2.1)$$

for $k = 1, \dots, K_{\max}$ and $n = 1, \dots, N_i$. Averaging over all the users and the time slots, we compute the average physical resource load

$$\bar{\ell}_i = \frac{1}{N_i} \sum_{n=1}^{N_i} \left(\frac{1}{K_{\max}} \sum_{k=1}^{K_{\max}} \frac{Q_i[k, n]}{Q_{\max}} \right), \quad i \in \{\text{DL}, \text{UL}\}. \quad (2.2)$$

To ease the presentation, the set of Active and Non-active users is represented by ordered indexes $\{1, \dots, K_i\}$ and $\{K_i + 1, \dots, K_{\max}\}$, respectively, while the set of active and non-active time slots is represented by ordered indexes $\{1, \dots, N_{a,i}\}$

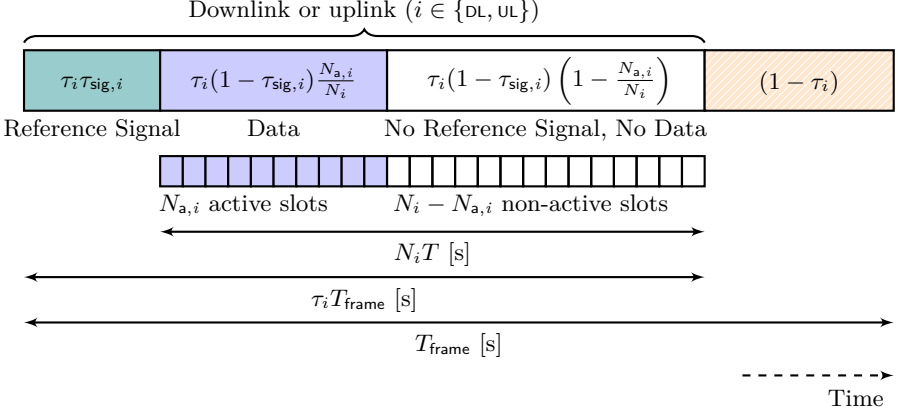


Figure 2.2: Time-domain (frame) structure considered in this thesis, where the quantities inside each rectangle are time ratios relative to the frame duration T_{frame} . In FDD, $\tau_{\text{DL}} = \tau_{\text{UL}} = 1$ and the rightmost rectangle has zero duration.

and $\{N_{a,i} + 1, \dots, N_i\}$, respectively. In this thesis, we assume that when a user is active in an active time slot, i.e., $k \in \{1, \dots, K_i\}$ and $n \in \{1, \dots, N_{a,i}\}$, it is allocated all the available subcarriers giving $Q_i[k, n] = Q_{\max}$. In other words, we do not consider orthogonal frequency-division multiple access (OFDMA). Given this assumption, the average load in equation (2.2) simplifies to

$$\bar{\ell}_i = \frac{N_{a,i}}{N_i} \frac{K_i}{K_{\max}}, \quad i \in \{\text{DL}, \text{UL}\}. \quad (2.3)$$

When the maximum number of users is active, the average load is simply given by the ratio of active to total number of time slots.

2.1 Communication Model

We now describe the communication at one data subcarrier ($q \in \{1, \dots, Q_{\text{DL}}\}$ in DL or $q \in \{1, \dots, Q_{\text{UL}}\}$ in UL) and one active OFDM symbol ($n \in \{1, \dots, N_{a,\text{DL}}\}$ in DL or $n \in \{1, \dots, N_{a,\text{UL}}\}$ in UL) to ease the notation. In the following, we specify when needed the dependency on q and n . All the quantities are expressed in the frequency domain as we consider them before the inverse fast Fourier transform (IFFT) at the transmitter side and after the fast Fourier transform (FFT) at the receiver side.

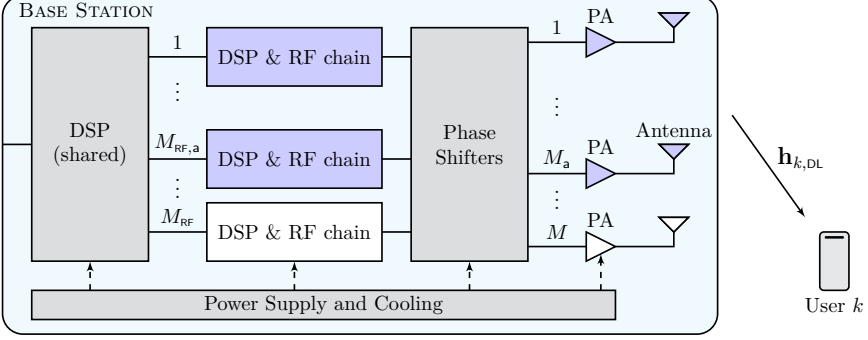


Figure 2.3: DL system model of this thesis, where the UL is symmetric except for the PAs being replaced by the LNAs and the channel being $\mathbf{h}_{k,\text{UL}}$. Purple color indicates an activated resource, while white indicates a deactivated resource.

In the thesis, we consider that M_a over M PAs and antennas are active in DL, and M_a over M LNAs and antennas are active in UL.⁵ Similarly, $M_{\text{RF},a}$ over M_{RF} are active in both DL and UL. Fig. 2.3 illustrates the DL system model. As assumed for the users and the time slots, the set of active antennas (RF chains) is represented by the first M_a ($M_{\text{RF},a}$) indexes over M (M_{RF}).

2.1.1 Downlink

The DL channel between the active BS antennas and the user k is denoted by $\mathbf{h}_{k,\text{DL}} \in \mathbb{C}^{M_a \times 1}$, hence the channel matrix is given by

$$\mathbf{H}_{\text{DL}} = [\mathbf{h}_{1,\text{DL}}, \dots, \mathbf{h}_{K_{\text{DL}},\text{DL}}]^T \in \mathbb{C}^{K_{\text{DL}} \times M_a}. \quad (2.4)$$

We assume a block-fading channel where the channel is constant over a coherence block that spans the coherence bandwidth and the coherence time [48]. The symbols of the users in DL are represented by the vector $\mathbf{s}_{\text{DL}} = [s_{1,\text{DL}}, \dots, s_{K_{\text{DL}},\text{DL}}]^T \in \mathbb{C}^{K_{\text{DL}} \times 1}$. The symbols are zero mean, uncorrelated among users, and with covariance equal to $\mathbf{D}_{\text{pDL}} = \text{diag}(p_{1,\text{DL}}, \dots, p_{K_{\text{DL}},\text{DL}})$, where $p_{k,\text{DL}}$ is the DL power allocation of user k . This can be compactly expressed as

$$\mathbb{E}\{\mathbf{s}_{\text{DL}}\mathbf{s}_{\text{DL}}^H\} = \mathbf{D}_{\text{pDL}}. \quad (2.5)$$

⁵We consider the antennas to be passive elements, hence we do not model their power consumption. With the term ‘active’ or ‘activated’ antenna, we indicate an antenna that is utilized for transmission or reception during the frame, in contrast to a ‘non-active’ or ‘deactivated’ antenna.

The BS performs hybrid analog-digital precoding and transmits the signal

$$\mathbf{x} = \mathbf{W}_{\text{ana}} \mathbf{W}_{\text{dig}} \mathbf{s}_{\text{DL}} \in \mathbb{C}^{M_a \times 1} \quad (2.6)$$

where $\mathbf{W}_{\text{ana}} \in \mathbb{C}^{M_a \times M_{\text{RF},a}}$ and $\mathbf{W}_{\text{dig}} = [\mathbf{w}_{\text{dig},1}, \dots, \mathbf{w}_{\text{dig},K_{\text{DL}}}] \in \mathbb{C}^{M_{\text{RF},a} \times K_{\text{DL}}}$ are the analog beamforming and digital precoding matrices, respectively. The vector $\mathbf{w}_{\text{dig},k} \in \mathbb{C}^{M_{\text{RF},a} \times 1}$ is the digital precoding vector of user k and has both a magnitude and a phase component. The analog beamforming matrix $\mathbf{W}_{\text{ana}}$, instead, contains phase-only elements and is selected from a codebook $\mathcal{W}_{\text{ana}}$.⁶ Hybrid analog-digital beamforming is a technique employed to reduce the number of RF chains compared to fully-digital beamforming [49]. More antennas share the same RF signal in hybrid beamforming, with reduction of BS power consumption. When there is no analog beamforming, the precoding is fully digital with $M_{\text{RF}} = M$, $M_{\text{RF},a} = M_a$ and $\mathbf{W}_{\text{ana}} = \mathbf{I}_{M_a}$, and the transmitted signal becomes

$$\mathbf{x} = \mathbf{W}_{\text{dig}} \mathbf{s}_{\text{DL}} \in \mathbb{C}^{M_a \times 1}. \quad (2.7)$$

We indicate the total transmit power at the BS, across all subcarriers, as $P_{\text{T,DL}}$. Considering that the cyclic prefix is larger than the maximal channel delay spread, the signal received by user k is expressed as

$$\begin{aligned} r_{k,\text{DL}} &= \mathbf{h}_{k,\text{DL}}^\top \mathbf{x} + \nu_{k,\text{DL}} = \mathbf{h}_{k,\text{DL}}^\top \mathbf{W}_{\text{ana}} \mathbf{W}_{\text{dig}} \mathbf{s}_{\text{DL}} + \nu_{k,\text{DL}} \\ &= \underbrace{\mathbf{h}_{k,\text{DL}}^\top \mathbf{W}_{\text{ana}} \mathbf{w}_{\text{dig},k} s_{k,\text{DL}}}_{\text{desired signal}} + \underbrace{\sum_{k'=1, k' \neq k}^{K_{\text{DL}}} \mathbf{h}_{k,\text{DL}}^\top \mathbf{W}_{\text{ana}} \mathbf{w}_{\text{dig},k'} s_{k',\text{DL}}}_{\text{inter-user interference}} + \underbrace{\nu_{k,\text{DL}}}_{\text{noise}} \end{aligned} \quad (2.8)$$

where $\nu_{k,\text{DL}}$ is zero-mean circularly-symmetric complex-Gaussian (ZMCSCG) noise with variance $\sigma_{k,\text{DL}}^2$. The noise is considered uncorrelated at the different users, and the received signal vector collecting the individual received signals is

$$\mathbf{r}_{\text{DL}} = \mathbf{H}_{\text{DL}} \mathbf{x} + \boldsymbol{\nu}_{\text{DL}} = \mathbf{H}_{\text{DL}} \mathbf{W}_{\text{ana}} \mathbf{W}_{\text{dig}} \mathbf{s}_{\text{DL}} + \boldsymbol{\nu}_{\text{DL}} \in \mathbb{C}^{K_{\text{DL}} \times 1} \quad (2.9)$$

where $\boldsymbol{\nu}_{\text{DL}} \sim \mathcal{CN}(\mathbf{0}_{K_{\text{DL}}}, \mathbf{D}_{\boldsymbol{\sigma}_{\text{DL}}}^2)$ and $\mathbf{D}_{\boldsymbol{\sigma}_{\text{DL}}}^2 = \text{diag}(\sigma_{1,\text{DL}}^2, \dots, \sigma_{K_{\text{DL}},\text{DL}}^2)$. To express the signal-to-interference-plus-noise ratio (SINR) of user k , we need to compute the power of the desired signal and divide it by the sum of the power of the inter-user interference (IUI) and of the noise. By using identity (2.5), the power of the desired signal is given by

$$\mathbb{E} \left\{ \left| \mathbf{h}_{k,\text{DL}}^\top \mathbf{W}_{\text{ana}} \mathbf{w}_{\text{dig},k} s_{k,\text{DL}} \right|^2 \right\} = p_{k,\text{DL}} \left| \mathbf{h}_{k,\text{DL}}^\top \mathbf{W}_{\text{ana}} \mathbf{w}_{\text{dig},k} \right|^2 \quad (2.10)$$

⁶Note that, while digital precoding is a pre-IFFT operation and it can therefore be applied per subcarrier, analog beamforming (i.e., the phase shifting) is a post-IFFT operation and is the same for each subcarrier.

while the power of the inter-user interference is expressed as

$$\mathbb{E}\left\{\left|\sum_{k'=1, k' \neq k}^{K_{\text{DL}}} \mathbf{h}_{k', \text{DL}}^{\text{T}} \mathbf{W}_{\text{ana}} \mathbf{w}_{\text{dig}, k'} s_{k', \text{DL}}\right|^2\right\} = \sum_{k'=1, k' \neq k}^{K_{\text{DL}}} p_{k', \text{DL}} \left|\mathbf{h}_{k', \text{DL}}^{\text{T}} \mathbf{W}_{\text{ana}} \mathbf{w}_{\text{dig}, k'}\right|^2. \quad (2.11)$$

Therefore, the DL SINR of user k can be written as

$$\text{SINR}_{k, \text{DL}} = \frac{p_{k, \text{DL}} \left|\mathbf{h}_{k, \text{DL}}^{\text{T}} \mathbf{W}_{\text{ana}} \mathbf{w}_{\text{dig}, k}\right|^2}{\sum_{k'=1, k' \neq k}^{K_{\text{DL}}} p_{k', \text{DL}} \left|\mathbf{h}_{k', \text{DL}}^{\text{T}} \mathbf{W}_{\text{ana}} \mathbf{w}_{\text{dig}, k'}\right|^2 + \sigma_{k, \text{DL}}^2}. \quad (2.12)$$

The above SINR is an instantaneous SINR for user k at one active subcarrier and active OFDM symbol. An achievable instantaneous rate per Hz can be expressed as

$$R_{k, \text{DL}} = \log_2 (1 + \text{SINR}_{k, \text{DL}}) \quad [\text{bit/s/Hz}]. \quad (2.13)$$

The sum rate is given by $\sum_{k=1}^{K_{\text{DL}}} R_{k, \text{DL}}$. The average rate over the frame (in bit/s) can be computed by averaging (empirically or probabilistically) expression (2.13) over the subcarriers and OFDM symbols, and multiplying it by the bandwidth B and the share of time over the frame duration in which data transmission occurs. This last multiplicative term is the usual pre-log factor that appears in the delivered rate expressions. In the following section and chapters, we adapt the rate or SINR expression to the considered scenario.

2.1.2 Uplink

In UL, the channel between the active BS antennas and the user k is given by $\mathbf{h}_{k, \text{UL}} \in \mathbb{C}^{M_{\text{a}} \times 1}$, therefore the channel matrix is denoted by

$$\mathbf{H}_{\text{UL}} = [\mathbf{h}_{1, \text{UL}}, \dots, \mathbf{h}_{K_{\text{UL}}, \text{UL}}] \in \mathbb{C}^{M_{\text{a}} \times K_{\text{UL}}}. \quad (2.14)$$

The users transmit the signal $\mathbf{s}_{\text{UL}} = [s_{1, \text{UL}}, \dots, s_{K_{\text{UL}}, \text{UL}}]^{\text{T}} \in \mathbb{C}^{K_{\text{UL}} \times 1}$, with covariance equal to $\mathbb{E}\{\mathbf{s}_{\text{UL}} \mathbf{s}_{\text{UL}}^{\text{H}}\} = \mathbf{D}_{\mathbf{p}_{\text{UL}}} = \text{diag}(p_{1, \text{UL}}, \dots, p_{K_{\text{DL}}, \text{UL}})$. The variable $p_{k, \text{UL}}$ represents the power transmitted by user k . The BS performs hybrid analog-digital combining via the matrices $\mathbf{V}_{\text{dig}} = [\mathbf{v}_{\text{dig}, 1}, \dots, \mathbf{v}_{\text{dig}, K_{\text{UL}}}]^{\text{T}} \in \mathbb{C}^{K_{\text{UL}} \times M_{\text{RF}, \text{a}}}$ and $\mathbf{V}_{\text{ana}} \in \mathbb{C}^{M_{\text{RF}, \text{a}} \times M_{\text{a}}}$, where $\mathbf{V}_{\text{ana}}$ is selected from a phase-only codebook $\mathcal{V}_{\text{ana}}$. The digital combining vector of user k is denoted by $\mathbf{v}_{\text{dig}, k} \in \mathbb{C}^{M_{\text{RF}, \text{a}} \times 1}$. After the users' transmission took place, the BS first receives the signal

$$\mathbf{r}_{\text{UL}} = \mathbf{H}_{\text{UL}} \mathbf{s}_{\text{UL}} + \tilde{\boldsymbol{\nu}}_{\text{UL}} \in \mathbb{C}^{M_{\text{a}} \times 1} \quad (2.15)$$

and then obtains the symbol estimate of user k as

$$y_{k, \text{UL}} = \mathbf{v}_{\text{dig}, k}^{\text{T}} \mathbf{V}_{\text{ana}} \mathbf{H}_{\text{UL}} \mathbf{s}_{\text{UL}} + \nu_{k, \text{UL}} \quad (2.16)$$

where $\nu_{k,\text{DL}} = \mathbf{v}_{\text{dig},k}^T \mathbf{V}_{\text{ana}} \tilde{\mathbf{v}}_{\text{UL}}$, with variance $\sigma_{k,\text{UL}}^2$, and $\tilde{\mathbf{v}}_{\text{UL}} \sim \mathcal{CN}(\mathbf{0}_{K_{\text{UL}}}, \mathbf{D}_{\tilde{\sigma}_{\text{UL}}}^2)$ is ZMCSCG noise with covariance matrix $\mathbf{D}_{\tilde{\sigma}_{\text{UL}}}^2 = \text{diag}(\tilde{\sigma}_{1,\text{UL}}^2, \dots, \tilde{\sigma}_{K_{\text{UL}},\text{UL}}^2)$. Following the same steps as for the DL, the instantaneous UL SINR for user k can be written as

$$\text{SINR}_{k,\text{UL}} = \frac{p_{k,\text{UL}} |\mathbf{v}_{\text{dig},k}^T \mathbf{V}_{\text{ana}} \mathbf{h}_{k,\text{UL}}|^2}{\sum_{k'=1, k' \neq k}^{K_{\text{UL}}} p_{k',\text{UL}} |\mathbf{v}_{\text{dig},k}^T \mathbf{V}_{\text{ana}} \mathbf{h}_{k',\text{UL}}|^2 + \sigma_{k,\text{UL}}^2} \quad (2.17)$$

and the corresponding achievable rate per Hz is

$$R_{k,\text{UL}} = \log_2(1 + \text{SINR}_{k,\text{UL}}) \quad [\text{bit/s/Hz}]. \quad (2.18)$$

As for the DL communication, we adapt in the following the rate or SINR expression to the analyzed scenario.

2.2 Power Consumption Model

In this section, we describe the power model used in the thesis. To evaluate the energy performance of BSs and design algorithms to optimize it, accurate power models are essential. A review of the models published in the literature until 2023 is given in [50]. The authors propose to classify the models in (i) linear models if they express the power consumption as linearly dependent on the BS output power, (ii) subcomponent models if they estimate the consumption of individual processing blocks, and (iii) massive multiple-input multiple-output (MIMO) models when they are specific to BSs with many antennas (e.g., 32 or more). The models typically target a specific frequency band (FR1 or FR2) and are used by both academic researchers and companies to perform evaluations, also at a standardized level [30]. Among more recent works, model [15] is a parametric power model that captures the power consumption of FR1 fourth-generation (4G) and fifth-generation (5G) BSs in Belgium. By utilizing measurements from a Belgian mobile network operator (MNO) and datasheets of BS manufacturers, the authors model the consumption of PAs, analog front-end (AFE), digital baseband (DBB), and power supply and cooling systems. The model applies to both massive MIMO and small-dimension MIMO BSs, and its parameters allow the inclusion of different energy-saving features such as micro-discontinuous transmission (μDTX). The PA model includes a non-constant PA efficiency and is discussed in Section 2.2.1. The model does not include subcomponent modeling of AFE and DBB, which are also considered independent from the DL physical resource load $\overline{\ell}_{\text{DL}}$ (differently from the PA) and the UL physical resource load $\overline{\ell}_{\text{UL}}$. The average BS consumption over the frame is computed by averaging the consumption of PA, AFE, and DBB over different operating modes, from working to sleep mode.

The model in [51], instead, is specific to FR2 and THz BSs and builds upon previous FR1 models in [19] and related papers. The main difference with [15] lies in the subcomponent modeling of AFE and DBB. Scaling laws are provided for analog HW devices such as digital-to-analog converters (DACs) and analog-to-digital converters (ADCs), as well as for digital HW modules such as MIMO precoders and combiners. In the model proposed hereinafter, we combine both approaches and build a power model that has component-level frame averaging over different operating modes with subcomponent modeling during working mode. We then showcase its application to BSs operating at FR3, which is receiving growing interest toward 6G networks [43].

Remark: We adopt the PA power model and the frame-averaging over different operating modes—including the use of power-reduction factors and implying the load dependency—from the model in [15], while we adopt the power modeling approaches for digital and analog subcomponents from the model in [51]. The fundamental differences with respect to [15] are the power modeling of digital and analog subcomponents, their load dependency, and the link with communication performance (i.e., ergodic rates) considering hybrid beamforming. The main novelty is in the analysis of prospective FR3 BSs, which we expect to combine the different features of power and communication models presented here.

We subdivide the BS in *components* and *subcomponents*, where we consider three different components: PAs, analog signal processing, and digital signal processing. Each component has an associated supply and cooling module. The total BS power consumption averaged over the frame is expressed as

$$P_{\text{cons}} = \frac{\overline{P_{\text{PA}}}}{\eta_{\text{PA},s/c}} + \frac{\overline{P_{\text{ana}}}}{\eta_{\text{ana},s/c}} + \frac{\overline{P_{\text{dig}}}}{\eta_{\text{dig},s/c}} \quad (2.19)$$

where $\overline{P_{\text{PA}}}$, $\overline{P_{\text{ana}}}$, and $\overline{P_{\text{dig}}}$ are the average consumptions by the PAs, the analog processing, and the digital processing. The quantities $\eta_{\text{PA},s/c} \in (0, 1]$, $\eta_{\text{ana},s/c} \in (0, 1]$, and $\eta_{\text{dig},s/c} \in (0, 1]$ are the supply and cooling efficiencies of each component. In case of no active cooling, the supply and cooling efficiency equals 1. Fig. 2.4 summarizes the BS architecture that we consider in the thesis.

Every component comprises one or more subcomponents that can be *active* or *non-active* for the full frame duration, where an active subcomponent can be in (i) working mode, (ii) micro-sleep mode, or (iii) idle mode, while a non-active subcomponent is in sleep mode. We employ reduction factors $\delta_{\text{mode}} \in [0, 1]$ to quantify the power consumption reduction of a subcomponent in a non-working mode compared to its working mode consumption [15].⁷ We emphasize that

⁷These power-reduction factors are considered to be a fixed feature of the HW, determined at production time, hence do not depend on the load and their values are not optimized during operation. An ideal HW would have all reduction factors equal to zero (e.g., $\delta_{\text{PA},\text{micro}} = 0$ meaning that the PA consumes no power during micro-sleep mode).

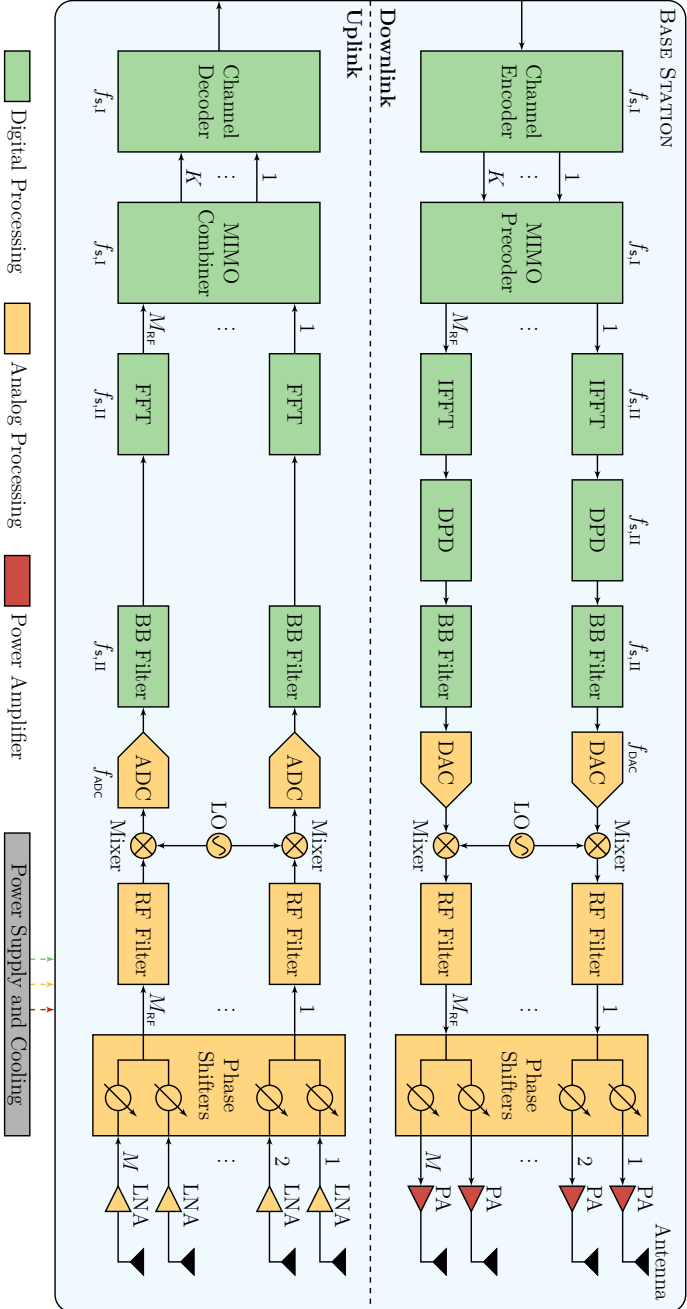


Figure 2.4: BS architecture considered in this thesis, where components are color coded, and a component comprises multiple subcomponents. Note that with fully-digital beamforming, the phase shifters are not present and $M_{\text{RF}} = M$. In this figure, active and non-active antennas or RF/DSP chains are not highlighted, and the BS serves K users in both DL and UL. The variables above or below some subcomponents are the sampling frequencies at which they operate.

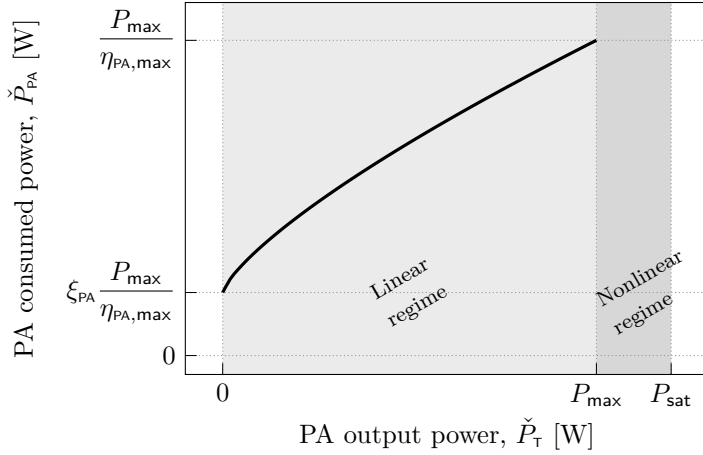


Figure 2.5: Illustration of PA power consumption versus PA output power, where in our model we consider $\check{P}_T \leq P_{\max}$ so that the PA is in the linear input-power output-power regime.

given the frame averaging of power consumption, the energy consumed by the BS in the frame is linearly proportional to the power consumed by the BS and given by $T_{\text{frame}}P_{\text{cons}}$. In the following, we describe the power consumption models of each component.

2.2.1 Power Amplifier

The PA has been and still is one of the devices consuming most of the power in cellular macro BSs due to the large BS-user distances and the consequent high transmit powers required. Evaluations on Long-Term Evolution (LTE) macro BSs have shown that the PA consumes more than 50% of the total BS consumption [14]. Model [15] has evaluated the PA power share to be at least 42% in New Radio (NR) macro BSs at full load. In the considered BS, there are M_a active PAs (and antennas) and $M - M_a$ non-active PAs (and antennas). The active PAs can be in three operating modes for the entire frame duration: working mode, micro-sleep mode, and idle mode. The non-active PAs are instead in sleep mode for the full frame duration.

The power consumption of a PA can be expressed as its output power divided by its efficiency. It is known that the PA efficiency, and hence the PA consumption, depends on the PA output power [40]. The larger the PA output power, the larger the PA efficiency. For an instantaneous output power $\check{P}_T$, the instantaneous PA

consumption is expressed via [15, eq. (4)] as

$$\check{P}_{\text{PA}}(\check{P}_{\text{T}}) = \xi_{\text{PA}} \frac{P_{\text{max}}}{\eta_{\text{PA},\text{max}}} + (1 - \xi_{\text{PA}}) \frac{P_{\text{max}}^{1-\alpha} \check{P}_{\text{T}}^{\alpha}}{\eta_{\text{PA},\text{max}}} \quad (2.20)$$

where P_{max} is the maximal PA output power considering an output power back-off from the PA saturation power P_{sat} , $\xi_{\text{PA}} \in [0, 1]$ controls the weight of static consumption to maximum consumption, $\eta_{\text{PA},\text{max}} \in (0, 1]$ is the maximum PA efficiency, and $\alpha \in [0, 1]$. Fig. 2.5 illustrates the above function, where the back-off is usually represented by the ratio $P_{\text{max}}/P_{\text{sat}}$ and is in the range from -6 dB to -12 dB in OFDM systems due to the high peak-to-average power ratio (PAPR) [14]. Note that the maximal PA efficiency $\eta_{\text{PA},\text{max}}$ is achieved when $\check{P}_{\text{T}} = P_{\text{max}}$. The parameters ξ_{PA} , α , and $\eta_{\text{PA},\text{max}}$ can be tailored to specific PA architectures. Moreover, the maximal efficiency $\eta_{\text{PA},\text{max}}$ and the saturation power P_{sat} typically decrease with the frequency [15, 52]. A PA power model similar to (2.20) is proposed in [25]. The works in [53, 54, 55] considered the same model with parameters $\xi_{\text{PA}} = 0$ and $\alpha = \frac{1}{2}$ for class B PAs, where the authors of [53, 54] further set $\eta_{\text{PA},\text{max}} = \frac{\pi}{4}$, which is the theoretical maximal efficiency. The authors of [15] propose, for the sub-6 GHz band, to set $\xi_{\text{PA}} = 0$, $\alpha = \frac{1}{2}$, and $\eta_{\text{PA},\text{max}} = 0.31$ to model class B PAs, and $\xi_{\text{PA}} \in (0, 1)$, $\alpha \in (0.5, 1)$, and $\eta_{\text{PA},\text{max}} \in (0.08, 0.79)$ to model Doherty PAs. Based on two recent industrial PAs for 5G massive MIMO [56, 57], $\alpha = \frac{1}{2}$ proves to be a good fit to model their consumption in the linear regime⁸ (with a back-off of 6 dB), and the maximal efficiencies are $\eta_{\text{PA},\text{max}} \in [0.21, 0.24]$ for the PA in the 3.3–3.8 GHz range [56], and $\eta_{\text{PA},\text{max}} \in [0.34, 0.41]$ for the PA in the 4.4–5.0 GHz range [57]. In this thesis, we adapt the PA parameters in each chapter based on the considered PA technology and frequency band.

Let us consider that the instantaneous transmit power of a PA m in working mode during data transmission is denoted by $P_{\text{T},m}$. The consumption of the PA m is then given by

$$P_{\text{PA},m} = \check{P}_{\text{PA}}(P_{\text{T},m}) = \xi_{\text{PA}} \frac{P_{\text{max}}}{\eta_{\text{PA},\text{max}}} + (1 - \xi_{\text{PA}}) \frac{P_{\text{max}}^{1-\alpha} P_{\text{T},m}^{\alpha}}{\eta_{\text{PA},\text{max}}} \quad (2.21)$$

and the consumption of the M_{a} active PAs when in working mode is

$$P_{\text{PA}} = \sum_{m=1}^{M_{\text{a}}} P_{\text{PA},m}. \quad (2.22)$$

We instead consider that the M_{a} active PAs transmit at power of $\zeta_{\text{DL},\text{sig}} P_{\text{max}}$ when they are in working mode during reference signaling, where $\zeta_{\text{DL},\text{sig}} \in [0, 1]$.

⁸The linear regime refers to the range of output powers for which the input signal can be amplified without introducing nonlinear distortions.

The corresponding power consumption is $\check{P}_{\text{PA}}(\zeta_{\text{DL},\text{sig}} P_{\text{max}})$. The transmit power of an active PA during micro-sleep and idle mode is zero, and its consumption is therefore given by $\check{P}_{\text{PA}}(0)$ if no additional power-saving features are enabled. By recalling the frame structure in Fig. 2.2, the average PA consumption over the frame is computed by adding the consumption in each phase weighted by the duration of that phase, giving

$$\begin{aligned} \overline{P}_{\text{PA}} = & \underbrace{\overline{\ell}_{\text{DL}} \frac{K_{\text{max}}}{K_{\text{DL}}} \tau_{\text{DL}} (1 - \tau_{\text{DL},\text{sig}}) P_{\text{PA}}}_{\text{working mode, data transm.}} + M_{\text{a}} \underbrace{\left(\tau_{\text{DL}} \tau_{\text{DL},\text{sig}} \check{P}_{\text{PA}}(\zeta_{\text{DL},\text{sig}} P_{\text{max}}) \right)}_{\text{working mode, ref. signal transm.}} \\ & + \underbrace{\left(1 - \overline{\ell}_{\text{DL}} \frac{K_{\text{max}}}{K_{\text{DL}}} \right) \tau_{\text{DL}} (1 - \tau_{\text{DL},\text{sig}}) \check{P}_{\text{PA}}(0) \delta_{\text{PA},\text{micro}}}_{\text{micro-sleep mode}} + \underbrace{(1 - \tau_{\text{DL}}) \check{P}_{\text{PA}}(0) \delta_{\text{PA},\text{idle}}}_{\text{idle mode}} \quad (2.23) \\ & + (M - M_{\text{a}}) \underbrace{\check{P}_{\text{PA}}(0) \delta_{\text{PA},\text{sleep}}}_{\text{sleep mode}} \end{aligned}$$

where P_{PA} is given in (2.22), and $\delta_{\text{PA},\text{micro}} \in [0, 1]$, $\delta_{\text{PA},\text{idle}} \in [0, 1]$, and $\delta_{\text{PA},\text{sleep}} \in [0, 1]$ quantify the power savings when a PA is in micro-sleep (DTX), idle mode, and sleep mode, respectively. We refer to them as *reduction factors*. A reduction factor of 1 (symmetrically 0) indicates no (symmetrically full) power saving with respect to working mode. Note that we expressed $(N_{\text{a,DL}}/N_{\text{DL}}) = \overline{\ell}_{\text{DL}}(K_{\text{max}}/K_{\text{DL}})$ to emphasize the dependency of the power consumption on the load $\overline{\ell}_{\text{DL}}$. In the case where we consider an equal transmit power across PAs in working mode and data transmission, denoted by P_{a} , expression (2.23) simplifies to

$$\begin{aligned} \overline{P}_{\text{PA}} = & M_{\text{a}} \left(\overline{\ell}_{\text{DL}} \tau_{\text{DL}} (1 - \tau_{\text{DL},\text{sig}}) \check{P}_{\text{PA}}(P_{\text{a}}) + \tau_{\text{DL}} \tau_{\text{DL},\text{sig}} \check{P}_{\text{PA}}(\zeta_{\text{DL},\text{sig}} P_{\text{max}}) \right) \\ & + (1 - \overline{\ell}_{\text{DL}}) \tau_{\text{DL}} (1 - \tau_{\text{DL},\text{sig}}) \check{P}_{\text{PA}}(0) \delta_{\text{PA},\text{micro}} + (1 - \tau_{\text{DL}}) \check{P}_{\text{PA}}(0) \delta_{\text{PA},\text{idle}} \quad (2.24) \\ & + (M - M_{\text{a}}) \check{P}_{\text{PA}}(0) \delta_{\text{PA},\text{sleep}}. \end{aligned}$$

It can be noticed as the average PA consumption has a linear dependence on the DL load $\overline{\ell}_{\text{DL}}$, and a nonlinear dependence on the transmit power P_{a} or $\check{P}_{\text{T},m}$ except in the case where the PA exponent α is equal to one. The case $\alpha = 1$ represents an ideal PA, whose consumption scales linearly with the transmit power. More details on this aspect are discussed in Chapter 3.

2.2.2 Analog Signal Processing

We consider that the analog signal processing includes the RF chains, comprising also DACs in DL and ADCs in UL, that process the signals after baseband processing until the PA input in DL and after antenna reception until baseband processing in UL. The phase shifters for analog beamforming are part of the analog processing. The subcomponents that are part of analog processing are illustrated in orange in Fig. 2.4. Note that the local oscillator (LO) is shared among the RF chains and the LNA is included in the UL subcomponents. In the evaluations, we will focus on hybrid partially-connected beamforming architectures with $M_{\text{PS}} = M/M_{\text{RF}}$ phase shifters per RF chain (equal to the number of antennas per subarray) and fully-digital beamforming architectures with $M_{\text{PS}} = 0$ phase shifters. We recall that we consider M_{a} RF chains in DL, of which $M_{\text{RF,a}}$ are active, and M_{a} RF chains in UL, of which $M_{\text{RF,a}}$ are active. The average consumption by analog processing over the frame is given by

$$\overline{P_{\text{ana}}} = \overline{P_{\text{ana,DL}}} + \overline{P_{\text{ana,UL}}} + P_{\text{ana,misc}} \quad (2.25)$$

where $\overline{P_{\text{ana,DL}}}$ and $\overline{P_{\text{ana,UL}}}$ are the average consumptions in DL and UL. To compute the consumption in DL or UL (represented by the variable $i \in \{\text{DL}, \text{UL}\}$), we follow the same approach as for the PA, giving

$$\begin{aligned} \overline{P_{\text{ana},i}} &= M_{\text{RF,a}} \left(\overline{\ell}_i \frac{K_{\text{max}}}{K_i} \tau_i (1 - \tau_{i,\text{sig}}) P_{\text{ana},i} + \tau_i \tau_{i,\text{sig}} P_{\text{ana},i} \right. \\ &\quad \left. + \left(1 - \overline{\ell}_i \frac{K_{\text{max}}}{K_i} \right) \tau_i (1 - \tau_{i,\text{sig}}) P_{\text{ana},i} \delta_{\text{ana,micro}} + (1 - \tau_i) P_{\text{ana},i} \delta_{\text{ana,idle}} \right) \\ &\quad + (M_{\text{RF}} - M_{\text{RF,a}}) P_{\text{ana},i} \delta_{\text{ana,sleep}}, \quad i \in \{\text{DL}, \text{UL}\} \end{aligned} \quad (2.26)$$

where $\delta_{\text{ana,micro}} \in [0, 1]$, $\delta_{\text{ana,idle}} \in [0, 1]$, and $\delta_{\text{ana,sleep}} \in [0, 1]$ are reduction factors when the analog subcomponents are in micro-sleep, idle, and sleep mode, respectively. The term $P_{\text{ana,misc}}$ is a fixed consumption term that accounts for miscellaneous contributions other than those of the considered subcomponents. The average analog consumption depends linearly on both the DL and UL loads $\overline{\ell}_{\text{DL}}$ and $\overline{\ell}_{\text{UL}}$, and also linearly on $M_{\text{RF,a}}$. The total analog processing consumption during working mode in DL and UL is computed as

$$P_{\text{ana,DL}} = \frac{P_{\text{LO}}}{M_{\text{RF,a}}} + 2P_{\text{DAC}} + 2P_{\text{filter,RF}} + 2P_{\text{mixer}} + M_{\text{PS}} P_{\text{PS}} \quad (2.27)$$

$$P_{\text{ana,UL}} = \frac{P_{\text{LO}}}{M_{\text{RF,a}}} + 2P_{\text{ADC}} + 2P_{\text{filter,RF}} + 2P_{\text{mixer}} + M_{\text{PS}} P_{\text{PS}} + \frac{M_{\text{a}}}{M_{\text{RF,a}}} P_{\text{LNA}} \quad (2.28)$$

where P_{LO} , P_{DAC} , P_{ADC} , $P_{\text{filter,RF}}$, P_{mixer} , P_{PS} , and P_{LNA} are the power consumptions of an LO, DAC, ADC, RF filter, mixer, phase shifter, and LNA, respectively.

We include two DACs, ADCs, RF filters, and mixers per RF chain, processing the in-phase (I) and quadrature (Q) channels. Note that the LO consumption is divided by $M_{\text{RF},a}$ because the LO is shared among RF chains, while in $P_{\text{ana,UL}}$ the LNA contribution scales as the number of antennas M_a because there are as many LNAs as antennas (cf. Fig. 2.4).

Let us now describe the power models of subcomponents. In the first part of the evaluations, we will consider that the DAC and the ADC operate at large sampling frequencies (in the order of Gsample/s) to enable full flexibility in the digital processing module. This follows the recent trend of RF-DACs and RF-ADCs [58, 59, 60]. A model for the DAC power consumption is given by [61]

$$P_{\text{DAC}} = \Xi_{\text{DAC},1} 2^{b_{\text{DAC}}} + \Xi_{\text{DAC},2} b_{\text{DAC}} f_{\text{DAC}} \quad (2.29)$$

where b_{DAC} is the effective number of bits, f_{DAC} is the sampling frequency, and $\Xi_{\text{DAC},1}$, $\Xi_{\text{DAC},2}$ are parameters. This well models state-of-the-art high-speed DACs such as [62]. The ADC power consumption can be generally modeled as [63]

$$P_{\text{ADC}} = \Xi_{\text{ADC}} f_{\text{ADC}} 2^{b_{\text{ADC}}} \quad (2.30)$$

where Ξ_{ADC} is the Walden's figure-of-merit, b_{ADC} is the effective number of bits, and f_{ADC} is the sampling frequency. We express the consumption of a mixer, phase shifter, and LNA as [64, 65, 66]

$$P_{\text{mixer}} = \Xi_{\text{mixer}} f_{c,i}, \quad P_{\text{PS}} = \Xi_{\text{PS}} B, \quad P_{\text{LNA}} = \Xi_{\text{LNA}} B \quad (2.31)$$

where Ξ_{mixer} , Ξ_{PS} , and Ξ_{LNA} are figure-of-merit parameters that depend on gain, noise factor, and resolution. The consumptions of the LO and of the RF filter, represented by P_{LO} and $P_{\text{filter,RF}}$, are considered to be constants. In Section 2.3, we use manufacturers' datasheets to set their values.

2.2.3 Digital Signal Processing

The digital signal processing part of a BS receives the information bits and returns the input to the DACs in DL, while it receives the ADCs output and returns the information bits in UL. The digital subcomponents that we consider in this thesis are illustrated in green in Fig. 2.4. The frame-averaged digital consumption in both DL and UL is quantified as

$$\overline{P_{\text{dig}}} = \overline{P_{\text{dig,DL}}} + \overline{P_{\text{dig,UL}}} + P_{\text{dig,link}} \quad (2.32)$$

where the digital consumption in DL or UL averaged over the frame is

$$\begin{aligned} \overline{P_{\text{dig},i}} &= M_{\text{RF},a} \left(\overline{\ell}_i \frac{K_{\text{max}}}{K_i} \tau_i (1 - \tau_{i,\text{sig}}) P_{\text{dig},i} + \tau_i \tau_{i,\text{sig}} P_{\text{dig},i} \right. \\ &\quad \left. + \left(1 - \overline{\ell}_i \frac{K_{\text{max}}}{K_i} \right) \tau_i (1 - \tau_{i,\text{sig}}) P_{\text{dig},i} \delta_{\text{dig},\text{micro}} + (1 - \tau_i) P_{\text{dig},i} \delta_{\text{dig},\text{idle}} \right) \quad (2.33) \\ &\quad + (M_{\text{RF}} - M_{\text{RF},a}) P_{\text{dig},i} \delta_{\text{dig},\text{sleep}}, \quad i \in \{\text{DL}, \text{UL}\} \end{aligned}$$

with $\delta_{\text{dig},\text{micro}} \in [0, 1]$, $\delta_{\text{dig},\text{idle}} \in [0, 1]$, and $\delta_{\text{dig},\text{sleep}} \in [0, 1]$ being the reduction factors when the digital subcomponents are in micro-sleep, idle, and sleep mode, respectively. The term $P_{\text{ref},\text{link}}$ is a fixed consumption term accounting for link-layer consumption. As for the analog processing, the digital processing consumption scales linearly with the loads and the number of active and total RF chains, and does not depend on the transmit power. The consumption of an active DL or UL DSP chain is given by

$$P_{\text{dig},\text{DL}} = P_{\text{encoder}} + P_{\text{precoder}} + P_{\text{IFFT}} + P_{\text{DPD}} + P_{\text{filter},\text{BB}} \quad (2.34)$$

$$P_{\text{dig},\text{UL}} = P_{\text{decoder}} + P_{\text{combiner}} + P_{\text{FFT}} + P_{\text{filter},\text{BB}} \quad (2.35)$$

where P_{encoder} , P_{decoder} , P_{precoder} , P_{combiner} , P_{IFFT} , P_{FFT} , $P_{\text{filter},\text{BB}}$ are the power consumptions of channel encoder and decoder, MIMO precoder and combiner, IFFT and FFT for OFDM modulation and demodulation, digital pre-distortion (DPD), and baseband (BB) filter.

To quantify the power consumption of a subcomponent, we include a static (signal-independent) and a dynamic (signal-dependent) term, which are influenced by the HW architecture, circuit, and semiconductor technology implementation. The static term is indicated by the subscript ‘s’, while following the approach in [19, 51], the dynamic term is computed as the number of complex giga operations per second (GOPS) divided by the computational efficiency (indicated by $\eta_{\text{component}}$) in complex GOPS/W. The channel encoding/decoding and the MIMO precoding/combining operate at the constellation rate, denoted as $f_{s,\text{I}}$, while the IFFT, FFT, DPD, and BB filtering operate at a sampling rate larger than $f_{s,\text{I}}$, denoted as $f_{s,\text{II}}$.⁹

Let us express the power consumption of each subcomponent when in active mode, i.e., when transmitting or receiving data or reference signals. The consumption of the channel encoder and decoder is computed as

$$P_j = \frac{P_{j,s}}{M_{\text{RF},a}} + \frac{f_{s,\text{I}}}{\eta_j} \frac{\Xi_j}{M_{\text{RF},a}}, \quad j \in \{\text{encoder}, \text{decoder}\} \quad (2.36)$$

⁹The two sampling rates are equal to $f_{s,\text{I}} = \mu_s B$, where $\mu_s \in [0, 1]$ and B is the bandwidth, and $f_{s,\text{II}} = Q_{\text{IFFT}} \Delta_f$, where Q_{IFFT} is the IFFT size and Δ_f is the subcarrier spacing.

where $P_{j,s}$ is the static consumption among all the $M_{\text{RF},a}$ active DSP chains (i.e., all the digital signal paths that are not switched off), η_j is the computational efficiency, and Ξ_j is the number of complex GOPS per sample among all the $M_{\text{RF},a}$ active DSP chains. Similarly, the consumption of MIMO precoder and combiner is quantified by

$$P_j = \frac{P_{j,s}}{M_{\text{RF},a}} + \frac{f_{s,I}}{\eta_j} \Xi_j, \quad j \in \{\text{precoder, combiner}\} \quad (2.37)$$

where this time Ξ_{precoder} and Ξ_{combiner} give the number of complex GOPS per sample and per active DSP chain. The same formula as above applies for the IFFT, FFT, DPD, and BB filter, with the only difference being the different sampling frequency $f_{s,\text{II}}$. This gives

$$P_j = \frac{P_{j,s}}{M_{\text{RF},a}} + \frac{f_{s,\text{II}}}{\eta_j} \Xi_j, \quad j \in \{\text{IFFT, FFT, DPD, filter, BB}\}. \quad (2.38)$$

To give an example, let us consider that a zero-forcing (ZF) scheme is used for MIMO precoding and combining. Assuming channel reciprocity between DL and UL, i.e., $\mathbf{H}_{\text{DL}} = \mathbf{H}_{\text{UL}}^H$, one can set $\mathbf{W}_{\text{ana}} = \mathbf{V}_{\text{ana}}^H$. The digital precoding and decoding matrix are given by $\mathbf{W}_{\text{dig}} = \mathbf{V}_{\text{dig}}^H = \mathbf{H}_{\text{eff}}^H (\mathbf{H}_{\text{eff}} \mathbf{H}_{\text{eff}}^H)^{-1}$, where $\mathbf{H}_{\text{eff}} = \mathbf{H}_{\text{DL}} \mathbf{W}_{\text{ana}}$ is the effective channel matrix seen by the digital precoder/combiner. Considering that the matrix inversion is computed via Cholesky decomposition and backward-forward substitution, the number of complex GOPS per sample and per DSP chain required by the precoder and the combiner are [21]

$$\Xi_{\text{precoder}} = 2K + \frac{1}{\iota_{\text{coh}}} \left(\frac{K^3}{3M_{\text{RF},a}} + 3K^2 + K \right), \quad \Xi_{\text{combiner}} = 2K \quad (2.39)$$

where ι_{coh} is the number of samples per coherence block.

2.3 Application to Upper Mid-Band Base Stations

We present here an application of the power and communication models to prospective upper mid-band BSs. This band, also known as FR3, is expected to trade off the benefits of FR1 (or sub-7 GHz band) in terms of propagation and coverage, with the large bandwidths offered by FR2 (or mmWave band) [44]. Recent works are characterizing FR3 with focus on aspects such as channel propagation [67], measurements of indoor channels [68], prototyping and channel sounders [69], and new fully-digital MIMO arrays having (much) more antenna elements than in current massive MIMO [70]. Evaluating the power consumption of FR3 BSs constitutes a fundamental task to link the

communication performance with environmental and economic sustainability. In this regard, the authors of [71] have studied the capacity and power consumption of networks using both FR1 and FR3. The power consumption model that they consider does not include non-working modes of components, hence it does not quantify the load dependency, nor subcomponents modeling. Further, we point out that applying models [15] and [51] to FR3 BSs is not straightforward. Models [15], although based on measurements at FR1, relies on fixed reference values for aggregate analog and digital consumption. Scaling these figures to a BS array with, e.g., 1024 antennas would introduce significant uncertainty. Model [51], which targets FR2, assumes RF technologies and system parameters that differ from those envisioned for FR3 (e.g., Indium phosphide (InP) PAs and bandwidths above 1 GHz). In the following, we present the HW architectural choices and the evaluations in the considered FR3 scenarios.

2.3.1 Architectural Choices

Let us focus on an upper mid-band BS with 1024 antennas and PAs in DL, and 1024 antennas and LNAs in UL. The number of RF chains is varied from 16 to 1024, where the latter case represents fully-digital beamforming. All the former cases represent hybrid partially-connected beamforming. MIMO configurations with hundreds to thousands of antenna elements and RF chains are being proposed to provide cell capacity boost in the 6G upper mid-band [45]. We consider a case in which the PAs/LNAs and RF/DSP chains are active for the full frame duration, giving $M_a = M$ and $M_{\text{RF},a} = M_{\text{RF}}$. This reduces the number of parameters and allows us to more clearly compare the selected BS architectures. We also assume, as simplification, that $P_{\text{ana},\text{misc}} = P_{\text{dig},\text{link}} = 0$. The bandwidth is fixed to $B = 400$ MHz and the number of users is $K_{\text{max}} = K_{\text{DL}} = K_{\text{UL}} = K = 8$. We consider that the BS operates in time-division duplexing (TDD) with $\tau_{\text{DL}} = 0.75$ and $\tau_{\text{UL}} = 0.25$. The carrier frequencies are set as $f_{\text{c,DL}} = f_{\text{c,UL}} = 10$ GHz, the subcarrier spacing as $\Delta_f = 120$ kHz, the IFFT size as $Q_{\text{IFFT}} = 4096$, and the number of data subcarriers as $Q_{\text{max}} = 3000$.

Power Amplifier

Concerning the PA, the four main technologies for implementing PAs at FR3 are gallium nitride (GaN), gallium arsenide (GaAs), silicon-germanium (SiGe), and complementary metal-oxide-semiconductor (CMOS) [46]. Each one gives a different maximal output power P_{max} and efficiency $\eta_{\text{PA},\text{max}}$, among other factors. GaN PAs are typically characterized by larger maximal powers at the expense of lower efficiencies. We select a recent SiGe implementation for FR3, proposed in [72], and set the model parameters to $P_{\text{max}} = 0.1$ W, $\eta_{\text{PA},\text{max}} = 0.15$, $\alpha = 0.75$,

and $\xi_{\text{PA}} = 0.1$. We consider an average per-PA (and per-antenna) transmit power of $P_a = P_{\text{T,DL}}/M$ W, where $P_{\text{T,DL}}$ is the total transmit power. The average PA consumption is therefore computed via equation (2.24).

Analog Processing

In the analog processing, we consider that both the DAC and the ADC operate at sampling frequencies of $f_{\text{DAC}} = f_{\text{ADC}} = 5$ Gsample/s and with effective number of bits equal to $b_{\text{DAC}} = b_{\text{ADC}} = 8$ [62, 73]. The model [61] quantifies the DAC's parameters as $\Xi_{\text{DAC},1} = 1.5 \cdot 10^{-5}$ and $\Xi_{\text{DAC},2} = 1.5 \cdot 10^{-12}$. Based on a state-of-the-art high-speed ADC [73], we set the Walden's figure-of-merit to $\Xi_{\text{ADC}} = 70$ fJ/conversion step. Recent implementations of a mixer, phase shifter, and LNA for FR3 give the respective parameters $\Xi_{\text{mixer}} = 2.5 \cdot 10^{-13}$, $\Xi_{\text{PS}} = 3.5 \cdot 10^{-11}$, and $\Xi_{\text{LNA}} = 2.7 \cdot 10^{-11}$ [64, 65, 66]. The datasheets of an LO [74] and RF filter [75] are used to set $P_{\text{LO}} = 40$ mW and $P_{\text{filter,RF}} = 5$ mW.

Remark: We caution that the power consumption of commercial HW may be larger than the one estimated here (resulting from designs published in research works). In a commercial HW solution, there could be other HW modules that consume power, in addition to the module under study. This extra power consumption can be included in the model as part of the (constant) parameter $P_{\text{ana,misc}}$ in equation (2.25).

Digital Processing

The two most common HW solutions for the baseband processing of cellular wireless BSs are field-programmable gate array (FPGA) and application-specific integrated circuit (ASIC). The first provides more flexibility in terms of reconfiguration at the cost of larger power consumption and lower speed. Each digital subcomponent can be associated with a specific HW platform in the power model by tailoring its parameters $P_{\text{subcomponent,s}}$ (static consumption) and $\eta_{\text{subcomponent}}$ (computational efficiency), cf. Section 2.2.3. To estimate the parameters of FPGA solutions, we synthesized, via high-level synthesis in the AMD VivadoTM environment, a MIMO precoder based on LDL decomposition on an AMD ZynqTM UltraScale+TM RF system-on-chip (RFSoc) ZCU216 [58]. With 32 antennas, 8 users, and a clock frequency of 200 MHz, the computations of precoding matrices and precoded signals are performed at 0.8 MHz and 400 MHz, respectively, where we estimate $\eta_{\text{precoder}} = 0.2 \cdot 10^3$ complex GOPS/W and a static consumption of 1 W. For ASIC implementations, we consider a ten times larger efficiency and a ten times lower static power consumption [76].

Remark: The $10\times$ difference in GOPS/W between ASIC and FPGA, although in line with a class of recent HW accelerators for neural networks [77], is an assumption and must be taken with care. Synthesis and implementation of ASIC solutions for communication tasks of FR3 BS are necessary to advance and refine this comparison.

By using the values provided in [51] for low-density parity-check (LDPC) codes, we set the parameters of the channel encoder and decoder to $\Xi_{\text{encoder}} = 0.59(R_{\text{DL}}/0.9B)$ and $\Xi_{\text{decoder}} = 29.17(R_{\text{UL}}/0.9B)$, where R_{DL} and R_{UL} are the DL and UL sum rates. We consider that both subcomponents are implemented in ASIC with $P_{\text{encoder},s} = P_{\text{decoder},s} = 0.1$ W and $\eta_{\text{encoder}} = \eta_{\text{decoder}} = 0.2 \cdot 10^4$ GOPS/W. As regards MIMO precoding and combining, we consider that ZF is used, requiring the GOPS per sample and per RF chain specified in equation (2.39). Due to the flexibility required to change the beamforming scheme with the scenario (e.g., interference-limited or noise-limited), we select FPGA as HW architecture, giving $\eta_{\text{precoder}} = \eta_{\text{combiner}} = 0.2 \cdot 10^3$ GOPS/W. If we consider that a single FPGA is shared among 32 RF chains, the static consumption is $P_{\text{precoder},s} = P_{\text{combiner},s} = \lceil M_{\text{RF},a}/32 \rceil \cdot 1$ W. For OFDM (de)modulation, we assume a 5G-like operation where the number of subcarriers is fixed and different subcarrier spacings determine the bandwidth [32]. We consider that they are implemented in ASIC with $P_{\text{IFFT},s} = P_{\text{FFT},s} = 0.1$ W and $\eta_{\text{IFFT}} = \eta_{\text{FFT}} = 0.2 \cdot 10^4$ GOPS/W. The GOPS of both IFFT and FFT are computed as $\Xi_{\text{IFFT}} = \Xi_{\text{FFT}} = \frac{3}{2} \log_2 Q_{\text{IFFT}}$. The DPD follows in the DL processing chain. It is used to linearize the overall input-output PA characteristic, and is typically characterized by large GOPS both for filtering and computing its coefficients. We utilize reference [78] to set $\Xi_{\text{DPD}} = 50$ and consider it is implemented in ASIC, giving $P_{\text{DPD},s} = 0.1$ W and $\eta_{\text{DPD}} = 0.2 \cdot 10^4$ GOPS/W. Last, the BB filter performs interpolation in DL and decimation in UL. Given that FR3 subbands are expected to have different spectral requirements, we opt for the FPGA solution [79].¹⁰ This yields $P_{\text{filter,bb},s} = \lceil M_{\text{RF},a}/32 \rceil \cdot 1$ W if we consider a single FPGA implementing 32 BB filters, and $\eta_{\text{filter,bb}} = 0.2 \cdot 10^3$ GOPS/W. Its GOPS is computed as $\Xi_{\text{filter,bb}} = N_{\text{filter,bb}} O_{\text{filter,bb}}$, where $N_{\text{filter,bb}}$ and $O_{\text{filter,bb}}$ are the number of filter taps and oversampling coefficient, which we set to $N_{\text{filter,bb}} = 20$ and $O_{\text{filter,bb}} = 4$.

2.3.2 Evaluation

In this section, we evaluate the power model and architectural choices together with the communication performance. We consider a 3GPP 39.801 Urban

¹⁰Considering the precoders, combiners, and BB filters to be implemented in FPGA induces a significant penalty in the power consumption (factor of 10 in our evaluations). In Section 2.3.3, we quantify the impact of this choice.

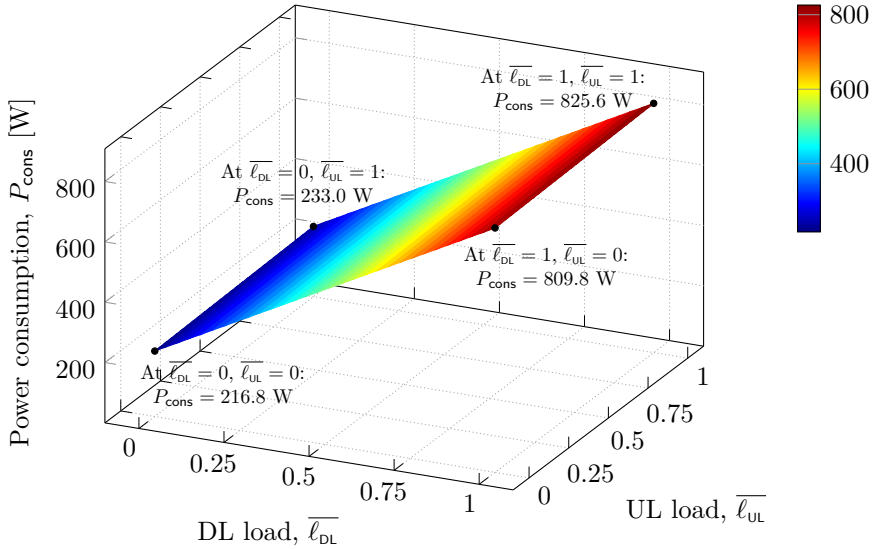


Figure 2.6: In the considered FR3 scenario and under the architectural choices of Section 2.3.1, BS power consumption versus DL and UL average physical resource loads. The main parameters are $M = 1024$ antennas, $M_{\text{RF}} = 256$ RF chains, $B = 400$ MHz, $K = 8$ users, $P_{\text{T,DL}} = 100$ W, $p_{k,\text{UL}} = 100$ mW $\forall k$, $\tau_{\text{DL}} = 0.75$, and $\tau_{\text{UL}} = 0.25$.

Micro (UMi) wideband channel model generated by the Sionna simulator [80]. The BS array is a uniform planar array (UPA) with 2048 antenna elements and half-wavelength antenna spacing in both directions. This gives $M = 1024$ in DL and $M = 1024$ in UL. The codebook for analog beamforming is the same in DL and UL, i.e., $\mathcal{W}_{\text{ana}} = \mathcal{V}_{\text{ana}}$, and is generated via a 2D-DFT. We set the total DL transmit power to $P_{\text{T,DL}} = 100$ W such that the PA output power is $P_{\text{a}} \approx 0.1 = P_{\text{max}}$ W. Further, we consider a uniform DL power allocation, giving $p_{k,\text{DL}} = 1$ $\forall k$. The UL transmit power is $p_{k,\text{UL}} = 100$ mW for all the users. The noise powers are set as $\sigma_{k,\text{DL}}^2 = \sigma_{k,\text{UL}}^2 = -123$ dBm $\forall k$. The parameters of the TDD protocol are $\tau_{\text{DL,sig}} = \tau_{\text{UL,sig}} = 1/14$ and $\zeta_{\text{DL,sig}} = 1/12$ [15]. We consider that channel reciprocity between DL and UL holds. We further set $\eta_{\text{dig,s/c}} = \eta_{\text{ana,s/c}} = \eta_{\text{PA,s/c}} = 0.8$. According to equation (2.3), the average physical loads are bounded as $\bar{\ell}_{\text{DL}} \in [0, 1]$ and $\bar{\ell}_{\text{UL}} \in [0, 1]$. Last, from equation (2.13) and equation (2.18), the ergodic sum

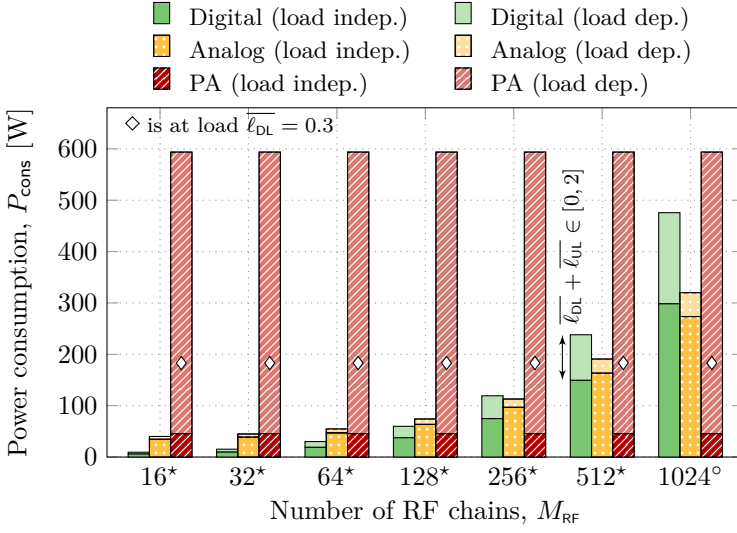
rates are computed as

$$\overline{R}_i = \overline{\ell}_i \tau_i (1 - \tau_{i,\text{sig}}) \tilde{B} \mathbb{E} \left\{ \sum_{k=1}^K R_{k,i} \right\} \quad [\text{bit/s}], \quad i \in \{\text{DL}, \text{UL}\} \quad (2.40)$$

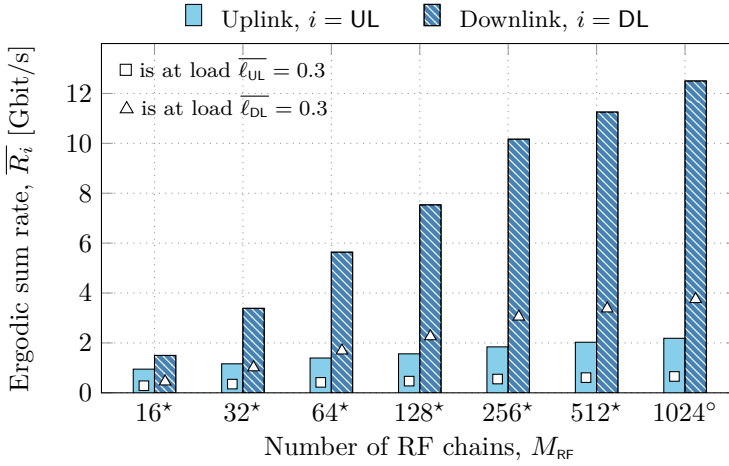
via Monte Carlo simulation, where $\tilde{B} = \mu_s B = 0.9 \cdot 400$ MHz is the effective bandwidth, accounting for the fact that 90% of the subcarriers contained in B are effectively used to transmit/receive the users' data. We consider that the PAs and the digital processing implement more time-domain power-saving features compared to the analog processing. Hence, we set the reduction factors of the PAs to $\delta_{\text{PA},\text{micro}} = 0.5$ and $\delta_{\text{PA},\text{idle}} = 0.25$, equally for the digital processing giving $\delta_{\text{dig},\text{micro}} = 0.5$ and $\delta_{\text{dig},\text{idle}} = 0.25$, while for the analog processing they are set to $\delta_{\text{ana},\text{micro}} = 0.75$ and $\delta_{\text{ana},\text{idle}} = 0.5$.

Fig. 2.6 illustrates the power consumption as a function of the DL and UL loads. The power consumption is an affine function of the loads. The DL load has a much higher impact on the consumption than the UL load, mainly due to the PAs consumption (that depends on $\overline{\ell}_{\text{DL}}$). At zero loads, the consumption is $P_{\text{cons}} = 216.8$ W, while it equals $P_{\text{cons}} = 825.6$ W at full loads. At typical loads (up to 30%), however, the consumption goes up to approximately 400 W, indicating that approximately 50% of the consumption is due to the static (load-independent) part. Note that, even though we do not analyze them here, energy-saving techniques would typically scale down the number of active antennas M_{a} , active RF chains $M_{\text{RF},\text{a}}$, and transmit powers $P_{\text{T,DL}}$ and $p_{k,\text{UL}}$ as a function of the loads, resulting in a non-affine (nonlinear) scaling of the power consumption at low loads.

Let us now quantify the impact of each BS component on the total consumption while varying the number of RF chains. In Fig. 2.7a, we show the power consumption and ergodic rates while varying the number of RF chains from 16 to 1024. The PA consumption is constant because it does not depend on M_{RF} but only on the number of antennas (and PAs) M , which is fixed to $M = 1024$. We split the consumption of each BS component in load dependent (lighter bars) and load independent (darker bars). The load-dependent terms are the ones in equation (2.23), (2.26), or (2.33) that multiply $\overline{\ell}_{\text{DL}}$ or $\overline{\ell}_{\text{UL}}$, and vice versa for the load-independent terms. We plot the full range of $\overline{\ell}_{\text{DL}} + \overline{\ell}_{\text{UL}}$, ranging from 0 to 2 and indicated with an arrow in one case. More realistic loads, up to $\overline{\ell}_{\text{UL}} = \overline{\ell}_{\text{DL}} = 0.3$, are highlighted with a marker. At full loads, the total BS consumption ranges from 643 W (when $M_{\text{RF}} = 16$) to 1389 W (when $M_{\text{RF}} = 1024$), while at $\overline{\ell}_{\text{UL}} = \overline{\ell}_{\text{DL}} = 0.3$, it varies from 253 W (when $M_{\text{RF}} = 16$) to 849 W (when $M_{\text{RF}} = 1024$). We observe that at full loads, the PA consumption is the largest among the BS components. At more realistic loads, however, the PA consumption becomes comparable to digital and analog consumptions approximately from $M_{\text{RF}} = 128$. The analog and digital processing have larger



(a)



(b)

*Hybrid analog-digital beamforming °Fully-digital beamforming

Figure 2.7: In the considered FR3 scenario and under the architectural choices of Section 2.3.1, (a) BS power consumption versus number of RF chains, (b) ergodic rates versus number of RF chains. For both figures, the main parameters are $M = 1024$ antennas, $B = 400$ MHz, $K = 8$ users, $P_{\text{T,DL}} = 100$ W, $p_{k,\text{UL}} = 100$ mW $\forall k$, $\tau_{\text{DL}} = 0.75$, and $\tau_{\text{UL}} = 0.25$.

Table 2.1: Energy efficiency versus number of RF chains at two load levels, referring to the architectural choices of Section 2.3.1.

M_{RF}	16	32	64	128	256	512	1024
EE at full loads [Mbit/J]	3.79	6.95	10.36	12.50	14.54	12.99	10.57
EE at 30% loads [Mbit/J]	2.89	5.21	7.50	8.51	9.01	7.15	5.19

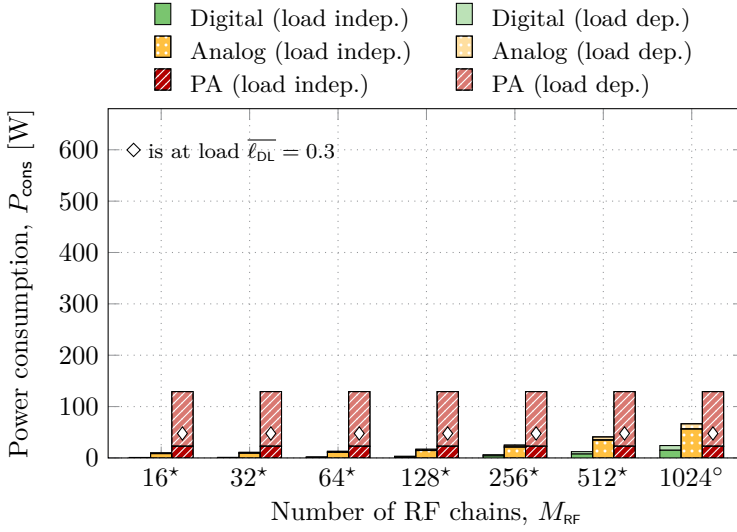
load-independent consumption compared to the load-dependent one, while it is the opposite for the PA. The largest contributors to the analog consumption are the LNAs at low values of M_{RF} , and the DACs and ADCs at high values of M_{RF} . In the digital consumption, the MIMO precoder/combiner and the BB filter, considered to be implemented in FPGA, have the largest share.

In Fig. 2.7b, we plot the ergodic rates in DL and UL for the full range of $\overline{\ell}_{\text{DL}}$ and $\overline{\ell}_{\text{UL}}$ (corresponding to the height of each bar), and highlight the values at loads of 30%. We point out that for small numbers of RF chains, the rates do not drastically drop because there are $M/M_{\text{RF}} = 1024/M_{\text{RF}}$ subarrays that perform analog beamforming. The DL sum rate reaches more than 12 Gbit/s and more than 2 Gbit/s in UL for the fully-digital beamforming at full load. We observe that hybrid beamforming with 128 or more RF chains provides rates larger than 50% of the ones achieved in both DL and UL by the fully-digital counterpart. The overall energy efficiency (EE), computed as $\text{EE} = (\overline{R}_{\text{DL}} + \overline{R}_{\text{UL}})/P_{\text{cons}}$, measured in bit per Joule and dependent on the loads, is shown in Table 2.1. The maximal EE of 12.5 Mbit/J is achieved by the hybrid beamforming configuration with 256 RF chains at full loads. The same configuration also improves the EE by a factor of $1.74\times$ at loads of 30%.

2.3.3 Impact of Architectural Choices

The HW architectural choices presented in Section 2.3.1 can have a significant impact on the final results. In this section, we focus on some critical assumptions, with consequent choice of parameters, and evaluate their effect on the evaluations. In particular, we consider a BS operating at a lower bandwidth and with a more energy-efficient choices for the DACs, ADCs, digital modules, and PAs. The introduced differences compared to Section 2.3.1 are the following:

- The bandwidth is scaled down to $B = 100$ MHz, instead of $B = 400$ MHz. The IFFT size is kept fixed at $Q_{\text{IFFT}} = 4096$ and the subcarrier spacing is reduced to $\Delta_f = 30$ kHz (instead of $\Delta_f = 120$ kHz). The lower bandwidth impacts both the analog and digital consumption. Accordingly,



*Hybrid analog-digital beamforming °Fully-digital beamforming

Figure 2.8: In the considered FR3 scenario and varying the architectural choices as explained in Section 2.3.3, BS consumption versus number of RF chains.

the transmit power is decreased to $P_{T,DL} = 25$ W and $p_{k,UL} = 25$ mW $\forall k$ (from $P_{T,DL} = 100$ W and $p_{k,UL} = 100$ mW $\forall k$).

- The DACs and ADCs operate at much lower sampling rates, i.e., $f_{DAC} = f_{ADC} = 500$ Msample/s, instead of $f_{DAC} = f_{ADC} = 5$ Gsample/s. This follows a more conventional operation of energy-optimized DACs and ADCs.
- The digital precoders, digital combiners, and BB filters are implemented in ASIC instead of FPGA. This gives a ten-times larger efficiency and ten-times lower static consumption, i.e., $\eta_{precoder} = \eta_{combiner} = \eta_{BB,filter} = 0.2 \cdot 10^4$ complex GOPS/W, and $P_{precoder,s} = P_{combiner,s} = P_{BB,filter,s} = 0.1$ W. As for the DACs and ADCs, this reduces the flexibility during operation but decreases the energy consumption.
- The PA efficiency doubles, giving $\eta_{PA,max} = 0.3$, instead of $\eta_{PA,max} = 0.15$. This foresees more energy-efficient PA implementations for FR3.

In Fig. 2.8, we plot the consumption against M_{RF} . Compared to Fig. 2.7a, there is a significant decrease in the power consumption by each BS component. Under fully-digital beamforming at full loads, the consumption of the PAs, analog processing, and digital processing decrease of 78%, 79%, and 95%, respectively.

Table 2.2: Energy efficiency versus number of RF chains at two load levels, referring to the architectural choices of Section 2.3.3.

M_{RF}	16	32	64	128	256	512	1024
EE at full loads [Mbit/J]	4.35	8.06	12.22	15.22	18.71	18.20	16.70
EE at 30% loads [Mbit/J]	2.85	5.23	7.80	9.44	11.03	9.93	8.36

These percentages of reduction are the same in the hybrid beamforming case with $M_{\text{RF}} = 256$. The bandwidth decrease has a larger impact on the digital consumption than on the analog consumption, which is instead more impacted by the reduced sampling frequencies of DACs and ADCs. In terms of EE (reported in Table 2.2) the hybrid beamforming configuration with $M_{\text{RF}} = 256$ RF chains remains the best option at both full loads and loads of 30%.

2.4 Conclusion

In this chapter, we have discussed the communication and power models used in the thesis. The BS architecture is split into PA, analog, and digital components, where we consider a number of active PAs as well as RF/DSP chains. Further, we consider that a number of active users are served by all the available data subcarriers during a number of active time slots, leaving the remaining time slots empty when not allocated for reference signaling. To provide a reply to **RQ1** on the power modeling of next-generation BSs, we proposed a generalization of existing power models by combining the frame-averaging of the components over different operating modes (from working to sleep) with subcomponent modeling that allows us to quantify the impact of individual HW devices or modules. Also, the load dependency of PAs is extended to analog and digital components to reflect upcoming BS implementations with load-dependent power-saving features in the RF/DSP chains other than in the PAs. We applied the power model to prospective upper mid-band or FR3 BSs and underlined the importance of considering communication performance to make the trade-off between power consumption and capacity transparent. Given the many open design questions on FR3 BSs, e.g., whether they should implement hybrid analog-digital or fully-digital beamforming, our analysis focused on quantifying both their impact on delivered rates and power consumption. In the set of evaluations, relative to an array of 1024 antenna elements using 400 MHz of bandwidth, we found that the hybrid analog-digital beamforming configuration with 256 RF chains provides the largest EE, reaching 18.71 Mbit/J.

Chapter 3

Deactivation of Antennas and Allocation of Transmit Power

Bibliographic Reference

This chapter is based on: E. Peschiera and F. Rottenberg, “Energy-saving precoder design for narrowband and wideband massive MIMO,” *IEEE Transactions on Green Communications and Networking*, vol. 7, no. 4, pp. 1793–1806, December 2023 [81].¹

In this chapter, we study the allocation of power and number of utilized antennas to reduce the base station (BS) consumption under users’ rate constraints. The target quality-of-service (QoS) is therefore fixed and the use of resources at the BS (antennas and transmit power) is optimized based on the traffic load level. The analysis centers on the downlink (DL) of a single-cell BS in a single time slot. We therefore discard the time slot index n in the following. The BS utilizes fully-digital beamforming, which is readily obtained from the DL communication model described in Section 2.1.1. The power model follows Section 2.2, with the power amplifier (PA) model adapted to class B PAs and the radio frequency (RF)/digital signal processing (DSP) chain consumption written as a unique variable at an aggregate level. A fixed consumption term represents the power that does not depend on the transmit power or on the number of

¹The Python simulation code is available in open access at the following link: github.com/dramco-research/energy-saving-precoder-epeschiera-tgcn23.

activated antennas. Note that given the use of fully-digital beamforming, the number of (activated) antennas corresponds to the number of (activated) PAs and (activated) RF chains. We recall that with activated (or, equivalently, active) antenna/PA/RF chain, we indicate an antenna/PA/RF chain that is utilized for transmission and allocated non-zero power.

State of the Art

Let us focus on a fifth-generation (5G) massive multiple-input multiple-output (MIMO) BS, equipped with M antennas and PAs to provide array gain and spatial multiplexing [20]. Following the discussion in Section 2.2.1, it is important to accurately model the PA consumption given that it is among the main contributors to the energy consumption of a BS. Despite this, the conventional massive MIMO precoders, that combine the users' symbols to determine the signal transmitted by the BS array, are typically found by optimizing a certain figure of merit at the receivers, e.g., rate, signal-to-interference-plus-noise ratio (SINR), or mean square error (MSE), under a transmit power constraint [82]. Indeed, the maximal transmit power is commonly fixed by regulations (e.g., by the Belgian Institute for Postal Services and Telecommunications (BIPT) in Belgium). Considering only transmit power would be optimal—in terms of PAs consumption—if the PA efficiency was fixed, but it actually depends on the output power level [40], making power consumption and transmit power not linearly proportional. Considering the model in equation (2.20) for the PA consumption and zero static consumption term, setting $\alpha = 1$ gives a constant efficiency and represents an ideal PA

$$\check{P}_{\text{PA}}^{(\text{ideal})} = \frac{P_{\text{max}} \check{P}_{\text{T}}}{\eta_{\text{PA}, \text{max}}} \quad (3.1)$$

where $\check{P}_{\text{PA}}^{(\text{ideal})}$ is the power consumption of an ideal PA, $\check{P}_{\text{T}}$ is its output power, while P_{max} and $\eta_{\text{PA}, \text{max}}$ are the maximal output power and efficiency. Model (3.1) is adopted by many works on the topic, e.g., [21, 53, 83]. A more realistic expression of PA consumption is obtained by setting $\alpha < 1$. For class B PAs, $\alpha = \frac{1}{2}$ is an accurate fit [15, 25, 54, 84], giving

$$\check{P}_{\text{PA}}^{(\text{class B})} = \frac{P_{\text{max}}^{\frac{1}{2}} \check{P}_{\text{T}}^{\frac{1}{2}}}{\eta_{\text{PA}, \text{max}}}. \quad (3.2)$$

Considering two industrial PAs for sub-6 GHz 5G massive MIMO [56, 57], their efficiency (actual efficiency, not maximal one) depends on the square-root of the output power when far from saturation (in this case, with a back-off of 6 dB). This efficiency reaches its maximal value $\eta_{\text{PA}, \text{max}}$ when $\check{P}_{\text{T}} = P_{\text{max}}$. This

yields equation (3.2) when dividing the output power by the efficiency in order to compute the PA consumption.

Despite the extensive research in massive MIMO signal processing, only a few works have considered (3.2) in the precoders design, especially for a system adopting orthogonal frequency-division multiplexing (OFDM) with multiple utilized subcarriers. In [38, 85], the authors analyzed the cases of multiple-input single-output (MISO) and MIMO with orthogonal channels. In MISO, the authors considered the minimization of (3.2) subject to a data rate constraint and per-antenna power constraints. The solution corresponds to antenna selection, saturating the antennas with the largest channel gains and not utilizing the remaining antennas. The application to point-to-point MIMO, where the receive antennas are colocated, is studied in [84]. The multi-user MIMO system is introduced but not investigated in detail. The work in [53] assumes a general form of the precoder as maximum ratio transmission (MRT) and zero-forcing (ZF),² and finds a posteriori the quantities (number of BS antennas, number of users, etc.) that minimize (3.2). They optimize the number of utilized antennas in a low-traffic regime, as in high traffic their choice is to always use all the BS antennas to guarantee the coverage in the cell. The solution (when considering only the power consumed by the PAs) is to activate all the BS antennas, in contrast to [38, 85]. The reason is that [38, 85] consider the minimization of (3.2) from the precoding problem definition, not assuming any particular precoding scheme. When other BS components than the PA are included in the power model, the optimal number of BS antennas to be utilized decreases.

Similarly to [53], several studies have investigated the whole BS consumption for energy-efficient massive MIMO. A holistic power model for massive MIMO is developed in [21], where the total consumed power is the sum of the contributions from the PAs and from the BS circuits (analog and digital), plus a fixed consumption term. The circuit power consumption considers the radio frequency transceivers, the baseband processing, the backhauling, the cooling, and the power supply, and except for some choices of processing schemes, scales linearly with M [21]. In [83], the authors jointly minimize the total transmit power and the circuit power consumption for MRT and ZF precoders. Different from [53], they consider different traffic situations and they show that the optimal number of antennas to use increases as a function of the traffic. Recently, dynamic antenna and power adaptation has been studied based on 3rd Generation Partnership Project (3GPP)-defined power model [86].

Remarkably, the multicarrier scenario has not been widely investigated in the literature. Considering a fixed PA efficiency, (3.1) allows one to decouple

²The MRT precoder maximizes the signal-to-noise ratio (SNR) at the receiver by discarding the inter-user interference (IUI), while the ZF precoder forces the IUI to zero. Depending on the regime (noise-limited or interference-limited), one or the other gives better performance [48].

and optimally solve the precoding problem per subcarrier. When optimizing model (3.2), instead, the problem is coupled between subcarriers as the PA efficiency depends on the total power it handles. The work [53] considers (3.1) assuming uniform power among the antennas, given that each antenna serves many subcarriers. Nonetheless, the form of the precoder is fixed a priori and this somewhat limits the analysis. The authors in [84] consider point-to-point MIMO and optimize the sum capacity under consumed power and per-antenna power constraints. The power model takes into account the multicarrier structure of the system. Their conclusion is in line with one we will present (all the BS antennas are equally good, hence random antenna selection performs well), but the analysis of the massive MIMO scenario and further elaborations on how to optimize the parameters of the system are not given.

Contributions

In this chapter, we propose massive MIMO precoders and number of utilized BS antennas that minimize consumed power under zero-IUI constraint in a multi-user multicarrier system. We analyze as a special case a single-user system, and we differentiate between *wideband* and *narrowband* systems. By assuming a fixed subcarrier spacing and the channel to be frequency flat at each subcarrier, we define a system to be narrowband when it utilizes only one subcarrier, and to be wideband when more than one subcarrier is utilized. We show that in the case with many subcarriers and considering the whole BS consumption, the precoding design problems boil down to resorting to the conventional precoder with a uniform power allocated among a number of utilized antennas, and finding the optimal number of antennas to utilize. In particular,

- We derive closed-form solutions to the problem of minimizing PA consumption in wideband systems (Section 3.2), where the transmit powers allocated to the BS antennas are computed via an iterative fixed-point algorithm. The focus is on the low-power-load scenario, where the antenna output powers are lower than the maximal output power.
- We characterize analytically the solutions to the same problem in narrowband systems and under low power load (Section 3.3), and we show that the single-user solution does not require the iterative algorithm to be retrieved.
- We analyze the asymptotic behavior of wideband systems (Section 3.4), solving the problem of minimizing the BS consumption under per-antenna power constraints. We derive the optimal number of antennas to utilize, which depends on the traffic load.

In this chapter, we define as *traffic load* the number of active users K , to differentiate it with the physical resource load defined in the previous chapter. In Section 3.5, the performance of the proposed solutions is evaluated numerically, and their computational complexity is discussed.

3.1 System Model and Assumptions

In this section, we briefly recall and adapt the DL scenario described in Section 2.1.1, the power model of Section 2.2, and give the main assumptions used in one or more sections of this chapter. We analyze a MIMO-OFDM BS serving single-antenna users using space-division multiple access (SDMA). The BS implements fully-digital beamforming using $M = M_{\text{RF}}$ antennas and RF chains and $M_{\text{a}} = M_{\text{RF,a}}$ active antennas and RF chains.

Remark: In this and the following chapters, the focus is on sub-6 GHz BSs that implement fully-digital beamforming. While hybrid analog-digital beamforming can be used in sub-6 GHz BSs, its application is somewhat limited by the typical number of antennas utilized in arrays at this band (e.g., 64 antennas, limiting the hybrid-beamforming configurations) compared to higher frequency bands (e.g., 1024 antennas). Furthermore, the optimization of phase shifts and number of RF chains for energy reduction would introduce another level of complexity in the analysis, and is left out of scope.

To lighten the notation, we define $Q \triangleq Q_{\text{DL}}$ as the number of data subcarriers, $K \triangleq K_{\text{DL}}$ as the number of active users, $\mathbf{H} \triangleq \mathbf{H}_{\text{DL}}$ as the channel matrix, $\mathbf{W} \triangleq \mathbf{W}_{\text{dig}}$ as the digital precoding matrix, and $P_{\text{T}} \triangleq P_{\text{T,DL}}$ as the total transmit power. We express the dependency on subcarrier q in square brackets, where $q \in \{1, \dots, Q\}$ is one data subcarrier. The symbols of the users for the subcarrier q are denoted by $\mathbf{s}_{\text{DL}}[q]$, where we consider equal power allocation among users giving $p_{k,\text{DL}} = 1 \forall k$. Indicating with $\mathbf{W}[q]$ the precoding matrix for the subcarrier q , the precoded signal in the frequency domain for the subcarrier q is given by

$$\mathbf{x}[q] = \mathbf{W}[q]\mathbf{s}_{\text{DL}}[q]. \quad (3.3)$$

The received signal for the subcarrier q is therefore expressed as

$$\mathbf{r}_{\text{DL}}[q] = \mathbf{H}[q]\mathbf{W}[q]\mathbf{s}_{\text{DL}}[q] + \boldsymbol{\nu}_{\text{DL}}[q] \quad (3.4)$$

where $\boldsymbol{\nu}_{\text{DL}} \sim \mathcal{CN}(\mathbf{0}_K, \sigma_{\nu}^2 \mathbf{I}_K)$ represents the thermal noise, which is assumed independently and identically distributed (i.i.d.) among subcarriers and users. We express the single entries of the precoded signal vector, channel matrix, and precoding matrix for one subcarrier as

$$x_m[q] = [\mathbf{x}[q]]_m, \quad h_{k,m}[q] = [\mathbf{H}[q]]_{k,m}, \quad w_{m,k}[q] = [\mathbf{W}[q]]_{m,k} \quad (3.5)$$

where, e.g., $h_{k,m}[q]$ is the channel coefficient between user k and antenna m for the subcarrier q . The above description represents a wideband multi-user scenario. The particular cases of narrowband and single-user systems are obtained for $Q = 1$ and $K = 1$, respectively. We recall that we assume a fixed subcarrier spacing, therefore the system bandwidth is inversely proportional to Q (hence the term ‘narrowband’ to describe the case $Q = 1$). The interest in analyzing a narrowband system is motivated, e.g., by the inclusion of services as narrowband IoT (NB-IoT) in the 5G standard [32].

Remark: From the perspective of the 5G standard, the minimal allocation unit is the so-called resource block (RB), corresponding to 12 consecutive subcarriers [32]. In the following, we do not consider RB allocation and let Q take any value. This allows us to study the behavior of particular cases such as $Q = 1$ and draw insights. If RB allocation is taken into account, the minimal granularity is $Q = 12$.

Considering the transmit signal in equation (3.3) and the inverse fast Fourier transform (IFFT) for OFDM modulation to be unitary, the transmit power of antenna m equals

$$P_{\tau,m} = \frac{1}{Q} \sum_{q=1}^Q \mathbb{E} \left\{ |x_m[q]|^2 \right\} = \frac{1}{Q} \sum_{k=1}^K \sum_{q=1}^Q |w_{m,k}[q]|^2. \quad (3.6)$$

Consequently, the total transmit power of the BS is given by

$$P_{\tau} = \frac{1}{Q} \sum_{m=1}^M \sum_{q=1}^Q \mathbb{E} \left\{ |x_m[q]|^2 \right\} = \frac{1}{Q} \sum_{m=1}^M \sum_{k=1}^K \sum_{q=1}^Q |w_{m,k}[q]|^2. \quad (3.7)$$

Under the PA power model in equation (2.21) explained in Section 2.2.1, with $\alpha = \frac{1}{2}$ and $\xi_{\text{PA}} = 0$ representing class B PAs, the total power consumed by the PAs in working mode during data transmission is given by

$$P_{\text{PA}} = \underbrace{\frac{P_{\text{max}}^{\frac{1}{2}}}{\eta_{\text{PA,max}}}}_{\gamma} \sum_{m=1}^M \left(\frac{1}{Q} \sum_{k=1}^K \sum_{q=1}^Q |w_{m,k}[q]|^2 \right)^{\frac{1}{2}}. \quad (3.8)$$

We recall that P_{max} is the maximal PA output power, which takes into account a back-off from the saturation power, and $\eta_{\text{PA,max}}$ is the maximal PA efficiency achieved when $P_{\tau,m} = P_{\text{max}}$.

Remark: Reducing the back-off could allow the PA to achieve higher efficiencies (as the it enters the nonlinear or saturation regime), but would significantly complicate the signal model by introducing nonlinear distortions in the

transmitted signal (3.3). This would render the problems at hand much more challenging to be solved in closed form. Depending on the design criteria, one could choose to force the distortion to zero, thus maintaining the target QoS, or accept a certain QoS degradation by leaving the distortion present. In the second case, an initial approach could use the solutions of this chapter as they are, and quantify the tradeoff between potentially reducing consumption (by operating the PAs at higher efficiencies) and potentially reducing the delivered QoS. Alternatively, the precoder proposed in [87] (for a single-user system and third-order nonlinear distortion) allocates a certain number of antennas to cancel the distortion. If the same strategy proves to be applicable to a multi-user system as the one studied here, we could have a reduction of the number of non-activated antennas (as a subset is allocated to cancel the distortion), but larger PA efficiencies that can be achieved by the activated antennas. Again, this would lead to a tradeoff.

Expression (3.8) only considers the PAs contribution, while other BS components also consume power as widely discussed in Chapter 2. By considering a fixed DL load ℓ_{DL} as we look at a single time slot, we can rewrite the BS consumption in (2.19) and subsequent expressions as

$$P_{\text{cons}} = P_{\text{PA}} + \mathcal{C}M_{\text{a}} + P_{\text{fixed}} \quad (3.9)$$

where $\mathcal{C}$ is a positive constant and P_{fixed} is a positive fixed consumption.³ The latter accounts for all the terms that are not depending on the transmit powers or on number of active antennas, while the term $\mathcal{C}M_{\text{a}}$ includes all the terms in the power consumption of analog and digital processing that scale with $M_{\text{a}} = M_{\text{RF,a}}$, cf. expressions (2.26) and (2.33) of Section 2.2. The terms of the average PA consumption over the frame ($\overline{P_{\text{PA}}}$, given in equation (2.23)) that do not depend on the transmit power are also part of $\mathcal{C}M_{\text{a}}$ or P_{fixed} (e.g., the idle-mode consumption of the M_{a} active PAs is part of the constant $\mathcal{C}$).

Some assumptions are made in certain sections of this chapter:

(As3.1): Low-power-load scenario, $P_{\text{max}} \rightarrow \infty$. We point out that, by letting P_{max} go to infinity, we practically mean that the constraints on the maximal power per antenna are not binding, as it usually happens in a scenario with few users or low path loss. Indeed, in this scenario we expect that $P_{\text{T},m} \ll P_{\text{max}} \forall m$.

(As3.2): Asymptotic wideband regime, $Q \rightarrow \infty$. In the following, we will show that the results obtained by letting Q go to infinity approximate well systems with finite and realistic values of Q (e.g., $Q = 256$ subcarriers).

³The load dependency of the components' consumption does not play a role in this chapter. In Chapter 5, we will study the optimization of resources over multiple time slots, and the load dependency will be considered.

(**As3.3**): Uncorrelated Rayleigh fading, giving $\mathbf{H}[q] = \mathbf{D}_\beta^{\frac{1}{2}} \mathbf{G}[q]$, where $\mathbf{H}[q]$ is the channel matrix at subcarrier q , $\mathbf{D}_\beta = \text{diag}(\beta_1, \dots, \beta_K)$ is the large-scale fading matrix and β_k is the large-scale fading coefficient of user k . The matrix $\mathbf{G}[q]$ models the small-scale fading and is composed of i.i.d. elements, where the single entry is $[\mathbf{G}[q]]_{k,m} = g_{k,m}[q] \sim \mathcal{CN}(0, 1)$. The i.i.d. assumption is made over space and not over frequency, thus the channels of different subcarriers can be correlated. In space (i.e., among BS antennas), several experimental studies have demonstrated that the i.i.d. assumption mostly does not hold [88, 89]. Still, this assumption is often used in the analysis of multi-antenna systems to provide theoretical foundations and results [90].

3.2 Precoders Design for Wideband System

This section covers the design of precoders that minimize the PA consumption subject to SNR and per-antenna power constraints in a wideband system (based on our terminology, with more than 1 subcarrier). The per-user SNR constraints are expressed as ZF constraints that force the IUI to zero and scale the desired signal amplitude at the receivers. Given the use of ZF constraints, the SNR coincides with the SINR.

3.2.1 Conventional Transmit Power Solution

Let us denote by ψ_k/Q the target (per-subcarrier) DL SNR for the user k . This affects the linear ZF constraints (the first constraints of the problem below), where the factor $\frac{1}{Q}$ is due to the transmit power normalization in equation (3.6). These per-subcarrier ZF constraints can be converted into rate constraints for the users, e.g., an achievable rate per Hz for the user k is $Q \log_2(1 + \frac{\psi_k}{Q})$ bits per second per Hz. This rate per Hz can be further converted into a rate in bits per second by multiplying by the subcarrier spacing and by the penalty term accounting for reference signaling and uplink duration. Considering both ZF constraints and per-antenna power constraints, the precoder minimizing the total transmit power is the solution to the following problem

$$\begin{aligned}
 & \underset{\{w_{m,k}[q]\}}{\text{minimize}} & P_\tau &= \sum_{m=1}^M P_{\tau,m} \\
 & \text{subject to} & \mathbf{H}[q] \mathbf{W}[q] &= \mathbf{D}_\beta^{\frac{1}{2}} \sigma_\nu, \quad q = 1, \dots, Q \\
 & & P_{\tau,m} &\leq P_{\max}, \quad m = 1, \dots, M
 \end{aligned} \tag{3.10}$$

where $P_{\tau,m}$ is the transmit power at antenna m given by (3.6), $\mathbf{H}[q] \in \mathbb{C}^{K \times M}$ and $\mathbf{W}[q] \in \mathbb{C}^{M \times K}$ are the channel and precoding matrices for the subcarrier q , respectively, $\mathbf{D}_\psi = \text{diag}(\psi_1, \dots, \psi_K)$ contains the target SNRs of the users, and σ_ν is the noise standard deviation.⁴ The first constraints are the Q ZF constraints, while the second constraints are the M maximal power constraints. In the majority of the next sections, we will rely on (As3.1) and relax the maximal power constraints.

The above problem is convex and differentiable. It can be solved, e.g., with the Lagrange multipliers method. Under (As3.1), the solution at the subcarrier q is given by

$$\mathbf{W}[q] = \underbrace{\mathbf{H}[q]^H (\mathbf{H}[q] \mathbf{H}[q]^H)^{-1}}_{\text{pseudo-inverse of } \mathbf{H}[q]} \underbrace{\mathbf{D}_\psi^{\frac{1}{2}} \sigma_\nu}_{\text{amplitude scaling}} \quad (3.11)$$

which corresponds to the per-subcarrier ZF [91]. Note that here and in the following, we consider that instantaneous channel state information (CSI) is available at the BS for the computation of the precoding matrices.

3.2.2 PA Consumed Power Solution

The precoder minimizing the total power consumed by the PAs while in working mode during data transmission, which differs from the PAs consumption while in working mode during reference signal transmission and in non-working mode (e.g., idle), is found by solving the following problem

$$\begin{aligned} \underset{\{w_{m,k}[q]\}}{\text{minimize}} \quad & P_{\text{PA}} = \gamma \sum_{m=1}^M P_{\tau,m}^{\frac{1}{2}} \\ \text{subject to} \quad & \mathbf{H}[q] \mathbf{W}[q] = \mathbf{D}_\psi^{\frac{1}{2}} \sigma_\nu, \quad q = 1, \dots, Q \\ & P_{\tau,m} \leq P_{\max}, \quad m = 1, \dots, M. \end{aligned} \quad (3.12)$$

As previously assumed to derive the conventional ZF, we consider (As3.1), thus neglecting the constraints on $P_{\tau,m}$. With this assumption, the transmit power at the individual antennas could become larger than the maximum allowed.

⁴One could consider a q -dependent target SNR per user, $\psi_k[q]$, and compute the rate of each user by summing over the subcarriers. This would give greater flexibility in following the instantaneous variations in frequency of the channel, but would introduce another level of complexity in determining the individual $\{\psi_k[q]\}$ starting from the user target SNR ψ_k . To simplify the analysis, we consider the SNR to be independent of q . This is further motivated by the fact that we will evaluate the proposed algorithms in terms of average power consumption on i.i.d. channels in frequency.

In practice, one can apply a rescaling of the rows of the precoding matrices corresponding to the antennas that exceed the maximum allowed power value (i.e., projecting the obtained unbounded solution onto the feasible set). This operation has the unavoidable effect of reducing the SNR delivered to the users. We focus here on the PA consumption because (i) as discussed, the PA has been and continues to be one of the main drivers of BS consumption [14, 15], and (ii) the addition of other consumption terms would render the problem less tractable at this stage. However, in Section 3.4 we will show how the asymptotic analysis allows us to consider the BS consumption in equation (3.8) while remaining very accurate in situations with a finite and realistic number of subcarriers (e.g., $Q = 256$ subcarriers).

The solution to the above problem can be found, e.g., by applying the Lagrange multipliers method, and is given in Theorem 3.1.⁵

Theorem 3.1. *Under (As3.1), the solution to problem (3.12) for the subcarrier q is given by*

$$\mathbf{W}[q] = \mathbf{D}_{\mathbf{p}_\tau}^{\frac{1}{2}} \mathbf{H}[q]^H \left(\mathbf{H}[q] \mathbf{D}_{\mathbf{p}_\tau}^{\frac{1}{2}} \mathbf{H}[q]^H \right)^{-1} \mathbf{D}_{\psi}^{\frac{1}{2}} \sigma_\nu \quad (3.13)$$

where $\mathbf{D}_{\mathbf{p}_\tau} = \text{diag}(P_{\tau,1}, \dots, P_{\tau,M}) = \text{diag}(\mathbf{p}_\tau)$ is the diagonal matrix containing the transmit powers of the BS antennas, which are found by solving the fixed-point equations⁶

$$P_{\tau,m} = \frac{1}{Q} \sum_{k=1}^K \sum_{q=1}^Q \left| \left[\mathbf{D}_{\mathbf{p}_\tau}^{\frac{1}{2}} \mathbf{H}[q]^H \left(\mathbf{H}[q] \mathbf{D}_{\mathbf{p}_\tau}^{\frac{1}{2}} \mathbf{H}[q]^H \right)^{-1} \mathbf{D}_{\psi}^{\frac{1}{2}} \sigma_\nu \right]_{m,k} \right|^2 \quad (3.14)$$

for $m = 1, \dots, M$.

Proof. See Appendix A.1. □

The system of fixed-point equations can be solved through the fixed-point iteration method, which is given in Algorithm 3.1. Once $\mathbf{D}_{\mathbf{p}_\tau}$ is known, it can be substituted in equation (3.13) to find the precoding matrices. Note that even though the matrix $\mathbf{D}_{\mathbf{p}_\tau}$ scales the precoding matrix in equation (3.13), its entries $\{P_{\tau,m}\}$ remains unbounded; a possible solution is to rescale the precoding matrix such that the transmit powers are within the allowed range.

Let us now analyze the single-user case ($K = 1$).

⁵The choice of $\alpha = \frac{1}{2}$ in the PA power model implies convexity of the cost function, which can be expressed as an $L_{2,1}$ -norm. Generalizations to the case $\alpha \in]0.5, 1]$ (that preserves convexity) are possible but are not treated here.

⁶Given a function $\mathbf{f}: \mathcal{X} \mapsto \mathcal{X}$, $\mathbf{x} \in \mathcal{X}$ is a fixed point of $\mathbf{f}$ if it satisfies $\mathbf{x} = \mathbf{f}(\mathbf{x})$.

Algorithm 3.1 Fixed-point method to retrieve the powers of the BS antennas

```

1: Input:  $\varepsilon, I_{\max}$  ▷ Tolerance and maximum number of iterations
2:  $i \leftarrow 0$ 
3: for  $m \in \{1, \dots, M\}$  do ▷ Initialize powers of the BS antennas
4:    $P_{\tau,m}^{(0)} \leftarrow 1$ 
5:    $P_{\tau,m}^{(-1)} \leftarrow \infty$ 
6: end for
7: while  $i < I_{\max}$  and  $\max_m \{|P_{\tau,m}^{(i)} - P_{\tau,m}^{(i-1)}|\} > \varepsilon$  do
8:    $\mathbf{D}_{\mathbf{p}_T} \leftarrow \text{diag}(P_{\tau,1}^{(i)}, \dots, P_{\tau,M}^{(i)})$ 
9:   for  $m \in \{1, \dots, M\}$  do ▷ Update powers of the BS antennas
10:     $P_{\tau,m}^{(i+1)} \leftarrow (3.14)$ 
11:   end for
12:    $i \leftarrow i + 1$ 
13: end while
14: Output:  $\mathbf{D}_{\mathbf{p}_T} = \text{diag}(P_{\tau,1}^{(i+1)}, \dots, P_{\tau,M}^{(i+1)})$ 
    
```

Corollary 3.1. Under (As3.1), the particularization of Theorem 3.1 when $K = 1$ user gives the solution for the subcarrier q

$$\mathbf{w}[q] = \sigma_\nu \psi^{\frac{1}{2}} \frac{1}{\sum_{m'=1}^M |h_{m'}[q]|^2 P_{\tau,m'}^{\frac{1}{2}}} \mathbf{D}_{\mathbf{p}_T}^{\frac{1}{2}} \mathbf{h}[q]^* \quad (3.15)$$

where $\mathbf{w}[q] \in \mathbb{C}^{M \times 1}$ and $\mathbf{h}[q] \in \mathbb{C}^{M \times 1}$ are the precoding and channel vectors at the subcarrier q , respectively. The fixed-point equations in the powers per antenna are given by

$$P_{\tau,m} = \frac{\sigma_\nu^2 \psi}{Q} P_{\tau,m} \sum_{q=1}^Q \frac{|h_m[q]|^2}{(\sum_{m'=1}^M |h_{m'}[q]|^2 P_{\tau,m'}^{\frac{1}{2}})^2} \quad (3.16)$$

for $m = 1, \dots, M$.

In line-of-sight (LoS) channels the problem simplifies, allowing us to draw additional insights.

Corollary 3.2. Under (As3.1), in pure LoS channels (i.e., characterized by $|h_{m,q}| = 1 \forall m, q$), and considering at least one $P_{\tau,m}$ to be different from zero, the fixed-point equation (3.16) reduces to

$$\sum_{m=1}^M P_{\tau,m}^{\frac{1}{2}} = \sigma_\nu \psi^{\frac{1}{2}}. \quad (3.17)$$

Every transmit power allocation satisfying the condition (3.17) is, therefore, an optimal solution to the wideband single-user problem in the low-power-load

case, and thus also the narrowband single-user solution. Remarkably, one now has freedom in the precoder design, and possible choices include:

- Allocate all the power to antenna m , which gives $P_{\tau,m} = \sigma_\nu^2 \psi$.
- Allocate the power uniformly across antennas, $P_{\tau,m} = (\sigma_\nu^2 \psi) / M^2 \forall m$.
- Randomly allocate the power, as long as equation (3.17) is fulfilled.

In pure LoS channels—where all the antennas share the same channel gain—the multiple-antenna gain is entirely traded off with the increased PA efficiency. By considering instead the whole BS consumption (as done in Section 3.4.2), however, one would prefer to activate the minimum number of antennas such that the RF chains of the non-activated antennas can be turned off to save power. In other words, the multiple-antenna gain is lost in circuit power.

3.3 Narrowband System

3.3.1 Conventional Transmit Power Solution

When $Q = 1$ subcarrier, the system reduces to a narrowband one. The subcarrier index q can then be discarded. The conventional solution, minimizing the total transmit power, corresponds to

$$\mathbf{W} = \mathbf{H}^H (\mathbf{H}\mathbf{H}^H)^{-1} \mathbf{D}_{\psi}^{\frac{1}{2}} \sigma_\nu \quad (3.18)$$

where $\mathbf{D}_{\psi} = \text{diag}(\psi_1, \dots, \psi_K)$, while $\mathbf{H} \in \mathbb{C}^{K \times M}$ and $\mathbf{W} \in \mathbb{C}^{M \times K}$ are the single-carrier channel and precoding matrices, respectively.

3.3.2 PA Consumed Power Solution

Corollary 3.3. *Under (As3.1), it directly follows from Theorem 3.1 that the precoding matrix for a narrowband system is*

$$\mathbf{W} = \mathbf{D}_{\mathbf{p}_\tau}^{\frac{1}{2}} \mathbf{H}^H \left(\mathbf{H} \mathbf{D}_{\mathbf{p}_\tau}^{\frac{1}{2}} \mathbf{H}^H \right)^{-1} \mathbf{D}_{\psi}^{\frac{1}{2}} \sigma_\nu. \quad (3.19)$$

The fixed-point equation for the antenna m is given by

$$P_{\tau,m} = \sum_{k=1}^K \left| \left[\mathbf{D}_{\mathbf{p}_\tau}^{\frac{1}{2}} \mathbf{H}^H \left(\mathbf{H} \mathbf{D}_{\mathbf{p}_\tau}^{\frac{1}{2}} \mathbf{H}^H \right)^{-1} \mathbf{D}_{\psi}^{\frac{1}{2}} \sigma_\nu \right]_{m,k} \right|^2. \quad (3.20)$$

The solution can be retrieved in the same way as for the wideband system, by first computing the powers at the antennas and then substituting them in the precoder expression (3.19).

In the following, we discuss the single-user case ($K = 1$).

Corollary 3.4. *Under (As3.1), by assuming different channel gains among the antennas and by defining $\hat{m} = \arg \max_m |h_m|$, the particularization of Corollary 3.1 when $Q = 1$ gives the single-user solution*

$$w_m = \begin{cases} \sigma_\nu \psi^{\frac{1}{2}} \frac{1}{|h_m|^2} h_m^* & \text{if } m = \hat{m} \\ 0 & \text{otherwise.} \end{cases} \quad (3.21)$$

All the power is therefore allocated to the antenna with the largest channel gain.

Proof. From Corollary 3.1, when $Q = 1$ we obtain

$$\mathbf{w} = \sigma_\nu \psi^{\frac{1}{2}} \frac{1}{\sum_{m'=1}^M |h_{m'}|^2 P_{\tau, m'}^{\frac{1}{2}}} \mathbf{D}_{\mathbf{P}_\tau}^{\frac{1}{2}} \mathbf{h}^* \quad (3.22)$$

where $\mathbf{w} \in \mathbb{C}^{M \times 1}$ and $\mathbf{h} \in \mathbb{C}^{M \times 1}$ are the single-carrier precoding and channel vectors, respectively. The fixed-point equation for the antenna m is given by

$$\sum_{m'=1}^M |h_{m'}|^2 P_{\tau, m'}^{\frac{1}{2}} = \sigma_\nu \psi^{\frac{1}{2}} |h_m|. \quad (3.23)$$

For all the activated antennas, there is a fixed-point equation. However, when the channel gains $|h_m|$ are different, (3.23) cannot be solved since the left-hand side does not depend on m , while the right-hand side depends on m . This actually shows that the optimal solution corresponds to using only one antenna, the one with the strongest channel gain.⁷ $\square$

The intuition behind the above solution is that the maximal per-antenna power constraint is not considered here, hence from the PA perspective it is best to activate a single PA with maximal efficiency and completely deactivate the remaining ones. If several antennas share the same maximal channel gains, they can all be activated while still solving equation (3.23). Equation (3.23) reduces to equation (3.17) among the antennas sharing the same maximal channel gain. In pure LoS channels, we obtain again (3.17), this time among all the antennas. In both cases, one has freedom of choice in the precoder design.

⁷When only one antenna is activated, the consumed power equals $(\sigma_\nu \psi^{\frac{1}{2}})/|h_m|$, which is minimized when using the antenna with the highest channel gain.

When the per-antenna power constraints are active, instead, the optimal strategy is to progressively saturate the antennas with the highest channel gains until the QoS is achieved (i.e., a power-pouring solution) [38]. Note that reference [38] does not elaborate on the multi-user results, which are discussed in this chapter.

3.4 Asymptotic Wideband System

We now assume a large value of Q , which represents well fourth-generation (4G) and 5G systems given the use of a number of active subcarriers from 72 to 1320 and from 132 to 3300, respectively [32]. The asymptotic results are validated numerically, and we show that they prove to be accurate even for a finite and relatively low number of subcarriers. While the results presented before apply for any channel, the closed-form expressions of the consumed powers in this section are obtained for the i.i.d. Rayleigh fading channel given in (As3.3). We will show that transmit power allocation tend to be uniform among antennas, and that it can be expressed in closed form. In a different channel, it is reasonable to expect that the power will not be uniformly allocated among antennas. In Chapter 4, we will consider a Kronecker channel model, with correlation among BS antennas, and show that more power is allocated to the least correlated antennas. However, numerical simulations will show that the uniform allocation still provides a good approximation, with negligible performance loss in the considered set of simulations.

3.4.1 PA Consumed Power

The following theorem gives the asymptotic expression of the PAs consumed power induced by the precoder (3.13). This characterization is possible because, in the asymptotic wideband regime, all the BS antennas tend to be allocated the same power under the assumption of i.i.d. channels.

Theorem 3.2. *Under (As3.1) – (As3.3),⁸ the transmit power allocated to each antenna by the precoder (3.13) converges to*

$$P_{\tau,m} \rightarrow \widetilde{P}_{\tau} = \frac{1}{M(M-K)} \text{tr} \left(\mathbf{D}_{\beta}^{-1} \mathbf{D}_{\psi} \sigma_{\nu}^2 \right) \quad (3.24)$$

implying that the total power consumed by the PAs satisfies

$$P_{\text{PA}} \rightarrow \widetilde{P}_{\text{PA}} = \gamma \left(\frac{M}{M-K} \text{tr} \left(\mathbf{D}_{\beta}^{-1} \mathbf{D}_{\psi} \sigma_{\nu}^2 \right) \right)^{\frac{1}{2}}. \quad (3.25)$$

⁸With (As1) – (AsN), we indicate the assumptions from 1 to N.

The per-subcarrier ZF precoder is therefore found

$$\mathbf{W}[q] = \mathbf{H}[q]^H (\mathbf{H}[q]\mathbf{H}[q]^H)^{-1} \mathbf{D}_\psi^{\frac{1}{2}} \sigma_\nu. \quad (3.26)$$

Proof. See Appendix A.2. $\square$

Looking at expression (3.25), which is plotted in blue in Fig. 3.3b, the power consumed by the PAs is a monotonically decreasing function of M : activating more antennas is then always beneficial. However, the marginal gain of activating more antennas decreases as M increases since $\frac{M}{M-K}$ converges to a unit constant.

3.4.2 BS Consumed Power

Considering the power consumption of the whole BS, the optimal precoder does not necessarily activate all the antennas. Indeed, the power consumed by the BS circuits (DSP and RF chains) induces a penalty on the number of active antennas M_a . We consider the BS consumption in equation (3.9) as objective function and the precoders $\{w_{m,k}[q]\}$ and number of active antennas M_a as optimization variables. The general problem, including the per-antenna power constraints, consists of solving

$$\begin{aligned} & \underset{\substack{\{w_{m,k}[q]\}, M_a \\ m=1, \dots, M_a}}{\text{minimize}} & P_{\text{cons}} &= \gamma \sum_{m=1}^{M_a} P_{\tau,m}^{\frac{1}{2}} + \mathcal{C}M_a + P_{\text{fixed}} \\ & \text{subject to} & \mathbf{H}[q]\mathbf{W}[q] &= \mathbf{D}_\psi^{\frac{1}{2}} \sigma_\nu, \quad q = 1, \dots, Q \\ & & P_{\tau,m} &\leq P_{\text{max}}, \quad m = 1, \dots, M \\ & & M_a &\leq M. \end{aligned} \quad (3.27)$$

Without loss of generality, one can first optimize with respect to the precoding coefficients considering a generic number of active antennas M_a , and then find the optimal integer number of active antennas.

Let us consider the power allocation defined in Theorem 3.2 among the M_a active antennas. By combining the BS consumption model in (3.9) with the asymptotic PAs consumption in (3.25), the asymptotic BS consumption can be rewritten as

$$P_{\text{cons}} \rightarrow \widetilde{P_{\text{cons}}} = \gamma \left(\frac{M_a}{M_a - K} \text{tr} \left(\mathbf{D}_\beta^{-1} \mathbf{D}_\psi \sigma_\nu^2 \right) \right)^{\frac{1}{2}} + \mathcal{C}M_a + P_{\text{fixed}}. \quad (3.28)$$

The above quantity is the solution, under **(As3.1)** – **(As3.3)**, to the minimization problem (3.27) with respect to the precoding coefficients and considering a fixed number of active antennas M_a . The corresponding precoder is a per-subcarrier ZF precoder among the M_a active antennas.

We can now proceed to optimize the number of active antennas M_a .

Optimal M_a with No Power Constraints

In this case, we can directly minimize expression (3.28) and check that the solution is within the allowed bounds.

Lemma 3.1. *Under **(As3.1)** – **(As3.3)**, the optimal integer number of active antennas $\{M_a^* \in \mathbb{N} \mid K + 1 \leq M_a^* \leq M\}$, minimizing $\widetilde{P}_{\text{cons}}$ is given by*

$$M_a^* = \left\lfloor \min \left\{ \max \left\{ \tilde{M}_a, K + 1 \right\}, M \right\} \right\rfloor \quad (3.29)$$

where $\tilde{M}_a = x$, and x is the solution to the quartic equation

$$x(x - K)^3 - \frac{\gamma \left(\text{tr} \left(\mathbf{D}_\beta^{-1} \mathbf{D}_\psi \sigma_\nu^2 \right) \right)^{\frac{1}{2}} K}{2\mathcal{C}} = 0 \quad (3.30)$$

for $x > K$. Closed-form solutions to equation (3.30) can be found via the method described in [92].

Proof. See Appendix A.3. □

Two corner cases can occur at this stage: $M_a^* = K + 1$ (circuits power-limited regime) and $M_a^* = M$ (PAs power-limited regime). Note that the ceil-floor operator $\lfloor \cdot \rfloor$ is evaluated after having checked the bounds. Indeed, if M_a^* is equal to one of the two extremes, the ceil-floor operator becomes trivial.

Optimal M_a with Power Constraints

Up to this point, there is no guarantee that the antenna powers by using M_a^* antennas satisfy the per-antenna power constraints. One has to solve problem (3.27) by relaxing **(As1.1)**. In Appendix A.4, we show that under **(As3.2)** and **(As3.3)**, the per-antenna power constraints reduce to $\widetilde{P}_\tau \leq P_{\text{max}}$, where $\widetilde{P}_\tau$ is the asymptotic transmit power at each active antenna given in equation (3.24). It is therefore sufficient to compute the continuous number

of antennas that gives exactly $P_{\max}$ as transmit power, and then apply the ceiling operator to prevent the constraint from being violated. This value is then included as a lower bound on the final solution.

In the following, we assume that problem (3.27) is feasible, i.e., activating all antennas is sufficient to meet the QoS constraints. This is expressed, using expression (3.24), by the following condition:

(As3.4): $\frac{1}{M(M-K)} \text{tr}(\mathbf{D}_{\beta}^{-1} \mathbf{D}_{\psi} \sigma_{\nu}^2) \leq P_{\max}$, which ensures that activating $M_a = M$ antennas does not violate the per-antenna power constraints.

Theorem 3.3. *Under (As3.2) – (As3.4), the optimal integer number of active antennas solving problem (3.27) is given by*

$$M_a^{\dagger} = \left\lceil \min \left\{ \max \left\{ \tilde{M}_a, K + 1, \hat{M}_a \right\}, M \right\} \right\rceil \quad (3.31)$$

where $\tilde{M}_a = x$, x is the solution to equation (3.30), and

$$\hat{M}_a = \left\lceil \frac{1}{2} \left(K + \left(K^2 + \frac{4 \text{tr}(\mathbf{D}_{\beta}^{-1} \mathbf{D}_{\psi} \sigma_{\nu}^2)}{P_{\max}} \right)^{\frac{1}{2}} \right) \right\rceil. \quad (3.32)$$

The quantity $\hat{M}_a$ is the minimal integer number of active antennas required to satisfy the per-antenna power constraint and achieve the users' QoS.

Proof. See Appendix A.4. □

We emphasize that all the involved quantities depend only on the large-scale fading coefficients, i.e., the second-order statistics of the channel. Therefore, instantaneous CSI is not required, and $M_a^{\dagger}$ does not need to be computed every channel coherence time. As in Lemma 3.1, the ceil-floor operator is evaluated at last to avoid unnecessary operations. Algorithm 3.2 illustrates the different steps that are performed to retrieve $M_a^{\dagger}$.

3.5 Performance Evaluation

In this section, we evaluate the proposed solutions via numerical experiments. In the experiments, the large-scale fading coefficient (in dB) for the user k is computed as [21]

$$10 \log_{10}(\beta_k) = -35.3 - 37.6 \log_{10}(u_k) \quad [\text{dB}] \quad (3.33)$$

Algorithm 3.2 Find the optimal number of active BS antennas in asymptotic wideband system

```

1: Input:  $\psi_1, \dots, \psi_K, \beta_1, \dots, \beta_K$   $\triangleright$  Set target SNRs and large-scale fading
2:  $\hat{M}_a \leftarrow (3.32)$   $\triangleright$  Minimal number of BS antennas required
3:  $\triangleright$  by per-antenna power constraints
4:  $\tilde{M}_a \leftarrow$  Solution of (3.30)  $\triangleright$  Number of BS antennas in  $[K, \infty)$  minimizing
5:  $\triangleright$  asymptotic BS consumption
6:  $y \leftarrow \max \{ \hat{M}_a, \tilde{M}_a \}$ 
7: if  $y \leq K + 1$  then
8:    $M_a^\dagger \leftarrow K + 1$   $\triangleright$  Use as few antennas as possible
9: else if  $y \geq M$  then
10:   $M_a^\dagger \leftarrow M$   $\triangleright$  Use as many antennas as possible
11: else
12:   $M_a^\dagger \leftarrow \lfloor y \rfloor$   $\triangleright$  Compute ceil-floor operation on
13:   $\triangleright$  intermediate number of antennas
14: end if
15: Output:  $M_a^\dagger$ 

```

where u_k is the distance in meters between the user k and the BS.⁹ We consider, as a simplification, that the users are uniformly distributed within a circular cell delimited by $u_{\min}$ and $u_{\max}$. The cumulative density function of u_k is given by the ratio between the area of the annulus bounded by u_k and the area of the largest possible annulus, i.e., $F_{u_k}(v) = (v^2 - u_{\min}^2) / (u_{\max}^2 - u_{\min}^2)$. Therefore, the probability density function of u_k is given by $f_{u_k}(v) = dF_{u_k}(v)/dv = (2v) / (u_{\max}^2 - u_{\min}^2)$. The target SNR (in dB) of user k is computed as

$$\psi_k = 5 \log_{10} \left(\frac{\beta_k}{4.86 \cdot 10^{-14}} \right) \quad [\text{dB}]. \quad (3.34)$$

Using equation (3.34), $\psi_k \in [4, 20]$ dB. The channel model is given in **(As3.3)**. The remaining parameters are listed in Table 3.1. As performance metrics, we consider the gain in P_{PA} and the gain in P_{cons} . The gain in P_{PA} is defined as the ratio between the PAs consumed power resulting from the transmit power-based precoder and the PAs consumed power resulting from the proposed precoder. In the same way, the gain in P_{cons} is defined as the ratio between the BS consumed power resulting from the transmit power-based precoder and the BS consumed power resulting from the proposed precoder. Therefore, we

⁹The considered large-scale fading model does not depend on the antenna index, and hence does not include the shadowing that could affect certain antennas of the array. Shadowing could bring a scenario similar to the one in Chapter 4 for distributed BSs. Having a different large-scale fading among the antennas of different distributed BSs leads to the deactivation of all the antennas of certain BSs (e.g., because their pathloss to the users is too large). In the case analyzed here, one could have that the shadowed antennas of the array are not activated.

Table 3.1: Parameters of numerical experiments.

Parameter	Value
PAs maximal output power: $P_{\max}$	1 W
PAs maximal efficiency: $\eta_{\text{PA},\max}$	0.22 [56]*
Noise power: σ_v^2	−96 dBm
Fixed power consumption: P_{fixed}	15 W
Circuit power consumption scaling per antenna: $\mathcal{C}$	0.7 W
Minimum distance from user to BS: $u_{\min}$	35 m
Maximum distance from user to BS: $u_{\max}$	250 m

*In our simulations for a single channel realization, we do not observe a clear difference in the results by letting $\eta_{\text{PA},\max}$ take larger values (up to $2\times$).

are using as a benchmark the conventional ZF precoder, which optimizes the transmit power and activates all the BS antennas.

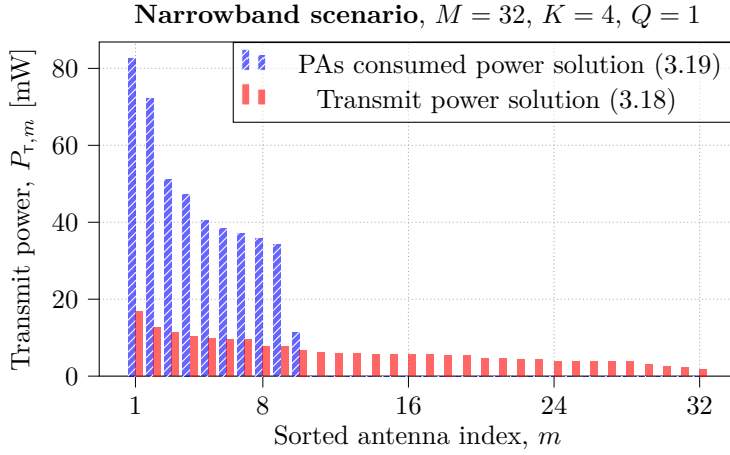
Remark: The simulations of the narrowband and wideband systems analyzed in Sections 3.3 and 3.2 assume no maximal per-antenna power constraints, as the precoders are obtained under $P_{\max} \rightarrow \infty$. However, $P_{\max}$ has to be used to quantify the consumed powers. Therefore, in the simulations, the results exceeding the $P_{\max}$ values are discarded.

When assessing the performance of asymptotic wideband systems, we run simulations by fixing $M = 64$ antennas and varying the value of K from 1 (low traffic) to 40 (high traffic). Using the parameters in Table 3.1 and considering the conventional case $M_a = M$, the average contributions of the terms P_{PA} , $\mathcal{C}M_a$, and P_{fixed} are found to be 7%, 70%, and 23% for $K = 1$, then 31%, 52%, and 17% for $K = 20$, while they are 46%, 40%, and 14% for $K = 40$.

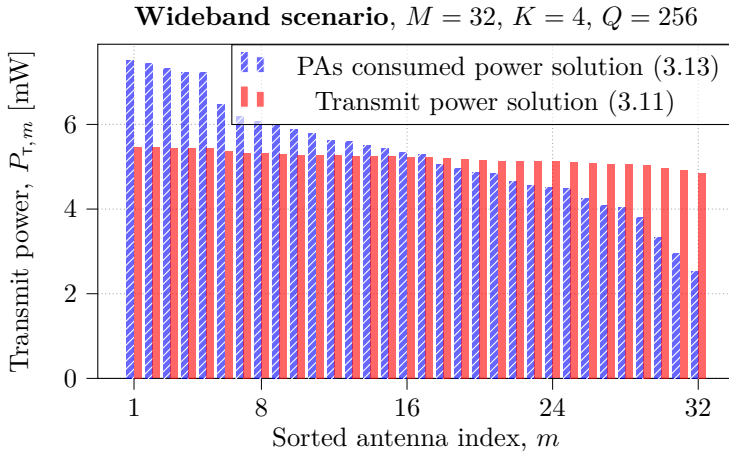
3.5.1 Precoding Solutions for a Channel Realization

Fig. 3.1a shows the solutions, in terms of transmit powers at the different antennas, for the narrowband system and a specific channel realization, $K = 4$ and $M = 32$. As discussed in [93], the solution based on the PAs consumed power activates only a few antennas, while the conventional ZF solution uses all the available antennas. Using expression (3.8) as the objective function induces sparsity in the power allocation. Note that, differently from the single-user case where only one antenna is used, the number of active antennas must be greater than K to enable spatial multiplexing.

When moving to a wideband system, the precoder based on the PAs consumed



(a)



(b)

Figure 3.1: Transmit power as a function of the antenna index with decreasing allocated power, for $M = 32$ antennas and $K = 4$ users in i.i.d. Rayleigh fading. (a) Narrowband case, $Q = 1$ subcarrier. The gain in PAs consumption is 1.19, while the gain in BS consumption is 1.51. (b) Wideband case, $Q = 256$ subcarriers. The gains in PAs and BS consumption are both equal to 1.00.

power gradually uses more antennas as Q increases. Fig. 3.1b shows the solutions for $Q = 256$ subcarriers. The solution based on the PAs consumed power, although presenting more variability in the powers per antenna than the conventional solution, still activates all the BS antennas.¹⁰ Achieving the SNR constraints for all the subcarriers—when Q grows large and while minimizing the consumed power by the PAs—requires the use of all the available antennas. In short, Fig. 3.1b shows that the per-subcarrier ZF, given in equation (3.11), is efficient in terms of power consumed by the PAs.

3.5.2 Average Power Consumption Gains over Conventional ZF

Following the previous observations, Fig. 3.2a–3.2d show the relative differences in the consumed power for the different systems. In the narrowband system and considering the PAs consumed power to evaluate the performance (Fig. 3.2a), the gain of the novel precoder (3.19) over the conventional one (3.18) is always greater than 1.5 for $K = 1$. When more users are present, the difference significantly decreases. For $K = 8$, the ratio is below 1.2 for every value of M . Instead, when the BS consumed power is evaluated, the achievable gains remain large even when more users are communicating (Fig. 3.2b). With $M = 64$, the gain in the BS power consumption ranges from 1.4 to 2.4, depending on K .

In a wideband system with $Q = 128$ subcarriers, the differences between the conventional and the novel precoder remain significant for $K = 1$ and considering the BS consumption (Fig. 3.2c). However, the presence of $K = 2$ users already requires the activation of all the antennas, and both the PAs and BS power consumption gains tend to the minimum value of 1. Note that the situation could be different if the two users are multiplexed in time, with a larger SNR provided to each user in its individual time slot. In this case, it is reasonable to expect that the proposed precoder would keep a low number of activated antennas (as it happens for $K = 1$) but with a larger transmit power (to satisfy the larger target SNR), and these two factors would compete in determining whether consumption gains are achieved or not. An advantage of this time-multiplexing scheme is not to switch off the hardware too frequently in time (e.g., as it happens when serving 2 users in a time slot and no users in the following time slot) to avoid potential latencies caused by reactivation times.

Considering the asymptotic wideband case, one can compute the gains achieved by using $M_a^\dagger$ antennas (the optimal number) over M . Fig. 3.2d shows that significant savings in the BS consumption can be obtained, up to a factor of

¹⁰Already with $Q = 12$, which is the minimal granularity of 5G systems, we observe that almost all antennas are usually activated. This suggests that the gains of antenna deactivation based on PA consumption-minimization are mostly attained for the specific case $Q = 1$.

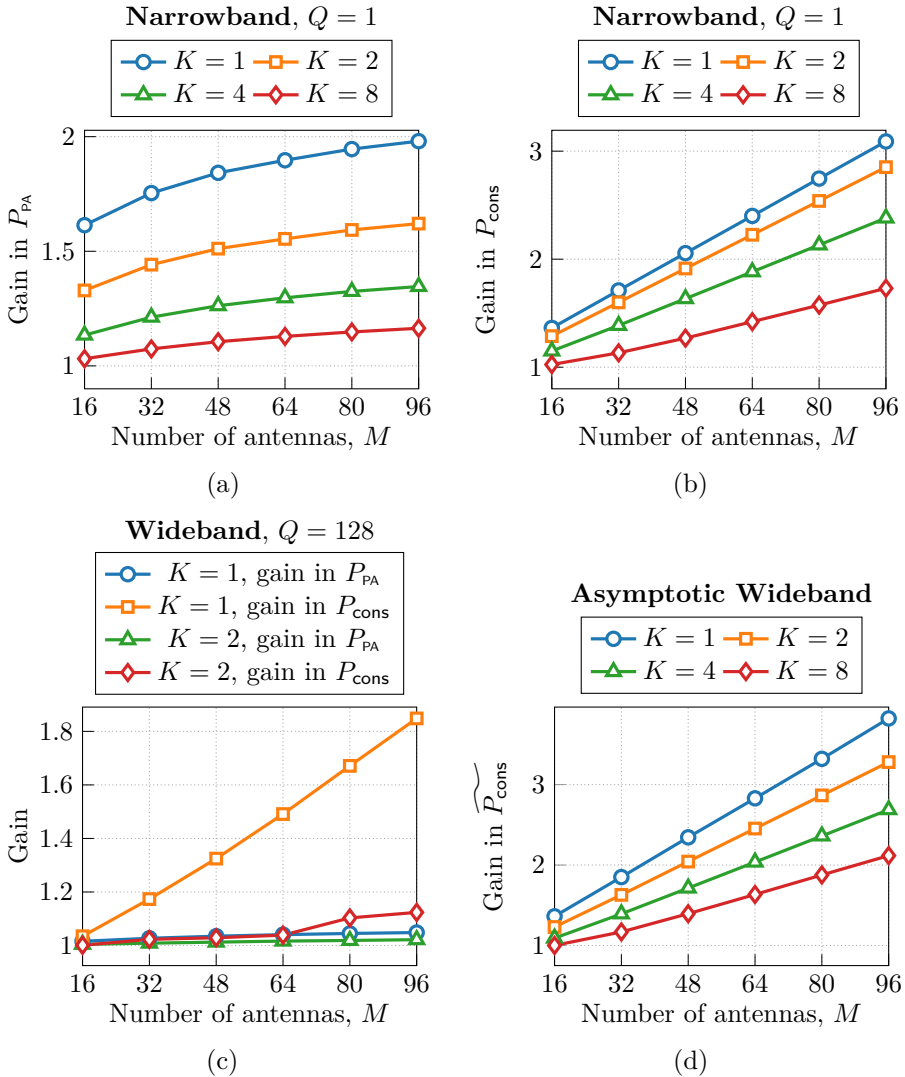


Figure 3.2: Gains (ratios) in power consumption, compared to conventional ZF, versus number of antennas for narrowband and wideband systems in i.i.d. Rayleigh fading ($2 \cdot 10^3$ realizations), for four number of users K . (a) Narrowband, gain in PA consumption. (b) Narrowband, gain in BS consumption. (c) Wideband with $Q = 128$, gain in PA and BS consumption. (d) Asymptotic wideband, gain in asymptotic BS consumption.

$3.8\times$. The benefits are similar to the ones achieved in a narrowband system when evaluating the BS consumption. For instance, when $K = 4$ users and $M = 48$ antennas, the BS driven by the conventional precoder consumes 1.7 times more power than the BS driven by the optimized precoder.

3.5.3 Asymptotic Wideband Regime

Fig. 3.3a illustrates the validity of the asymptotic results in Section 3.4 for $M = 32$ antennas and different values of K . The simulated PAs consumed powers converge to the asymptotic values, and $Q = 128$ subcarriers are already sufficient (in i.i.d. channels) to allow the asymptotic result to be a tight approximation (i.e., the average errors are below 10^{-1}). The trend of the asymptotic PAs and BS consumed powers as a function of M_a is shown in Fig. 3.3b, for $K = 12$ users. The PAs consumed power monotonically decreases with M_a , while the BS consumed power first decreases for small values of M_a (it is beneficial to use more antennas to reduce the PAs consumed power) and then increases for large values of M_a (it is detrimental to use more antennas due to the larger circuit power consumption). The convexity of expression (3.28) implies the existence of one global minimum.

The achievable power consumption savings by activating $M_a^\dagger$ antennas can also be characterized as a function of the number of users. Fig. 3.4 illustrates the comparison for $M = 64$ antennas. We consider the number of users to be an indication of the system's traffic load. Indeed, the per-antenna power constraints are not binding when K is small, and they gradually start to be active when K increases. The conventional precoder always utilizes all the M antennas, while the optimized precoder adapts the number of utilized antennas to the traffic. The value of $M_a^\dagger$ grows proportionally to K , as shown in Fig. 3.4b.¹¹ When the system becomes more loaded, $M_a^\dagger$ increases due to the larger value of P_{PA} and due to the transmit power constraint (i.e., more antennas need to be saturated to achieve the QoS constraints). The BS consumptions induced by activating M and $M_a^\dagger$ antennas eventually converge (Fig. 3.4a). We also make a comparison with a precoder that always uses the minimum number of antennas ($K + 1$). This precoder minimizes the BS consumption when $K = 1$, and then starts to diverge from the optimal precoder because it does not counterbalance the increase in P_{PA} with the utilization of more antennas.

In terms of achievable savings (i.e., the ratio between the asymptotic BS consumptions resulting from the benchmark and optimized precoders), Fig. 3.4b shows that the optimized precoder reduces up to a factor of $2.8\times$ the consumption

¹¹The same trend is observed in [83], with the difference that in [83] the authors considered P_{cons} to be the sum of the transmit power (not PA power) and the circuit power consumption.

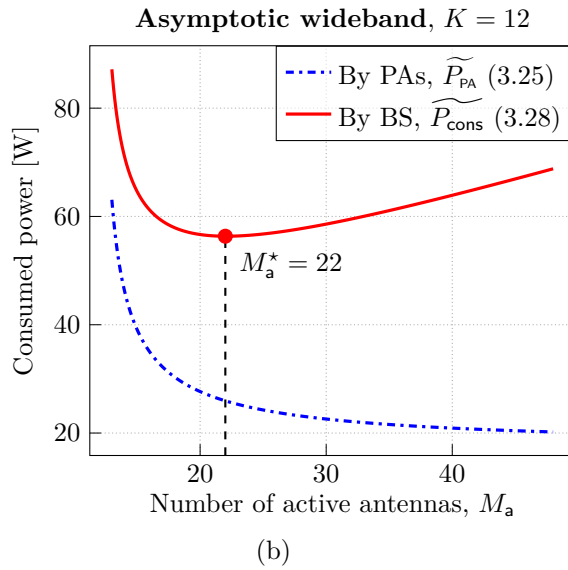
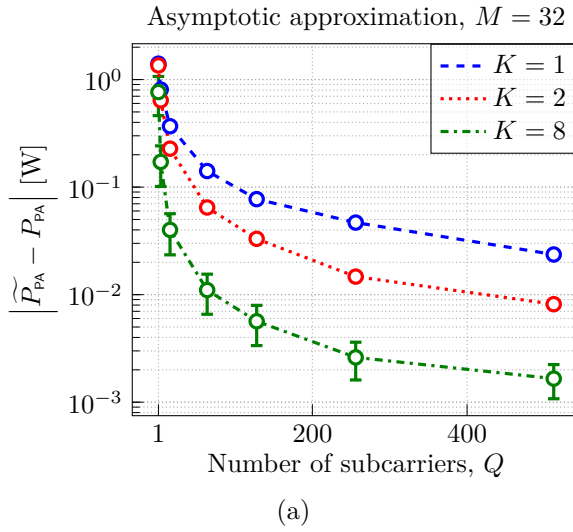
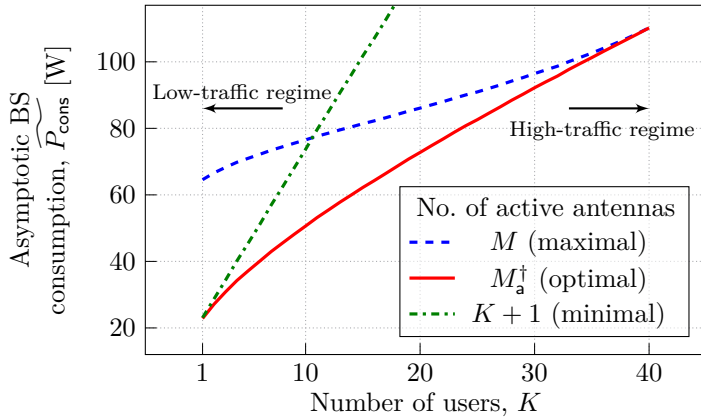
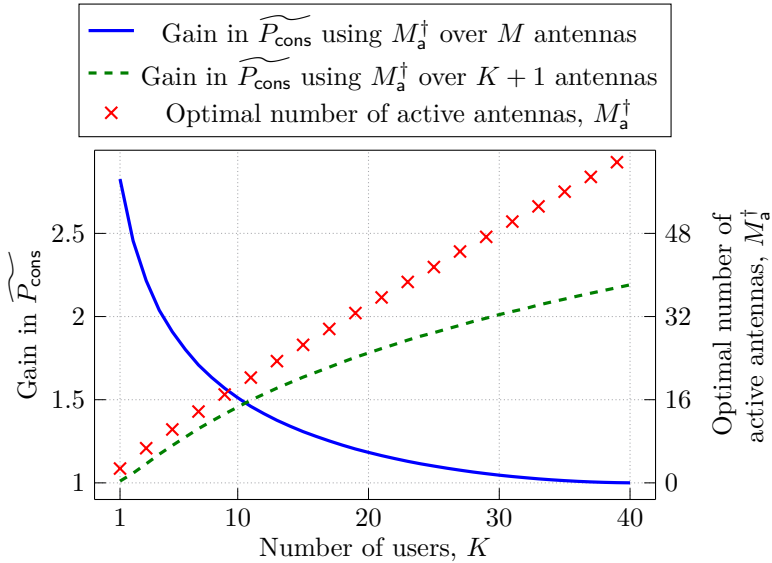


Figure 3.3: (a) Difference magnitude between asymptotic and non-asymptotic PA consumptions versus number of subcarriers for $M = 32$ and different values of K . (b) Asymptotic PAs and BS power consumption versus number of active antennas for $M = 48$ and $K = 12$. The number of antennas M_a^* is also shown.

In both (a) and (b): **Asymptotic wideband, $M = 64$**



(a)



(b)

Figure 3.4: Characterization of asymptotic wideband system versus K for $M = 64$, averaged over $2 \cdot 10^3$ channel realizations. (a) Asymptotic BS consumption (3.28) by using M , $K+1$, and $M_a^\dagger$ antennas. (b) Referring to the same simulations, gain in $\widetilde{P_{\text{cons}}}$ and optimal number of active antennas $M_a^\dagger$ (crosses).

in low traffic with respect to the conventional ZF precoder. When $K = 10$, a gain equal to 1.5 is still achieved. When there are more users, the gain reduces and eventually vanishes (all antennas are activated in both cases). The savings over the precoder that activates $K + 1$ antennas follow the opposite trend, increasing in high traffic up to a factor of $2.2\times$.

3.5.4 Complexity Analysis

To quantify the computational complexity of the proposed precoders, we calculate the number of complex floating-point operations (FLOPs). This method is often used among signal processing experts, yet also contested as the complexity of the hardware (HW) implementation does not depend only on the FLOPs count. As a reference for the number of FLOPs required by standard linear algebraic operations, we consider [94, Appendix C.1.1].

Wideband System

In the wideband system, we need to calculate the precoding matrix (3.13) for every subcarrier q . The computation of the next three matrices, as well as of the m -th row of the fourth matrix using Cholesky factorization and forward-backward substitution

$$\mathbf{H}[q]\mathbf{D}_{\mathbf{p}_T}^{\frac{1}{2}}, \mathbf{D}_{\mathbf{p}_T}^{\frac{1}{2}}\mathbf{H}[q]^H, \mathbf{H}[q]\mathbf{D}_{\mathbf{p}_T}^{\frac{1}{2}}\mathbf{H}[q]^H, \mathbf{D}_{\mathbf{p}_T}^{\frac{1}{2}}\mathbf{H}[q]^H(\mathbf{H}[q]\mathbf{D}_{\mathbf{p}_T}^{\frac{1}{2}}\mathbf{H}[q]^H) \quad (3.35)$$

require KM FLOPs, KM FLOPs (corresponding to a scalar multiplication for each element of $\mathbf{H}[q]$), K^2M FLOPs exploiting the symmetry of the matrix, and $K^3/3 + 2K^2$ FLOPs. The forward and backward substitutions need to be performed M times (for every row of the above matrix), hence the number of FLOPs becomes $K^3/3 + 2K^2M$. At the end, the scaling by $\mathbf{D}_{\psi}$ requires K FLOPs. All that needs to be repeated for every subcarrier. By adding the terms, we obtain $K^3Q/3 + 3K^2MQ + 2KMQ + KQ$ FLOPs, which scales as $\mathcal{O}(K^3Q + K^2MQ)$. Before computing (3.13), the powers at the antennas need to be retrieved. The computation of (3.14) requires, other than the necessary FLOPs to compute $\mathbf{W}[q]$, KQ multiplications to calculate the magnitudes and $K + Q - 2$ additions to calculate the sums. Multiplying by M , we obtain $KQM + KM + QM - 2M$ additional FLOPs. The trend, however, remains $\mathcal{O}(K^3Q + K^2MQ)$. By using Algorithm 3.1, the computation of powers is repeated until convergence. If the algorithm converges in I iterations, the total number of FLOPs is $\mathcal{O}(K^3QI + K^2MQI)$. The conventional ZF, instead, requires $\mathcal{O}(K^3Q + K^2MQ)$ FLOPs. In short, the proposed precoder is I times as complex compared to the conventional one.

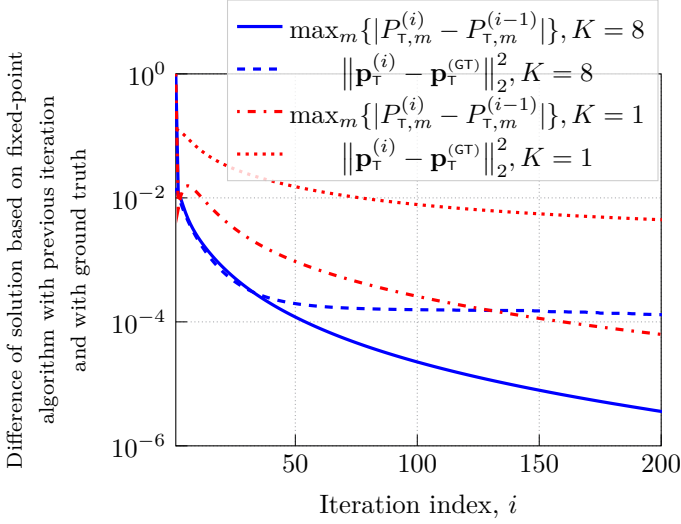


Figure 3.5: Performance of Algorithm 3.1 when $Q = 1$ and $M = 32$, using a tolerance $\varepsilon = 10^{-4}$. The ground truth $\mathbf{p}^{(\text{GT})}$ is computed with a convex solver. Results are averaged over 10^3 executions of the algorithm.

Table 3.2: Complexity of computing precoders in the three analyzed systems.

System	Proposed solution	Conventional solution
Wideband	$\mathcal{O}(K^3 QI + K^2 M QI)$	$\mathcal{O}(K^3 Q + K^2 M Q)$
Narrowband	$\mathcal{O}(K^3 I + K^2 M I)$	$\mathcal{O}(K^3 + K^2 M)$
Asympt. wideband	$\mathcal{O}(K^3 Q + K^2 M_a Q)$	$\mathcal{O}(K^3 Q + K^2 M Q)$

Fig. 3.5 illustrates the convergence of the iterative algorithm. We show two narrowband cases because, when Q increases, the convergence is faster (i.e., in tens of iterations), likely due to the more uniform power allocation that might be associated with fewer changes from one iteration to the other. We display the maximum absolute variation among the powers between iterations and the squared norm of the difference between the current solution and the ground truth, which is obtained through a convex solver. Given that the tolerance ε is set to 10^{-4} , the algorithm converges in $I \approx 200$ iterations for $K = 1$, and in $I \approx 50$ iterations for $K = 8$. Moreover, it can be seen as, considering this tolerance, the squared norm stays below 10^{-2} .

Narrowband System

The same reasoning used for the wideband system applies to the narrowband one, which is then associated with a complexity of $\mathcal{O}(K^3I + K^2MI)$ FLOPs for the PAs consumed-power solution, while it becomes $\mathcal{O}(K^3 + K^2M)$ for the transmit-power solution.

Asymptotic Wideband System

In the asymptotic wideband scenario, the precoding matrices have M_a rows, then are computed with complexity $\mathcal{O}(K^3Q + K^2M_aQ)$. Before executing Algorithm 3.2, which involves only comparisons, we need to solve the quartic equation (3.30). The solution of quartics can be computed with a fixed number of operations. Calculating the last coefficient in (3.30) requires the computation of $\text{tr}(\mathbf{D}_\beta^{-1}\mathbf{D}_\psi)$, which has complexity $\mathcal{O}(K)$. Therefore, retrieving $M_a^\dagger$, which needs to be recomputed when the large-scale fading coefficients change, has lower complexity than retrieving the precoding matrices, which instead need to be retrieved every time the small-scale fading coefficients change. We can thus consider the complexity to be $\mathcal{O}(K^3Q + K^2M_aQ)$. In the conventional case, the complexity remains $\mathcal{O}(K^3Q + K^2MQ)$.

Table 3.2 summarizes the complexities of the precoders for the three investigated systems. Given the behavior of Algorithm 3.1, the proposed wideband and narrowband precoders have complexities of 1 or 2 orders of magnitude larger than the conventional precoders. However, the larger complexity is justified by the large savings shown in Fig. 3.2, especially for narrowband systems. Instead, the asymptotic wideband precoder has better performance and lower complexity in low-to-medium traffic scenarios.

3.6 Conclusion

In this chapter, we have analyzed the allocation of antennas and transmit power in narrowband and wideband massive MIMO with the goal of reducing the PA or BS consumption while guaranteeing the targets QoS of the users. The initial problems involved precoding design to minimize PA consumption (targeting a class B PA), where a ZF constraint is considered. The analysis is then extended to the BS consumption in the case where the number of subcarriers grows large, and the results well approximate, under i.i.d. Rayleigh fading, current broadband systems using hundreds to thousands of subcarriers. The findings provide the

first replies of this thesis to **RQ2**, related to transmit power allocation and antenna (de)activation.

In narrowband systems, the proposed precoder activates a few BS antennas, with consequent savings—under the considered parameters—up to a factor of $3\times$ in the BS consumption by deactivating the RF chains of the non-active antennas. In wideband systems, more BS antennas are progressively activated as the number of subcarriers increases, hence less and less RF/DSP chains can be switched off. The asymptotic analysis of wideband systems subject to per-antenna power constraints revealed, however, how a simple optimization on the number of active antennas has the potential to achieve significant BS consumption savings. The theoretical savings are estimated to reach a factor of $2.8\times$ in low-traffic scenarios. The complexity analysis suggests the viability of the proposed solutions, with the optimized number of active antennas that needs to be computed only when the large-scale fading changes.

Chapter 4

Generalization to Distributed Base Stations

Bibliographic Reference

This chapter is based on: E. Peschiera, X. Mestre and F. Rottenberg, “Energy-saving cell-free massive MIMO precoders with a per-AP wideband Kronecker channel model,” in *Proceedings of 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2024)*, Seoul, Republic of Korea, 2024, pp. 9026-9030 [95].¹

In this chapter, we generalize the allocation of power and number of active antennas to distributed multi-antenna base stations (BSs). The objective is to reduce the total consumption across all the BSs under quality-of-service (QoS) constraints (expressed as signal-to-noise ratio (SNR) constraints). In distributed systems, geographically-separated BSs cooperate to serve the users via, e.g., joint phase-coherent beamforming. These BSs are typically referred to as access points (APs) and have the form factor of micro BSs [96]. A network of distributed and cooperating APs is no longer a cellular network and becomes instead a *cell-free network*. Cell-free networks provide macro-diversity gains and reduced interference compared to cellular ones, and different cell-free topologies are being proposed and studied in the literature (e.g., [97]). In this

¹The Python simulation code is available in open access at the following link: github.com/dramco-research/energy-saving-precoder-cellfree-epeschiera-icassp24.

chapter, we focus on a cell-free network with APs having multiple antennas and communicating with all the single-antenna users in the analysed service area.

State of the Art

In a centralized (or co-located) multi-antenna system, the transmitting or receiving antennas share the same macro-scale location (e.g., a BS site) and are separated by a distance in the order of the wavelength, typically half a wavelength between adjacent antennas. Centralized multiple-input multiple-output (MIMO) is highly resilient to small-scale fading in wireless channels, due to multipath propagation, because the phase-coherent transmission or reception over a large number of antennas reduces the randomness of the fading [90]. This phenomenon is also known as channel hardening or micro-scale diversity [90]. A BS equipped with an antenna array in a cellular network communicating with multiple users is an example of centralized MIMO. One of the major drawbacks of centralized MIMO is the high sensitivity to the large-scale fading, mainly due to pathloss, shadowing, and blockage. When the channel magnitude between the BS and a user is low because of a large BS-user distance or shadowing/blockage by large objects, centralized MIMO can suffer from QoS degradation. Macro-scale diversity is needed to provide good QoS in these situations, and can be achieved by distributed MIMO systems [96]. In distributed MIMO, the antennas are deployed such that they are spread out in space, with macro-scale separation. In a distributed MIMO system, multiple BSs cooperate and simultaneously serve one or more users. This concept dates back to the “network MIMO” literature and to standardization activities that led to the inclusion of coordinated multi-point (CoMP) in fourth-generation (4G) and fifth-generation (5G) networks [98, 99, 32]. However, the industry has not seen a widespread commercial adoption of distributed MIMO.

In the last ten years, there has been a renewed interest in distributed MIMO thanks to the introduction of the cell-free MIMO concept [100]. In its primal form, cell-free MIMO considered distributed single-antenna BSs, referred to as APs, which communicate with all the single-antenna users in a certain area. Later, multi-antenna APs and/or multi-antenna users have been analysed (e.g., [101]). The concept of user-centric cell-free networks, where a subset of APs serve a subset of users, has been proposed in [102]. In general, the APs are connected to one or multiple centralized units (CUs) via fronthaul (FH) links, while the CUs are connected to the core network via backhaul links. The CUs perform APs control, synchronisation, and signal processing tasks such as channel estimation, precoding, and decoding. The baseband signal processing can be fully centralized at the CUs, fully distributed at the APs, or a combination of them. Several studies suggested the superiority of cell-free

MIMO compared to centralized MIMO in terms of both spectral efficiency and energy efficiency [96, 103]. Currently, distributed and cell-free MIMO is proposed for the sixth-generation (6G) standard [104, 105].

Despite the advantages, there remain fundamental challenges to be addressed toward the deployment of cell-free networks. Achieving phase-synchronous transmission among the APs in downlink (DL), distributing the massive amount of data over potentially capacity-constrained backhaul links, and keeping the network's energy consumption under control are among the main technical challenges [106]. Other network-related challenges are discussed in [107]. Early evaluations have shown that the energy efficiency (EE) of cell-free systems can be one order of magnitude higher compared to cellular systems [96]. Subsequent studies (e.g., [108]) pointed out that the absolute energy consumption of cell-free systems may increase compared to cellular systems due to the large consumption of FH links. The optimal orchestration of distributed MIMO networks (including federations of APs, clustering of users, multi-domain resource allocation, handover strategies, energy-saving techniques, etc.) is an active area of research. Studies reported in literature have addressed the problem of switching on/off APs to minimize absolute power consumption [109, 110, 111]. In most cases, they rely on conventional precoders and optimize the power allocations according to the power consumption. Similarly, works on AP selection typically optimize the EE by fixing the precoder structure [103, 112]. Moreover, the impact of multicarrier transmission is often overlooked in the current literature.

Contributions

In this chapter, we study the optimal precoding and power allocation, in terms of power amplifiers (PAs) consumption, in a cell-free massive MIMO network where distributed multi-antenna APs jointly serve the users. The problem is formulated as a precoding problem aiming at minimizing the power consumed by the PAs under DL QoS constraints. These constraints are expressed as per-subcarrier zero-forcing (ZF) constraints, which can be equivalently converted to per-user rate constraints in bits per second. We consider the case of processing centralized at the CU, hence the problem's form closely recalls the cellular problem of Chapter 3, with the crucial difference being the channel model. The channel incorporates the characteristics of a cell-free massive MIMO scenario, where the spatial distribution of the APs implies a channel having a different covariance matrix for each AP. Leveraging random matrix theory (RMT) and the structure of the precoding solution, we derive low-complexity solutions to the addressed problem. The CU determines which APs to activate prior to computing the precoding matrices, and the set of active APs changes only with

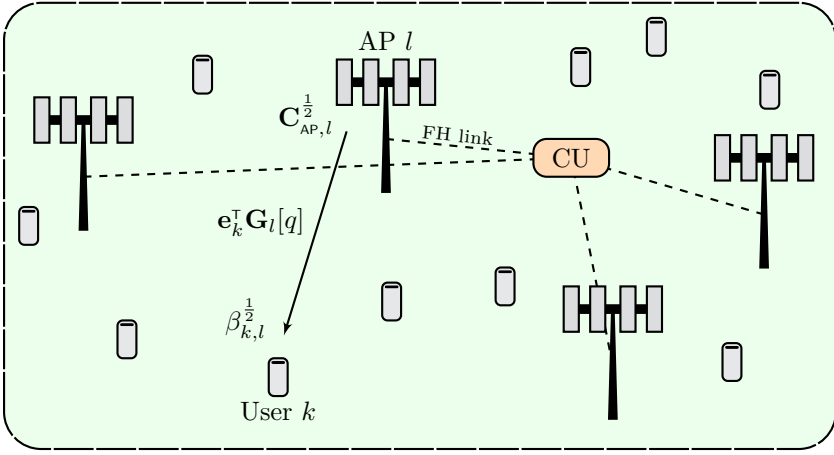


Figure 4.1: Cell-free MIMO system model considered in this chapter, where all APs serve all users. AP l has M_l antennas, and $\mathbf{e}_k^T \mathbf{G}_l[q]$ is the $M_l \times 1$ small-scale fading term of the channel of user k .

the second-order channel statistics. This prevents a significant growth in the power consumed by the FH links, the baseband processing at the CU, and the radio frequency (RF) circuitry and power supply of the APs. We show that the proposed solution approaches the optimal one when the number of subcarriers increases (referred to as wideband regime since we consider a fixed subcarrier spacing). The energy savings are evaluated in terms of network consumption, given by the sum of the PA consumption at each PA, the analog and digital circuit consumption at each AP, and a fixed network-wide consumption.

4.1 System Model and Assumptions

In this section, we adapt the DL MIMO-orthogonal frequency-division multiplexing (OFDM) communication model of Section 2.1.1 to the distributed scenario. We consider that there are L APs, where AP l is equipped with M_l antennas. The total number of transmit antennas (across all APs) is denoted by $\tilde{M} = \sum_{l=1}^L M_l$. The computation of beamforming coefficients is fully centralized at the CU, and FH links connect each AP to the CU. All the APs serve all the users in the analyzed area. Each AP performs fully-digital beamforming, giving M_l RF chains at AP l . The number of active (or, equivalently, utilized)

antennas corresponds to the number of active RF chains. Given that we consider the DL, we simplify the notation by setting $Q \triangleq Q_{\text{DL}}$ as the number of data subcarriers, $K \triangleq K_{\text{DL}}$ as the number of active users, $\mathbf{H} \triangleq \mathbf{H}_{\text{DL}}$ as the channel matrix, $\mathbf{W} \triangleq \mathbf{W}_{\text{dig}}$ as the precoding matrix, and $P_{\text{T}} \triangleq P_{\text{T,DL}}$ as the total transmit power. We express the dependency on subcarrier q in square brackets. The symbols of the users for the subcarrier q , transmitted by all the APs, are denoted by $\mathbf{s}_{\text{DL}}[q] \in \mathbb{C}^{K \times 1}$. We assume an equal power allocation among users giving $p_{k,\text{DL}} = 1 \forall k$. By expressing with $\mathbf{H}_l[q] \in \mathbb{C}^{K \times M_l}$ the channel matrix of AP l for the subcarrier q , the aggregated channel matrix (across all APs) for the subcarrier q is expressed by juxtaposing the individual channel matrices

$$\mathbf{H}[q] = [\mathbf{H}_1[q], \dots, \mathbf{H}_L[q]] \in \mathbb{C}^{K \times \tilde{M}}. \quad (4.1)$$

The precoded (transmitted) signal by AP l for the subcarrier q is $\mathbf{x}_l[q] \in \mathbb{C}^{M_l \times 1}$, hence the transmitted signal by all APs is written as

$$\mathbf{x}[q] = [\mathbf{x}_1[q]^T, \dots, \mathbf{x}_L[q]^T]^T = \mathbf{W}[q]\mathbf{s}_{\text{DL}}[q] \in \mathbb{C}^{\tilde{M} \times 1} \quad (4.2)$$

where $\mathbf{W}[q] \in \mathbb{C}^{\tilde{M} \times K}$ is the aggregated digital precoding matrix for the subcarrier q . The received signal vector for the subcarrier q is expressed as

$$\mathbf{r}_{\text{DL}}[q] = \mathbf{H}[q]\mathbf{x}[q] + \boldsymbol{\nu}[q] \in \mathbb{C}^{K \times 1} \quad (4.3)$$

where $\boldsymbol{\nu}[q] \sim \mathcal{CN}(\mathbf{0}_K, \sigma_{\nu}^2 \mathbf{I}_K)$ is the thermal noise that is assumed independently and identically distributed (i.i.d.) across subcarriers and users. We consider a ZF constraint of the type

$$\mathbf{H}[q]\mathbf{W}[q] = \mathbf{D}_{\psi}^{\frac{1}{2}} \sigma_{\nu} \quad \forall q \quad (4.4)$$

where user k has a target SNR given by ψ_k , and all the target SNRs are collected in the matrix $\mathbf{D}_{\psi} = \text{diag}(\psi_1, \dots, \psi_K)$.

In this chapter, we model the DL channel at the AP l through a Kronecker channel model. This model extends the i.i.d. Rayleigh fading model by introducing a correlation at both ends of the communication link. In DL, there can be correlation among the transmitted signals at the different AP antennas, and between the received signals at the different users. Although this model has been fitted to measurements of both indoor and outdoor channels, it assumes that the local scattering processes around the transmitter and the receiver are decoupled, and this can lead to overestimation of channel capacity [113]. Under the Kronecker model, the channel is given by

$$\mathbf{H}_l[q] = \underbrace{\mathbf{D}_{\beta,l}^{\frac{1}{2}}}_{\text{correlation at receiver}} \overbrace{\mathbf{G}_l[q]}^{\text{i.i.d. term}} \underbrace{\mathbf{C}_{\text{AP},l}^{\frac{1}{2}}}_{\text{correlation at transmitter}} \quad (4.5)$$

where $\mathbf{D}_{\beta,l} = \text{diag}(\beta_{1,l}, \dots, \beta_{K,l}) \in \mathbb{R}^{K \times K}$ is the matrix collecting the large-scale fading coefficients at the AP l , $\beta_{k,l} > 0$ is the large-scale fading coefficient between user k and AP l , $\mathbf{G}_l[q] \in \mathbb{C}^{K \times M_l}$ models the small-scale fading and has i.i.d. entries drawn from a $\mathcal{CN}(0, 1)$ distribution, and $\mathbf{C}_{\text{AP},l} \in \mathbb{C}^{M_l \times M_l}$ expresses the correlation among the antennas of AP l . Note that the correlation at receiver is diagonal because we assume no correlation among the channels of the users. Considering a per-AP Kronecker channel model leads to an aggregated channel $\mathbf{H}[q]$ following a Weichselberger channel model [114]. An illustration of the described system model and channel model variables is given in Fig. 4.1.

Using $\tilde{m}$ as the index of the total (across all APs) number of transmit antennas, the transmit power at antenna $\tilde{m}$ is given by

$$P_{\tau, \tilde{m}} = \frac{1}{Q} \sum_{q=1}^Q \mathbf{e}_{\tilde{m}}^{\text{T}} \mathbf{W}[q] \mathbf{W}[q]^{\text{H}} \mathbf{e}_{\tilde{m}} \quad (4.6)$$

where $\mathbf{e}_{\tilde{m}}$ is the $\tilde{m}$ -th canonical basis vector of $\mathbb{R}^{\tilde{M}}$, i.e., an $\tilde{M} \times 1$ vector with only one non-zero value, equal to 1, at position $\tilde{m}$. The individual transmit powers are collected in the vector $\mathbf{p}_{\tau} \in \mathbb{R}^{\tilde{M} \times 1}$ and in the matrix $\mathbf{D}_{\mathbf{p}_{\tau}} = \text{diag}(\mathbf{p}_{\tau}) = \text{diag}(P_{\tau,1}, \dots, P_{\tau, \tilde{M}}) \in \mathbb{R}^{\tilde{M} \times \tilde{M}}$. Let us denote the total power transmitted by AP l (across its M_l antennas) as $M_l \rho_{\tau,l}$. The total power transmitted by the L APs is expressed via

$$P_{\tau} = \frac{1}{Q} \sum_{\tilde{m}=1}^{\tilde{M}} \sum_{q=1}^Q \mathbf{e}_{\tilde{m}}^{\text{T}} \mathbf{W}[q] \mathbf{W}[q]^{\text{H}} \mathbf{e}_{\tilde{m}} = \frac{1}{Q} \sum_{q=1}^Q \text{tr}(\mathbf{W}[q] \mathbf{W}[q]^{\text{H}}). \quad (4.7)$$

Under the PA consumption model in equation (2.21) explained in Section 2.2.1, with $\alpha = \frac{1}{2}$ and $\xi_{\text{PA}} = 0$ representing class B PAs, the total power consumed by the $\tilde{M}$ PAs during working mode and data transmission is computed as

$$P_{\text{PA}} = \frac{P_{\text{max}}^{\frac{1}{2}}}{\eta_{\text{PA}, \text{max}}} \sum_{\tilde{m}=1}^{\tilde{M}} \left(\frac{1}{Q} \sum_{q=1}^Q \mathbf{e}_{\tilde{m}}^{\text{T}} \mathbf{W}[q] \mathbf{W}[q]^{\text{H}} \mathbf{e}_{\tilde{m}} \right)^{\frac{1}{2}} \quad (4.8)$$

where P_{max} and $\eta_{\text{PA}, \text{max}}$ are the maximal PA transmit power and efficiency, respectively. Based on the discussion of Chapter 3 showing that optimizing P_{PA} leads in some cases to sparsity in the utilized antennas/RF chains (non-zero antenna powers), we expect that the solutions that optimize (4.8) could activate only part of the L APs. Let us identify the set of APs that are activated by the optimal solution (minimizing P_{PA}) as $\mathcal{L}_{\text{a}} \subseteq \{1, \dots, L\}$, where the number of active APs L_{a} is the cardinality of the set $\mathcal{L}_{\text{a}}$, i.e., $L_{\text{a}} = |\mathcal{L}_{\text{a}}|$. An active AP transmits a non-zero power. Similarly, an antenna is active when it transmits a non-zero power, and the number of active antennas at AP l is denoted by

$M_{a,l}$. To quantify the consumed power at the network level, which considers the consumption of the L APs (of which L_a are activated) and of the FH links, we use the following model

$$P_{\text{net}} = P_{\text{PA}} + \sum_{l \in \mathcal{L}_a} (\mathcal{C}M_{a,l} + P_{\text{fixed}}) \quad (4.9)$$

where $\mathcal{C}$ is the cost (in terms of analog and digital circuit power consumption) to activate a single antenna, and P_{fixed} is the fixed consumption associated with the activation of an entire AP. As for the power model used in Chapter 3, the power model in (4.9) is a particularization of the power model of Section 2.2 when looking only at the dependency on the transmit power (in P_{PA}) and on the number of active antennas (and RF chains). The parameters $\mathcal{C}$ and P_{fixed} aggregate the different terms of the overall power model. Note that the consumption of the FH links can be considered as part of the term $\sum_{l \in \mathcal{L}_a} \mathcal{C}M_{a,l}$, scaling then with the number of active APs [103]. Other parameters that are typically considered to scale the FH consumption are the bandwidth and the data rate [103], but these are not explicitly considered in this chapter.

In this chapter, we study the precoders that optimize expression (4.8) under SNR constraints in the form of ZF constraints.

Remark: The solution obtained in Section 3.2 of Chapter 3 applies to this scenario, despite the different structure of the channel matrix compared to centralized systems. However, the large dimension of a complete cell-free MIMO observation can easily lead to significant increase of the power consumed by the FH links and by the baseband processing at the CU. Therefore, we consider an asymptotic regime that allows us to greatly simplify the solution while maintaining high accuracy. This asymptotic regime corresponds to the following assumptions.

(As4.1): Number of antennas per AP and number of users going to infinity, giving $M_l \rightarrow \infty \forall l \in \{1, \dots, L\}$ and $K \rightarrow \infty$. Moreover, defining the variable $c_l = \frac{K}{M_l}$, there exist $b > a > 0$ for which $a < \min_l \liminf_K c_l < \max_l \limsup_K c_l < b$. Thanks to the hardening effect that happens in this regime, the random quantities introduced later in equation (4.15) behave as deterministic ones [115]. Nonetheless, the solution to the system of equations in (4.15) has $\tilde{M}$ components, hence **(As4.1)** implies that the dimension of the solution grows infinitely large. The number of APs L is instead fixed. This motivates the introduction of the following assumption.

(As4.2): Equal power transmitted by the antennas of an individual AP, giving

$$\mathbf{D}_{\mathbf{P}_T} = \text{diag}(\mathbf{p}_T) = \text{diag}(P_{T,1}, \dots, P_{T,\tilde{M}}) = \text{blkdiag}\left(\{\rho_{T,l} \mathbf{I}_{M_l}\}_{l=1}^L\right) \quad (4.10)$$

where $\text{blkdiag}(\cdot)$ is the block-diagonal operator. Thanks to this, the effective dimension of the solution is fixed and equal to L . This assumption is an approximation, indeed one can retrieve the exact solution to equation (4.15), explained in the following, and observe that more power is allocated to the least correlated antennas. However, the numbers of unknowns (the transmit power at each of the $\sum_{l=1}^L M_l$ antennas) grows to infinity when $M_l \rightarrow \infty$, and this prevents the application of the RMT method. With the above assumption, the number of unknowns remains finite and equal to L . The numerical simulations will demonstrate that the results obtained under (As4.2) have a negligible performance loss compared to the optimal ones.

(As4.3): Rayleigh small-scale fading, giving $\forall l \in \{1, \dots, L\}$, $\mathbf{G}_l[q] \in \mathbb{C}^{K \times M_l}$ made of i.i.d. entries with zero mean and unit variance.

(As4.4): $\mathbf{D}_l = \sigma_\nu^{-2} \tilde{\mathbf{D}}_\psi^{-1} \mathbf{D}_{\beta,l} \in \mathbb{R}^{K \times K}$, $\mathbf{C}_{\text{AP},l} \in \mathbb{C}^{M_l \times M_l}$, and $\mathbf{D}_{\mathbf{p}_T} \in \mathbb{R}^{\tilde{M} \times \tilde{M}}$, $\forall l \in \{1, \dots, L\}$, are deterministic matrices with bounded spectral norms. Moreover, the matrix $\mathbf{D}_{\mathbf{p}_T}$ has at least $K + 1$ non-zero entries (giving a total number of active antennas larger than the number of users, as required by the ZF constraint).

4.2 Precoder Design and Asymptotic Analysis

In this section, we briefly recall the solutions to the transmit power-minimization and PA power-minimization problems discussed in Chapter 3, and we then propose a low-complexity deterministic equivalent suited for distributed BSs and based on RMT. Under centralized processing performed by the CU, the system model (neglecting the structure of the channel) is equivalent to that of a single AP having $\tilde{M}$ antennas. The conventional ZF precoder is the solution to

$$\underset{\{\mathbf{W}[q]\}}{\text{minimize}} \quad P_T = \frac{1}{Q} \sum_{\tilde{m}=1}^M \sum_{q=1}^Q \mathbf{e}_{\tilde{m}}^T \mathbf{W}[q] \mathbf{W}[q]^H \mathbf{e}_{\tilde{m}} \quad (4.11)$$

$$\text{subject to} \quad \mathbf{H}[q] \mathbf{W}[q] = \mathbf{D}_\psi^{\frac{1}{2}} \sigma_\nu, \quad q = 1, \dots, Q$$

which is the minimization of total transmit power under ZF constraints. The matrix $\mathbf{D}_\psi = \text{diag}(\psi_1, \dots, \psi_K)$ contains the target SNRs of the users, while σ_ν^2 is the noise power. The ZF constraints can be easily translated into rate constraints, e.g., an achievable rate for user k for the subcarrier q is $\log_2(1 + \frac{\psi_k}{Q})$ bits per channel use. The solution to problem (4.11) is the per-subcarrier ZF

$$\mathbf{W}[q] = \mathbf{H}[q]^H (\mathbf{H}[q] \mathbf{H}[q])^{-1} \mathbf{D}_\psi^{\frac{1}{2}} \sigma_\nu. \quad (4.12)$$

If we look instead at the precoder that minimizes the power consumed by the PAs under a ZF constraint, we obtain the following problem

$$\begin{aligned}
 & \underset{\{\mathbf{W}[q]\}}{\text{minimize}} & P_{\text{PA}} &= \frac{P_{\text{max}}^{\frac{1}{2}}}{\eta_{\text{PA,max}}} \sum_{\tilde{m}=1}^{\tilde{M}} \left(\frac{1}{Q} \sum_{q=1}^Q \mathbf{e}_{\tilde{m}}^{\top} \mathbf{W}[q] \mathbf{W}[q]^{\text{H}} \mathbf{e}_{\tilde{m}} \right)^{\frac{1}{2}} \\
 & \text{subject to} & \mathbf{H}[q] \mathbf{W}[q] &= \mathbf{D}_{\psi}^{\frac{1}{2}} \sigma_{\nu}, \quad q = 1, \dots, Q.
 \end{aligned} \tag{4.13}$$

Let us define $\tilde{\mathbf{H}}[q] = \sigma_{\nu}^{-1} \mathbf{D}_{\psi}^{-\frac{1}{2}} \mathbf{H}[q]$ as the normalized channel matrix. Recalling Theorem 3.1, the solution to problem (4.11) has the following form

$$\mathbf{W}[q] = \mathbf{D}_{\mathbf{p}_{\text{T}}}^{\frac{1}{2}} \mathbf{H}[q]^{\text{H}} \left(\mathbf{H}[q] \mathbf{D}_{\mathbf{p}_{\text{T}}}^{\frac{1}{2}} \mathbf{H}[q]^{\text{H}} \right)^{-1} \tilde{\mathbf{D}}_{\psi}^{\frac{1}{2}} \sigma_{\nu} \tag{4.14}$$

where $\mathbf{D}_{\mathbf{p}_{\text{T}}}$ contains the powers transmitted by the antennas. The transmit power at antenna $\tilde{m}$ can be written as the solution to the set of equations (for $\tilde{m} = 1, \dots, \tilde{M}$)

$$P_{\text{T},\tilde{m}} = \frac{1}{Q} \sum_{q=1}^Q \mathbf{e}_{\tilde{m}}^{\top} \mathbf{D}_{\mathbf{p}_{\text{T}}}^{\frac{1}{2}} \tilde{\mathbf{H}}[q]^{\text{H}} \left(\tilde{\mathbf{H}}[q] \mathbf{D}_{\mathbf{p}_{\text{T}}}^{\frac{1}{2}} \tilde{\mathbf{H}}[q]^{\text{H}} \right)^{-2} \tilde{\mathbf{H}}[q] \mathbf{D}_{\mathbf{p}_{\text{T}}}^{\frac{1}{2}} \mathbf{e}_{\tilde{m}} \tag{4.15}$$

which can be retrieved via iterative fixed-point algorithm.

The structure of the precoder in equation (4.14) allows one to allocate more power to some antennas, which can work closer to the PA saturation level and hence at larger efficiencies. Turning off some antennas or APs brings further savings in the network consumption P_{net} , given in equation (4.9), thanks to the deactivation of the RF and digital signal processing (DSP) chains (in case of single antennas) or even of the full AP. Note that the CU needs to (i) compute the antenna powers in equation (4.15) every time the small-scale fading changes (i.e., every channel coherence time), and (ii) know the instantaneous channel coefficients of all the APs, which correspond to $\tilde{M}KQ$ elements. These operations impose challenging requirements on the FH links' capacity of the cell-free MIMO system due to the large number of antennas and/or APs. For this reason, we propose an RMT approach that reduces the complexity of retrieving the powers at the APs, resulting in a smaller number of channel and precoding coefficients to be exchanged between the CU and the APs.

While in the previous part of this section we did not rely on any specific channel model, we now consider the per-AP Kronecker channel model given in equation (4.5). We also make use of the assumptions **(As4.1)** – **(As4.4)**. Let us recall that **(As4.2)** imposes the following structure on the transmit-powers

matrix $\mathbf{D}_{\mathbf{p}_\tau}$

$$\mathbf{D}_{\mathbf{p}_\tau} = \text{diag} \left(\underbrace{\rho_{\tau,1}, \dots, \rho_{\tau,1}}_{M_1 \text{ times}}, \dots, \underbrace{\rho_{\tau,l}, \dots, \rho_{\tau,l}}_{M_l \text{ times}}, \dots, \underbrace{\rho_{\tau,L}, \dots, \rho_{\tau,L}}_{M_L \text{ times}} \right). \quad (4.16)$$

Exploiting the structure of the aggregated channel matrix, we can write

$$\tilde{\mathbf{H}}[q] \mathbf{D}_{\mathbf{p}_\tau}^{\frac{1}{2}} \tilde{\mathbf{H}}[q]^H = \sum_{l=1}^L \rho_{\tau,l}^{\frac{1}{4}} \mathbf{D}_l^{\frac{1}{2}} \mathbf{G}_l[q] \mathbf{C}_{\text{AP},l} \mathbf{G}_l[q]^H \mathbf{D}_l^{\frac{1}{2}} \rho_{\tau,l}^{\frac{1}{4}} \quad (4.17)$$

where $\mathbf{D}_l = \sigma_\nu^{-2} \tilde{\mathbf{D}}_\psi^{-1} \mathbf{D}_{\beta,l}$. The above model is analyzed by the authors in [116]. In our case, we want to characterize the following quantity

$$\mathbf{e}_{\tilde{m}}^\tau \mathbf{D}_{\mathbf{p}_\tau}^{\frac{1}{2}} \tilde{\mathbf{H}}[q]^H \left(\tilde{\mathbf{H}}[q] \mathbf{D}_{\mathbf{p}_\tau}^{\frac{1}{2}} \tilde{\mathbf{H}}[q]^H \right)^{-2} \tilde{\mathbf{H}}[q] \mathbf{D}_{\mathbf{p}}^{\frac{1}{2}} \mathbf{e}_{\tilde{m}} \quad (4.18)$$

which appears in the right-hand side of (4.15). It can be shown that this quantity is asymptotically equivalent to a deterministic one (almost surely) as the system dimensions increase to infinity. Let us now introduce the variables b_l , $l \in \{1, \dots, L\}$, which form the unique non-negative solutions to the L equations

$$b_l = \frac{1}{K} \text{tr} \left(\rho_{\tau,l}^{\frac{1}{2}} \mathbf{D}_l \left(\sum_{l'=1}^L \frac{1}{M_{l'}} \sum_{m=1}^{M_{l'}} \frac{\xi_{l',m}}{1 + c_{l'} \xi_{l',m} b_{l'}} \rho_{\tau,l'}^{\frac{1}{2}} \mathbf{D}_{l'} \right)^{-1} \right) \quad (4.19)$$

where $\xi_{l,m}$ is the m -th eigenvalue of $\mathbf{C}_{\text{AP},l}$. The variables b_l , $l \in \{1, \dots, L\}$, are defined by the following equation

$$\begin{bmatrix} \dot{b}_1 \\ \vdots \\ \dot{b}_L \end{bmatrix} = (\mathbf{I}_L - \mathbf{A})^{-1} \begin{bmatrix} \frac{1}{K} \text{tr} \left(\rho_{\tau,1}^{\frac{1}{2}} \mathbf{D}_1 \mathbf{B}^{-2} \right) \\ \vdots \\ \frac{1}{K} \text{tr} \left(\rho_{\tau,L}^{\frac{1}{2}} \mathbf{D}_L \mathbf{B}^{-2} \right) \end{bmatrix} \quad (4.20)$$

with $\mathbf{A} \in \mathbb{R}^{L \times L}$, whose single entry is $[\mathbf{A}]_{l,l'} = \frac{1}{K} \rho_{\tau,l}^{\frac{1}{2}} \text{tr} \left(\mathbf{D}_l \mathbf{B}^{-2} \rho_{\tau,l'}^{\frac{1}{2}} \mathbf{D}_{l'} \right)$, and

$$\mathbf{B} = \sum_{l=1}^L \frac{1}{M_l} \sum_{m=1}^{M_l} \frac{\xi_{l,m}}{1 + c_l \xi_{l,m} b_l} \rho_{\tau,l}^{\frac{1}{2}} \mathbf{D}_l \quad (4.21)$$

$$\mathbf{D}'_l = \frac{1}{M_l} \sum_{m=1}^{M_l} \frac{c_l \xi_{l,m}^2}{(1 + c_l \xi_{l,m} b_l)^2} \mathbf{D}_l. \quad (4.22)$$

Let us now proceed to express the deterministic equivalent of the individual transmit powers at the APs.

Theorem 4.1. *Under (As4.1) – (As4.4), for a fixed number of APs L and considering the Kronecker channel model in equation (4.5), it holds that $(\rho_{\tau,l} - \widetilde{\rho}_{\tau,l}) \xrightarrow{\text{a.s.}} 0$, where*

$$\widetilde{\rho}_{\tau,l} = \frac{1}{M_l^2} \text{tr} \left(\mathbf{D}_{\text{sel},l} \mathbf{D}_{\mathbf{p}_\tau}^{\frac{1}{2}} \mathbf{V} \right) \quad (4.23)$$

with the matrix $\mathbf{V}$ given by

$$\mathbf{V} = \text{blkdiag} \left(\left\{ (\mathbf{I}_{M_l} + c_l b_l \mathbf{C}_{\text{AP},l})^{-2} c_l \dot{b}_l \mathbf{C}_{\text{AP},l} \right\}_{l=1}^L \right) \quad (4.24)$$

and the matrix $\mathbf{D}_{\text{sel},l} = \text{blkdiag} \left(\left\{ \delta_{l,l'} \mathbf{I}_{M_{l'}} \right\}_{l'=1}^L \right)$, where $\delta_{l,l'}$ is the Kronecker delta function, selecting the antenna indexes of the AP l .

Proof. The proof can be derived from [116, Theorem 1 and Corollary 1], in particular the expressions of $e_i(z)$ and $\delta_i(-x)$, and [115, Lemma 3.1]. $\square$

Equation (4.23) is a fixed-point equation in the APs powers and can be solved by using a standard fixed-point method (e.g., Algorithm 3.1). The variables $\{b_l\}$ can also be computed via fixed-point algorithm, while $\{\dot{b}_l\}$ are computed in closed form as a function of $\{b_l\}$. Note that the computation of $\{\widetilde{\rho}_{\tau,l}\}$ requires knowing only the second-order channel statistics (i.e., large-scale fading coefficients and the correlation coefficients between the AP antennas), which are much more stable over time compared to the instantaneous channels. In our overall proposed strategy, the CU performs the steps depicted in Fig. 4.2. Note that the value of the threshold ϵ relates to the performance of the fixed-point algorithm and is set empirically. Notably, the number of instantaneous channel coefficients received by the CU is reduced from $\tilde{M}KQ$ to $\sum_{l \in \mathcal{L}^*} M_l KQ$.

4.3 Performance Evaluation

In this section, we present the numerical evaluation of the proposed cell-free energy-saving algorithm. We assume a square deployment area of $1 \times 1 \text{ km}^2$ with 16 possible AP locations on a regular grid, where the positions of the L APs are randomly selected at every channel realization. A channel realization generates new values for both small-scale and large-scale fading coefficients. The users are uniformly distributed within the scenario, with a minimum distance

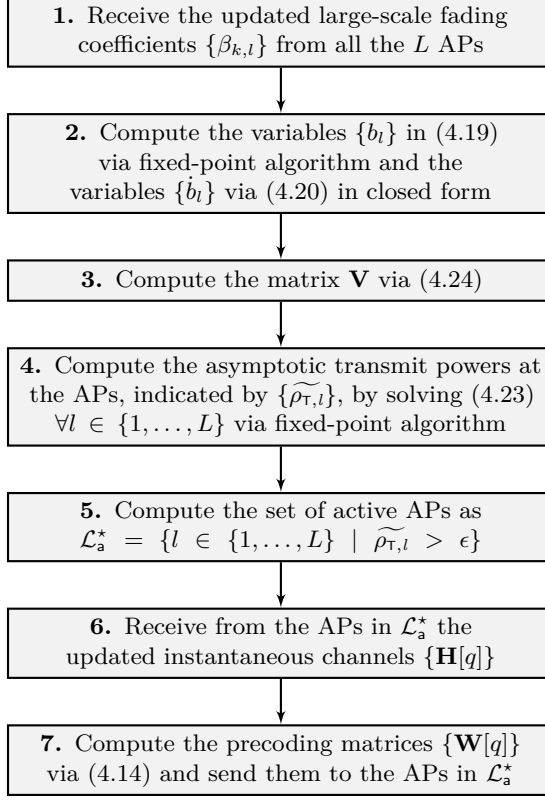


Figure 4.2: Algorithm to be executed at the CU based on deterministic equivalent to retrieve the powers at the APs, determining the active APs, and compute the precoders among these active APs.

of 10 meters to an AP, and their locations are randomly generated at every channel realization. We consider a constant number of antennas per AP, set to $M_l = M = 8 \forall l$. When characterizing the average gains in power consumption, we consider a number of users varying from $K = 1$ to $K = 19$, as well as a number of APs ranging from $L = 2$ to $L = 10$. The target SNRs $\{\psi_k\}$ are randomly generated between 1 dB and 20 dB with a uniform distribution, while the noise power is $\sigma_\nu^2 = -96$ dBm. The large-scale fading coefficients are computed (in dB) as [117]

$$10 \log_{10}(\beta_{k,l}) = -30.5 - 37.6 \log_{10}(u_{k,l}) + F_{k,l} \quad [\text{dB}] \quad (4.25)$$

where $u_{k,l}$ is the distance (in meters) between the user k and the AP l , and $F_{k,l} \sim \mathcal{N}(0, 16)$ represents the shadow fading. The correlation between the

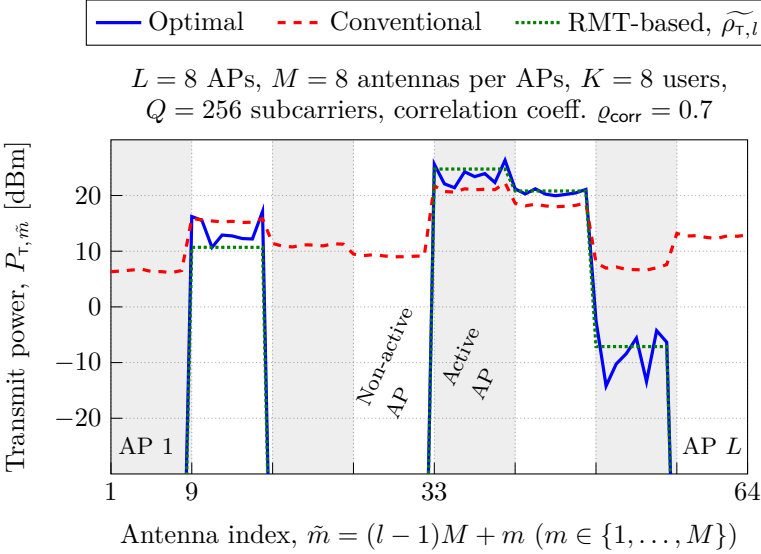


Figure 4.3: Transmit power as a function of the antenna index for a single realization of the per-AP Kronecker channel model. The graph shows the transmit power per antenna for the optimal solution (4.14), the conventional one (4.12), and the asymptotic one based on RMT (Theorem 4.1).

antennas of an AP follows the exponential model [118], which yields

$$[\mathbf{C}_{\text{AP}, l}]_{m, m'} = \varrho_{\text{corr}}^{|m-m'|} \quad (4.26)$$

where ϱ_{corr} is the correlation coefficient that is set to $\varrho_{\text{corr}} = 0.7$. We consider a PA maximal transmit power and efficiency of $P_{\text{max}} = 3$ W and $\eta_{\text{PA}, \text{max}} = 0.34$, respectively [57]. The fixed power consumption is set to $P_{\text{fixed}} = 15$ W, while the circuit power consumption per antenna is set to $\mathcal{C} = 0.7$ W [14].

4.3.1 Single Channel Realization

Fig. 4.3 shows the transmit powers at the different antennas for a single channel realization, $K = 8$ users, $L = 8$ APs, and $Q = 256$ subcarriers. We observe that the powers among the antennas of an AP share similar values, with a certain variability that is likely caused by the correlation among antennas. The first and last antennas, being the least correlated given that we are considering a uniform linear array at each AP, are allocated more power. The RMT-based

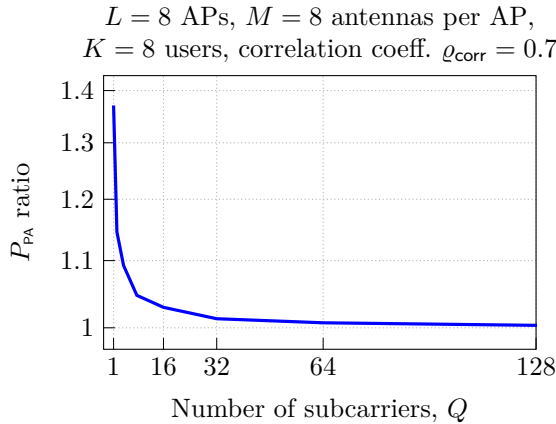


Figure 4.4: Ratio of PA consumption by using the asymptotic solution and PA consumption by using the optimal solution as a function of the number of subcarriers, averaged over 10^2 channel realizations (note: y -axis is in log scale).

solution uses a uniform power per AP, and this power approximates the average of the optimal solution among the M antennas of the AP. Fig. 4.3 indicates also that the results in Theorem 4.1, even though derived for $M, K \rightarrow \infty$, are accurate for finite and relatively low values of M and K (in this case $M = 8$ antennas per AP and $K = 8$ users).

It is interesting to notice that the solution that minimizes the PAs consumed power turns on/off entire APs. All the antennas of the active APs are utilized, while all the antennas of the other APs (which will then not be active) are not active. Instead, the solution that optimizes transmit power always activates all the APs. This happens because minimizing transmit power is equivalent to maximizing the antenna array gain. The consumed power solution, on the other hand, utilizes $L_a = 4$ APs, corresponding to halving the FH requirements. The gain in network consumption (i.e., the ratio between P_{net} by using the proposed solution and P_{net} by using the conventional solution), equals 1.78 when both using the optimal solution and the RMT-based one, while the gains in PAs consumption are equal to 1.28 and 1.27, respectively.

4.3.2 Impact of Number of Subcarriers

We observe, via numerical experiments, that the uniform power allocation per AP in (As4.2) approximates well the optimal solution when Q becomes large.

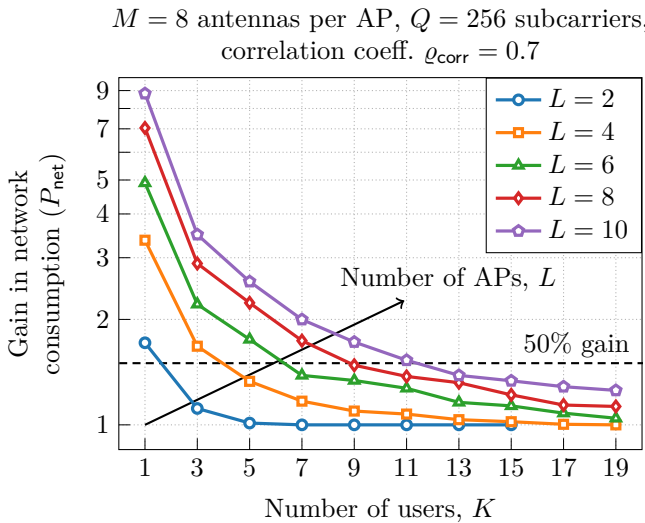


Figure 4.5: Gain in network power consumption (by utilizing the RMT-based solution against the conventional one), averaged over 10^2 channel realizations, as a function of the number of users for different number of APs, $M = 8$ antennas, and $Q = 256$ subcarriers.

This is in line with the findings in Chapter 3 for cellular systems, where uniform allocation among all the BS antennas is optimal (in terms of PAs consumption) when Q is large and there is no spatial correlation. Fig. 4.4 depicts the ratio between the values of P_{PA} by using the optimal and asymptotic transmit powers (in equations (4.15) and (4.23), respectively) as a function of Q . For a few subcarriers, the optimal solution activates only a part of the antennas of certain APs. When Q is sufficiently large, instead, the two strategies achieve the same performance. This tells us that the essential information is given by the set of active APs $\mathcal{L}_a^*$. Once we know which APs to activate, we can use a uniform power allocation among them with a negligible performance loss.²

4.3.3 Achievable Gains in Network Consumption

Fig. 4.5 illustrates the potentially achievable (averaged) savings in network consumption by using the asymptotic solution $\{\widetilde{\rho}_{\text{T},l}\}$ over the conventional one for $Q = 256$ subcarriers. As expected, the larger the number of APs, the higher the gains, as the consumed power solution activates only a fraction of the L

²This holds under the assumption of a Kronecker channel model.

APs in low and medium traffic scenarios. The savings decrease as K increases because (i) more users require the activation of more antennas and (ii) it is more likely that a user will be in the proximity of each AP. For $L = 10$ APs, the gain reaches a factor of $9\times$ when there is $K = 1$ user and remains large also for $K = 7$, when it equals 2. For $L = 8$ APs and $K = 9$ users, the saving is still equal to 1.5 (corresponding to a 50% less consumed power). In general, the ratios become lower than or equal to 1.5 when $K > L$.

Remark: The gains (or savings) in network consumption achieved by the RMT-based solution over the conventional one (that optimizes the transmit power) depend, among other factors, on the consumption of the FH links. This FH consumption can be part of the term $\sum_{l \in \mathcal{L}_a} \mathcal{C}M_{a,l}$ in the power model (4.9). We point out that the FH may not always be the bottleneck of consumption (e.g., if the FH is implemented via highly-efficient fiber links as in the connection between baseband and radio units in cellular deployments). Even in this case, the proposed solution has energy-saving potential over the conventional one thanks to the lower number of activated APs. This implies (a) less computations required at the CU, and (b) zero or reduced consumption in the APs that are not activated. Compared to the ‘exact’ solution of previous chapter, given in equation (4.14) and equation (4.15), the RMT-based solution activates the same number of APs. In the RMT-based solution, however, the CU computes the power allocations only when the large-scale fading changes—in contrast to the solution of previous chapter. This brings energy-saving potential in terms of reduced CU (computing) consumption, even with ideal FH links not consuming power (hence the less frequent use of FH links does not affect consumption).

4.4 Conclusion

In this chapter, we addressed the problem of designing cell-free MIMO precoders that minimize the PAs consumption subject to QoS constraints, with the baseband beamforming processing centralized at the CU. Under a per-AP Kronecker channel model that incorporates correlation among the antennas of each AP, we applied RMT to derive simple fixed-point equations in the antenna powers that depend only on the channel covariances and are accurate when the number of subcarriers grows. The rate of change of the channel covariances (i.e., the channel’s second-order statistics) is lower than the rate of change of the channel’s instantaneous coefficients, and this implies that the CU needs a less frequent exchange of information with all the APs. The optimized strategy consists of activating all the antennas of certain APs with uniform transmit powers, and deactivating all the antennas of the remaining APs. This replies to **RQ3** on the generalization of the transmit-power allocation and (de)activation

of antennas framework for centralized BSs to distributed BSs. By switching off the PAs, analog and digital circuits of the non-active APs, as well as reducing the FH links' consumption and requirements, we have shown the significant potential in network power consumption savings (up to a factor of $9\times$) in low-to-medium traffic scenarios.

Extensions of this work include the consideration of local processing (instead of centralized ZF, the precoding can be computed partly or fully at the APs) and user-centric implementations, where each user is served by a (potentially different) subset of the APs. This per-user AP selection could follow similar mechanisms to the one proposed in this chapter. The integration of deactivation of resources in the time domain, i.e., transmitting only in a subset of the available OFDM symbols, could also unlock larger energy savings, with APs deactivation jointly determined in space and time.

Chapter 5

Integration of Deactivation of Transmit Symbols

Bibliographic Reference

This chapter is based on: E. Peschiera, Y. Agram, F. Quitin, L. Van der Perre, and F. Rottenberg, “On optimizing time-, space- and power-domain energy-saving techniques for sub-6 GHz base stations,” accepted for publication in *IEEE Transactions on Communications*, 2026 [119].¹

This chapter focuses on the downlink (DL) operation of a single base station (BS) and integrates the time dimension by allowing the resource allocation scheme to activate a subset of the transmit orthogonal frequency-division multiplexing (OFDM) symbols in a frame without compromising the delivered DL quality-of-service (QoS). A transmit OFDM symbol is referred to as a time slot. The BS utilizes fully-digital beamforming. To deal with the increased dimensionality of the problem, we consider the use of a zero-forcing (ZF) precoder and optimize the number of active antennas, the transmit power, and the number of active time slots under fixed target users’ rates and maximal power constraints. The power model follows Section 2.2 and is kept more general than in the previous chapters (e.g., it includes other power amplifier (PA) architectures in addition to class B PAs). The goal of this chapter is to provide a unified framework to

¹The Python simulation code is available in open access at the following link: github.com/dramco-research/energy-saving-time-space-power-epeschiera-tcom26.

explore the following question: to minimize the BS energy usage while meeting the users' QoS, is it best to minimize the active time slots (*rush-to-sleep*), the active antennas (*rush-to-mute*), the transmit power (*awake-but-whisper*), or to use a combination of these?

State of the Art

The introductory discussion in Section 1.1.3 mentioned that energy savings can be achieved by optimizing the BS resource usage in different domains. In the time domain, micro-discontinuous transmission (μ DTX) has been used to switch off the PAs when not needed on an OFDM symbol timescale (e.g., 71 μ s for a subcarrier spacing of 15 kHz) [120, 33]. The fifth-generation (5G) standard has been designed with a lean approach, requiring less frequent control signaling and therefore enabling longer sleep durations [32]. The impact of using sleep modes during both data transmission and control signaling phases has been evaluated at the standard level [30] and via system-level simulations [34, 121]. The tradeoff between energy saving and latency due to reactivation times has been discussed in [122]. In [123], the authors introduce a two-dimensional modeling of the energy efficiency (EE) that incorporates the time dimension. However, the fundamental works in this field have often considered that the transmission takes place at a constant data rate per time slot [124, 125], thereby precluding the assessment of the optimal transmit powers and data rates in different time slots. The work in [25] fills this theoretical gap in the case of a single-input single-output (SISO) system by considering an average data rate constraint over multiple time slots in a frame. It is shown that transmission at constant power among a number of active time slots is optimal and that a rush-to-sleep approach, using as few time slots as possible at maximum transmit power, is optimal in the low-signal-to-noise ratio (SNR) regime.

In the space domain, many works have focused on the load-aware (de)activation of antennas in massive multiple-input multiple-output (MIMO) BSs [17]. In Chapter 3, we have shown that antenna deactivation provides energy savings with respect to the PA or BS consumed power, depending on the scenario. The authors of [21] find the optimal number of active antennas and users that maximize the EE by building on a detailed circuit consumption model. A similar analysis is done in [35] by considering a multi-cell network and traffic load variations. The work [83] addresses the problem of minimizing power consumption under per-user signal-to-interference-plus-noise ratio (SINR) constraints. The precoder that minimizes BS consumption under SINR constraints for a multicarrier centralized massive MIMO system is derived in Chapter 3, where we have shown that conventional ZF among a subset of antennas is optimal. The work [22] specifically targets energy consumption

reduction by maximizing the number of muted (deactivated) antennas subject to QoS constraints, and solves the problem using neural networks. Space-domain resource adaptation becomes even more important with the increasing interest in cell-free MIMO, and we have shown in Chapter 4 that it is essential to keep the total consumption under control by turning on/off access points (APs) and their antennas.

Power-domain energy-saving techniques can instead optimize either the total BS transmit power as is the case in classic precoding designs [126], or the individual transmit powers at the BS antennas in order to improve the PAs efficiency [39]. This has also been discussed in Chapter 3. Among the other different techniques, constant-envelope precoding has been proposed in [127]. The authors in [84] find the optimal point-to-point MIMO precoder that maximizes the data rate under a PA consumption constraint. More recently, flat-ZF precoding has been proposed with the same objective of reducing the PA consumption [128].

Recent works have looked at multi-domain resource allocation to reduce the BS energy usage or to increase the EE. The joint optimization of spectral and spatial resources in a multi-cell network under ZF precoding is addressed in [129], while the optimal precoders and scheduling of resources are found in [130] via convex solvers. In [131], the authors optimize the sleep mode configuration, number of active antennas, and user-BS associations in a multi-cell network using reinforcement learning. However, obtaining accurate while simple and insightful solutions to the resource allocation problem in the domains of time (symbol), space (antenna), and transmit power remains an open question, and is the focus of this chapter.

In this thesis, we consider OFDM but do not consider the deactivation of specific subcarriers for energy reduction. Indeed, we estimate that lower energy savings can be achieved by subcarrier deactivation compared to PAs or radio frequency (RF) chains deactivation, symbols deactivation, and transmit power reduction. Based on the power model evaluations of Section 2.3.2 scaled to consider 100 MHz of bandwidth and 64 BS antennas, we quantify that at 30% of loads, MIMO precoding/combining and OFDM modulation/demodulation (which are the main modules impacted by subcarrier deactivation) account for 4.6% of the total BS consumption. Therefore, we estimate that at most, subcarrier deactivation can provide 4.6% of BS energy savings, where this percentage will be lower in practice as a minimum number of subcarriers must be activated to guarantee the user rates. Different considerations apply to the deactivation of entire frequency bands in multi-band systems. In this thesis, however, we limit the scope of the analysis to one frequency band.

Contributions

In this chapter, we investigate the optimal combination of active antennas, active time slots, and transmit power levels such that the average BS consumption over a time frame is minimized, the users' rates are guaranteed, and the maximal transmit power is not exceeded. Three particular allocations are identified: a rush-to-sleep that minimizes the number of active time slots, a rush-to-mute that minimizes the number of active antennas, and an awake-but-whisper that minimizes the transmit power at active antennas. Under the main assumptions of this chapter, which are the use of a ZF precoder and independently and identically distributed (i.i.d.) Rayleigh fading channels, our specific contributions are the following:

- We first show that the initial non-convex three-dimensional optimization problem can be simplified into a two-dimensional problem in the number of active antennas and number of active time slots.
- We prove the convexity of the two-dimensional objective function, relaxed to be continuous and not discrete, and show a sufficient condition for the convexity of the optimization domain.
- We propose a low-complexity approximation of the final discrete solution, involving simple rounding operations, and provide the asymptotic decay of the approximation error.
- We showcase an efficient algorithm based on unconstrained optimizations that leverage convexity to retrieve the proposed solution without the need of using a black-box convex solver.

5.1 System Model

In this section, we describe the transmission and power models considered throughout this chapter. Both models are derived from Chapter 2.

5.1.1 Transmission Model

Following Section 2.1.1, we consider a BS with M antennas and K single-antenna users communicating in DL using OFDM. The users are multiplexed via space-division multiple access (SDMA) implying that they are being served at the same time and frequency, and we consider that $M \geq K$. The BS

utilizes fully-digital beamforming with $M = M_{\text{RF}}$ antennas and RF chains and $M_{\text{a}} = M_{\text{RF},\text{a}}$ active antennas and RF chains. In the following, antenna index m also represents the PA m and the RF chain m , with $m \in \{1, \dots, M\}$. Given that we fix the target QoS in DL, we consider the DL transmission and lighten the notation by defining $Q \triangleq Q_{\text{DL}}$, $K \triangleq K_{\text{DL}}$, $\mathbf{H} \triangleq \mathbf{H}_{\text{DL}}$, $\mathbf{W} \triangleq \mathbf{W}_{\text{dig}}$, $P_{\text{T}} \triangleq P_{\text{T,DL}}$, $N \triangleq N_{\text{DL}}$, $N_{\text{a}} \triangleq N_{\text{a,DL}}$, and $p_k \triangleq p_{k,\text{DL}}$ such to represent the number of data subcarriers, number of active users, channel matrix, digital precoding matrix, total transmit power, total number of time slots, number of active time slots, and power allocation of user k , respectively.

The time-domain structure follows Fig. 2.2, with the transmission divided into frames lasting T_{frame} seconds. In every frame, there are N OFDM symbols available to carry data symbols. Of these N OFDM symbols, only N_{a} are active, i.e., used to communicate. With the perspective of DL transmission, a frame is divided in four phases: (i) DL and reference signal transmission, (ii) DL and data transmission, (iii) DL and no reference signal and no data transmission, and (iv) no DL. This last phase is either the uplink (UL) in time-division duplexing (TDD) or has zero duration in frequency-division duplexing (FDD) given that the UL takes place on a different frequency. The duration of each phase, relative to T_{frame} , is given inside the blocks of Fig. 2.2 by setting $i = \text{DL}$, where we set in the following $\tau_{\text{sig}} \triangleq \tau_{\text{DL,sig}}$ for simplicity. Without loss of generality, we consider a single frame in the rest of the chapter.

We consider that only a subset of M_{a} antennas out of M is active, with $K \leq M_{\text{a}} \leq M$. Under the model in equation (2.5), the transmitted symbols at each subcarrier and OFDM symbol are zero mean and uncorrelated among the users. There is one symbol per user on a given subcarrier and active OFDM symbol, with variance p_k that is the power allocation of user k . The transmitted symbol vector on one subcarrier and OFDM symbol is $\mathbf{s}_{\text{DL}} \in \mathbb{C}^{K \times 1}$, where $\mathbb{E}\{\mathbf{s}_{\text{DL}}\mathbf{s}_{\text{DL}}^{\text{H}}\} = \text{diag}(p_1, \dots, p_K)$. This is different from previous chapters, where we considered a uniform power allocation among users. The corresponding channel matrix between the active antennas and the users is denoted by $\mathbf{H} \in \mathbb{C}^{K \times M_{\text{a}}}$. We consider the use of a ZF precoder on any subcarrier and OFDM symbol, fixing $\mathbf{W} = \mathbf{H}^{\text{H}}(\mathbf{H}\mathbf{H}^{\text{H}})^{-1}$ and giving the precoded symbols

$$\mathbf{x} = \mathbf{H}^{\text{H}}(\mathbf{H}\mathbf{H}^{\text{H}})^{-1} \mathbf{s}_{\text{DL}} \in \mathbb{C}^{M_{\text{a}} \times 1}. \quad (5.1)$$

Remark: Differently from the previous chapters where we optimized the precoders under a ZF constraint, we hereinafter fix the precoder to a conventional ZF and optimize the power allocation (equal at any utilized antenna) and number of utilized resources. This choice follows the observations in Chapter 3, where in a system with many subcarriers such as a broadband-OFDM one, conventional ZF—which uniformly allocates the power among the utilized antennas—has a

negligible performance loss compared to the proposed precoder—which has the freedom of allocating a different power to each of the utilized antennas.

The large-scale fading coefficient of user k is denoted by β_k . The small-scale fading is assumed to be i.i.d. Rayleigh, where any entry of $\mathbf{H}$ follows a $\mathcal{CN}(0, 1)$ distribution. While i.i.d. Rayleigh fading is commonly considered in the massive MIMO literature [90], several experimental studies have shown that it is often not confirmed in practice [88, 89]. Still, this assumption is made here as being useful for the purpose of finding mathematically tractable resource-allocation strategies. Considering a unitary inverse fast Fourier transform (IFFT) for OFDM modulation, the average power of the precoded symbols in the frequency domain is the same as in the time domain (after the IFFT for OFDM demodulation). Under i.i.d. Rayleigh fading, the average total transmitted power is given by

$$\begin{aligned} P_{\tau} &= \mathbb{E} \left\{ \text{tr}(\mathbf{H}^H(\mathbf{H}\mathbf{H}^H)^{-1} \mathbf{s}_{\text{DL}} \mathbf{s}_{\text{DL}}^H (\mathbf{H}\mathbf{H}^H)^{-1} \mathbf{H}) \right\} \\ &\stackrel{(*)}{=} \text{tr}(\mathbb{E} \{ (\mathbf{H}\mathbf{H}^H)^{-1} \} \mathbb{E} \{ \mathbf{s}_{\text{DL}} \mathbf{s}_{\text{DL}}^H \}) \stackrel{(**)}{=} \frac{1}{M_{\text{a}} - K} \sum_{k=1}^K \frac{p_k}{\beta_k} \end{aligned} \quad (5.2)$$

where $(*)$ follows from the trace cyclic property and the fact that the symbols and channel coefficients are uncorrelated, while $(**)$ uses the expression of the mean of a complex inverse-Wishart distribution [132].

An array gain of a factor $M_{\text{a}} - K$ is present in (5.2). To minimize P_{τ} , the maximal number of antennas should be active, i.e., $M_{\text{a}} = M$. In terms of consumed power, denoted by P_{cons} , this is often not optimal as P_{cons} is not linearly proportional to P_{τ} because of the PA non-constant efficiency and the load-independent consumption of activating each antenna. This is discussed in Section 2.2.1. Given that the channels of the different antennas are identically distributed, the average transmit power is the same at each antenna and equal to

$$P_{\text{a}} = \frac{P_{\tau}}{M_{\text{a}}} = \frac{1}{M_{\text{a}}(M_{\text{a}} - K)} \sum_{k=1}^K \frac{p_k}{\beta_k}. \quad (5.3)$$

The selection of the M_{a} active antennas can be arbitrary given their identical distribution. In the following, we consider a maximal per-antenna transmit power constraint $P_{\text{a}} \leq P_{\text{max}}$. After going through the channel and OFDM demodulation at the receiver, the transmitted symbols are recovered with the addition of zero-mean circularly-symmetric complex-Gaussian (ZMCSCG) noise, where the noise variance at user k is denoted by σ_k^2 . Therefore, the total bit rate delivered to user k averaged over the frame duration is given by

$$R_{k,\text{deliv}} = \frac{Q}{T} \tau_{\text{DL}} (1 - \tau_{\text{sig}}) \frac{N_{\text{a}}}{N} \log_2 \left(1 + \frac{p_k}{\sigma_k^2} \right) \quad [\text{bit/s}] \quad (5.4)$$

expressed in bits per second, where T is the OFDM symbol period and Q is the number of utilized data subcarriers. The rate loss due to the “Reference Signal” and “No Reference Signal, No Data” phases is quantified by the pre-log factor $\tau_{\text{DL}}(1 - \tau_{\text{sig}}) \frac{N_a}{N}$ that is the duration of the “Data” phase relative to T_{frame} (cf. Fig. 2.2). To simplify notations in the following, we define

$$R_k = \frac{R_{k,\text{deliv}}}{\frac{Q}{T} \tau_{\text{DL}}(1 - \tau_{\text{sig}})} \quad [\text{bit}/(\text{OFDM symbol} \cdot \text{subcarrier})] \quad (5.5)$$

as the normalized average rate expressed in bits per OFDM symbol and per subcarrier. We consider that each user has a target average rate R_k , which then fixes its QoS. Given our frame and average rate models, we are constraining the delivery of $NT R_{k,\text{deliv}}$ bits (payload) to user k in T_{frame} seconds.²

5.1.2 Power Consumption Model

In this section, we adapt the power model of Section 2.2 to the scenario of this chapter. The model is mapped to the one in [15] that is validated on operator measurements and manufacturer documentation of sub-6 GHz fourth-generation (4G) and 5G BSs in Belgium. In particular, it predicts the consumption of the analyzed BSs within 10–20% of uncertainty and can be applied to radio unit (RU) types including remote radio unit (RRU) and active antenna unit (AAU), working either in FDD or TDD, and running either 4G Long-Term Evolution (LTE) or 5G New Radio (NR). Let us recall that we express the BS consumption averaged over the frame duration as

$$P_{\text{cons}} = \frac{\overline{P_{\text{PA}}}}{\eta_{\text{PA},\text{s/c}}} + \frac{\overline{P_{\text{ana}}}}{\eta_{\text{ana},\text{s/c}}} + \frac{\overline{P_{\text{dig}}}}{\eta_{\text{dig},\text{s/c}}} \quad (5.6)$$

where $\overline{P_{\text{PA}}}$, $\overline{P_{\text{ana}}}$, and $\overline{P_{\text{dig}}}$ are the average powers consumed by the PAs, analog, and digital components during the frame, while $\eta_{\text{PA},\text{s/c}} \in (0, 1]$, $\eta_{\text{ana},\text{s/c}} \in (0, 1]$, and $\eta_{\text{dig},\text{s/c}} \in (0, 1]$ are the supply and cooling efficiencies of each component. Both DL and UL phases are considered in the consumptions $\overline{P_{\text{ana}}}$ and $\overline{P_{\text{dig}}}$. Each of the three components contains subcomponents (cf. Fig. 2.4). In the model [15], the analog component is termed analog front-end (AFE), and the digital component is termed digital baseband (DBB). The differences between the model of Section 2.2 and the model [15] are:

1. The signaling and data transmission phases are not overlapped in time in our model, while [15] considers that a part of signaling is performed at the

²The exact maximum delivery time is $\tau_{\text{DL}} \tau_{\text{sig}} T_{\text{frame}} + NT$ seconds. If all the reference signals are sent first, followed by all the active OFDM symbols, the payload of each user is sent in $\tau_{\text{DL}} \tau_{\text{sig}} T_{\text{frame}} + N_a T$ seconds.

same time as data transmission (carried out on different subcarriers). The time-domain structure is therefore slightly different. The choice in [15] reflects more closely the 5G-NR implementation. However, we made a different choice to decouple the transmit power dedicated to signaling from the transmit power dedicated to data transmission. Otherwise, a term $(P'_a)^\alpha$, where P'_a is the sum of the power for signaling and data transmission, would appear in the objective function and render the problem coupled.

2. There is no (average physical) load dependency of analog and digital consumption in [15] (in both DL and UL), equivalent to setting the reduction factors $\delta_{\text{ana,micro}} = 0$ and $\delta_{\text{dig,micro}} = 0$ in our model. The fact that the PA is the only load-dependent component is commonly observed in sub-6 GHz BSs [14, 15, 50, 133].
3. The model [15] does not express the consumption of analog and digital subcomponents, but uses aggregated values for AFE and DBB consumption. In our model, we express the power consumption of analog and digital subcomponents as shown in Section 2.2.2 and Section 2.2.3. This subcomponents' modeling was applied to upper mid-band BSs in Section 2.3, while is not applied here as the aggregate values provided in [15] are suited for sub-6 GHz BSs—which are the focus of this chapter.
4. We consider an arbitrary PA output power (P_a) during data transmission in our model, while in [15] the active PAs always transmit at maximum power ($P_{\max}$) during data transmission. The use of $P_a \leq P_{\max}$ in the following does not alter the correctness of the power modeling and allows us to study different transmit-power allocations.
5. Our model considers a single cell, i.e., single sector and frequency band, while model [15] considers multiple cells. This choice is made here to limit the presentation complexity, but multiple cells can be easily included by considering the sum of the consumptions in each cell.

The only approximation when mapping our model to model [15] is the difference in point 1. of the above list. We recall that in our framework, when a time slot is active, we assume that all its data subcarriers are used to serve all the users using SDMA. Hence, the average physical loads in DL and UL are given by $\overline{\ell}_{\text{DL}} = (K_{\text{DL}}/K_{\max})(N_{\text{a,DL}}/N_{\text{DL}}) = (K/K_{\max})(N_{\text{a}}/N)$ and $\overline{\ell}_{\text{UL}} = (K_{\text{UL}}/K_{\max})(N_{\text{a,UL}}/N_{\text{UL}})$. Also, we consider that the number of active and total spatial layers is equal to the number of active users and total number of users K and $K_{\max}$.

When expressing and rearranging the terms of equation (5.6), we can write the consumed power as a function of the number of active time slots N_{a} , number of

active antennas M_a , and transmit power at active antennas P_a , giving

$$P_{\text{cons}}(N_a, M_a, P_a) = \frac{N_a}{N} M_a \left(\frac{P_0}{M} + \gamma P_a^\alpha \right) + \frac{M_a}{M} P_1 + P_{\text{sleep}} \quad (5.7)$$

where P_0 , P_1 , P_{sleep} , γ are non-negative parameters, and $\alpha \in [0.5, 1]$ is the PA exponent. The term multiplying P_0 depends on both the ratio of active time slots and active antennas. The term depending on γP_a^α models the power consumed by the PAs. The third term scales with the ratio of active antennas, while the last term is constant. Equation (5.7) is a compact representation of the model presented in Section 2.2, mapped to replicate the sub-6 GHz model in [15]. In the following, we detail the complete expressions of the model parameters P_0 , P_1 , P_{sleep} , and γ to highlight their dependencies.

Power Amplifier

To model the PA consumption, we use the model (2.24) presented in Section 2.2.1, which corresponds to the case where the PA output power is constant among active antennas and equal to P_a . By expressing $\overline{\ell_{\text{DL}}} = (K/K_{\text{max}})(N_a/N)$ and grouping the terms depending on N_a and M_a , we obtain

$$\begin{aligned} \overline{P_{\text{PA}}} &= \frac{N_a}{N} M_a \left[\tau_{\text{DL}} (1 - \tau_{\text{DL, sig}}) (\check{P}_{\text{PA}}(P_a) - \check{P}_{\text{PA}}(0) \delta_{\text{PA, micro}}) \right] \\ &+ M_a \left[(\tau_{\text{DL}} \tau_{\text{sig}} \check{P}_{\text{PA}}(\zeta_{\text{DL, sig}} P_{\text{max}}) + \tau_{\text{DL}} (1 - \tau_{\text{sig}}) \check{P}_{\text{PA}}(0) \delta_{\text{PA, micro}} \right. \\ &\left. + (1 - \tau_{\text{DL}}) \check{P}_{\text{PA}}(0) \delta_{\text{PA, idle}} - \check{P}_{\text{PA}}(0) \delta_{\text{PA, sleep}} \right] + M \check{P}_{\text{PA}}(0) \delta_{\text{PA, sleep}} \end{aligned} \quad (5.8)$$

where we recall that $\check{P}_{\text{PA}}(\check{P}_{\text{T}}) = \xi_{\text{PA}} (P_{\text{max}}/\eta_{\text{PA, max}}) + (1 - \xi_{\text{PA}}) (P_{\text{max}}^{1-\alpha} \check{P}_{\text{T}}^\alpha / \eta_{\text{PA, max}})$, therefore the static PA consumption (at zero output power) is given by $\check{P}_{\text{PA}}(0) = \xi_{\text{PA}} (P_{\text{max}}/\eta_{\text{PA, max}})$. We can identify in $\overline{P_{\text{PA}}}$ the terms that are part of γ and P_0/M in equation (5.7), as well as the terms that are part of P_1/M and P_{sleep} .

Analog Signal Processing

Concerning the power model of analog signal processing, [15] uses an aggregated value among subcomponents, considers that the consumption in DL and UL is equal,³ and assumes there is no load dependency (in both DL and UL). If

³This assumption, even though quite drastic, allows the authors to limit the power model's complexity while remaining accurate in predicting the power consumed by the analyzed sub-6 GHz BSs configurations.

we set $\delta_{\text{ana,micro}} = 0$ in equation (2.26), the addition of the first and third terms removes the load dependency. Let us recall that $M_{\text{RF,a}} = M_{\text{a}}$ in this chapter. Following the assumption of equal DL and UL consumptions of [15] and using an aggregated value for the subcomponents' consumption during working mode, given by P_{TRX} , we can express

$$\begin{aligned} \overline{P_{\text{ana}}} = & M_{\text{a}}(\tau_{\text{DL}}P_{\text{TRX}} + \tau_{\text{UL}}P_{\text{TRX}} + (1 - \tau_{\text{DL}})P_{\text{TRX}}\delta_{\text{ana,idle}} + (1 - \tau_{\text{UL}})P_{\text{TRX}}\delta_{\text{ana,idle}}) \\ & + 2(M - M_{\text{a}})P_{\text{TRX}}\delta_{\text{ana,sleep}} + P_{\text{ana,misc}}. \end{aligned} \quad (5.9)$$

The above directly follows from equation (2.25) and equation (2.26) when considering that there are M_{a} active transmitting RF chains and M_{a} active receiving RF chains that are in working mode for $\tau_{\text{DL}}T_{\text{frame}}$ and $\tau_{\text{UL}}T_{\text{frame}}$ seconds, respectively, and are in idle mode for $(1 - \tau_{\text{DL}})T_{\text{frame}}$ and $(1 - \tau_{\text{UL}})T_{\text{frame}}$ seconds, respectively. We can rearrange the different terms and write $\overline{P_{\text{ana}}}$ as

$$\begin{aligned} \overline{P_{\text{ana}}} = & M_{\text{a}} \left[P_{\text{TRX}}((\tau_{\text{DL}} + \tau_{\text{UL}}) + (2 - \tau_{\text{DL}} + \tau_{\text{UL}})\delta_{\text{ana,idle}} - 2\delta_{\text{ana,sleep}}) \right] \\ & + 2MP_{\text{TRX}}\delta_{\text{ana,sleep}} + P_{\text{ana,misc}}. \end{aligned} \quad (5.10)$$

Equation (5.10) corresponds to [15, equation (10)]. Moreover, the authors of [15] propose to set P_{TRX} and $P_{\text{ana,misc}}$ such that they scale with the bandwidth and the transmit power as follows

$$P_{\text{TRX}} = P_{\text{ref,TRX}} \left(\frac{B}{B^*} \right)^{\varsigma_1} \left(\frac{P_{\text{max}}}{P^*} \right)^{\varsigma_2} \quad (5.11)$$

$$P_{\text{ana,misc}} = P_{\text{ref,ana,misc}} \left(\frac{B}{B^*} \right)^{\varsigma_3} \left(\frac{MP_{\text{max}}}{P^*} \right)^{\varsigma_4} \quad (5.12)$$

where $P_{\text{ref,TRX}}$ and $P_{\text{ref,ana,misc}}$ are reference values for the consumption by the RF chains and miscellaneous functions, B^* and P^* are reference values for the bandwidth and transmit power, and ς_1 , ς_2 , ς_3 , and ς_4 are scaling exponents. The analog processing consumption is therefore (physical) load independent and part of P_1/M and P_{sleep} in equation (5.7).

Digital Signal Processing

Similarly to the analog consumption, model [15] quantifies the consumption of digital processing by using an aggregated value among subcomponents, considering equal consumption in DL and UL, and assuming that there is no load dependency (in both DL and UL). In the model presented in Section 2.2.3, there is both a dependency on the number of (active) antennas, in equation (2.33), and

on the number of (active) users in the individual expressions of subcomponents in equation (2.34). Model [15] considers only a dependency in the number of (active) users (i.e., the number of (active) spatial layers). Defining P_{phy} as the aggregated consumption of digital subcomponents in working mode, [15, equation (11)] gives the average digital consumption

$$\begin{aligned}\overline{P_{\text{dig}}} &= K(\tau_{\text{DL}}P_{\text{phy}} + \tau_{\text{UL}}P_{\text{phy}} + (1 - \tau_{\text{DL}})P_{\text{phy}}\delta_{\text{dig,idle}} + (1 - \tau_{\text{UL}})P_{\text{phy}}\delta_{\text{dig,idle}}) \\ &\quad + 2(K_{\text{max}} - K)P_{\text{phy}}\delta_{\text{dig,sleep}} + P_{\text{dig,link}} \\ &= KP_{\text{phy}}((\tau_{\text{DL}} + \tau_{\text{UL}}) + (2 - \tau_{\text{DL}} - \tau_{\text{UL}})\delta_{\text{dig,idle}} - 2\delta_{\text{dig,sleep}}) \\ &\quad + 2K_{\text{max}}P_{\text{phy}}\delta_{\text{dig,sleep}} + P_{\text{dig,link}}.\end{aligned}\tag{5.13}$$

The quantity P_{phy} is set in [15] as

$$P_{\text{phy}} = P_{\text{ref,phy}} \left(\frac{B}{B^*} \right)^{\varsigma_5}\tag{5.14}$$

where $P_{\text{ref,phy}}$ and B^* are reference values for the physical layer consumption and for the bandwidth, while ς_5 is a scaling exponent. As for the analog processing, the digital processing consumption is (physical) load independent and this time is only part of P_{sleep} in equation (5.7).

We can now provide the full expressions of the parameters of the power model in equation (5.7). Plugging equations (5.8), (5.10), and (5.13) into equation (5.6), we obtain the expressions given in the Table 5.1. The power model's parameters γ and P_1 depend on PA-related parameters, where $\delta_{\text{PA,micro}} = 1$ (no μDTX) gives $P_1 = 0$. The PA and analog consumption have an effect on the parameter P_1 , while all the BS components (PA, analog processing, and digital processing) affect the parameter P_{sleep} . The values of the parameters are set according to [15, Table II] and will be discussed in Chapter 6.

5.2 Optimal Allocation of Time, Spatial, and Power Resources

This section considers the problem of minimizing the BS consumed power P_{cons} by optimizing the number of active time slots N_{a} , active antennas M_{a} , and the transmit power at active antennas P_{a} . The optimization is subject to a target rate constraint R_k for each user and a maximal transmit power constraint.

Table 5.1: Expressions of parameters of power model in equation (5.7).

Parameter	Expression
γ	$\frac{\tau_{\text{DL}}(1 - \tau_{\text{sig}})}{\eta_{\text{PA},\text{s/c}}}(1 - \xi_{\text{PA}})\frac{P_{\text{max}}^{1-\alpha}}{\eta_{\text{PA},\text{max}}}$
$\frac{P_0}{M}$	$\frac{\tau_{\text{DL}}(1 - \tau_{\text{sig}})}{\eta_{\text{PA},\text{s/c}}}\xi_{\text{PA}}\frac{P_{\text{max}}}{\eta_{\text{PA},\text{max}}}(1 - \delta_{\text{PA},\text{micro}})$
$\frac{P_1}{M}$	$\begin{aligned} &\frac{1}{\eta_{\text{PA},\text{s/c}}}\left(\tau_{\text{DL}}\tau_{\text{sig}}(1 - \xi_{\text{PA}})\frac{\zeta_{\text{DL},\text{sig}}^\alpha P_{\text{max}}}{\eta_{\text{PA},\text{max}}} + \xi_{\text{PA}}\frac{P_{\text{max}}}{\eta_{\text{PA},\text{max}}}(\tau_{\text{DL}}\tau_{\text{sig}} + \right. \\ &\left. \tau_{\text{DL}}(1 - \tau_{\text{sig}})\delta_{\text{PA},\text{micro}} + (1 - \tau_{\text{DL}})\delta_{\text{PA},\text{idle}} - \delta_{\text{PA},\text{sleep}})\right) + \\ &\frac{1}{\eta_{\text{ana},\text{s/c}}}\left(P_{\text{TRX}}((\tau_{\text{DL}} + \tau_{\text{UL}}) + (2 - \tau_{\text{DL}} + \tau_{\text{UL}})\delta_{\text{ana},\text{idle}} - 2\delta_{\text{ana},\text{sleep}})\right) \end{aligned}$
P_{sleep}	$\begin{aligned} &\frac{1}{\eta_{\text{PA},\text{s/c}}}M\xi_{\text{PA}}\frac{P_{\text{max}}}{\eta_{\text{PA},\text{max}}}\delta_{\text{PA},\text{sleep}} + \frac{1}{\eta_{\text{ana},\text{s/c}}}\left(2MP_{\text{TRX}}\delta_{\text{ana},\text{sleep}} + P_{\text{ana},\text{misc}}\right) + \\ &\frac{1}{\eta_{\text{dig},\text{s/c}}}\left(KP_{\text{phy}}((\tau_{\text{DL}} + \tau_{\text{UL}}) + (2 - \tau_{\text{DL}} - \tau_{\text{UL}})\delta_{\text{dig},\text{idle}} - 2\delta_{\text{dig},\text{sleep}}) + \right. \\ &\left. 2K_{\text{max}}P_{\text{phy}}\delta_{\text{dig},\text{sleep}} + P_{\text{dig},\text{link}}\right) \end{aligned}$

Mathematically, the problem reads

$$\begin{aligned}
 &\underset{N_a, M_a, P_a, \{p_k\}}{\text{minimize}} && P_{\text{cons}} \\
 &\text{subject to} && R_k = \frac{N_a}{N} \log_2 \left(1 + \frac{p_k}{\sigma_k^2} \right), \quad k = 1, \dots, K \quad (5.15) \\
 &&& P_a \leq P_{\text{max}}
 \end{aligned}$$

where P_{cons} is given in equation (5.7). Moreover, as detailed in Section 5.1, the optimization variables have additional constraints, not directly shown in (5.15), given their physical meaning, i.e., $0 \leq N_a \leq N$ and $M_a \leq M$, N_a and M_a are integers while P_a is continuous. Given our realistic power modeling, this problem is a mixed-integer non-convex programming problem, thus challenging to solve. This section adopts a step-by-step approach to solve (5.15) and shows that it can be efficiently solved through a two-dimensional differentiable convex problem. More specifically, in Section 5.2.1 we first give the optimal transmit power P_a as a function of N_a and M_a . In Section 5.2.2, we find the optimal time

and spatial resources N_a and M_a to allocate and we identify four benchmark schemes. Section 5.3.1 proposes an approximation of the optimal solution and Section 5.3.2 details an algorithm to retrieve it. We emphasize that the optimizations and allocation schemes presented hereinafter are the main novelty compared to [15]. Indeed, this and the next chapter go beyond the evaluations of fixed power-saving features (e.g., scaling the number of active antennas with the physical load) by optimizing the joint allocation of transmit power and (de)activation of antennas and time slots in a unified framework. The framework considers per-user QoS constraints and per-antenna power constraints.

5.2.1 Optimal Transmit Power at Active Antennas

As a first step to solve problem (5.15), we can find that the transmit power P_a can be expressed in terms of N_a and M_a through the user rate constraint. Indeed, solving for p_k

$$R_k = \frac{N_a}{N} \log_2 \left(1 + \frac{p_k}{\sigma_k^2} \right) \Leftrightarrow p_k = \sigma_k^2 \left(2^{R_k \frac{N}{N_a}} - 1 \right) \quad (5.16)$$

and substituting the expression of p_k into equation (5.3) gives

$$P_a = \frac{1}{M_a(M_a - K)} \sum_{k=1}^K \frac{\sigma_k^2}{\beta_k} \left(2^{R_k \frac{N}{N_a}} - 1 \right). \quad (5.17)$$

The term $R_k \frac{N}{N_a}$ can be interpreted as the user rate per active time slot, where R_k gives the frame-averaged number of bits per subcarrier and per time slot. Substituting (5.17) into problem (5.15), it simplifies to the two-dimensional problem in M_a and N_a

$$\begin{aligned} \underset{\substack{0 \leq N_a \leq N \\ K \leq M_a \leq M}}{\text{minimize}} \quad & P_{\text{cons}} = \frac{N_a}{N} \frac{M_a}{M} P_0 + \frac{M_a}{M} P_1 + P_{\text{sleep}} \\ & + \frac{N_a}{N} M_a \gamma \left(\frac{1}{M_a(M_a - K)} \sum_{k=1}^K \frac{\sigma_k^2}{\beta_k} \left(2^{R_k \frac{N}{N_a}} - 1 \right) \right)^\alpha \end{aligned} \quad (5.18)$$

subject to $(\mathcal{C}_{P_{\max}})$.

Note that, given the way we fixed p_k in equation (5.16), the user rate constraints are implicitly taken into account and thus always satisfied. Moreover, the maximal power constraint $P_a \leq P_{\max}$ can be converted in terms of M_a and N_a

$$(\mathcal{C}_{P_{\max}}): \quad \frac{1}{M_a(M_a - K)} \sum_{k=1}^K \frac{\sigma_k^2}{\beta_k} \left(2^{R_k \frac{N}{N_a}} - 1 \right) \leq P_{\max}. \quad (5.19)$$

Clearly, P_a is minimized when $M_a = M$ and $N_a = N$, giving

$$P_{a,\min} = \frac{1}{M(M-K)} \sum_{k=1}^K \frac{\sigma_k^2}{\beta_k} (2^{R_k} - 1). \quad (5.20)$$

Hence, problem (5.15) is feasible if $P_{a,\min} \leq P_{\max}$. In other words, activating all resources ($M_a = M$, $N_a = N$) allows the system to meet the target users' rates within the power budget. Note that this feasibility constraint is equivalent to (A3.4) of Chapter 3. As shown in Appendix A.5, inequality (5.19) can be further simplified as

$$M_a \geq \frac{K}{2} + \frac{1}{2} \sqrt{K^2 + 4 \sum_{k=1}^K \varrho_k^{-1} (2^{R_k \frac{N}{N_a}} - 1)} \quad (5.21)$$

where $\varrho_k = P_{\max} \beta_k / \sigma_k^2$, which implies $M_a \geq K$. Two useful quantities can be derived from this inequality. Firstly, the minimum amount of active antennas, defined as $M_{a,\min}$, is obtained by finding the minimum value of M_a that satisfies (5.21) when $N_a = N$

$$M_{a,\min} = \left\lceil \frac{K}{2} + \frac{1}{2} \sqrt{K^2 + 4 \sum_{k=1}^K \varrho_k^{-1} (2^{R_k} - 1)} \right\rceil. \quad (5.22)$$

Secondly, the minimum amount of active slots, defined as $N_{a,\min}$, is obtained by finding the minimum value of N_a that satisfies (5.21) when $M_a = M$

$$M \geq \frac{K}{2} + \frac{1}{2} \sqrt{K^2 + 4 \sum_{k=1}^K \varrho_k^{-1} (2^{R_k \frac{N}{N_{a,\min}}} - 1)}. \quad (5.23)$$

In general, $N_{a,\min}$ does not have a closed-form solution.

5.2.2 Optimal Space and Time Resources

Let us consider a continuous relaxation of problem (5.18). Making the change of variable $x = N/N_a$ and $y = M_a$, and removing the constant term P_{sleep} not impacting the optimization, the objective function becomes

$$f(x, y) = \frac{y}{x} \frac{P_0}{M} + \frac{y}{x} \gamma \left(\frac{1}{y(y-K)} \sum_{k=1}^K \frac{\sigma_k^2}{\beta_k} (2^{R_k x} - 1) \right)^\alpha + \frac{y}{M} P_1. \quad (5.24)$$

In Appendix A.6, we show that $f(x, y)$ is convex in (x, y) .⁴ Moreover, the constraints $0 \leq N_a \leq N$, $K \leq M_a \leq M$, and (5.21) imply that x and y must

⁴As a note, defining x as N_a or N_a/N instead of N/N_a would not have led to a convex function $f(x, y)$.

belong to the feasibility domain $\mathcal{D}$, defined as

$$\mathcal{D} = \left\{ 1 \leq x, y \leq M, y \geq \frac{K}{2} + \frac{1}{2} \sqrt{K^2 + 4 \sum_{k=1}^K \varrho_k^{-1} (2^{R_k x} - 1)} \right\}. \quad (5.25)$$

In a convex problem, both the objective function and the constraints need to be convex. In Appendix A.7, we show that a sufficient condition for the convexity of the set $\mathcal{D}$ is

$$\varrho_k = \frac{P_{\max} \beta_k}{\sigma_k^2} \geq \frac{2}{K} \quad \text{for } k = 1, \dots, K. \quad (5.26)$$

This can be seen as a minimal average SNR constraint of each user when using a single transmit antenna at maximal power. It is a sufficient condition, which is required to make a strong general statement but can be neglected in certain regimes. For instance, for a small number of users and/or small target data rates, the $P_{\max}$ constraint is not binding and (5.26) can be left out. In the extreme opposite case, for a large number of users and/or large target data rates, a sufficiently large SNR for each user will be necessary to make the problem feasible, implying that (5.26) will be satisfied. We emphasize that (5.26) is derived from the transmit power constraint ($\mathcal{C}_{P_{\max}}$), also depicted in green in Fig. 5.1. If it is not convex, we may lose optimality in P_{cons} but still satisfy the user target rates given how we fixed the power allocation of each user.

An illustration of $\mathcal{D}$ is shown in Fig. 5.1, which resembles a triangle with a curved bottom right side. Each of the three vertices corresponds to a specific energy-saving regime:

- **Rush-to-sleep** (time-domain only): all spatial resources are active with full transmit power, giving $M_a = M$, $P_a = P_{\max}$ with minimum number of active time slots $N_a = N_{a,\min}$.
- **Rush-to-mute** (spatial-domain only): all time slots are active with full transmit power, giving $N_a = N$, $P_a = P_{\max}$ with minimum number of active antennas $M_a = M_{a,\min}$.
- **Awake-but-whisper** (power-domain only): all spatial and temporal resources are activated, giving $M_a = M$ and $N_a = N$ with minimum transmit power $P_{a,\min}$.

The rush-to-sleep can be seen as an extreme case of cell discontinuous transmission (DTX) with minimized data transmission interval. Cell DTX, which allows the PA to enter micro-sleep, is considered by 3rd Generation Partnership Project (3GPP) as part of “Adaptation of DTX/DRX” [30, Sec. 6.1.4] and has been previously studied in [33]. The rush-to-mute takes to

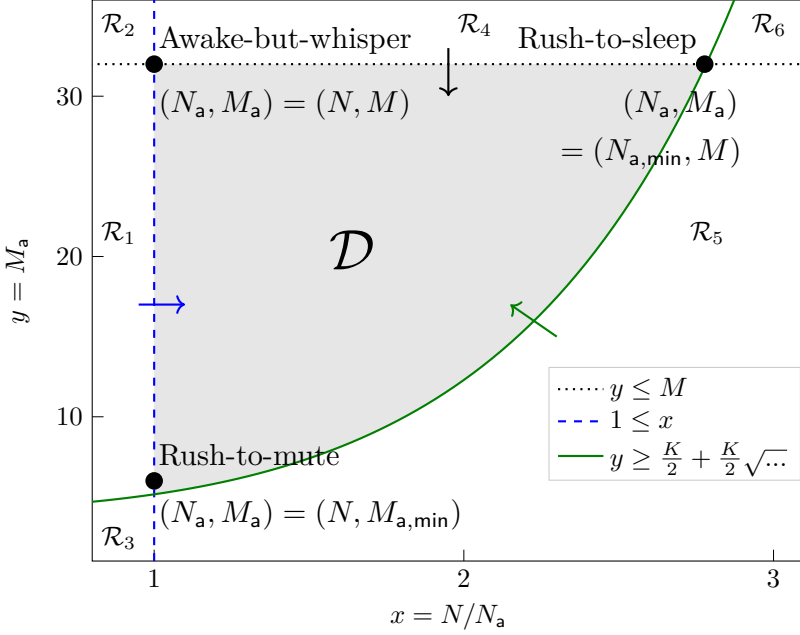


Figure 5.1: Example of feasibility domain $\mathcal{D}$ for $M = 32$, $K = 4$, $P_{\max} = 1$, $\beta_k = 1$, $\sigma_k^2 = 10^{-1}$, $R_k = 4$, for $k = 1, \dots, K$. The point $(N, M_{a,\min})$ has a small offset from the minimum y due to the ceiling operation in (5.22).

the extreme techniques such as “Adaptation of spatial elements” of 3GPP [30, Sec. 6.3.1] or antenna selection (e.g., [37] under bulk selection). The awake-but-whisper can be linked to “Adaptation of transmission power of signals and channels” studied by 3GPP [30, Sec. 6.4.1] when applied to the data channel with minimal transmit power, and is equivalent to using a conventional ZF precoder using all time slots and antennas (e.g., [126]). The awake-but-whisper and rush-to-mute approaches have a zero PA DTX-mode duration while the rush-to-sleep has the largest one. In other words, the rush-to-sleep has the smallest “latency-within-the-frame” (in downlink and proportional to $N_a T$, cf. Fig. 2.2). In the evaluations of Chapter 6, we introduce a fourth benchmark, proposed in [83] and jointly optimizing the number of active antennas and transmit power for a single-time slot system under ZF precoding. After reformulating in our framework problem (P2) addressed in [83], this can be seen as a particular case of problem (5.18) with $N_a = N$ active time slots and PA consumption exponent $\alpha = 1$. We refer to this as “Benchmark [83]”.

As a result of the convexity of $f(x, y)$ and $\mathcal{D}$, the problem

$$(\bar{x}, \bar{y}) = \arg \min_{(x, y) \in \mathcal{D}} f(x, y) \quad (5.27)$$

is convex and differentiable. We can thus use simple and efficient techniques to perform the minimization, with a guarantee to find the global minimum. Another favourable consequence of the convexity of (5.27) is that the solution of the original (discrete) problem can be found through the ceil-floor operator

$$(N_a, M_a) = \lfloor N/\bar{x}, \bar{y} \rfloor. \quad (5.28)$$

Indeed, for a convex problem, the optimal discrete point should be one of the nearest (feasible) neighbours.

5.3 Low-Complexity Solution

5.3.1 Approximation of Optimal Solution

The ceil-floor operation requires to evaluate and compare P_{cons} for four possible values as N_a is either $\lceil N/\bar{x} \rceil$ or $\lfloor N/\bar{x} \rfloor$, while M_a is either $\lfloor \bar{y} \rfloor$ or $\lceil \bar{y} \rceil$. In practice, if N is large, the ceil-floor operator can be approximated by a simpler rounding operation $N_a = \lfloor N/\bar{x} \rfloor$, not requiring any comparison (but just checking the feasibility). The resulting deviation from the optimal value of the cost function vanishes as $\mathcal{O}(1/N)$. The intuition is that P_{cons} depends on N_a through the ratio N_a/N , implying that the discretization error is at most $1/N$ and that asymptotically, the ratio N_a/N can take an infinite number (not only discrete) number of points between 0 and 1. A similar reasoning can be applied for the number of antennas: replacing the ceil-floor operation by a rounding operator $M_a = \lfloor \bar{y} \rfloor$. The resulting approximation error decays as $\mathcal{O}(1/M_a^{2\alpha})$ as M_a grows large. In summary, as shown in Appendix A.8, using

$$N_a = \lfloor N/\bar{x} \rfloor, \quad M_a = \lfloor \bar{y} \rfloor \quad (5.29)$$

provides the minimal P_{cons} up to an error term that decreases as $\mathcal{O}(1/N + 1/M + 1/M_a^{2\alpha})$ when the number of time slots $N \rightarrow +\infty$ and the number of active antennas $M_a \rightarrow +\infty$. In practice, the large N assumption is realistic as frames are usually made of many symbols. The impact of the second and third contributions to the error term depends on which BS configuration is selected because that fixes the number of antennas M and users K , and imposes a constraint $M_a \geq K$.

As a final remark, problem (5.18) only depends on the large-scale fading coefficients of the user-to-BS channels, not on the instantaneous channels.

These coefficients are simpler to estimate than the instantaneous channels as they vary more slowly in time. Moreover, the optimal allocation of time-space-power resources is to be performed once per channel stationary time, i.e., when the large-scale fading coefficients change, or when a different scheduling policy should be found. The optimization is thus already light given the problem's convexity, but on top, it has to be updated on a relatively slow time scale.

5.3.2 Practical Algorithm and Complexity

To solve problem (5.27), we utilize Algorithm 5.1 where every unconstrained problem is solved via Newton's method [94, Algorithm 9.5] with tolerance of 10^{-8} . After checking the problem's feasibility, we solve the unconstrained two-dimensional problem $\min_{x,y} f(x,y)$. Through simulations, we observe that 20 to 30 iterations are necessary to reach convergence. The gradient and Hessian of $f(x,y)$ depend on combinations of powers of

$$\begin{aligned}\phi(x) &= \sum_{k=1}^K \frac{\sigma_k^2}{\beta_k} (2^{R_k x} - 1), \quad \phi'(x) = \sum_{k=1}^K \frac{\sigma_k^2}{\beta_k} R_k (2^{R_k x} - 1) \\ \phi''(x) &= \sum_{k=1}^K \frac{\sigma_k^2}{\beta_k} R_k^2 (2^{R_k x} - 1)\end{aligned}\tag{5.30}$$

and of x and y , where their values can be reused among different terms in order to reduce complexity. Each iteration has a complexity of $\mathcal{O}(K)$, and this gives a total complexity of $\mathcal{O}(IK)$, where I is the number of iterations. The complexity expression does not directly depend on M and N , but lower values of M and N shrink the optimization domain. Also, smaller values of total BS antennas M decrease the maximum number of supported users $K_{\max}$, with $K \leq K_{\max}$.

If the solution of the two-dimensional problem lies in $\mathcal{D}$, the constraints are satisfied and the problem is solved. Otherwise, looking at the geometry of $\mathcal{D}$, either one or two constraints are not satisfied. We can identify six cases corresponding to as many domains $\mathcal{R}_i$, $i = 1, \dots, 6$, illustrated in Fig. 5.1. Let us define

$$y_{\min}(x) = \frac{K}{2} + \frac{1}{2} \sqrt{K^2 + 4 \sum_{k=1}^K \varrho_k^{-1} (2^{R_k x} - 1)}\tag{5.31}$$

as the value of y that satisfies the power constraint with equality. The feasibility domain can therefore be written as $\mathcal{D} = \{1 \leq x, y \leq M, y \geq y_{\min}(x)\}$. The six

Algorithm 5.1 Unconstrained optimization with inspection of the constraints to solve problem (5.27) and retrieve (5.29)

- 1: **Input:** $\{R_k\}, \{\beta_k\}, \{\sigma_k^2\}, M, N, K, \gamma, \alpha, P_0, P_1, P_{\max}$
- 2: Compute $P_{a,\min}$ via (5.20). If $P_{a,\min} > P_{\max}$, quit ▷ Check problem's feasibility
- 3: Set $(\bar{x}, \bar{y}) \leftarrow \arg \min_{(x,y)} f(x, y)$ ▷ Solve unconstrained 2D problem
- 4: **if** $(\bar{x}, \bar{y}) \in \mathcal{D}$ **then** ▷ All constraints are satisfied
- 5: Break, problem solved
- 6: **else if** $(\bar{x}, \bar{y}) \in \mathcal{R}_1$ **then** ▷ First constraint is not satisfied
- 7: Set $\bar{x} \leftarrow 1, \bar{y} \leftarrow \arg \min_y f(1, y)$ and make $\bar{y}$ within bounds
- 8: **else if** $(\bar{x}, \bar{y}) \in \mathcal{R}_2$ **then** ▷ First and second constraints are not satisfied
- 9: Set $\bar{x}_1 \leftarrow 1, \bar{y}_1 \leftarrow \arg \min_y f(1, y)$ and make $\bar{y}_1$ within bounds
- 10: Set $\bar{y}_2 \leftarrow M, \bar{x}_2 \leftarrow \arg \min_x f(x, M)$ and make $\bar{x}_2$ within bounds
- 11: Set $(\bar{x}, \bar{y}) \leftarrow \text{best}^*$ among $(\bar{x}_1, \bar{y}_1)$ and $(\bar{x}_2, \bar{y}_2)$
- 12: **else if** $(\bar{x}, \bar{y}) \in \mathcal{R}_3$ **then** ▷ First and third constraints are not satisfied
- 13: Set $\bar{x}_1 \leftarrow 1, \bar{y}_1 \leftarrow \arg \min_y f(1, y)$ and make $\bar{y}_1$ within bounds
- 14: Set $\bar{x}_2 \leftarrow \arg \min_x f(x, y_{\min}(x))$, make $\bar{x}_2$ within bounds and
 set $\bar{y}_2 \leftarrow y_{\min}(\bar{x}_2)$
- 15: Set $(\bar{x}, \bar{y}) \leftarrow \text{best}^*$ among $(\bar{x}_1, \bar{y}_1)$ and $(\bar{x}_2, \bar{y}_2)$
- 16: **else if** $(\bar{x}, \bar{y}) \in \mathcal{R}_4$ **then** ▷ Second constraint is not satisfied
- 17: Set $\bar{y} \leftarrow M, \bar{x} \leftarrow \arg \min_x f(x, M)$ and make $\bar{x}$ within bounds
- 18: **else if** $(\bar{x}, \bar{y}) \in \mathcal{R}_5$ **then** ▷ Third constraint is not satisfied
- 19: Set $\bar{x} \leftarrow \arg \min_x f(x, y_{\min}(x))$, make $\bar{x}$ within bounds and
 set $\bar{y} \leftarrow y_{\min}(\bar{x})$
- 20: **else if** $(\bar{x}, \bar{y}) \in \mathcal{R}_6$ **then** ▷ Second and third constraints are not satisfied
- 21: Set $\bar{y}_1 \leftarrow M, \bar{x}_1 \leftarrow \arg \min_x f(x, M)$ and make $\bar{x}_1$ within bounds
- 22: Set $\bar{x}_2 \leftarrow \arg \min_x f(x, y_{\min}(x))$, make $\bar{x}_2$ within bounds and
 compute $\bar{y}_2 \leftarrow y_{\min}(\bar{x}_2)$
- 23: Set $(\bar{x}, \bar{y}) \leftarrow \text{best}^*$ among $(\bar{x}_1, \bar{y}_1)$ and $(\bar{x}_2, \bar{y}_2)$
- 24: **end if**
- 25: **Output:** $(N_a, M_a) \leftarrow ([N/\bar{x}], [\bar{y}])$

*Best is the one that minimizes $f(x, y)$ in (5.24)

domains are defined as

$$\begin{aligned}\mathcal{R}_1 &= \{1 > x, y \leq M, y \geq y_{\min}(x)\}, \quad \mathcal{R}_2 = \{1 > x, y > M, y \geq y_{\min}(x)\} \\ \mathcal{R}_3 &= \{1 > x, y \leq M, y < y_{\min}(x)\}, \quad \mathcal{R}_4 = \{1 \leq x, y > M, y \geq y_{\min}(x)\} \quad (5.32) \\ \mathcal{R}_5 &= \{1 \leq x, y \leq M, y < y_{\min}(x)\}, \quad \mathcal{R}_6 = \{1 \leq x, y > M, y < y_{\min}(x)\}.\end{aligned}$$

When the first, second and third constraints are not satisfied we obtain the domains $\mathcal{R}_1$, $\mathcal{R}_4$, and $\mathcal{R}_5$, respectively. When only the third, second and first constraints are satisfied we obtain the domains $\mathcal{R}_2$, $\mathcal{R}_3$, and $\mathcal{R}_6$, respectively. If the solution of $\min_{x,y} f(x,y)$ lies in $\mathcal{R}_1$, the first constraint is not satisfied. Making it binding sets the value of x to 1, and we can then solve the one-dimensional unconstrained problem $\min_y f(1,y)$. If the obtained solution is either larger than M or smaller than $y_{\min}(1)$, we make the unsatisfied constraint binding and this gives the final value of y . In case the solution of $\min_{x,y} f(x,y)$ lies in $\mathcal{R}_2$, we let the first constraint bind fixing x to 1, solve the problem $\min_y f(1,y)$ and further check the other two constraints to make the obtained y within bounds. We also let the second constraint bind fixing y to M , solve the problem $\min_x f(x,M)$ and further check the other two constraints fixing x to either 1 or $x_{\max}$ if not satisfied, where $x_{\max}$ is the value solving $y_{\min}(x_{\max}) = M$. Among the two solutions, we select the one that minimizes $f(x,y)$. The same procedure is applied in the remaining four cases. Hence, we may need to solve two of the three one-dimensional unconstrained problems $\min_y f(1,y)$, $\min_x f(x,M)$ and $\min_x f(x, y_{\min}(x))$, with complexities of $\mathcal{O}(I_1)$, $\mathcal{O}(I_2K)$, and $\mathcal{O}(I_3K)$. Via simulations, we observe that 5 to 20 iterations (I_1 , I_2 , or I_3) are necessary to solve any of the three problems. Computing the first and second derivatives of the three functions involves similar computations to the ones required for the gradient and Hessian of $f(x,y)$, but with fewer terms given that the problems are one-dimensional.

5.4 Conclusion

In this chapter, we examined the joint allocation of number of active transmit symbols, number of active antennas, and transmit power for energy reduction in sub-6 GHz BSs under QoS and power constraints. The power model in Chapter 2 is mapped to the model [15], validated on operator measurements of sub-6 GHz BSs. As a reply to **RQ2**, we have shown that the formulated problem can be converted into a convex form with a sufficient condition on the domain convexity. This allows us to efficiently retrieve the number of active symbols and antennas, which then determines the transmit power per active antenna. Our proposed algorithm involves solving one, two, or three

unconstrained optimizations, followed by a computation of rounding operations. We further characterized analytically three benchmarks that minimize the use of one of the three BS resources: rush-to-sleep that uses the least possible number of time slots, rush-to-mute that uses the least possible number of antennas, and awake-but-whisper that uses the lowest possible transmit power. A fourth benchmark from the literature, jointly optimizing transmit power and (de)activation of antennas, is introduced.

Chapter 6

Data-Based Analysis

Bibliographic Reference

This chapter is based in part on: E. Peschiera, Y. Agram, F. Quitin, L. Van der Perre, and F. Rottenberg, “On optimizing time-, space- and power-domain energy-saving techniques for sub-6 GHz base stations,” accepted for publication in *IEEE Transactions on Communications*, 2026 [119].¹

In this chapter, we utilize data and fitted power model parameters from on-site measurements of a mobile network operator (MNO) in Belgium and evaluate the methods proposed in Chapter 5. The data from measurements are used in a twofold way: (i) indirectly to initialize the parameters of the power model in equation (5.7), based on [15] and explained in Section 5.1.2, and (ii) directly to compute the large-scale fading coefficients used in the simulations. To compute the large-scale fading coefficients, we use two measured channel quality indicator (CQI) distributions, one for fourth-generation (4G) and one for fifth-generation (5G) base stations (BSs). The derived large-scale fading values are representative of real-world deployments rather than synthesized based on analytical models. The evaluations cover the power consumed by the studied energy-saving schemes in a single time frame and averaged over several time frames at three network load levels, and the median power consumption and median energy efficiency (EE) for different numbers of active users at three network load levels. Two

¹The Python simulation code is available in open access at the following link: github.com/dramco-research/energy-saving-time-space-power-epeschiera-tcom26.

cases are analyzed: enabled and disabled time-domain power-saving modes in the BS hardware (HW).

Contributions

In the literature, not many works have used network measurements to evaluate or calibrate the BS energy-reduction algorithms. The study in [134] has used on-site measurements to evaluate deactivations of 4G and 5G radio units (RUs) in the time domain. The works [12] and [15] have presented a set of measurements-driven evaluations of fixed power-saving features (such as scaling the number of active power amplifiers (PAs) with the physical load). A recent use of deployment data is reported in [135] with the aim of evaluating the delivered capacity and energy consumption at Frequency Range 3 (FR3). In this chapter, by building on the framework developed in Chapter 5, we leverage data from on-site measurements to provide the evaluation of the baseline and proposed energy-saving algorithms in practical scenarios. We study three BS configurations at different *network loads*, where the network load is defined as the ratio of maximal sum rate (in bits per second). The specific contributions of this chapter are the following:

- We discuss the optimal energy-saving strategy, in terms of minimal energy consumption while meeting the target quality-of-service (QoS), for the 4G and 5G BSs under study in [15], not implementing time-domain power savings (specifically, PA micro-discontinuous transmission (μ DTX) and analog front-end (AFE) idle modes). A rush-to-mute approach provides quasi-optimal performance at any studied network load for massive multiple-input multiple-output (MIMO) BSs, and at low and medium network loads for BSs with fewer antennas.
- We consider the case when the analyzed 4G and 5G BSs utilize time-domain power savings in the form of PA μ DTX and AFE idle modes. It turns out that a combination of the three energy-saving techniques is the one providing minimal energy consumption. Massive MIMO BSs still tend more towards a rush-to-mute at any network load, while the BSs with fewer antennas steer towards a rush-to-sleep at low and medium loads.
- We consider the EE achieved by the different energy saving strategies, and show that the EE takes values in the range 10–300 kbit/J, with the proposed method achieving up to 50% improvement in the EE, depending on the network load. In general, we find that serving the maximum number of users is always beneficial in terms of EE.

6.1 Base Station Configurations

In this section, we discuss the initialization of the power model's parameters in equation (5.7), which we report here for convenience

$$P_{\text{cons}}(N_a, M_a, P_a) = \frac{N_a}{N} M_a \left(\frac{P_0}{M} + \gamma P_a^\alpha \right) + \frac{M_a}{M} P_1 + P_{\text{sleep}}$$

where P_0 , P_1 , P_{sleep} , γ are non-negative parameters, and $\alpha \in [0.5, 1]$ is the PA exponent parameter. These parameters determine the functional shape of P_{cons} , which is the objective function considered in the optimizations of Chapter 5. Let us recall that the power model in equation (5.7) corresponds (up to a simplification in the signaling structure) to the one proposed in [15], and the link is explained in Section 5.1.2. The variables α , γ , P_0 , P_1 , and P_{sleep} (that are actually aggregated parameters) depend on the other set of parameters as shown in Table 5.1. To set their values, we use the fitted values proposed in [15, Table II]. These are reported in Table 6.1. Every set of parameters corresponds to a different BS *configuration*. We select three BS configurations to analyze in the following: 4T4R, 8T8R, and 64T64R. The first one includes a remote radio unit (RRU) with 4 PAs and transmitting or receiving radio frequency (RF) chains, and works in frequency-division duplexing (FDD) under the 4G Long-Term Evolution (LTE) protocol. The second one includes a RRU with 8 PAs and transmitting or receiving RF chains, and works in time-division duplexing (TDD) under the 5G New Radio (NR) protocol. The third one includes instead an active antenna unit (AAU) with 64 PAs and transmitting or receiving RF chains, and works in TDD under the 5G NR protocol. We refer to the latter as the massive MIMO configuration. The 5G BSs operate at a carrier frequency of 3.5 GHz with a bandwidth of 100 MHz, while the 4G BS operates at a carrier frequency of 1.8 GHz with a bandwidth of 20 MHz. We consider a variation on two quantities (highlighted in gray in Table 6.1), which are the power-reduction factors $\delta_{\text{PA},\text{micro}}$ of PA in micro-sleep mode and $\delta_{\text{ana},\text{idle}}$ of AFE in idle mode, leading to two cases:

- Disabled time-domain hardware power-saving modes, with $\delta_{\text{PA},\text{micro}} = 1$ and $\delta_{\text{ana},\text{idle}} = 1$ as considered in [15]. This corresponds to BSs not using PA μDTX (no power is saved on the static PA consumption when there is no downlink (DL) transmission) and not using AFE idle power-saving modes (no power is saved by turning off, during the uplink (UL) phase, an RF chain that was used during the DL phase).
- Enabled time-domain hardware power-saving modes, with $\delta_{\text{PA},\text{micro}} = 0.25$ and $\delta_{\text{ana},\text{idle}} = 0.5$. The obtained model parameters reflect the BSs under study in [15] but implementing PA μDTX and AFE idle power-saving

modes. The values of $\delta_{\text{PA,micro}}$ and $\delta_{\text{ana,idle}}$ are set to be equal to $\delta_{\text{PA,idle}}$ and $\delta_{\text{ana,sleep}}$, respectively (i.e., the next active or sleep mode).

A summary of the aggregated parameters and main system model's quantities in the three configurations is provided in Table 6.2. The displayed values of P_{sleep} correspond to the case $K = K_{\text{max}}$. A smaller number of active users K implies a smaller sleep consumption P_{sleep} . We can observe that the value of P_0 is equal to zero when there are no time-domain HW power-saving modes. By enabling these time-domain power-saving modes, the value of P_0 is different from zero and the value of P_1 decreases compared to the previous case. In all BS configurations, the sleep consumption is the largest.

6.2 Large-Scale Fading and Network Load

In the evaluations, we consider a frame of $T_{\text{frame}} = 10$ ms, giving a number of available time slots equal to

$$N = \tau_{\text{DL}}(1 - \tau_{\text{sig}}) \frac{T_{\text{frame}}}{T} = \begin{cases} 1 \cdot (1 - \frac{2}{7}) \frac{10 \cdot 10^{-3}}{71.4 \cdot 10^{-6}} = 100 & \text{for 4G BSs} \\ 0.75 \cdot (1 - \frac{1}{14}) \frac{10 \cdot 10^{-3}}{35.6 \cdot 10^{-6}} = 195 & \text{for 5G BSs} \end{cases} \quad (6.1)$$

Unless otherwise specified, we set the number of active users to the maximum allowed, giving $K = K_{\text{max}}$. The evaluation scenario comprises the set of large-scale fading coefficients β_k and users' target normalized rates R_k , for $k = 1, \dots, K$. To compute the large-scale fading coefficients, we use the CQI distribution obtained from on-site measurements of sub-6 GHz BSs in Belgium [134], in particular from three angular sectors of two BSs:

1. 4G LTE RRU with 4 PAs, 4 transmitting RF chains, and 4 receiving RF chains, operating at $f_{\text{c,DL}} = 1.8$ GHz with $B = 20$ MHz. This corresponds to the 4T4R configuration.
2. 5G NR RRU with 8 PAs, 8 transmitting RF chains, and 8 receiving RF chains, operating at $f_{\text{c,DL}} = 3.5$ GHz with $B = 100$ MHz. This corresponds to the 8T8R configuration.

The raw data give the number of times each CQI index (from 0 to 15) is reported every hour over six days, and from those we compute the two CQI statistical distributions (one for 4G BSs and one for 5G BSs). The two CQI distributions are depicted in Fig. 6.1a. Given the use of different modulation orders, 4G and 5G systems use two separate CQI tables giving a different spectral efficiency for the same CQI index [136, Table 5.2.2.1-2 and Table 5.2.2.1-3]. Therefore,

Table 6.1: Fitted parameters of power consumption model in equation (5.7) for the three analyzed BS configurations, values from [15].

Parameter	BS configuration		
	4T4R	8T8R	64T64R
ξ_{PA}	0.12	0.12	0.12
$\eta_{PA,max}$	0.37	0.35	0.29
$\delta_{PA,micro}$	{1, 0.25}	{1, 0.25}	{1, 0.25}
$\delta_{PA,idle}$	0.25	0.25	0.25
$\delta_{PA,sleep}$	0	0	0
$P_{ref,ana,misc}$	22 W	22 W	17 W
$P_{ref,TRX}$	0.18 W	0.18 W	0.18 W
ς_1	0.5	0.5	0.5
ς_2	0.5	0.5	0.5
ς_3	0	0	0
ς_4	0.25	0.25	0.25
$\delta_{ana,idle}$	{1, 0.5}	{1, 0.5}	{1, 0.5}
$\delta_{ana,sleep}$	0.5	0.5	0.5
$P_{ref,link}$	90 W	90 W	90 W
$P_{ref,phy}$	0.07 W	0.07 W	0.07 W
ς_5	1	1	1
$\delta_{dig,idle}$	0.75	0.75	0.75
$\delta_{dig,sleep}$	0.75	0.75	0.75
$\eta_{PA,s/c}$	0.81	0.81	0.81
$\eta_{ana,s/c}$	0.81	0.81	0.81
$\eta_{dig,s/c}$	0.86	0.86	0.86
τ_{DL}	1	0.75	0.75
τ_{UL}	1	0.25	0.25
τ_{sig}	$\frac{2}{7}$	$\frac{1}{14}$	$\frac{1}{14}$
$\zeta_{DL,sig}$	$\frac{2}{6}$	$\frac{1}{12}$	$\frac{1}{12}$
B^*	1 MHz	1 MHz	1 MHz
P^*	1 W	1 W	1 W

we need to convert the CQI to signal-to-noise ratio (SNR) by using different relations. We use [137, Fig. 4 and Fig. 5] that provide these relations, based on simulations with a target block-error rate of 10^{-1} as defined by 3GPP [117].

Table 6.2: Summary of parameters of power consumption model in equation (5.7) for the three analyzed BS configurations.

Parameter	BS configuration		
	4T4R	8T8R	64T64R
RU type	RRU	RRU	AAU
Duplex	FDD	TDD	TDD
Protocol	4G LTE	5G NR	5G NR
M	4	8	64
$K_{\max}$	2	4	8
$f_{c,DL}$	1.8 GHz	3.5 GHz	3.5 GHz
B	20 MHz	100 MHz	100 MHz
Δ_f	15 kHz	30 kHz	30 kHz
T	71.4 μ s	35.6 μ s	35.6 μ s
$P_{\max}$	40 W	40 W	3.125 W
α	0.75	0.75	0.75
γ	5.33	5.38	3.50
P_0 ^a	{0, 34.69} W	{0, 69.98} W	{0, 53.92} W
P_1 ^a	{149.40, 114.71} W	{229.47, 103.26} W	{341.57, 161.95} W
P_{sleep} ^b	233.55 W	363.78 W	550.23 W

^a Left value is obtained with $\delta_{PA,micro} = 1$ and $\delta_{ana,idle} = 1$ as in [15],
right value is obtained with $\delta_{PA,micro} = 0.25$ and $\delta_{ana,idle} = 0.5$.

^b Shown for the case $K = K_{\max}$.

The resulting SNR distributions are given in Fig. 6.1b. As observed in [134] that worked on the same data from on-site measurements, users served by 5G BSs have statistically larger SNRs than users served by 4G BSs. The 5G users have an average SNR of 12.2 dB, while the 4G users have an average SNR of 10.3 dB. This could be due to the fact that the users with better channels were scheduled more often to be served by 5G BSs. While the SNR distributions are thus not the same in 4G and 5G, we do not want to adapt them as this is what really happens in the studied deployment area and the measurement period. Once the SNR of user k is known, denoted as SNR_k , our rate model (5.4) shows that $\text{SNR}_k = p_k/\sigma_k^2$. Hence, the power allocation at user k is given by $\sigma_k^2 \text{SNR}_k$. The noise power is computed as $\sigma_k^2 = T_\sigma k_B B F_\sigma$, where $T_\sigma = 290$ Kelvin and $F_\sigma = 9$ dB are the noise temperature and the noise figure, and k_B is the Boltzmann constant [96]. Last, we insert these data in our system model and consider $K = 1$ as a simplification, obtaining β_k by inverting the

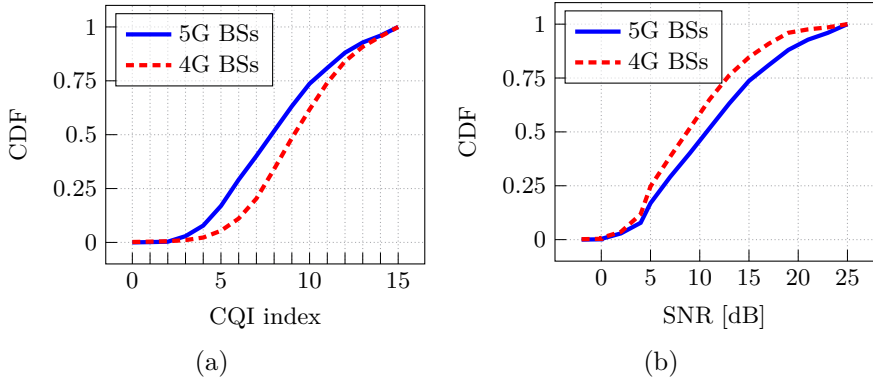


Figure 6.1: For 4G and 5G BSs, estimated CDF of CQI index (a) and of SNR (b) from on-site measurements.

expression of the total transmit power in equation (5.2)

$$\beta_k = \frac{\sigma_k^2 \text{SNR}_k}{P_T(M-1)}. \quad (6.2)$$

The value of β_k is therefore derived from a standard SNR expression of a multiple-input single-output (MISO) system under zero-forcing (ZF) precoding [90]. Based on the measurements and on the certificates of conformity that are publicly available upon request for deployed BSs [138], we set the total transmit power P_T to 160 W, 32 W, and 20 W for the 4T4R, 8T8R, and 64T64R configurations, respectively.

Concerning the users' target rates, let us consider some baseline rates $\{R_{k,0}\}$ normalized such that $\sum_{k=1}^K R_{k,0} = 1$, hence $\{R_{k,0}\}$ are the users' shares of the sum rate. The baseline rates are scaled by the variable κ , giving $R_k = \kappa R_{k,0}$. The maximum rate scaling $\kappa_{\max}$ is defined as the value of κ that solves equation (5.21) with equality when $M_a = M$ and $N_a = N$, i.e.,

$$M = \frac{K}{2} + \frac{1}{2} \sqrt{K^2 + 4 \sum_{k=1}^K \varrho_k^{-1} (2^{\kappa_{\max} R_{k,0}} - 1)} \quad (6.3)$$

for a fixed set of $\{\beta_k\}$ and $\{R_{k,0}\}$. With this choice and given the definition of $N_{a,\min}$ in equation (5.23), $N_{a,\min} = \lceil (\kappa/\kappa_{\max}) N \rceil$. This happens because $N_{a,\min}$ and $\kappa_{\max}$ are equally impacted by changes in the set of large-scale fading coefficients $\{\beta_k\}$ (changing $\varrho_k = P_{\max} \beta_k / \sigma_k^2$) and baseline rates $\{R_{k,0}\}$. Using equation (6.3), the maximal power allocation within the power budget for user k is $\sigma_k^2 (2^{\kappa_{\max} R_{k,0}} - 1)$, which is what the rush-to-sleep wants to achieve in order to

enable sleep for the largest possible amount of time slots. Indeed, the maximal power allocation for a given rate $\kappa R_{k,0}$ and minimal number of active slots $N_{a,\min}$ is $\sigma_k^2 (2^{\kappa R_{k,0}(N/N_{a,\min})} - 1)$. Equating the two power allocations and taking the floor to avoid exceeding the maximal power allocation, we obtain the above expression of $N_{a,\min}$. In the following, we generate $\{R_{k,0}\}$ as K realizations of a standard uniform distribution that are normalized to sum to 1, and we define the network load as the ratio $\kappa/\kappa_{\max}$. A larger κ indicates a more loaded system in terms of larger user rates. The median per-user delivered rates $\{R_{k,\text{deliv}}\}$ in bits per second are specified in the subtitles of Fig. 6.3 and Fig 6.4.

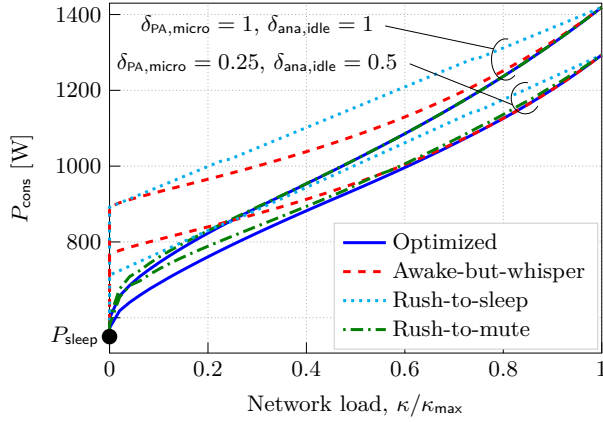
Remark: The target QoS is represented by the SNRs (the set of ψ_k) in Chapter 3 and Chapter 4, while it is represented by the rates (the set of R_k) here. The two are linked by $R_k = \log_2(1 + \psi_k) \implies \psi_k = (2^{R_k} - 1)$. In Chapter 3, ψ_k is proportional to the large-scale fading coefficient (equation (3.34)—simulating a scenario where a user requests a service depending on its channel conditions). In Chapter 4 and here, ψ_k and R_k follow a uniform distribution, hence the service requested by a user is independent from its channel conditions.

6.3 Single Channel Realization

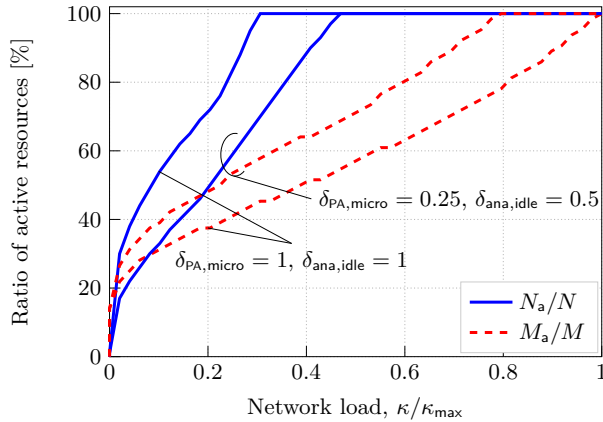
In this and the following sections, we evaluate the proposed solution (5.29), which we refer to as ‘optimized’, and compare it with the four introduced benchmarks. We consider a fixed set of large-scale fading coefficients $\{\beta_k\}$ and baseline normalized rates $\{R_{k,0}\}$, giving a unique value of $\kappa_{\max}$, and vary κ to observe the evolution of P_{cons} and (N_a, M_a) as the network load changes. We focus here on the massive MIMO configuration 64T64R. We notice for this and other configurations that with the computed values of β_k , the domain $\mathcal{D}$ is often not convex. This and the use of the rounding operator make the proposed solution an approximation of the optimal one. However, in the set of the numerical simulations that we performed, the solution often lies on the left-edge of $\mathcal{D}$ (see Fig. 5.1), hence the non-convexity of the domain has no effect on the approximated solution.

The left plot of Fig. 6.2 shows that at any network load, the rush-to-mute approach achieves similar or equivalent performance to the optimized solution when there are no time-domain HW power savings ($\delta_{\text{pa,micro}} = 1$ and $\delta_{\text{ana,idle}} = 1$). This is in line with the values in Table 6.2, where P_0 is zero while P_1 takes large values. All four strategies consume P_{sleep} at zero network load, while they consume the same amount of power at maximum network load as that point corresponds to the activation of all the time slots and antennas at power $P_{\max}$. The difference between the proposed or rush-to-mute scheme and the

Configuration: 64T64R ($M = 64$ antennas),
 $N = 195$ time slots, $K = K_{\max} = 8$ users,
 $\kappa_{\max} = 13.5$, $R_{k,\text{deliv}} = 880.2 \cdot (\kappa/\kappa_{\max}) R_{k,0}$ Mbit/s



(a)



(b)

Figure 6.2: For the 64T64R BS, (a) power consumption vs. network load for different energy-saving strategies, (b) optimal ratio of activated spatial and time resources vs. network load. Disabled time-domain hardware power-saving modes corresponds to $\delta_{\text{PA,micro}} = 1$ and $\delta_{\text{ana,idle}} = 1$, while enabled time-domain hardware power-saving modes to $\delta_{\text{PA,micro}} = 0.25$ and $\delta_{\text{ana,idle}} = 0.5$.

awake-but-whisper or rush-to-sleep scheme becomes smaller as the network load

increases. This is logical as most of the resources should be activated when there are large user target rates. When time-domain power-saving modes are enabled ($\delta_{\text{PA,micro}} = 0.25$ and $\delta_{\text{ana,idle}} = 0.5$), the gap between the optimized and rush-to-mute solutions increases at low network loads because the active PAs and RF chains consume less power. Also, the awake-but-whisper becomes optimal at very high network loads.

The right plot of Fig. 6.2 presents the change in the number of optimal resources with the network load. The value of N_a first increases and then reaches its maximum value N at a certain level of $\kappa/\kappa_{\max}$. On the other hand, M_a keeps increasing until the maximum network load when there are no time-domain power savings, while it reaches its maximum value M at a network load below 1 when time-domain power savings are enabled. We observe a similar trend for different $\{\beta_k\}$ and $\{R_{k,0}\}$. Therefore, when looking at the dependency on the network load in the massive MIMO 64T64R BS, we find that:

- With no time-domain HW power-saving modes, the optimal energy-saving strategy operates first in time, space and power, and then in space and power only.
- With time-domain HW power-saving modes, the optimal energy-saving strategy operates first in time, space and power, then in space and power only, and last in power only.

6.4 Average Power Consumption

We now evaluate the CDF of the power consumed by the five energy-saving schemes for the three BS configurations at three network loads: low, medium, and high, corresponding to $\kappa/\kappa_{\max} = 0.01$, $\kappa/\kappa_{\max} = 0.06$, and $\kappa/\kappa_{\max} = 0.18$, respectively. We consider 10^3 realizations of large-scale fading coefficients $\{\beta_k\}$ and target baseline rates $\{R_{k,0}\}$. The CDFs of the rush-to-sleep approach assume values only in one point (for a fixed BS configuration and network load) given that $N_{a,\min}$ is fixed as $\lceil (\kappa/\kappa_{\max}) N \rceil$. We observe that the CDFs obtained by using the optimized solution (solid blue) and the global optimum (black crosses), obtained by evaluating all the feasible solutions, present a negligible difference. We recall that absolute numbers of power consumption are subject to the power model uncertainty of 10–20% as detailed in [15, Fig. 1 and Fig. 2].

In Fig. 6.3, we investigate the performance when PA μDTX and AFE idle-mode power savings are disabled. At low network load, the optimized, rush-to-mute and benchmark [83] schemes perform significantly better than the rush-to-sleep and awake-but-whisper for any BS configuration. The optimized

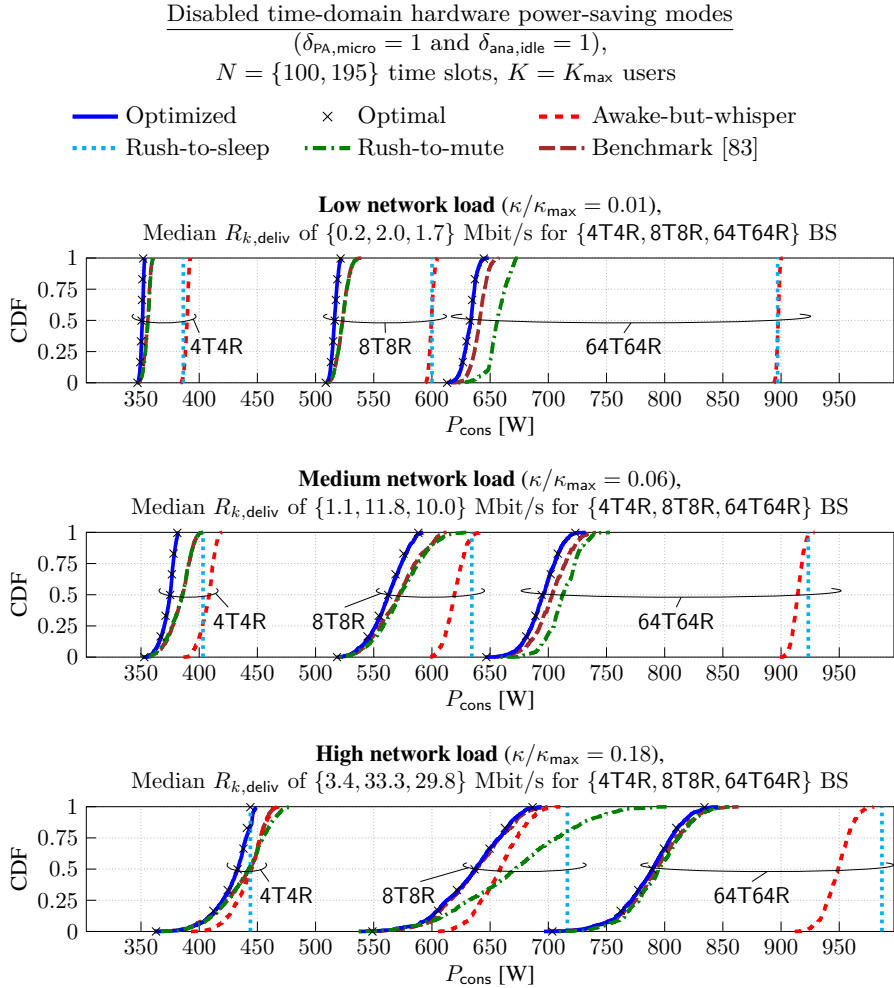


Figure 6.3: For disabled PA μ DTX and AFE idle-mode power savings, CDFs of power consumption for the three BS configurations at different network loads.

solution achieves 30% (64T64R), 14% (8T8R), and 10% (4T4R) median energy savings with respect to the rush-to-sleep and awake-but-whisper. At medium network load, the difference between the energy-saving schemes decreases but the optimized, rush-to-mute and benchmark [83] remain well separated from the other two schemes. This gap further decreases at high network load, as highlighted also in Section 6.3. Except for the 64T64R BS, the rush-to-mute approach has larger gaps to the proposed scheme while the awake-but-whisper

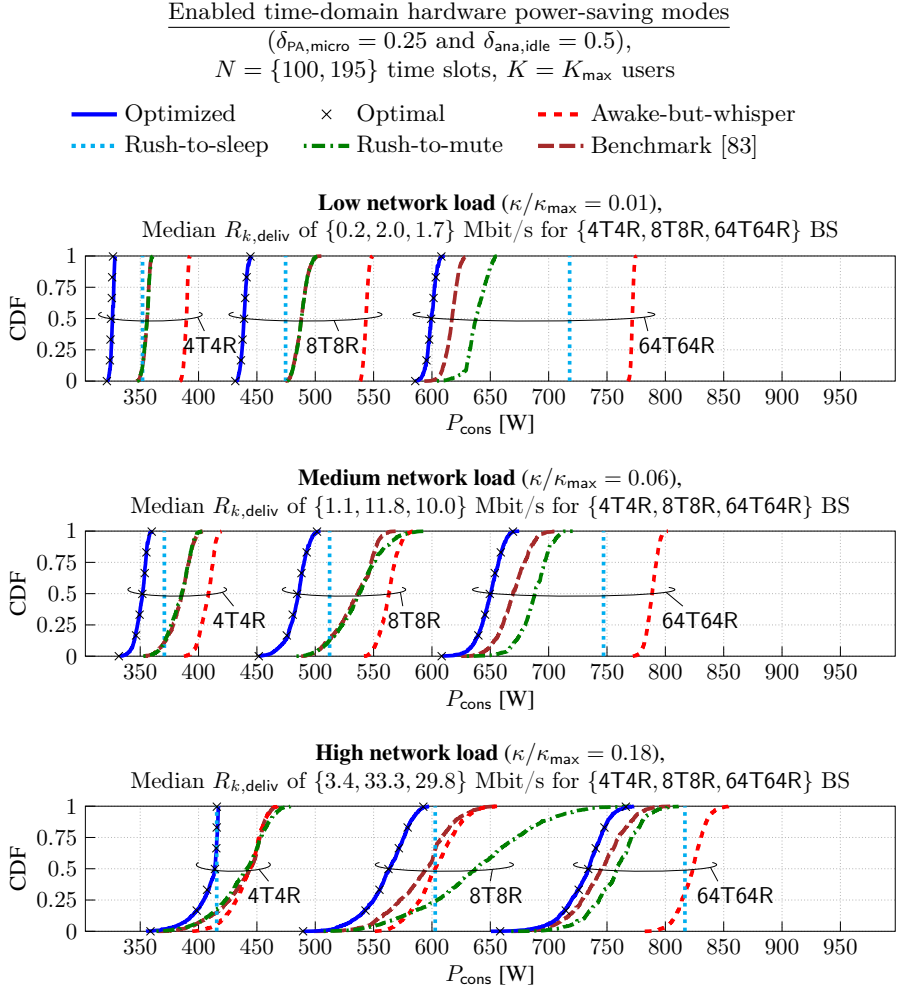


Figure 6.4: For enabled PA μ DTX and AFE idle-mode power savings, CDFs of power consumption for the three BS configurations at different network loads.

(for the 8T8R BS) and both the awake-but-whisper and rush-to-sleep (for the 4T4R BS) reduce their gap with the proposed scheme. The benchmark [83] jointly optimizes the number of antennas and the transmit power. Therefore, it has in this case similar performance to the rush-to-mute given that the awake-but-whisper is far from optimal. We highlight that the three BS configurations can be only compared with care or proper normalization as a larger M (more

antennas) is associated with a larger maximum rate scaling $\kappa_{\max}$ and more served users $K_{\max}$. Moreover, the 5G BSs utilize a larger bandwidth, with a direct effect on the delivered rates. These factors imply a different QoS provided by each BS configuration.

The performance of the rush-to-sleep technique comparatively improves when PA μ DTX and AFE idle-mode power savings are enabled, as shown in Fig. 6.4. For the 64T64R configuration at any network load, the rush-to-mute and benchmark [83] schemes remain the closest to the optimal one, with a larger difference than in Fig. 6.3 indicating that the optimal scheme uses a combination of time-, space-, and power-domain energy-saving techniques. The optimized scheme achieves 17%, 13%, 6%, and 3% median energy savings with respect to awake-but-whisper, rush-to-sleep, rush-to-mute, and benchmark [83] at medium network load. For the 8T8R configuration, the rush-to-sleep achieves the highest performance among the benchmarks at low and medium network loads, while at high network load its performance is similar to the awake-but-whisper and benchmark [83]. The rush-to-sleep is instead the closest approach to the proposed one for the 4T4R configuration at any network load. In general, even with time-domain HW power-saving modes, the cost of activating all the BS antennas in massive MIMO remains high and this still favors a rush-to-mute-like allocation. The case is different for the two other configurations, where the optimized solution exploits more the degrees of freedom in time.

6.5 Impact of Number of Users

In Fig. 6.5 and Fig. 6.6, we characterize the dependency of the power consumption on the number of users K , which is varied from 1 to $K_{\max}$. We analyze the 8T8R and 64T64R configurations with disabled time-domain HW power-saving modes (Fig. 6.5) and enabled time-domain HW power-saving modes (Fig. 6.6), and plot the median P_{cons} . As a complement to Fig. 6.3 and Fig. 6.4 that show the CDFs, one can directly compare the median energy performance of the five analyzed schemes. Let us consider, as an example, the optimized and rush-to-sleep scheme in the 8T8R BS with disabled PA μ DTX and AFE idle-mode power savings at high network load (bottom left plot in Fig. 6.5): the optimized scheme achieves savings in P_{cons} from 11% (when $K = 4$) to 32% (when $K = 1$).

Logically, the power consumption increases when K increases. Another trend that we observe is the diminishing gap between the rush-to-mute/benchmark [83] and the optimized scheme when K gets smaller. Let us consider the terms in the P_{cons} expression in problem (5.18): the smaller K , the smaller the impact

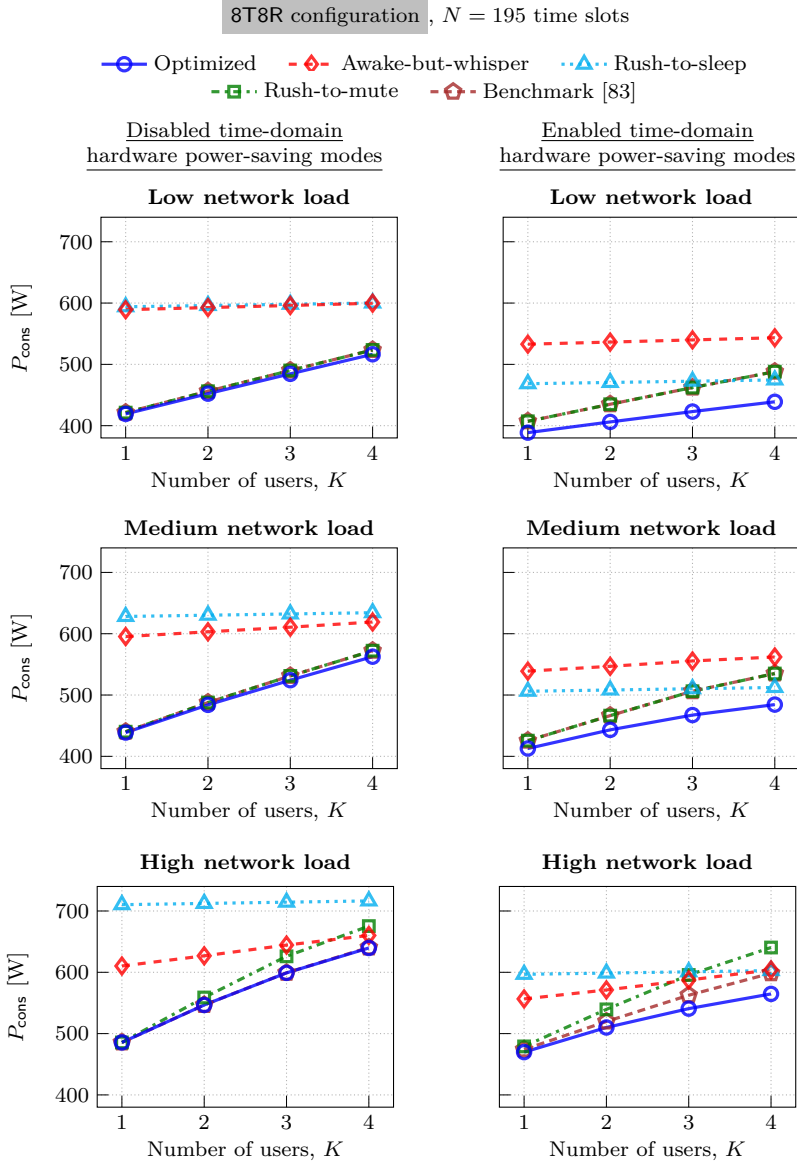


Figure 6.5: For the 8T8R BS, median power consumption vs. number of users at low, medium, and high network loads ($\kappa/\kappa_{\max} = 0.01$, $\kappa/\kappa_{\max} = 0.06$, and $\kappa/\kappa_{\max} = 0.18$, respectively).

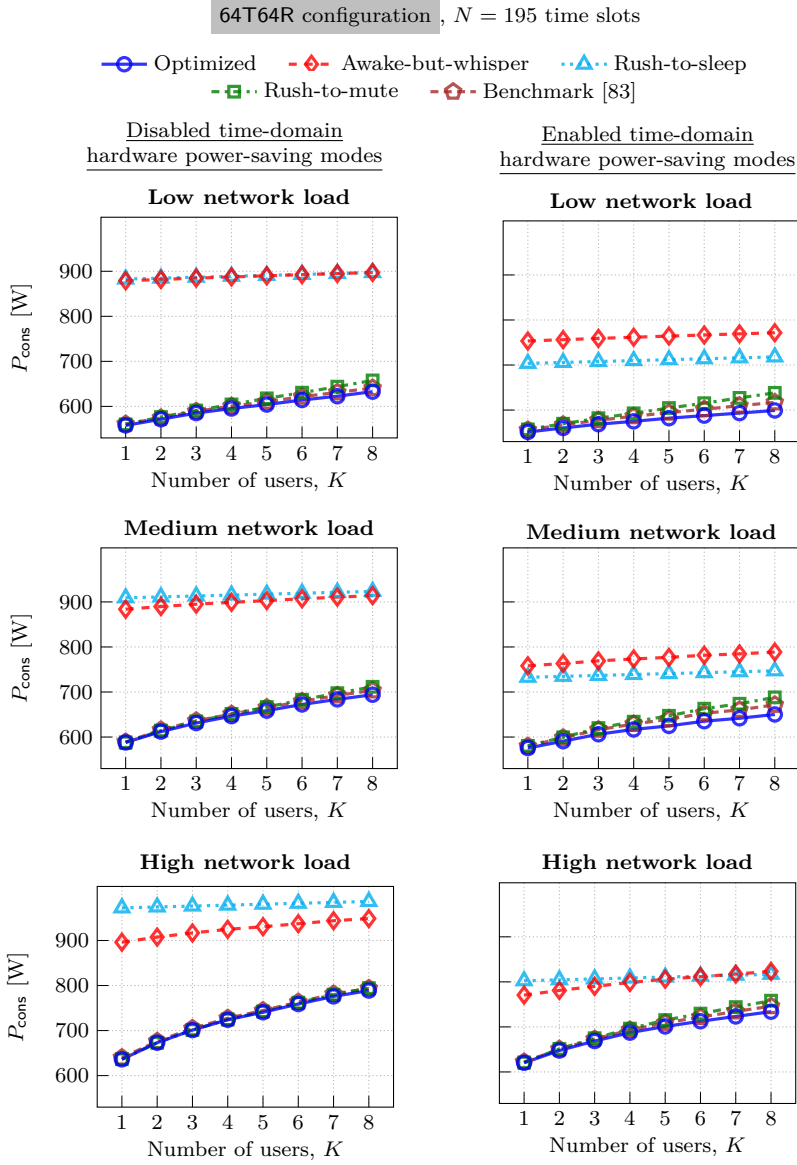


Figure 6.6: For the 64T64R BS, median power consumption vs. number of users at low, medium, and high network loads ($\kappa/\kappa_{\max} = 0.01$, $\kappa/\kappa_{\max} = 0.06$, and $\kappa/\kappa_{\max} = 0.18$, respectively).

of the last term (corresponding to the PA consumption, which depends on the inverse of the array gain $M_a - K$), while the larger the impact of the first and second terms. As we observed in Chapter 3, when K decreases, less spatial degrees of freedom (M_a) are utilized in favour of a lower circuit consumption. Notably, in the 8T8R configuration with time-domain HW power savings (first column of Fig. 6.6), the rush-to-mute and benchmark [83] allocations are the closest to the optimized one when $K = 1$ or $K = 2$, while this is not the case when $K = 4$, which is the focus of Fig. 6.4.

In the 64T64R configuration, depicted in the Fig. 6.6, it can again be noticed that a rush-to-mute-like allocation is the way forward in the 64T64R BS, with significant energy savings against the rush-to-sleep and awake-but-whisper schemes, up to 37% at low network load, $K = 1$ active user and with no time-domain HW power-saving modes. When there are $K = 8$ users at high network load, the median energy saving remains larger than 20%.

6.6 Energy Efficiency

The focus of the previous sections was on the absolute energy consumption. In this section, we analyze the EE achieved in the different BS configurations when changing the number of users. The EE is defined as

$$EE = \frac{\sum_{k=1}^K R_{k,\text{deliv}}}{P_{\text{cons}}} \quad [\text{bit/J}] \quad (6.4)$$

and is measured in bits per Joule, where $R_{k,\text{deliv}}$ is the delivered rate for user k given in equation 5.4. This EE metric has been extensively used in academia and industry [17, 139]. Nonetheless, we point out that other metrics may be more reasonable in some scenarios, for instance relating the provided coverage [140], which we do not explicitly consider in our analysis. The EE provides a way of comparing the different BS configurations because the delivered QoS is normalized by the power consumption. In Fig. 6.7 and Fig. 6.8, we plot the EE for the 8T8R and 64T64R configurations. The values of EE are approximately in the range 10–300 kbit/J, where 10 kbit/J are achieved at low network load and 300 kbit/J are achieved at high network load. We further find that the EE in the 4T4R configuration (not plotted in the following) is one order of magnitude lower than in the 5G configurations, i.e., it is in the range 1–20 kbit/J. These values are in line with recent evaluations based on operator measurements in Belgium [12], which reported an energy intensity (the inverse of the EE) in the range 0.01–1 kWh/Gbyte, corresponding to approximately 2–200 kbit/J. We observe that the EE monotonically increases with K , as it happens for the power consumption. When the BS serves more users, the benefit of delivering a larger sum rate is larger than the downside of consuming more power.

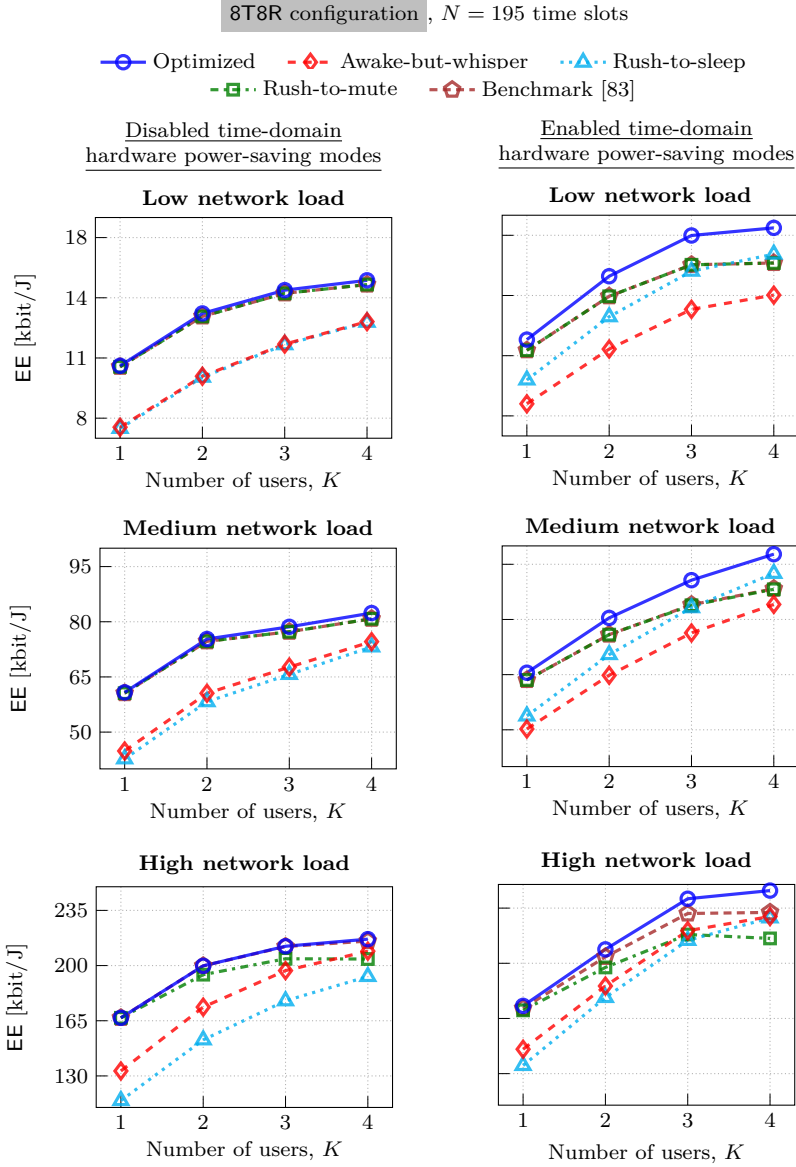


Figure 6.7: For the 8T8R BS, energy efficiency vs. number of users at low, medium, and high network loads ($\kappa/\kappa_{\max} = 0.01$, $\kappa/\kappa_{\max} = 0.06$, and $\kappa/\kappa_{\max} = 0.18$, respectively).

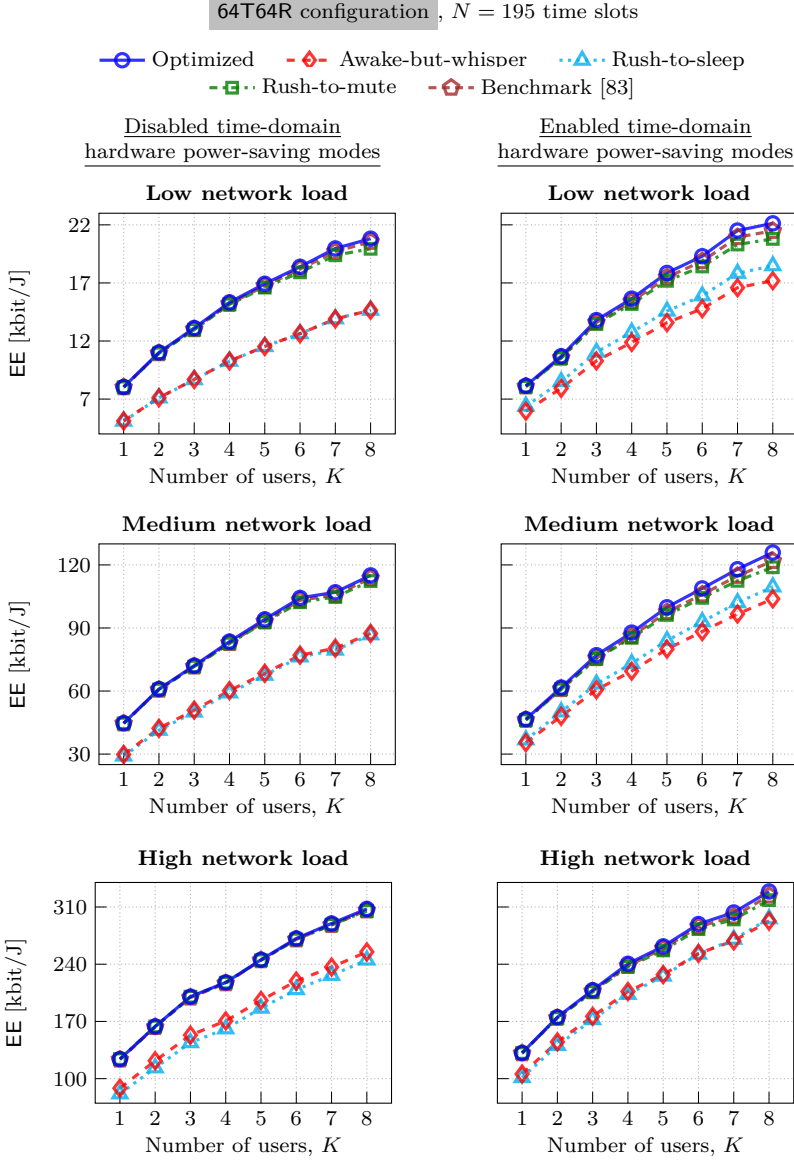


Figure 6.8: For the 64T64R BS, energy efficiency vs. number of users at low, medium, and high network loads ($\kappa/\kappa_{\max} = 0.01$, $\kappa/\kappa_{\max} = 0.06$, and $\kappa/\kappa_{\max} = 0.18$, respectively).

Let us quantify the EE improvements of the optimize scheme over the awake-but-whisper and rush-to-mute schemes in the 8T8R configuration at medium network load. The EE improves of more than 35% for $K = 1$ when time-domain HW power-saving modes are disabled, while it improves of more than 28% for $K = 1$ when the time-domain HW power-saving modes are enabled. For $K = 4$ users at the same medium network load, these EE improvements reduce to a minimum of 5%. In the 64T64R configuration, the maximal EE improvement of 57% over the awake-but-whisper and rush-to-sleep is achieved at low network load, $K = 1$ user, and disabled time-domain HW power-saving modes.

6.7 Conclusion

This chapter has focused on evaluating, using data and parameters resulting from on-site measurements of deployed networks, the energy-saving algorithms of Chapter 5. We provided practical insights for a 4G BS configuration (4T4R) and two 5G BS configurations (8T8R and 64T64R), which provide the reply of this thesis to **RQ4**. The rush-to-mute technique, using as few antennas as possible at maximum output power in any available time slot, provides quasi-optimal performance at low and medium network loads in the analyzed BSs when these are not implementing time-domain power-saving modes such as μ DTX. Median energy savings up to 30% are attained in the 64T64R BS at low network loads when the maximum number of users is active. When the BS HW implements time-domain power-saving modes, the optimal energy-saving strategy operates in two or more domains at the same time. Still, rush-to-mute has the smallest gap to the optimized scheme in the 64T64R BS at any examined network load. The rush-to-sleep, instead, provides the best performance among the benchmarks in the 4T4R and 8T8R BS at low and medium network loads. In general, the difference between the use of the three energy-saving domains decreases at high network loads. In terms of EE, the results highlight that serving the maximum number of users is always beneficial, with EE values up to 20 kbit/J in the 4G BSs and 300 kbit/J in the 5G BSs.

Chapter 7

Valorization

This chapter discusses the potential valorization of the research presented in this thesis. We start by discussing why energy reduction in the radio access network (RAN) is needed and what it brings to the industrial players. We continue by exploring the policies and technical solutions adopted by three leading network equipment manufacturers. We eventually provide energy-reduction insights for mobile network operators (MNOs) resulting from the findings of this thesis.

7.1 Need for Energy Reduction in RAN

The information and communications technology (ICT) sector has set a clear goal in reducing its carbon emissions by half before 2030 relative to the 2015 levels [2]. As part of ICT, the mobile communications sector considers the climate transition as an urgent need, evidenced by the fact that MNOs representing two-thirds of the industry by revenue have committed to these targets [141]. The GSM Association (GSMA) and the International Telecommunication Union (ITU) have recently published a document on the climate transition planning guidance for telecommunication companies, which reports as a forward [142]: *“By integrating climate considerations into their core business strategies, companies are not just taking climate action, they are reducing operational costs, unlocking new opportunities for innovation, and building confidence among investors and stakeholders.”* Accountability and ambition of companies are emphasized as requested factors. A breakdown of emissions indicates that one of the main hotspots for MNOs is the electricity usage from the operation of their RAN. The related decarbonization levers are

the increase in the use of electricity from renewable sources and the improvement of the RAN energy efficiency and modernization.

The GSMA has also recently published a report on the industry target of reaching net zero by 2050 [141]. It is estimated that the global operational emissions from MNOs decreased by 8% between 2019 and 2023 (with a large variability worldwide), while mobile connections increased by 9% and data traffic nearly quadrupled. Strong progress on energy efficiency and renewable energy has enabled this emissions reduction, and further progress across all regions is needed to accelerate the average annual reductions. A key recommendation for MNOs is to “*optimise energy use of networks by adopting energy-efficient hardware and best practices and retiring legacy networks.*”

The RAN energy efficiency is often highlighted in industrial reports, but care should be taken when the actual goal is to reduce the *absolute* energy use. A previous report from the same GSMA stated that fifth-generation (5G) networks were going to offer significant energy-efficiency improvements per gigabyte over previous generations. However, new 5G use cases and the deployment of more sites and antennas were prospecting a more efficient network that could paradoxically result in larger emissions without active intervention [10]. This led to suggestions on reducing the consumption of, e.g., a 5G active antenna unit (AAU) by focusing on maximizing the time spent in sleep states. The Next Generation Mobile Networks (NGNM) Alliance also highlights the need for these RAN energy-saving functionalities, with the objective of a base station (BS) power consumption very close to zero Watt for zero (carrier) load [143]. This serves as a basis for MNOs to face the upcoming environmental and economic challenges. It is emphasized that load-adaptive intelligent resource allocation at the BS can reduce its energy consumption without impacting the users’ quality-of-service (QoS). The 6G Smart Networks and Services Industry Association (6G-IA) suggests that next-generation sixth-generation (6G) networks should be deployed with environmental sustainability in mind from day one [144]. Enabling technologies for the RAN energy consumption reduction encompass both the radio unit (RU), with power amplifiers (PAs) and analog front-end, and the baseband unit.

7.2 Zero-Bit Zero-Watt Strategies

Let us now focus on what are the main (publicly available) RAN energy-reduction policies and strategies adopted by three major equipment manufacturers and implemented in their 5G equipment. Ericsson points out that the term *energy performance* broadens the focus beyond energy per bit to total energy

consumption [28]. The lean design is the key enabler, with BSs maximizing their idle-state intervals to achieve a load-proportional energy consumption. As part of Ericsson's RAN modernization guide [145], there are three steps to upgrade networks by balancing the performance gains with energy and emissions reductions:

1. **Holistic target evaluation**, which integrates the traditional indicators such as delivered rates or user experience with new metrics like carbon emissions. This enables to continue to expand the network capacity while optimizing the energy- and carbon-saving features.
2. **Identify modernization sites**, which uses holistic targets and on-site measurements to identify the traffic volume, delivered user experience, and energy consumption of different network sites and plan (per site) operational features such as cell sleep, massive multiple-input multiple-output (MIMO) sleep, or deep sleep during low-traffic periods without compromising the users' QoS.
3. **Define timing and scope**, which leverages opportunities like 5G deployments or capacity boosts to install new lean-designed equipment or refresh the legacy one while complying with power budgets and other constraints. Decisions on what sites should be first prioritized for modernization fall within this step.

The general approach of this thesis aligns with this view, from the inclusion of absolute energy consumption as a first design criterion to the comparison of fourth-generation (4G) and 5G BS configurations leveraging data from on-site measurements. Ericsson features several energy-saving functionalities in its radio products [145]. "Massive MIMO sleep mode" dynamically (de)activates transmit antenna branches to match the capacity with the actual traffic needs. "Booster carrier sleep" dynamically activates capacity carriers based on the traffic load, securing the user experience. The technique called "RAN compute deep sleep mode" enables the silicon in the RAN computational unit to go into deep sleep mode when the corresponding radio part is in deep sleep mode (at the timescale of seconds). "Micro sleep TX" is a software feature that automatically (de)activates the PA and other hardware blocks during fixed idle periods, reducing the energy consumption while not impacting performance. This technique also ensures that the PA is ready to be switched on when needed.

Huawei has published a white paper on green 5G in 2021 [146]. Despite placing the emphasis on the energy efficiency and not on the absolute energy consumption, energy-reduction software functionalities are discussed as a major characteristic of 5G as part of the so-called "network intelligence". It is stated

that maximization of both energy saving and network performance can be achieved by putting resources such as carriers, channels, and symbols to sleep and tweaking the transmit power during off-peak hours. This dynamic shutdown is implemented based on historical service-load characteristics. The use of artificial intelligence (AI) is foreseen to handle the problem's complexity and multidimensionality. More recently, Huawei has reported on the award-winning AI-based technology “zero bit zero Watt”, able to cut down power use in wireless networks [147]. At zero-load hours, the radio frequency (RF) equipment enters a switch-off depth level of 99%. Innovations at the PA and antenna level are used at other loads, and AI is utilized to improve the resource and power scheduling at the network and site levels. A similar technology commercialized by Huawei is called “zero bit zero Watt zero loss” to stress that the user experience is not compromised (zero loss) in any scenario [148].

Nokia is also involved in the development and deployment of RAN energy-saving technologies. In a white paper from 2022, the main technologies are classified with respect to the load level [133]. At low-to-medium load, Nokia suggests the use of micro-discontinuous transmission (μ DTX), working in the time domain, at the RF and digital level (coupled with baseband unit power management), as well as of cell switch off. At medium-to-high load, Nokia envisions the use of massive MIMO muting, working at the antenna level, and of scalable baseband computing. They further propose techniques to improve the PA efficiency (e.g., digital pre-distortion (DPD)), compute-in-memory in the digital unit, and network densification for any load level. In a more recent white paper [149], Nokia identifies hardware and software innovations towards “zero bit zero Watt”. More efficient gallium nitride (GaN)-based PA and system-on-chip (SoC) are discussed as main hardware novelties. At the software level, μ DTX and MIMO muting (turning off both PAs and RF chains) are indicated for low-to-medium load levels, while cell switch off, deep and extreme deep sleep modes are indicated for very low load levels. The extreme deep sleep mode is being tested on Orange's network, with the aim of achieving zero energy consumption in the absence of traffic [150].

7.3 Actionable Insights for Mobile Network Operators

We have contextualized the market needs and existing solutions on RAN energy-saving techniques, and we now leverage the results of this thesis to provide actionable insights to MNOs with the goal of reducing operational emissions.

7.3.1 Single-Carrier versus OFDM

Let us differentiate between a single-subcarrier system (termed ‘narrowband’ in Chapter 3 and Chapter 4, and referred to as single-carrier in the following for clarity) and a system with a hundred or more subcarriers (for brevity, orthogonal frequency-division multiplexing (OFDM) in the following). We have analyzed the allocation of transmit power and number of active antennas¹ for a sub-6 GHz BS, and our analysis suggests the following:

- In a single-carrier system, there is potential for energy savings by deactivation of antennas based on the computed digital precoder (used to spatially multiplex the users’ symbols in downlink (DL)). These precoder also allocates the transmit power needed to achieve the target QoS. This is done at each new channel coherence time (the time interval in which the channel can be seen as time-invariant). The number of activated antennas increases when there are more served users and if the maximal transmit power per antenna is reached. This approach contrasts with many reports from equipment manufacturers, which do not classify precoding among the energy-saving techniques, but rather consider it for data rate or coverage increase.
- In an OFDM system, potential for energy savings is identified in the selection of a number of active antennas per channel stationarity time (in which the long-term channel statistics such as the pathloss do not change, spanning multiple coherence times) depending on the target QoS and on the long-term channel statistics. Then, the precoders are computed among these active antennas at each new coherence time. The number of active antennas is selected such that the users’ QoS is not compromised and the maximal power per antenna is not exceeded.

7.3.2 Centralized and Distributed BSs

The majority of MNOs have not yet deployed systems of cooperating distributed BSs (e.g., a cell-free network). Analyses such as the one presented in Chapter 4 can serve as a basis to estimate the communication-energy performance tradeoff and provide pre-deployment insights. Let us focus on OFDM systems (with a hundred or more subcarriers). Our analysis identified a key difference between centralized BSs (not cooperating) and cooperating distributed BSs:

¹With the term antenna, we represent also the corresponding PA and associated RF chain in a fully-digital beamforming architecture.

- Individual antennas at a BS should be turned off in centralized BSs to reduce the energy consumption of the BS. The deactivation ratio depends on the scenario and is selected at each new channel stationarity time.
- Either all or no antennas of a distributed BS should be turned off to reduce the energy consumption of the network. This is reasonable because deactivating all antennas of a BS enables to completely deactivate the BS, thus saving power in other BS components, and keep the network complexity under control by e.g., reducing the number of utilized fronthaul (FH) links. This has connections with BS on/off strategies in cell-free networks studied in the literature [109, 110, 111]. As for centralized BSs, the deactivation ratio (this time of entire BSs over the total number of BSs) changes with the scenario and with a new channel stationarity time.

7.3.3 Changes per BS Configuration

The previous two guidelines are applicable to any BS configuration (e.g., massive MIMO and small-dimension MIMO) in which the active antennas transmit data in all the available DL time slots of the analyzed time frame. The number of degrees of freedom of the small-dimension MIMO is thus limited. When including time-domain deactivations (where the active antennas can transmit data in a subset of the available DL time slots), we can provide more specific insights per BS configuration in OFDM systems:

- In the small-dimension MIMO configuration (in our analysis, 4T4R and 8T8R), antenna deactivation should be preferred to transmit-power reduction and transmit-symbol deactivation when the PAs and RF chains do not implement power-saving modes (e.g., PA μ DTX). When these power-saving modes are implemented, transmit-symbol deactivation should instead be prioritized to achieve the largest energy savings. Using only transmit-power reduction is the least preferred strategy according to our analyses.
- In the massive MIMO configuration (64T64R), antenna deactivation should always be the preferred strategy to achieve the largest energy savings. This is in line with the measurements on which our optimizations are based [15], which highlight that the static consumption of PAs and RF chains is the largest part of the massive MIMO BS consumption.

In terms of energy efficiency (EE) (in bits per Joule), the 5G configurations should be preferred to the 4G configuration as they provide, under the same

conditions, a ten-fold improvement. The 5G massive MIMO configuration achieves a similar EE compared to the 5G small-dimension MIMO configuration.

7.3.4 Impact of Network Load Level

Considering different target QoS levels is essential when designing and evaluating network energy-saving techniques. In our analyses, we varied the so-called network load to represent the target QoS variation. In general, the optimized allocation and (de)activation of resources change with the network load. As an example, in the **64T64R** (massive MIMO) configuration, transmit-symbol deactivation is not used by the optimized strategy at very high network loads. However, at the selected low and medium network loads (e.g., corresponding to 1.7 and 10 Mbit per second per user in the **64T64R** configuration), our results identify that one of the three resource domains (antennas, symbols, transmit power) is clearly prioritized. This holds for all three BS configurations. When targeting energy reductions at low-to-medium network loads, one could therefore prioritize rush-to-mute in the **64T64R** configuration, and rush-to-mute or rush-to-sleep in the **4T4R** and **8T8R** configurations, depending on the implementation of time-domain hardware power-saving modes.

Chapter 8

Conclusions and Future Work

This thesis focuses on power allocations and (de)activation of antennas and symbols for energy reduction in sub-6 GHz base stations. The overall conclusions are provided in this chapter by replying to the five addressed research questions and highlighting the main contributions. Future work towards multi-domain resource allocation and (de)activation techniques for the sustainable operation of current and next-generation base stations (BSs) is discussed.

8.1 Conclusions

In Chapter 1, we have introduced the five research questions examined in this thesis. We hereby provide the one-by-one replies that serve as conclusions of this thesis and underline the contributions of each chapter.

RQ1: How can existing base station power consumption models be extended and refined to include next-generation base stations?

We have addressed this question in Chapter 2 by proposing a power consumption model that incorporates features of different state-of-the-art models. The multi-antenna BS is split in three main components (power amplifiers (PAs), analog processing, and digital processing), and each component comprises multiple subcomponents (e.g., the digital-to-analog converters (DACs) and analog-to-digital converters (ADCs) that are part of the analog processing). The average consumption of each component during a time frame is quantified by the weighted average of the consumption in different states, i.e., working and data transmission, working and reference signal transmission, micro-sleep, idle, and

sleep modes. The consumption of each subcomponent is further modeled by taking into account the specificities and scalings of that subcomponent. Differently from available models, the analog and digital processing components can enter micro-sleep mode, making their consumption physical-load dependent. This aligns with foreseen scalable analog and digital hardware (HW) that implements load-dependent deactivations (e.g., [133]). Moreover, we emphasized the importance of linking the power model with the communication model, which considers hybrid analog-digital beamforming and quantifies the achievable rates on a per-coherence block basis. By focusing on the Frequency Range 3 (FR3) (7–24 GHz) band, we evaluated the power consumption of prospective BSs with thousands of antenna elements and larger bandwidths than in the current sub-6 GHz band. It is shown that in the analyzed scenario, the hybrid analog-digital architecture (with one-fourth of radio frequency (RF) chains compared to the antennas, PAs, and low-noise amplifiers (LNAs)) provides the highest energy efficiency (EE) at both full loads and reduced loads.

RQ2: How can transmit power allocations and (de)activation of antennas and transmit symbols be optimized to cut base station energy use while still meeting quality-of-service targets?

This question has been examined in Chapter 3 and Chapter 5 under different perspectives. In Chapter 3, we considered the (de)activation of BS antennas and allocation of transmit power from a precoding design problem at first. We investigated the minimization of the PA consumption under a non-constant PA efficiency and a zero-forcing (ZF) constraint per subcarrier, which is linked to a constraint on the user rates. We derived fixed-point equations (that can be iteratively solved) for the antenna powers and closed-form expressions for the precoders. In a system with one subcarrier (termed narrowband), the proposed solution generally allocates all the transmit power to a few antennas, which can work at higher PA efficiencies, and allocates no power to the remaining antennas. This is in contrast to the baseline solution (a ZF precoder), which allocates the transmit power to all the BS antennas. Consequently, these antennas work at lower PA efficiencies. In a system with many subcarriers (in the order of a hundred or more), it is shown that all the antennas need to be activated to meet the users' quality-of-service (QoS). Considering the case with an infinite number of subcarriers (asymptotic wideband) and an independently and identically distributed (i.i.d.) Rayleigh fading channel, all the antennas tend to be allocated with the same transmit power. This allowed us to fix the QoS constraints among a number of active antennas, and optimize the number of active antennas based on the whole BS consumption. We have further shown how to retrieve this optimized active-antenna number in closed form. Numerical simulations illustrated the applicability of the asymptotic approximation to realistic orthogonal frequency-division multiplexing (OFDM) systems with a

finite number of subcarriers, as well as the potential for significant energy savings, especially when there is a low number of users. The complexity analysis testified the viability of the proposed solutions.

In Chapter 5, we have integrated the (de)activation of transmit (OFDM) symbols. By focusing on an OFDM system with many subcarriers using ZF precoding, we have looked at the problem of minimizing BS consumption subject to QoS (expressed as per-user target rate averaged over the analyzed time frame) and transmit power constraints. Under i.i.d. Rayleigh fading, the transmit power at active antennas is expressed in terms of the number of active antennas and active time slots (i.e., the number of active transmit symbols) via the QoS constraints. The convexity of the resulting two-dimensional objective function, which is the BS consumption when considering continuous instead of discrete optimization variables, is proven. A sufficient condition for the convexity of the optimization domain is derived. The convexity is exploited to propose a low-complexity algorithm based on unconstrained optimizations and iterative check of the constraints. Further, a simple rounding of the optimized continuous solution can be applied to obtain the discrete number of activated space-time resources, with an error that asymptotically vanishes. Three simpler allocations are identified, corresponding to corner cases in each of the three allocation domains: a “rush-to-sleep” that minimizes the number of activated time slots, a “rush-to-mute” that minimizes the number of activated antennas, and an “awake-but-whisper” that minimizes the transmit power.

RQ3: How can the RQ2 optimization framework be generalized to distributed and cell-free base stations?

In Chapter 4, we extended the allocation of transmit power and (de)activation of antennas to BSs that are distributed in space and cooperate to provide service to the users. Specifically, the analysis focuses on a cell-free network where all the active multi-antenna BSs (termed access points (APs)) communicate with all the single-antenna users in the considered service area. The optimization is formulated as PA-consumption minimization subject to ZF constraints, and the focus is on OFDM systems with many subcarriers. The key difference compared to Chapter 3 is the channel structure. It is shown that the optimal strategy, which follows from Chapter 3, consists in the activation of all the antennas of a certain number of APs, and in the complete deactivation of the remaining APs. This could bring significant energy savings in the network consumption, given by the sum of the consumption of the APs and a network-wide term including the consumption of the fronthaul (FH) links (that connect the APs with the centralized unit (CU)). Indeed, the CU needs to compute and send the precoding coefficients only relative to the active APs. However, this AP selection needs to be repeated every time the small-scale fading changes (i.e., every channel coherence time), and the CU needs to receive the instantaneous

channel coefficients of all the APs. By applying an asymptotic approximation based on random matrix theory (RMT), this AP selection can be performed based on the second-order channel statistics. These change much more slowly in time compared to the small-scale fading coefficients. Once the set of active APs is selected, the CU needs only to receive the instantaneous channel coefficients from those APs. The asymptotic solution has the potential to bring further reductions in the utilization of the FH links. Under a per-AP Kronecker channel model that considers correlation among the antennas of each AP, the asymptotic solution well approximates the optimal one when the number of subcarriers is large (in the order of a hundred or more).

RQ4: What practical gains and insights can be achieved when real-world measurements are used to design and evaluate transmit power allocations and (de)activation of antennas and transmit symbols?

Data from on-site measurements from a mobile network operator (MNO) in Belgium have been leveraged in Chapter 6 to initialize and assess the performance of the algorithms proposed in Chapter 5. The power model's parameters have been set according to the power model [15], validated on the on-site measurements of sub-6 GHz BSs. We examined three different BS configurations, i.e., a fourth-generation (4G)-Long-Term Evolution (LTE) 4T4R with 4 PAs and 4 transmitting and receiving RF chains, a fifth-generation (5G)-New Radio (NR) 8T8R configuration, and a 5G-NR 64T64R configuration. Reports of channel quality indicator (CQI) index from both 4G and 5G BSs have been used to compute the large-scale fading coefficients used in the simulations. Thanks to these choices, the energy performance resulting from the numerical experiments is linked to real-world measurements. We considered three different network loads (where the network load is the ratio of maximal sum rate) and a variation of parameters that represent two HW implementations: enabled or disabled time-domain HW power-saving modes (i.e., PA micro-discontinuous transmission (μ DTX) and idle power-saving modes in the analog processing). With disabled time-domain HW power-saving modes and maximal number of active users, the optimized allocation tends toward a rush-to-mute for the 64T64R BS at any of the three analyzed network loads (corresponding to 1.7, 10, and 29.8 Mbit/s/user). The energy savings of the optimized allocation compared to the rush-to-sleep and awake-but-whisper reach 30% at low network load. The rush-to-mute is still preferable for the 4T4R and 8T8R BSs at low and medium network loads, while the situation is more nuanced at high network loads. In these two BS configurations, the percentage energy savings are lower. Even with enabled time-domain HW power-saving modes, the rush-to-mute allocation has the smallest gap to the optimized one for the 64T64R BS at any analyzed network load. The energy savings of the optimized allocation compared to the rush-to-sleep and awake-but-whisper are equal to 17% at low

network load. The rush-to-sleep allocation becomes instead the allocation with the smallest gap to the optimized one for the 4T4R at any network load, and for the 8T8R BSs at low and medium network loads. In terms of EE, the results show that an EE in the range 10–300 kbit/J is achieved by the 5G BSs, while an EE in the range 1–20 kbit/J is achieved by the 4G BS. Moreover, the EE is maximized when the maximum number of users is served.

RQ5: Which concrete advice can be given to mobile network operators to systematically reduce the energy consumption of their networks?

As part of chapter 7, we have summarized the actionable insights for MNOs resulting from this thesis and discussed energy-savings techniques implemented by equipment manufacturers. A first insight is that in single-subcarrier systems, energy savings could be achieved by non-conventional digital precoding design. This non-conventional precoder, updated at a low time scale (i.e., every new channel coherence time), utilizes a few antennas at high transmit powers and hence high PA efficiencies, and completely deactivates the remaining antennas and associated PA and RF chain. In systems with many subcarriers, such as broadband OFDM systems, an advise is to use a conventional precoder among a subset of antennas (with uniform allocated transmit power), and optimize the number of activated antennas at a higher time scale than the one of precoder updates. We showcased a scenario in which this optimized number is retrieved in closed form. Moving towards next-generation BSs with even more antenna elements than currently deployed BSs, the key insight is that (in both single-carrier and OFDM systems) the benefit of a larger multi-antenna array gain should be carefully compared with the drawback of an increased circuit power. In systems of distributed cooperating BSs, the concrete advise follows a similar logic, with full BS deactivation that are preferred to deactivation of individual BS antennas. When comparing whether it is more beneficial, in terms of energy reduction for a fixed target QoS, to minimize the activated time slots (rush-to-sleep), the activated antennas (rush-to-mute), or the transmit power (awake-but-whisper), the insights depend on the BS configuration. In a massive multiple-input multiple-output (MIMO) BS (64T64R), a rush-to-mute should always be preferred. This is due to the large static consumption of the many PAs and RF chains. In small-dimension MIMO BSs (4T4R and 8T8R) implementing time-domain power-saving modes such as PA μ DTX, a rush-to-sleep often becomes the best option. The preferred strategy generally depends on the level of target delivered rates, but we find that it remains stable for the analyzed low and medium target delivered rates. In terms of EE measured in bits per Joule, our analysis suggests that 5G BS configurations should be preferred to 4G BS configurations.

8.2 Future Work

There are several directions that could be addressed in future work. These concern deployment trials, measurements, reconsideration of the main assumptions on which the algorithms are derived, and extensions to frequency-domain allocations and alternative deployment scenarios of distributed BSs. First, the application of the power model to FR3 BSs should be validated on hardware measurements. Even initial prototypes can aid in assessing the accuracy of the presented results and setting the parameters of the model. Recent prototypes are presented in [69, 151]. As discussed in Section 2.3.3, the power model has a high sensitivity to some HW parameters such as the DACs/ADCs sampling rates, hence linking their operation with the upcoming FR3 HW trends is an important task. Of equal importance is to tailor the BS array architecture, beamforming scheme, and channel models to the upcoming specifications at FR3. If the research on FR3 continues at the current rate, there will be more guidelines or insights on how to operate a BS in this band.

Second, the proposed energy-saving algorithms could be gradually adapted and tested in real-world scenarios, with the goal of conducting live network trials, for instance in a proof of concept (PoC). A first step in this direction would be to use operator data on traffic patterns, replacing the synthetic target QoS used in the simulations presented in this thesis. An experimental testbed would also provide significant insights. The testbed should preferably include multiple HW deactivation capabilities (e.g., deactivations of PAs with μ DTX, of RF chains, adaptation of transmit power) in order to evaluate the resource allocation and (de)activations in multiple domains. Other QoS metrics might also be considered alongside the delivered rates, e.g., the latency.

The impact of the main assumptions on which the proposed algorithms are derived should also be further investigated. More advanced channel models than i.i.d. Rayleigh or Kronecker models could be considered in the formulations, or could be considered as part of the evaluations. The effect of imperfect channel state information (CSI) could also be quantified. Moreover, the power back-off of the PAs might be reduced such that larger PA efficiencies are achievable, at the cost of introducing nonlinear distortions. Importantly, assumptions that deviate from 3rd Generation Partnership Project (3GPP) specifications, such as considering reference signaling to be fully separated in time from data transmission, should be relaxed to reduce the gap with real-world deployments.

In the analysis of distributed cooperating BSs, more advanced deployment scenarios could be considered. For instance, a user-centric cell-free network can be studied, where a user is served by a subset of the APs in the considered service area. Non-centralized beamforming processing (e.g., performed at the

APs based on their local available information) is also of great interest given the reduced complexity and consumption of FH links. Incorporating transmit symbols (de)activation to the distributed scenario is an important perspective. Last, frequency-domain allocations and (de)activations could be considered in both centralized BSs and distributed BSs scenarios. Deactivations of entire frequency bands have the potential of bringing larger energy savings compared to deactivations of single subcarriers, and an interesting future work is the optimized operation of multi-band BSs at Frequency Range 1 (FR1) and FR3, activating the capacity carrier(s) as a function of the load.

Appendix A

Collection of Proofs

A.1 Proof of Theorem 3.1

The Lagrangian function of problem (3.12) under **(As3.1)**, where the Lagrange multipliers are $\{\lambda_{k',k}[q] \in \mathbb{C}\}$ and the single element of $\mathbf{D}_{\psi}^{\frac{1}{2}}\sigma_{\nu}$ is $d_{k',k}$, is given by

$$\begin{aligned} & \mathcal{L}(w_{1,1}[1], \dots, w_{M,K}[Q], \lambda_{1,1}[q], \dots, \lambda_{K,K}[Q]) \\ &= \underbrace{\sum_{m=1}^M \left(\sum_{k=1}^K \sum_{q=1}^Q |w_{m,k}[q]|^2 \right)^{\frac{1}{2}}}_{(a)} \\ &+ \underbrace{\sum_{q=1}^Q \Re \left(\sum_{k'=1}^K \sum_{k=1}^K \lambda_{k',k}[q] \left(\sum_{m=1}^M h_{k',m}[q] w_{m,k}[q] - d_{k',k} \right) \right)}_{(b)}. \end{aligned} \tag{A.1}$$

By using the property that, for $z \in \mathbb{C}$, $\Re\{z\} = \frac{1}{2}(z + z^*)$, we expand the constraint as

$$\begin{aligned} (b) &= \frac{1}{2} \sum_{q=1}^Q \sum_{k'=1}^K \sum_{k=1}^K \lambda_{k',k}[q] \left(\sum_{m=1}^M h_{k',m}[q] w_{m,k}[q] - d_{k',k} \right) \\ &+ \frac{1}{2} \sum_{q=1}^Q \sum_{k'=1}^K \sum_{k=1}^K \lambda_{k',k}[q]^* \left(\sum_{m=1}^M h_{k',m}[q]^* w_{m,k}[q]^* - d_{k',k} \right). \end{aligned} \tag{A.2}$$

Let us now compute the Wirtinger derivative of the Lagrangian function using the properties that, for $z \in \mathbb{C}$, $\frac{\partial z^*}{\partial z^*} = 1$, $\frac{\partial(zz^*)}{\partial z^*} = z$, and $\frac{\partial z}{\partial z^*} = 0$ [152].¹ To differentiate (a), we note that for $z_1, \dots, z_N \in \mathbb{C}$,

$$\frac{\partial}{\partial z_n^*} \left(\left(\sum_{n=1}^N z_n z_n^* \right)^{\frac{1}{2}} \right) = \frac{1}{2} z_n \left(\sum_{n'=1}^N z_{n'} z_{n'}^* \right)^{-\frac{1}{2}}. \quad (\text{A.3})$$

The differentiation of (b) follows from the basic properties listed above. The Wirtinger derivative of $\mathcal{L}$ with respect to a precoding coefficient is then given by

$$\frac{\partial \mathcal{L}}{\partial w_{m,k}[q]^*} = \frac{1}{2} \frac{w_{m,k}[q]}{\left(\sum_{k'=1}^K \sum_{q'=0}^Q |w_{m,k'}[q']|^2 \right)^{\frac{1}{2}}} + \frac{1}{2} \sum_{k'=1}^K \lambda_{k',k}[q]^* h_{k',m}[q]^*. \quad (\text{A.4})$$

Recalling that $P_{\tau,m} = \sum_{k=1}^K \sum_{q=1}^Q |w_{m,k}[q]|^2$ and setting (A.4) to zero, we obtain

$$\frac{\partial \mathcal{L}}{\partial w_{m,k}[q]^*} = 0 \Leftrightarrow w_{m,k}[q] = -P_{\tau,m}^{\frac{1}{2}} \sum_{k'=1}^K \lambda_{k',k}[q]^* h_{k',m}[q]^*. \quad (\text{A.5})$$

The constraint gives

$$\sum_{m=1}^M h_{k',m}[q]^* w_{m,k}[q]^* - d_{k',k} = 0 \quad (\text{A.6})$$

which, using (A.5), can be rewritten as

$$\sum_{m=1}^M h_{k',m}[q]^* P_{\tau,m}^{\frac{1}{2}} \sum_{k''=1}^K \lambda_{k',k}[q] h_{k'',m}[q] = -d_{k',k}. \quad (\text{A.7})$$

We are left with the following system of equations

$$\begin{cases} w_{m,k}[q] = -P_{\tau,m}^{\frac{1}{2}} \sum_{k'=1}^K \lambda_{k',k}[q]^* h_{k',m}[q]^* \\ \sum_{m=1}^M h_{k',m}[q]^* P_{\tau,m}^{\frac{1}{2}} \sum_{k''=1}^K \lambda_{k',k}[q] h_{k'',m}[q] = -d_{k',k}. \end{cases} \quad (\text{A.8})$$

¹The Lagrangian function is not differentiable in the origin, i.e., $w_{m,k}[q] = 0$, due to the presence of the square root. However, we observe through simulations that the optimal solution approaches but not reaches zero for certain precoding coefficients, hence the non-differentiability in the origin is not an issue.

Making use of $\mathbf{D}_{\mathbf{p}_T} = \text{diag}(P_{T,1}, \dots, P_{T,M})$ and denoting with $\mathbf{\Lambda}[q] \in \mathbb{C}^{K \times K}$ the matrix containing the Lagrange multipliers for the subcarrier q , (A.8) can be written in matrix form as

$$\begin{cases} \mathbf{W}[q] = -\mathbf{D}_{\mathbf{p}_T}^{\frac{1}{2}} \mathbf{H}[q] \mathbf{\Lambda}[q]^* \\ \mathbf{H}[q]^* \mathbf{D}_{\mathbf{p}_T}^{\frac{1}{2}} \mathbf{H}[q]^T \mathbf{\Lambda}[q] = -\mathbf{D}_{\psi}^{\frac{1}{2}} \sigma_\nu \end{cases} \quad (\text{A.9})$$

The matrix $\mathbf{\Lambda}[q]$ is then given by

$$\mathbf{\Lambda}[q] = -\left(\mathbf{H}[q]^* \mathbf{D}_{\mathbf{p}_T}^{\frac{1}{2}} \mathbf{H}[q]^T\right)^{-1} \mathbf{D}_{\psi}^{\frac{1}{2}} \sigma_\nu \quad (\text{A.10})$$

therefore, by substituting it back to the first expression of (A.9), we obtain

$$\mathbf{W}[q] = \mathbf{D}_{\mathbf{p}_T}^{\frac{1}{2}} \mathbf{H}[q]^H \left(\mathbf{H}[q] \mathbf{D}_{\mathbf{p}_T}^{\frac{1}{2}} \mathbf{H}[q]^H\right)^{-1} \mathbf{D}_{\psi}^{\frac{1}{2}} \sigma_\nu. \quad (\text{A.11})$$

A.2 Proof of Theorem 3.2

Problem (3.12) is convex. In the asymptotic wideband regime, the power allocations in the neighborhood of the global optimum achieve substantially the same performance. Fig. 3.1b illustrates as (i) different power allocations consume approximately the same amount of power while achieving the users' QoS, and (ii) the power tends to be uniformly allocated among the antennas as Q increases. Among the possible allocations, let us consider a uniform allocation, i.e., $\mathbf{D}_{\mathbf{p}_T} = P_{\text{unif}} \mathbf{I}_M$. When validating the asymptotic results, we will prove that this allocation is optimal. Equation (3.14) becomes

$$P_{\text{unif}} = \frac{1}{Q} \sum_{q=1}^Q \sum_{k=1}^K \left| \left[\mathbf{H}[q]^H (\mathbf{H}[q] \mathbf{H}[q]^H)^{-1} \mathbf{D}_{\psi}^{\frac{1}{2}} \sigma_\nu \right]_{m,k} \right|^2. \quad (\text{A.12})$$

By summing over the antennas we can write

$$\begin{aligned} MP_{\text{unif}} &= \frac{1}{Q} \sum_{q=1}^Q \sum_{m=1}^M \sum_{k=1}^K \left| \left[\mathbf{H}[q]^H (\mathbf{H}[q] \mathbf{H}[q]^H)^{-1} \mathbf{D}_{\psi}^{\frac{1}{2}} \sigma_\nu \right]_{m,k} \right|^2 \\ &= \frac{1}{Q} \sum_{q=1}^Q \text{tr} \left(\mathbf{H}[q]^H (\mathbf{H}[q] \mathbf{H}[q]^H)^{-1} \mathbf{D}_{\psi} \sigma_\nu^2 (\mathbf{H}[q] \mathbf{H}[q]^H)^{-1} \mathbf{H}[q] \right). \end{aligned} \quad (\text{A.13})$$

By using the cyclic property of the trace

$$P_{\text{unif}} = \frac{1}{M} \frac{1}{Q} \sum_{q=1}^Q \text{tr} \left((\mathbf{H}[q] \mathbf{H}[q]^H)^{-1} \mathbf{D}_{\psi} \sigma_\nu^2 \right). \quad (\text{A.14})$$

Under (A3.2), the law of large numbers gives

$$\frac{1}{Q} \sum_{q=1}^Q (\mathbf{H}[q] \mathbf{H}[q]^H)^{-1} \mathbf{D}_\psi \sigma_\nu^2 \rightarrow \mathbb{E} \left\{ (\mathbf{H}[q] \mathbf{H}[q]^H)^{-1} \mathbf{D}_\psi \sigma_\nu^2 \right\} \quad (\text{A.15})$$

then (A.14) becomes

$$P_{\text{unif}} \rightarrow \frac{1}{M} \text{tr} \left(\mathbb{E} \left\{ (\mathbf{H}[q] \mathbf{H}[q]^H)^{-1} \mathbf{D}_\psi \sigma_\nu^2 \right\} \right). \quad (\text{A.16})$$

Under (A3.3), the Gram matrix $\mathbf{H}[q] \mathbf{H}[q]^H \sim \mathcal{CW}(\mathbf{D}_\beta, M, K)$, then its inverse $(\mathbf{H}[q] \mathbf{H}[q]^H)^{-1} \sim \mathcal{CW}^{-1}(\mathbf{D}_\beta^{-1}, M, K)$ [132]. We obtain a deterministic expression of the transmit power at each antenna

$$P_{\text{unif}} \rightarrow \frac{1}{M(M-K)} \text{tr}(\mathbf{D}_\beta^{-1} \mathbf{D}_\psi \sigma_\nu^2) \quad (\text{A.17})$$

thereby the total PAs consumed power can be computed as

$$P_{\text{PA}} = M \gamma P_{\text{unif}}^{\frac{1}{2}} \rightarrow \gamma \left(\frac{M}{M-K} \text{tr}(\mathbf{D}_\beta^{-1} \mathbf{D}_\psi \sigma_\nu^2) \right)^{\frac{1}{2}}. \quad (\text{A.18})$$

A.3 Proof of Lemma 3.1

The optimal number of active antennas lies in between $K+1$ (required by the ZF constraint) and M . By considering a continuous number of active antennas $x \in \mathbb{R}$, the function (3.28)

$$f(x) = t \left(\frac{x}{x-K} \right)^{\frac{1}{2}} + \mathcal{C}x + P_{\text{fixed}} \quad (\text{A.19})$$

where $t = \gamma \left(\text{tr}(\mathbf{D}_\beta^{-1} \mathbf{D}_\psi \sigma_\nu^2) \right)^{\frac{1}{2}}$, is convex in the domain $(K, M]$. To prove that, let us compute the first derivative of f

$$\frac{df(x)}{dx} = t \left(\frac{x^{-\frac{1}{2}}}{2(x-K)^{\frac{1}{2}}} - \frac{x^{\frac{1}{2}}}{2(x-K)^{\frac{3}{2}}} \right) + \mathcal{C} = -\frac{t}{2} \left(\frac{Kx^{-\frac{1}{2}}}{(x-K)^{\frac{3}{2}}} \right) + \mathcal{C}. \quad (\text{A.20})$$

By taking the second derivative

$$\frac{d^2 f(x)}{dx^2} = -\frac{tK}{2} \left(-\frac{x^{-\frac{3}{2}}}{2(x-K)^{\frac{3}{2}}} - \frac{3x^{-\frac{1}{2}}}{2(x-K)^{\frac{5}{2}}} \right) = \frac{tK}{4} \frac{4x-K}{x^{\frac{3}{2}}(x-K)^{\frac{5}{2}}} \quad (\text{A.21})$$

we can see that, for $x > K$, (A.21) is always positive. To minimize (A.19), one can compute its derivative with respect to x , given in (A.20), set it to zero, and isolate the constant term, obtaining the following polynomial

$$x(x - K)^3 - \frac{tK}{2\mathcal{C}} = 0. \quad (\text{A.22})$$

From Descartes' rule of signs, (A.22) has 2 positive real roots. Among them, there is only one bigger than K . This can also be seen by visualizing (3.28) in Fig. 3.3b. The $\min\{\cdot\}$ and $\max\{\cdot\}$ operations guarantee that the final value stays within the allowed range. Due to the convexity of the function (A.19), the result of the ceil-floor operation $\lfloor \cdot \rfloor$ is guaranteed to be the optimal integer value.

A.4 Proof of Theorem 3.3

Let us consider problem (3.27). Fixing the value of M_a , the minimization problem with respect to the precoding coefficients is

$$\begin{aligned} &\underset{\substack{\{w_{m,k}[q]\} \\ m=1,\dots,M_a}}{\text{minimize}} && P_{\text{cons}} = P_{\text{PA}} + \mathcal{C}M_a + P_{\text{fixed}} \end{aligned} \quad (\text{A.23a})$$

$$\text{subject to} \quad \mathbf{H}[q]\mathbf{W}[q] = \mathbf{D}_{\psi}^{\frac{1}{2}}\sigma_{\nu} \quad \forall q, \quad (\text{A.23b})$$

$$P_{\tau,m} \leq P_{\max} \quad \forall m. \quad (\text{A.23c})$$

From Theorem 3.2, we know that the solution to the above problem (in the asymptotic wideband regime and i.i.d. Rayleigh fading) when the per-antenna power constraints are not binding tends to allocate the power uniformly among the antennas,

$$P_{\tau,m} \rightarrow \widetilde{P}_{\tau} = \frac{1}{M_a(M_a - K)} \text{tr} \left(\mathbf{D}_{\beta}^{-1} \mathbf{D}_{\psi} \sigma_{\nu}^2 \right) \quad \forall m. \quad (\text{A.24})$$

Let us consider two cases, recalling that we are considering an arbitrary value of M_a :

1. $\widetilde{P}_{\tau} \leq P_{\max}$, which guarantees that the above solution is optimal. Indeed, solution (A.24) minimizes (A.23a) while satisfying (A.23b), and in this case it fulfils also (A.23c).
2. $\widetilde{P}_{\tau} > P_{\max}$, which indicates that the above solution is not feasible by using M_a antennas. Indeed, to satisfy the constraints (A.23c), we would need

to decrease every value of $P_{\tau,m}$ to $P_{\max}$ or a smaller quantity. By doing this, we would need a solution consuming less power (given that PAs consumption increases monotonically with $P_{\tau,m}$) while fulfilling (A.23b). This is not possible, otherwise this new solution would correspond to the optimal one for the unconstrained problem, i.e., the one allocating the power $\widetilde{P}_{\tau}$ to all antennas.

The above reasoning entails that, in the asymptotic wideband regime and i.i.d. Rayleigh fading, the per-antenna power constraints imply a lower bound on the number of active antennas M_a (by using (A.24) and assuming $M_a > K$)

$$\widetilde{P}_{\tau} \leq P_{\max} \iff M_a^2 - KM_a - \frac{\text{tr}(\mathbf{D}_{\beta}^{-1} \mathbf{D}_{\psi} \sigma_{\nu}^2)}{P_{\max}} \geq 0, \quad (\text{A.25})$$

$$M_a \geq \frac{1}{2} \left(K + \left(K^2 + \frac{4 \text{tr}(\mathbf{D}_{\beta}^{-1} \mathbf{D}_{\psi} \sigma_{\nu}^2)}{P_{\max}} \right)^{\frac{1}{2}} \right). \quad (\text{A.26})$$

Using this lower bound as a constraint in the original problem, we can drop the per-antenna power constraints and solve the equivalent problem

$$\begin{aligned} \underset{M_a}{\text{minimize}} \quad & \widetilde{P}_{\text{cons}} = \gamma \left(\frac{M_a}{M_a - K} \text{tr}(\mathbf{D}_{\beta}^{-1} \mathbf{D}_{\psi} \sigma_{\nu}^2) \right)^{\frac{1}{2}} + P_{\text{fixed}} + \mathcal{C}M_a \\ \text{subject to} \quad & M_a \geq \frac{1}{2} \left(K + \left(K^2 + \frac{4 \text{tr}(\mathbf{D}_{\beta}^{-1} \mathbf{D}_{\psi} \sigma_{\nu}^2)}{P_{\max}} \right)^{\frac{1}{2}} \right), \\ & M_a \leq M. \end{aligned} \quad (\text{A.27})$$

The condition in **(As3.4)** ensures that the power constraint is always satisfied by activating all the M antennas. It can be simply derived from (A.24) by setting $M_a = M$.

A.5 Simplification of Maximal Power Constraint

Let us use the definition $y = M_a$ and consider it as continuous. Using $\varrho_k = P_{\max} \beta_k / \sigma_k^2$, we can rewrite (5.19) as

$$0 \leq y^2 - Ky - \sum_{k=1}^K \varrho_k^{-1} \left(2^{R_k \frac{N}{N_a}} - 1 \right). \quad (\text{A.28})$$

The right term is a quadratic expression in y with two roots

$$y = \frac{K}{2} \pm \frac{1}{2} \sqrt{K^2 + 4 \sum_{k=1}^K \varrho_k^{-1} \left(2^{R_k \frac{N}{N_a}} - 1 \right)}. \quad (\text{A.29})$$

Given that $M_a \geq K$, also $y \geq K$ and we should only keep the one with positive sign. Given that the parabolic function is looking upward, we find the condition

$$y \geq \frac{K}{2} + \frac{1}{2} \sqrt{K^2 + 4 \sum_{k=1}^K \varrho_k^{-1} \left(2^{R_k \frac{N}{N_a}} - 1 \right)} \quad (\text{A.30})$$

which implies (5.21) going back to the discrete domain.

A.6 Convexity of the Objective Function

We aim at showing that the function (5.24) is jointly convex, considering that $x \geq 1$ and $K \leq y \leq M$. Clearly, the term in y is convex. To show the convexity of the term in $\frac{y}{x}$, we check that its Hessian is positive semidefinite. A sufficient condition is that its determinant and trace are non-negative. The determinant is given by $\frac{1}{x^4} \geq 0$, while the trace is given by $\frac{y}{x^3} \geq 0$. Therefore, $\frac{y}{x}$ is convex in the domain of interest. We are left with the term $g(y)h(x)$, where

$$g(y) = \frac{y}{(y(y-K))^\alpha}, \quad h(x) = \frac{\left(\sum_{k=1}^K \zeta_k (e^{R'_k x} - 1) \right)^\alpha}{x} \quad (\text{A.31})$$

with $R'_k = R_k \ln 2$ and $\zeta_k = \sigma_k^2 / \beta_k$. To show convexity, let us show that the determinant and the trace of its Hessian are non-negative. The determinant condition requires that

$$h''(x)h(x)g''(y)g(y) - (h'(x)g'(y))^2 \geq 0. \quad (\text{A.32})$$

Let us consider the two following conditions

$$\begin{cases} h''(x)h(x) \geq (h'(x))^2 \\ g''(y)g(y) \geq (g'(y))^2. \end{cases} \quad (\text{A.33})$$

When these hold, we directly have that $h''(x)h(x)g''(y)g(y) \geq (h'(x)g'(y))^2$, which is condition (A.32). Therefore, the two conditions in (A.33) are sufficient to show that (A.32) holds. The trace condition is $h''(x)g(y) + h(x)g''(y) \geq 0$ and given that $h(x) \geq 0$ and $g(y) \geq 0$, it is verified by checking that both

functions are convex, *i.e.*, $h''(x) \geq 0, g''(y) \geq 0$. Let us first study $g(y)$, with derivatives

$$\begin{aligned} g'(y) &= \frac{1}{y^\alpha(y-K)^{1+\alpha}} (y(1-2\alpha) - K(1-\alpha)) \\ g''(y) &= \alpha \frac{2y^2(2\alpha-1) + 4Ky(1-\alpha) - K^2(1-\alpha)}{y^{1+\alpha}(y-K)^{2+\alpha}}. \end{aligned} \quad (\text{A.34})$$

The denominator of $g''(y)$ is positive and given that $\alpha \geq 0.5$, $2\alpha - 1 \geq 0$. Moreover, $y \geq K$ implies $4Ky \geq K^2$, hence $g''(y) \geq 0$. By developing $g''(y)g(y) - (g'(y))^2$, we obtain

$$\frac{y^2(2\alpha-1) + y2K(1-\alpha) - K^2(1-\alpha)}{y^{2\alpha}(y-K)^{2+2\alpha}}. \quad (\text{A.35})$$

Again, the denominator is positive and given that $\alpha \geq 0.5$, we have $2\alpha - 1 \geq 0$. Moreover, $y \geq K$ implies $y2K \geq K^2$. Hence, $g''(y)g(y) - (g'(y))^2 \geq 0$. Let us now study $h(x)$ and let us define the function $k(x) = \ln h(x)$. Given that $h(x) = e^{k(x)}$, we have

$$\begin{aligned} h'(x) &= e^{k(x)}k'(x), \quad h''(x) = e^{k(x)}(k'(x))^2 + e^{k(x)}k''(x) \\ h''(x)h(x) - (h'(x))^2 &= e^{2k(x)}k''(x). \end{aligned} \quad (\text{A.36})$$

Therefore, it is sufficient to show that $k''(x) \geq 0$. After simplification and manipulation, $k''(x)$ can be expressed as

$$k''(x) = \alpha \frac{\sum_{k=1}^K \sum_{k'=1}^K \zeta_k \zeta_{k'} t_{k,k'}}{(\sum_{k=1}^K \zeta_k (e^{R'_k x} - 1))^2} + (1-\alpha) \frac{1}{x^2}. \quad (\text{A.37})$$

where

$$\begin{aligned} t_{k,k'} &= \frac{1}{2} R'_k{}^2 e^{xR'_k} (e^{xR'_{k'}} - 1) + \frac{1}{2} (e^{xR'_k} - 1) e^{xR'_{k'}} R'_{k'}{}^2 \\ &\quad - R'_k e^{R'_k x} e^{R'_{k'} x} R'_{k'} + \frac{1}{x^2} (e^{R'_k x} - 1) (e^{R'_{k'} x} - 1). \end{aligned} \quad (\text{A.38})$$

Given that $\alpha \in [0.5, 1]$, a sufficient condition for $k''(x) \geq 0$ is that $t_{k,k'} \geq 0 \forall k, k'$. Using $\sinh(x) \geq x$ for $x \geq 0$, we find the inequality

$$(e^{xR'_k} - 1) = 2e^{\frac{xR'_k}{2}} \sinh(xR'_k/2) \geq e^{\frac{xR'_k}{2}} xR'_k \quad (\text{A.39})$$

and thus

$$\begin{aligned}
t_{k,k'} &\geq \frac{1}{2} R_k'^2 e^{xR_k'} (e^{xR_{k'}} - 1) + \frac{1}{2} (e^{xR_k'} - 1) e^{xR_{k'}} R_k'^2 \\
&\quad - R_k' e^{xR_k'} e^{xR_{k'}} R_k' + R_k' e^{xR_k'/2} e^{xR_{k'}/2} R_k' \\
&= -\frac{1}{2} \left(R_k' e^{\frac{xR_k'}{2}} - e^{\frac{xR_{k'}}{2}} R_k' \right)^2 + \frac{1}{2} e^{xR_k' + xR_{k'}} (R_k' - R_{k'})^2.
\end{aligned} \tag{A.40}$$

We need to show that this term is greater or equal than zero. Multiplying by $x^2/2$ and defining $a = xR_k/2$ and $b = xR_{k'}/2$, it is equivalent to show that

$$e^{2a+2b}(a-b)^2 \geq (ae^a - be^b)^2. \tag{A.41}$$

Let us consider that $a > b$ (otherwise symmetric) and take the square root, we should then show that

$$e^{a+b}(a-b) \geq ae^a - be^b. \tag{A.42}$$

Subtracting $(a-b)e^a$ gives $(e^{a+b} - e^a)(a-b) \geq b(e^a - e^b)$, and dividing by e^a , we obtain $(e^b - 1)(a-b) \geq b(1 - e^{b-a})$. We can then exploit $b \leq e^b - 1$ and $1 - e^{b-a} \leq a - b$ to arrive at the result.

A.7 Convexity of the Optimization Domain

The linear constraints in the definition of $\mathcal{D}$ in (5.25) are clearly convex. Let us look further at the $P_{\max}$ inequality, which can be rewritten as $-y + \frac{K}{2} + m(x) \leq 0$, where

$$m(x) = \sqrt{\frac{K^2}{4} + \sum_{k=1}^K \varrho_k^{-1} (e^{R_k' x} - 1)}. \tag{A.43}$$

Given that $-y$ is convex, we need to check when $m''(x) \geq 0$. Let us further define

$$l(x) = \sum_{k=1}^K \varrho_k^{-1} \left(e^{R_k' x} + \frac{K}{4} \varrho_k - 1 \right) \tag{A.44}$$

so that $m(x) = \sqrt{l(x)}$ with derivatives

$$m'(x) = \frac{1}{2} \frac{l'(x)}{\sqrt{l(x)}}, \quad m''(x) = \frac{1}{4} \frac{2l''(x)l(x) - (l'(x))^2}{(l(x))^{3/2}}. \tag{A.45}$$

Given that $l(x) \geq 0$, we need to check when $2l''(x)l(x) - (l'(x))^2 \geq 0$. We have

$$2l''(x)l(x) - (l'(x))^2 = \sum_{k=1}^K \sum_{k'=1}^K \frac{R_k 2^{R_k x}}{\varrho_k \varrho_{k'}} \left[2R_k \left(\frac{K}{4} \varrho_{k'} - 1 \right) + 2^{R_{k'} x} (2R_k - R_{k'}) \right]. \quad (\text{A.46})$$

The double sum contains K^2 additive terms which can be partitioned in K terms where $k = k'$ and $K(K-1)$ terms where $k \neq k'$. In the following we show that the condition (5.26) which can be rewritten as

$$\varrho_k \geq \frac{2}{K} \Leftrightarrow \frac{K}{4} \varrho_k - 1 \geq -\frac{1}{2} \quad (\text{A.47})$$

for $k = 1, \dots, K$ is sufficient to guarantee that $l''(x) \geq 0$. Let us first consider the K terms of the double sum where $k = k'$. These are positive if

$$2R_k \left(\frac{K}{4} \varrho_k - 1 \right) + 2^{R_k x} (2R_k - R_k) \geq 0 \Leftrightarrow -2 \left(\frac{K}{4} \varrho_k - 1 \right) \leq 2^{R_k x} \quad (\text{A.48})$$

which is verified if (A.47) holds. Let us now consider the $K(K-1)$ terms where $k \neq k'$. For each term k, k' , there exists also a term k', k . The total set of $K(K-1)$ terms can then be subdivided in $K(K-1)/2$ pairs of distinct indices k, k' . For instance, if $K = 3$, there exists $3(3-1)/2 = 3$ pairs: 1, 2 and 2, 1; 1, 3 and 3, 1; 2, 3 and 3, 2. Let us consider the addition of one pair of terms k, k' and k', k . The contribution of these two terms is positive if $f_{k,k'} \geq 0$ where

$$\begin{aligned} f_{k,k'} &= R_k 2^{R_k x} \left[2R_k \left(\frac{K}{4} \varrho_{k'} - 1 \right) + 2^{R_{k'} x} (2R_k - R_{k'}) \right] \\ &\quad + R_{k'} 2^{R_{k'} x} \left[2R_{k'} \left(\frac{K}{4} \varrho_k - 1 \right) + 2^{R_k x} (2R_{k'} - R_k) \right]. \end{aligned} \quad (\text{A.49})$$

If (A.47) holds, we have

$$\begin{aligned} f_{k,k'} &\geq R_k 2^{R_k x} [-R_k + 2^{R_{k'} x} (2R_k - R_{k'})] \\ &\quad + R_{k'} 2^{R_{k'} x} [-R_{k'} + 2^{R_k x} (2R_{k'} - R_k)] \\ &= 2^{R_k x + R_{k'} x} \left[2R_k^2 + 2R_{k'}^2 - 2R_{k'} R_k - \frac{R_k^2}{2^{R_{k'} x}} - \frac{R_{k'}^2}{2^{R_k x}} \right] \end{aligned} \quad (\text{A.50})$$

and given that $-1/2^{R_k x} \geq -1$ and $-1/2^{R_{k'} x} \geq -1$, we find

$$\begin{aligned} f_{k,k'} &\geq 2^{R_k x + R_{k'} x} [2R_k^2 + 2R_{k'}^2 - 2R_{k'} R_k - R_k^2 - R_{k'}^2] \\ &= 2^{R_k x + R_{k'} x} (R_k - R_{k'})^2 \end{aligned} \quad (\text{A.51})$$

which is positive. Hence, the contribution from all pairs of indices where $k \neq k'$ gives a positive contribution to the double sum. The same holds for the terms where $k = k'$. The proof is thus complete.

A.8 Error Order of Using a Rounding Operation

Let us define the optimal allocation as $(N_a, M_a) = \lfloor N/\bar{x}, \bar{y} \rfloor$ and its approximation through the rounding operator as $\hat{N}_a = \lceil N/\bar{x} \rceil$, $\hat{M}_a = \lceil \bar{y} \rceil$. We can also write

$$\hat{N}_a = N_a + \epsilon_1, \quad \hat{M}_a = M_a + \epsilon_2 \quad (\text{A.52})$$

where $|\epsilon_1| \leq 1$, $|\epsilon_2| \leq 1$. The extra consumed power of the approximation with respect to the optimal allocation is defined

$$\epsilon_3 = P_{\text{cons}}(\hat{N}_a, \hat{M}_a) - P_{\text{cons}}(N_a, M_a) \quad (\text{A.53})$$

where the function $P_{\text{cons}}(N_a, M_a)$ is given in (5.18). Developing ϵ_3 , we find

$$\begin{aligned} \epsilon_3 = & \frac{\epsilon_2}{M} P_1 + (N_a \epsilon_2 + \epsilon_1 M_a + \epsilon_1 \epsilon_2) \frac{P_0}{NM} \\ & + \frac{N_a + \epsilon_1}{N} (M_a + \epsilon_2) \gamma \left(\frac{\sum_{k=1}^K \frac{\sigma_k^2}{\beta_k} \left(2^{R_k \frac{N}{N_a + \epsilon_1}} - 1 \right)}{(M_a + \epsilon_2)(M_a + \epsilon_2 - K)} \right)^\alpha \\ & - \frac{N_a}{N} M_a \gamma \left(\frac{\sum_{k=1}^K \frac{\sigma_k^2}{\beta_k} \left(2^{R_k \frac{N}{N_a}} - 1 \right)}{M_a(M_a - K)} \right)^\alpha. \end{aligned} \quad (\text{A.54})$$

As $M_a \rightarrow +\infty$ and $N \rightarrow +\infty$, the first two terms can be bounded as $\mathcal{O}(1/N + 1/M)$. For the two last ones, after performing a first-order Taylor expansion around $\epsilon_1 = 0$ and $\epsilon_2 = 0$, we find the bound $\mathcal{O}(1/N + 1/M_a^{2\alpha})$ so that $|\epsilon_3| = \mathcal{O}(1/N + 1/M + 1/M_a^{2\alpha})$.

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Statement on the use of Generative AI

I did use the generative AI assistance tools Perplexity and ChatGPT. Generative AI was used for language assistance, generating and debugging TikZ code for figures/diagrams, and generating and debugging Python plotting code. The text, code, and images in this thesis are my own (unless otherwise specified). Generative AI has only been used in accordance with the KU Leuven guidelines and appropriate references have been added. I have reviewed and edited the content as needed and I take full responsibility for the content of the thesis.

List of publications

Articles in Internationally Reviewed Academic Journals

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